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Physical Properties of Poly (n-alkyl acrylate) Copolymers

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Physical Properties of Poly (n-alkyl acrylate) Copolymers

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To
My Family

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Physical Properties of Poly (n-alkyl acrylate) Copolymers

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The physical properties of n-alkyl acrylate copolymers, including thermal characteristics, structure as determined by small angle X-ray scattering, and gas permeability as a function of temperature, were examined in detail and compared to the corresponding homopolymers. Two types of copolymers were examined: those with two crystalline comonomers and those with one crystalline and one non-crystalline comonomer. The crystalline / crystalline copolymers exhibit co-crystallization and, thus, for a given average side-chain length have comparable melting temperatures as the corresponding homopolymers. For a given side-chain length, the copolymers have somewhat lower heats of fusion than the corresponding homopolymers because of a reduction in crystallite size as revealed by SAXS. The crystalline / non-crystalline

copolymers do not co-crystallize and experience melting point depression in which the non-crystalline comonomer does not affect the T_m and ΔH_f as much as two crystalline comonomers do. Though not entering the lattice, the non-crystalline comonomers impede the formation of perfect crystals, also reducing the crystallite size, as indicated by SAXS. This depression in crystallinity is reflected in the permeability data for the copolymers. Poly (n-alkyl acrylates) exhibit a 'jump' in their gas permeability at the T_m of the side-chain lengths that is mainly caused by a switch in the side-chain morphology from crystalline to amorphous upon melting. The depression in crystallinity for both types of copolymers results in a smaller permeation jump. Interestingly, copolymers containing A10, a comonomer on the border of being crystalline, experience the broadest jump peak. The jump breadth of all copolymers examined correlate with the melting endotherms for these polymers as determined by DSC. Ultimately, the melting endotherms for these copolymer systems provides an excellent tool for predicting permeability changes across the melting region.

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Chapter 1

Introduction

1.1 Overview

In 1988, it was estimated that between 25 and 40% of the fruits and vegetables harvested in the United States would not make it to the consumer's table. This loss was due to mishandling and spoilage and it was huge since the produce market, at that time, was a 50-billion-dollar industry.[1] This substantial loss in potential revenues drove, and continues to drive, the industry to develop more produce-friendly technologies for shipping and packaging. From this need, controlled atmosphere packaging (CAP) and modified atmosphere packaging (MAP) technologies have been gaining tremendous momentum as possible solutions to the current produce packaging problems.

CAP is a more traditional approach to retard produce decay. The surrounding atmosphere is continuously monitored and regulated to maintain a gaseous environment that prolongs freshness. CAP is a capital intensive technology that is generally used for bulk storage and shipping of certain stationary fruits and vegetables and is not practical for marketing of produce. On the other hand, MAP is, by definition; a less controlled, and less expensive, method of packaging that is suitable for marketing.[2] In this technique, barrier materials are used to maintain the gas levels immediately surrounding the produce to broader, but still acceptable concentrations. Modified atmosphere packaging is the motivation for this proposal.[2-4]

Due to their flexibility and wide ranges in gas permeability characteristics, polymer membranes are ideal for modified atmosphere packaging. One of the major goals of MAP is to maintain the levels of O₂ and CO₂ within the package at values known

to slow the senescence, or aging, of the produce. This may be done by choosing a polymer tailored to the respiration rate, i.e., oxygen is consumed and carbon dioxide is released by the produce, of a given item of produce.

While a polymer membrane may be tailored to give exactly the desired O_2/CO_2 concentration levels for a particular produce at a given temperature, this balance will not be preserved at other temperatures. The respiration rate of produce increases rapidly with temperature. The desired concentration levels of O_2/CO_2 would also change. While the permeability of O_2 and CO_2 through currently available polymers do generally increase with temperature, the Arrhenius mechanism behind this does not lead to a large enough change in permeation rate to match the change in respiration rate; hence, these membranes cannot keep the O_2 and CO_2 levels constant in the package as temperature changes. Therefore, there is interest in polymer membranes that are more thermally responsive. Side-chain crystalline polymers pose a possible solution to these problems. More specifically, poly (n-alkyl acrylate) homo- and co-polymers exhibit very unique physical properties around their melting temperatures. Upon traversing the T_m , certain poly (n-alkyl acrylates) will reversibly ‘switch’ from a crystalline to a molten state and back with a large jump in properties like permeation. The size and breadth of the jump are direct functions of the side chain length.[5-7]

1.2 Research Objectives

Mogri and Paul conducted a study on the permeability properties of poly (n-alkyl acrylate) homopolymers, specifically examining the jump in permeability over the melting temperature as a function of the polymers side-chain length (n) and thermal

history. The non-Arrhenius relationship between gas permeability and temperature showed the potential for these polymers to be used as a membrane for modified atmospheric packaging purposes. The jumps in permeability at T_m for these homopolymers, however, were slightly larger and not broad enough to fully control the atmospheric conditions of produce over shipping conditions. Poly (n-alkyl acrylate) membranes may be better engineered for MAP purposes by making copolymer or physical blend combinations with more suitable gas permeability – temperature relationships. Thus, a major objective of this research program has been to measure the gas permeability properties of poly (n-alkyl acrylate) copolymers and establish relationships between copolymer composition and their transport properties for potential uses in MAP applications. This goal was pursued by first establishing the copolymer composition via ^{13}C -NMR and, secondly, by characterizing the thermal, structural, and transport properties of the copolymer systems. Two types of copolymers were studied, those with two crystallizeable comonomers and those with one crystallizeable and one non-crystallizeable comonomer. The thermal and structural properties of the systems were analyzed to understand the interactions between comonomer side chains and their affects on gas permeability in an effort to ultimately predict copolymer gas permeation behavior based on these properties.

1.3 Dissertation Organization

The first and second chapters of this dissertation provide background for the reader. Chapter 1 explains the motivation for the research project as well as its objectives and layout. Chapter 2 describes the established theories for semi-crystalline polymers

and copolymers. Chapter 3 describes all experimental techniques utilized during the research program.

Determining the exact composition of poly (n-alkyl acrylate) copolymers proved to be a tedious task that was accomplished using ^{13}C -NMR. These and other techniques used to determine the effects of conversion on copolymer composition are described in Chapter 4. The main body of work is presented in chapters 5-7. The thermal analyses of the copolymers as well as discussion about their crystallinity are listed in Chapter 5. Chapter 6 includes all structural analysis for the copolymer performed by small angle X-ray scattering (SAXS). The gas transport properties of the copolymers are described in Chapter 7. Chapter 8 provides the reader with conclusions and recommendations for future work.

There are also several Appendices included at the end of the dissertation. Appendix A contains the program written for computerized data acquisition of permeation data while Appendices B, C, and D have additional relevant figures not shown in the text. Appendix E contains tabulated permeability data.

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Chapter 2

Background and Theory

2.1 Introduction

This work has been built from a framework of observations and theories established for poly (n-alkyl acrylates) and semi-crystalline polymers by many scientists. Thus, this chapter is a review of much of the literature available for these polymers and copolymers. It will explain the relevant background and theories utilized in this research program including the crystallinity of homopolymers and copolymers as well as those governing the gas permeation through semi-crystalline polymers.

2.2 Crystallinity of Poly (n-alkyl acrylates)

2.2a Homopolymers

The physical properties of poly (n-alkyl acrylates) have been of continuing interest since they were first investigated in the 1940's by Rehberg and Fisher.[1] Unlike conventional crystalline polymers where the backbone crystallizes, it is the long n-alkyl side-chains of these polymers that crystallize.[2, 3] The melting - crystallization transition of the long side-chains, which occurs at the melting temperature (T_m) and can be controlled by side-chain length, causes significant changes in the physical properties of the polymer.[4-12] The longer the side-chain length (n) of the polymer, the more side-chain carbons are able to crystallize which increases the energy required to melt the polymer (ΔH_f) as well as the crystallite size and distribution which influences T_m and its breadth.

It was been well established via wide angle X-ray diffraction (WAXS) that poly (n-alkyl acrylates) are paraffin-like in the ability of their side chains to hexagonally pack.[3, 5, 13, 14] Figure 2.1 is a schematic taken from Plate illustrating the hexagonal packing structure for the comb-shaped polymer.[3] The side chains extend in an all-trans direction from the backbone. No matter the side-chain length of the polymer, a signature WAXS d-spacing peak always arises at approximately 4.2Å. This peak corresponds to the distances between side chains, α . Using Equation 2.1, this calculates to 4.85Å which is in good agreement with the diameter of the methylene chains.

$$\alpha = \frac{2d_{100}}{\sqrt{3}} \quad (2.1)$$

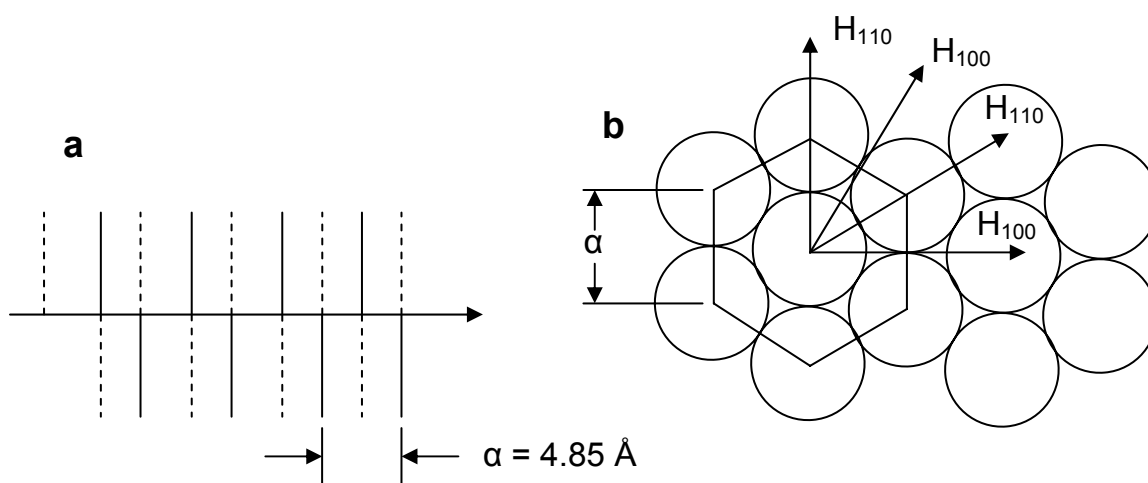


Figure 2.1 This diagram was taken from Plate [3] to illustrate the hexagonal packing structure for comb-shaped polymers. The side chains extending from the main chain in an all-trans conformation is illustrated in 2.1a where solid side-chain lines denote side chains extending from main chain and dashed side chains are those from other neighboring main chains alternating into the packing structure. The distance between side chains of the main chain extended in the same direction, α , is 4.85Å. 2.1b shows the hexagonal packing of the side chains perpendicular to the main chains. α is the same in both a and b and correlates to the distance measured using WAXS.

Limited small angle X-ray scattering (SAXS) studies have been performed on the homopolymers.[3, 13, 14] These studies were used to determine the packing formations of the side chains extended from the parallel main chains. Plate and Morawetz both observed three d-spacing peaks for crystalline PA 16 and PA 18. Two peaks were broad and diffuse while one peak was weak and sharp indicating two types of packing formations. The two broad diffuse peaks were orders of each other representing the end-to-end packing formation while the weak, sharp peak correlated with an interdigitating packing formation. Both packing structures are shown in Figure 2.2

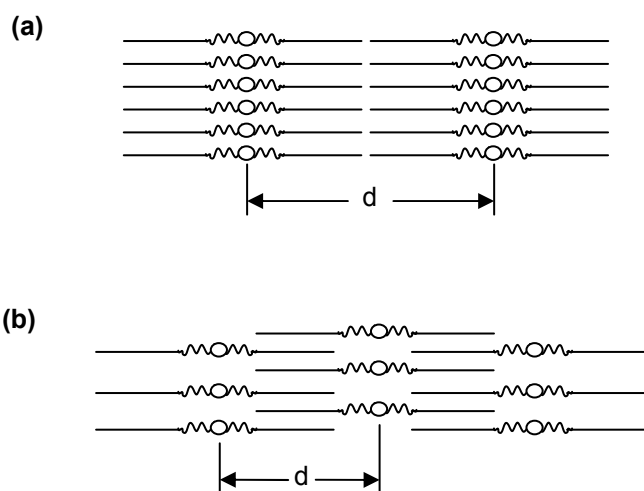


Figure 2.2 Illustrates the two side-chain packing formations revealed by SAXS include interdigitating packing (2.2a) and end-to-end side-chain packing (2.2b).

2.2 b Copolymers

Though poly (n-alkyl acrylates) were initially studied in the 1940's, the limited work on n-alkyl acrylate copolymers did not begin until the 1970's and has mainly focused on the effects of copolymerizing crystallizeable long side-chain alkyl acrylates

with short-chain alkyl acrylates or other monomers without crystallizeable side chains.[2, 3, 9, 15] Jordan and Hirabayashi found that incorporation of styrene or methyl styrene into the polymer increases the stiffness of the backbone and impedes the crystallization of the side chains. Greenberg, Hirabayashi, and Jordan also investigated the effects of non-crystallizeable units on the overall crystallinity of methacrylates and acrylates ultimately showing that non-crystallizeable units affect the crystallite size, degree of crystallinity, and physical properties of the copolymers and only affect the ability of the copolymers to crystallize when the non-crystallizeable composition exceeds about 90 mol %.[3, 15]

Two small studies were performed on copolymers of two crystallizeable comonomers.[5, 15] In his paper on thermal properties of poly (n-alkyl acrylate) copolymers, Jordan investigated one copolymer system comprised of two crystallizeable comonomers, P(A12-co-A18), where he showed that they formed a solid solutions as well as had melting temperatures and heats of fusion of the copolymers that fall between those of the homopolymer values.[15] Mogri and Paul also measured the thermal properties of several copolymers and reported similar findings.[5]

Plate performed WAXS and SAXS studies on copolymers of octadecyl acrylate and isopropyl acrylate.[3] He determined that the side chains of acrylate copolymers remain in a tight hexagonally packed lattice but that when copolymerized with amorphous comonomers, the packing formation switched to an all-interdigitating formation caused by the restriction of the conformational freedom of the backbone.

2.3 Gas Transport in Semi-Crystalline Rubbery Polymers

2.3a Michaels and Bixler's theory for semi-crystalline polymers

Permeation through dense polymer films occurs in three stages. First the penetrant molecules adsorb into the upstream surface of the membrane, then the molecules diffuse through the membrane, and finally desorb from the downstream surface of the membrane. The permeability coefficient (P) is then defined in Equation 2.2 as a combination of diffusion and sorption.

$$P = DS \quad (2.2)$$

For a membrane above its T_g , Henry's law is used to define the sorption coefficient (S) in Equation 2.3 where c is the equilibrium concentration and p is the pressure of the gas penetrant.

$$S = cp \quad (2.3)$$

$$S = S^* \alpha \quad (2.4)$$

For semi-crystalline polymers, however, sorption is defined as a function of the volume fraction of polymer in the amorphous phase (α) and the solubility of a purely amorphous polymer (S^*). The permeation of semi-crystalline poly (n-alkyl acrylates) were analyzed in terms of a two-phase model proposed by Michaels and Bixler.[16-18] The model describes the system in terms of two distinct and idealized phases, crystalline and amorphous.

Michaels also showed that the crystalline phase causes two effects on diffusion [16]. The crystallites are generally impermeable to the small penetrant molecules, forcing all permeation to occur along a tortuous path through the amorphous phase. The

crystallites also hinder the chain mobility within the amorphous phase. The model incorporates these effects in the overall diffusion coefficient D given as follows:

$$D = \frac{D^*}{\tau\beta} \quad (2.5)$$

where D^* is the diffusion coefficient through a purely amorphous polymer, τ is the tortuosity or geometric impedance factor, while β is the chain immobilization factor. Combining the diffusion and sorption equations, results in an expression for permeability as a function of the two phases.

$$P = DS = D^* S^* \frac{\alpha}{\tau\beta} \quad (2.6)$$

Defining the permeability coefficient for a purely amorphous polymer (P^*) and combining it with Equation 2.6 results in Equation 2.7.

$$P^* = D^* S^* \quad (2.7)$$

The permeation jump ratio between a purely amorphous polymer and the polymer in the two phase system as given by

$$\frac{P^*}{P} = \frac{\tau\beta}{\alpha} \quad (2.8)$$

2.3b Permeation of Poly (n-alkyl acrylates)

Figure 2.3 is a schematic illustration of the permeability jump for a poly (n-alkyl acrylate) adapted from Mogri and Paul.[5] Various parameters were extracted for comparison and analysis from this figure. The slopes, E_a and E_c , are the activation energies for permeation in the molten and crystalline states, respectively, while P_T^+ and P_T^- correspond to the gas permeabilities of the molten and crystalline polymers, respectively, extrapolated to some temperature T . As explained by Mogri and Paul, the

choice of T is arbitrary but important for calculating the jump ratio (P_T^+ / P_T^-) and is usually taken close to the melting temperature.[5, 6, 19] The temperature used greatly affects the magnitude of the jump ratios, or calculated jump

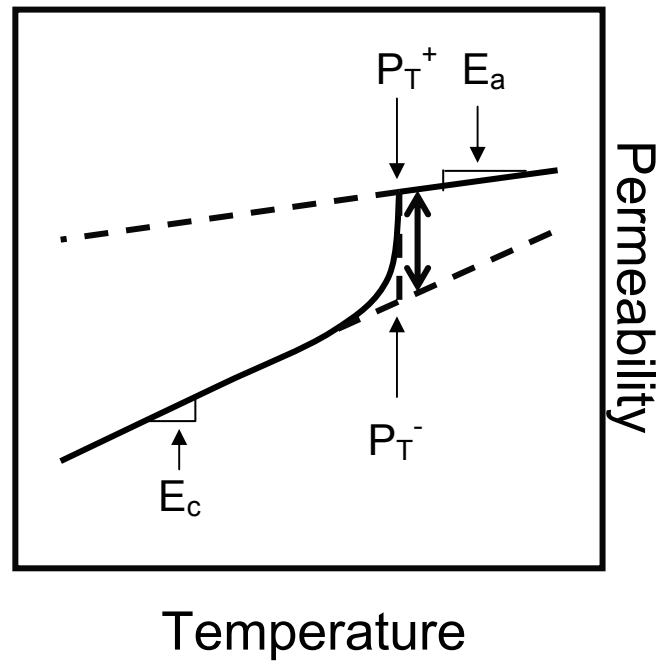


Figure 2.3 This illustration of the typical permeation jump for an n-alkyl acrylate homopolymer was taken from Mogri and Paul. The permeabilities extrapolated to the jump temperature from the amorphous and crystalline phases are marked along with their slopes.

heights, since the activity energies of the molten and crystalline polymers are not the same.

Mogri and Paul measured the gas permeability for poly (n-alkyl acrylate) homopolymers of different side-chain lengths and recorded many trends. Overall, they saw an increase in permeability and jump ratios, as well as a decrease in the E_a as the side-chain length of the homopolymers increased.[5, 6, 19] Longer side-chain length

polymers have greater crystallinity than shorter side-chain length polymers. Polymers with increased crystallinity because of longer side-chain lengths have a greater change in morphology at T_m than the shorter, less crystalline polymers producing a larger change in jump ratio. They attributed part of the jump ratio to the change in tortuosity at T_m . While crystalline, the gas molecules follow a tortuous path around the crystallites; whereas, in the amorphous state, the tortuosity disappears.

Another aspect of the jump ratios, though not as dominant as the change in permeability with side-chain length caused by crystallinity, is the change in amorphous permeability with increased side-chain length. Overall, as the side-chain length increases, so does the permeability through the amorphous polymer caused by an increased amorphous volume fraction, α . [19]

Mogri and Paul also looked at the effects of penetrant gas diameter on the permeation jump ratios. [5, 6, 19] Observing an increase in jump ratios with penetrant diameter, they recognize that the tortuosity did not tell the entire story and that the chain immobilization factor, β , also played a large part in determining the jump ratios. The β -term reflects change in segmental dynamics and is not the same for all penetrants. This strong jump dependence on penetrant size is unique to side-chain crystalline polymers; permeability jump ratios of main-chain polymers seem to have a much weaker dependence on penetrant size.

They also initiated a small study on 50/50 mol % of P(A14-co-A18) where they found the melting temperature and all the permeability properties lie in between those of PA 14 and PA 18, similar to the thermal behavior of the copolymers. [5] This became the starting point for the current copolymer studies.

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Chapter 3

Experimental Techniques

3.1 Introduction

Several experimental techniques were utilized during the course of this project. The copolymers were initially synthesized and then cast for permeation experiments using methods developed by Mogri and Paul.[1, 2] All polymers were characterized using several techniques including DSC, GPC, ^{13}C -NMR, and small angle X-ray scattering (SAXS) in addition to the constant pressure and volume permeation experiments.

^{13}C -NMR was used to determine copolymer composition. Though this chapter will briefly discuss the experimental conditions and techniques used to perform ^{13}C -NMR, Chapter 4 will contain an in depth explanation of the methods utilized in order to determine composition of the copolymers. Understanding the side-chain packing formations were also imperative to this study and involved SAXS. This chapter will also include a brief description of the SAXS apparatus and computer program used to analyze the data; however, all actual data analysis will be explained in Chapter 6.

3.2 Polymer Synthesis

All copolymers were prepared in a three-step process including monomer purification, polymer synthesis, and polymer purification. Dodecyl (A12), tetradecyl (A14), octadecyl (A18), and behenyl (A22) acrylate monomers were generously donated by Landec Corporation, while hexyl (A6) and decyl (A10) acrylate was purchased from Scientific Polymer Products. The liquid monomers, A6, A10, A12, and A14, were purified by stirring 2-5 grams of alumina oxide (Aldrich Chemicals) into the monomer

for several hours, allowing the oxide to settle to the bottom of the container, and syringe filtering (Whatman 0.2 μm PTFE membrane filters) the purified monomer from the mixture. The solid monomers, A18 and A22, were purified by a method provided by Landec Corporation that involved heating the monomers to 60°C in an oven over night until the monomer solutions were thoroughly melted and then adding approximately 5 grams of alumina oxide. The mixture was agitated by hand several times and allowed to sit in an oven for 2 hours until the oxide settled to the bottom of the flask. The purified monomer was then carefully decanted. [3, 4]

Solution polymerizations were performed in toluene and initiated by azobisisobutyronitrile (AIBN). The reactions were conducted at 60°C ranging from 30 minutes for <10% conversion to 24 hours for 100% conversion. The monomer mixtures were prepared at fixed mole percents while initiator and solvent concentrations were fixed at approximately 0.2 and 65 wt. % of monomer, respectively.[3]

The <10% conversion polymers were quenched rapidly in a large volume of ethanol immediately following the time sensitive synthesis there by instantaneously ending the reaction; the 100% conversion polymers were slowly dripped into a large volume of ethanol after synthesis to maximize contact between the polymer and ethanol. Both the <10 and 100% conversion polymers were stirred continuously during the quenching process. After continued stirring for approximately one hour, the polymers that were solid at room temperature were filtered using a 0.45 μm Durapore membrane filter (Millipore) and left to dry for approximately one hour; the polymers that were molten at room temperature settled to the bottom of the container and the ethanol mixture was decanted from the polymer. The molten polymer was stirred for an hour allowing

the remaining ethanol to evaporate. After being separated from the ethanol, all the polymers were heated and dissolved in toluene and quenched again in ethanol. On the second quenching, the <10% conversion samples were slowly dripped into the ethanol like the 100% conversion polymers maximizing contact between the polymers and the ethanol. The purification process continued for two more cycles for the 100% conversion polymers and three more cycles for the <10% conversion polymers.[3]

All glassware, stir bars, and filter paper used in the synthesis were carefully weighed before each polymerization. The polymerizations were performed as described for an amount of time to reach <10% conversion as determined by an iterative trial and error method. The reactions were quenched, polymers rinsed and collected, and all polymer product as well as labware used were first air dried in a hood for 4-5 days and then placed in a vacuum oven to dry for another 4-5 days. Everything was then reweighed and the new weights subtracted from the pre-polymerization weights. The difference was the mass of the polymer product. This was divided by the mass of monomer used in the polymerization to calculate the percent conversion. This process was repeated for each copolymer examined until all were polymerized to less than 10% of the initial monomer weight, though the samples were dried for about 10 days before weighing to remove as much residual solvent as possible it was still assumed minimal solvent was present; therefore, all were polymerized to less than 10% as a precaution. It should be noted that all copolymers were allowed to dry for weeks to eliminate any possible residual solvent present before any physical characterizations were made.[3]

3.3 Permeation Sample Construction

Permeability measurements were made on films formed from various compositions of P(A6-co-A12), P(A6-co-A22), P(A12-co-A22), P(A10-co-A14), P(A10-co-A18), P(A14-co-A18) using techniques described previously. [1-6] Films of these polymers were supported during the permeation measurements by ceramic Anopore® discs (Whatman) coated with poly(2,6-dimethyl-1,4-phenylene oxide) (PPO), generously donated by GE Plastics ($M_w = 46,000$, intrinsic viscosity = 0.46 dl/g).

Polymer films were cast by two different methods depending on their morphology at room temperature.[2, 3] Polymers that were solid at room temperature, listed in Tables 3.1 and 3.2, were solution cast from toluene (10% solids) onto a flat glass mirror using a stainless steel casting ring to contain the solution. The samples were protected from dust by an inverted glass funnel covered with a KimWipe (Fisher) and then dried in a fume hood for two weeks at room temperature. The films were removed from the glass by submerging them into an ice water bath and allowing the polymer to slowly peel off the mirror on its own. The floating polymer films were collected from the water and carefully dried with KimWipes. Polymers that were molten at room temperature, also listed in Tables 3.1 and 3.2, were melt cast in a 60°C oven directly onto the coated ceramic support.[3] The 75/25 % samples of P(A6-co-A22) and P(A14-co-A18) had melting temperatures at room temperature, which caused their morphologies to change with slight temperature changes in the lab. These polymers were solution cast using the

Table 3.1
Homopolymer Morphology at Room Temperature

Homopolymer	Morphology at room temperature
PA 10	amorphous
PA 12	amorphous
PA 14	amorphous
PA 18	crystalline
PA 22	crystalline

Table 3.2
Copolymer Morphology at Room Temperature

Copolymer	Mol % monomer 1	Morphology at room temperature
P(A10-co-A14)	75	amorphous
	50	amorphous
	25	amorphous
P(A10-co-A18)	75	amorphous
	50	amorphous
	25	crystalline
P(A14-co-A18)	75	crystalline/amorphous
	50	crystalline
	25	crystalline
P(A6-co-A12)	75	amorphous
	50	amorphous
	25	amorphous
P(A6-co-A22)	75	crystalline/amorphous
	50	crystalline
	25	crystalline
P(A12-co-A22)	75	amorphous
	50	crystalline
	25	crystalline

same technique used for other solid polymers except that after the solvent evaporated, the samples were placed into a laboratory freezer to insure they were crystalline before being

placed in the ice bath for removal from the silicon wafer. They were then stored in the freezer to maintain their crystalline morphology.[3]

Solid film thicknesses were measured using an Ames Micrometer prior to permeation experimentation while the molten film thickness was determined by performing a simple mass balance calculation involving the measured mass of polymer, the estimated density of the amorphous polymer (0.986 g/cm^3), and the known diameter of the ceramic support (47 mm).[2, 3]

Accurate permeation experiments require the film to have an uniform thickness and a known area for permeation. The permeation sample construction developed by Mogri and Paul was utilized here.[1, 6-8] A porous ceramic Anopore® disc was used to provide mechanical support for the polymer film. A constant thickness of the film was achieved by coating the ceramic disc with PPO to impede the molten polymer from flowing into the pores when running permeation experiments at elevated temperatures. The pores in the ceramic disk were sealed by quickly coating its surface with 1.5 ml of PPO solution (15 wt % solids in trichloroethylene) via a syringe. The disc was immediately scraped with a razor removing nearly all the PPO solution, leaving a very thin layer of PPO solution coating the disc. The disc remained in the hood covered for 24 hours and was then held under vacuum for another 24 hours at 100°C until it was finally placed in a convection oven to age at 60°C for a minimum of 10 days.[1, 3] The bottom of the support was masked with a piece of aluminum tape that had a hole of known diameter in order to define the area for the permeation experiment. The cast films were then placed on the top of the ceramic PPO coated support over the hole determined by the aluminum tape. The samples were given a defined thermal history by placing them on a

Linkham TMS91 hot stage where they were heated well above the T_m of the P(A14-co-A18) copolymer and subsequently cooled to room temperature at 1°C /min.[1, 3]

The PPO support, though cast as thin as possible with a razor, still contributed to the total permeation resistance for the composite assembly.[3] The permeance, $\left(\frac{l}{P}\right)$, of the composite is the sum of the permeance for the PPO layer and the poly (n-alkyl acrylate) film, i.e.,

$$\left(\frac{l}{P}\right)_{composite} = \left(\frac{l}{P}\right)_{Poly(n-alkyl\ acrylate)} + \left(\frac{l}{P}\right)_{PPO} \quad (3.1)$$

where P is the permeability and l the thickness[2, 3] The permeance of the PPO layer was measured before adding the acrylate copolymer film to the disk. The quality of the PPO sample was determined by comparing the selectivity of the membrane for various gasses to literature values.[9] A layer of known thickness of the poly (n-alkyl acrylate) polymer was added on top of the PPO support. The permeance of the composite was measured as a function of temperature above and below the T_m of the acrylate sample. Using the permeance measurements for the composite and PPO along with the thickness of the acrylate sample, the permeability of the poly (n-alkyl acrylate) sample could be calculated from Equation 3.1. Slight aging and temperature corrections for the PPO layer were also factored into the permeance of the PPO support. The gases used, He, H₂, O₂, N₂, CH₄, and CO₂, were purchased from Matheson Tri Gas with at least 99.9% purity and were run at a 2 atm upstream pressure for all temperatures.

3.4 Computerized DAQ System

A computerized data acquisition system was implemented to collect permeation data more effectively. The system includes a PCI-6013 I/O board, CB68LP multi-channel input, and LabView software all purchased through National Instruments. The downstream pressure for each permeation cell was measured with an MKS Baratron absolute pressure transducer type 627B which was powered by an MKS PDR-5B 5 channel power supply readout. The analogue pressure readings were converted into voltage by the PDR and a signal was sent to the CB68LP multi-channel input. From there, the signal, along with signals from up to eight other permeation cells, were sent to the PCI-6013 I/O DAQ board to be digitized and recognized by the computer. Using the LabView programming software, a program was written to record and store the voltage data as a function of time for each of the nine permeation cells attached to the computer. The program, shown in Appendix A, was written with the help of Pavlos Tsiartas. Data files were saved under individualized names for each cell and were opened using Microsoft Excel. Voltage, which was directly correlated with downstream pressure, was plotted against time to determine the flux and used to calculate the steady state permeability of the membrane.

3.5 DSC Experiments

Differential scanning calorimetry (DSC) was conducted using a Perkin Elmer DSC 7 with polymer samples weighing approximately 15 mg. All samples were initially heated to well above their melting temperatures at 20°C /min, cooled to below their T_m at 1°C /min, and reheated at 10°C /min. All values reported in this work were taken on the second heating.[3]

3.6 Gel Permeation Chromatography

Gel Permeation Chromatography (GPC) was conducted using an Agilent 1100 chromatograph with 20 μ l of polymer solution made from approximately 10 mg of polymer per 2 ml GPC grade THF. They were filtered with a Whatman 0.2 μ m PTFE membrane syringe filter and run at room temperature. The samples were analyzed using a calibration with polystyrene standards.[4]

3.7 ^{13}C -NMR

^{13}C -NMR was performed with a Varian Inova 500 spectrometer operating at a frequency of 125.7 MHz. Samples were prepared using approximately 10 mg of polymer sample dissolved in 1 ml of deuterated chloroform (Fisher Scientific). All spectra were obtained at a flip angle of 90° using a pulse interval of approximately 30 seconds, or 5 times the determined relaxation time T_1 for each sample run, with an acquisition time of 4 seconds over 128 repetitions. Though generally used for quantitative analysis with ^{13}C -NMR, gated decoupling was not appropriate for these samples.[3]

3.8 SAXS Characterization

Small angle X-ray scattering (SAXS) was performed using a MolMelt SAXS equipped with a liquid sample holder. Samples consisted of about 50 mg of polymer and as many as 4 samples could be loaded into the sample holder at a given time. The temperature was controlled with a Fisher Brand refrigerated circulator that circulated fluid through the sample holder, around the samples. Though only the sample holder temperature could be measured, all temperature adjustments were held for 12 hours

before any experiments were run to ensure thermal equilibrium between the sample holders and the polymer samples. Upon loading the samples, the temperature of the sample holder was heated to some temperature 10°C above the T_m of the polymer with the highest melting temperature. This temperature was held for 6 hours and then cooled to the appropriate experimental temperature ensuring a uniform thermal history for all samples characterized. All data were collected over a real time of 2 hours and analyzed using the FIT2D software program written by A.P. Hammersley available on the FIT2D-ESRF website.[10]

3.9 References

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Chapter 4

Effects of Copolymer Conversion on Composition

4.1 Introduction

It is the purpose of this section to establish that copolymers of poly (n-alkyl acrylates) of different side-chain lengths are similar enough in structure not to exhibit composition drift resulting in a copolymer product that is identical in composition to the monomer mixture. It is imperative that the composition of copolymers be uniform in order to attribute correct properties to structure and composition. All copolymer studies to date for alkyl acrylate systems have assumed the compositions of the copolymers were the same as that of the monomer reaction mixtures, i.e., the monomer reactivity was assumed not to vary among n-alkyl acrylates of different side-chain length. [1-3] To date, there have not been any studies confirming this assumption. This chapter examines these issues using two methods, first by determining the reactivity ratios of different alkyl acrylate copolymers and second by comparing the physical properties of copolymers polymerized to <10% and 100% conversion.

4.2 Reactivity Ratios for Poly(n-alkyl acrylate) Copolymers

Two sets of copolymers were synthesized to less than <10% conversion to evaluate the reactivity ratios. The first set included P(A10-co-A14), P(A10-co-A18), and P(A14-co-A18); while the second set consisted of P(A6-co-A12), P(A6-co-A22), and P(A12-co-A22). Copolymers were formed from monomer mixtures at 25 mol % intervals ranging from 0 through 100 mol %. The first set of copolymers have similar side-chain lengths as well as crystalline properties. As has been reported extensively in the literature, poly (n-alkyl acrylates) only crystallize after the ninth or so carbon from

the backbone; the backbone and the nine proximal carbons exist in an amorphous state. [2, 4, 5] The first set includes acrylates of similar side-chain length while the second set involves more extreme differences in crystallinity as well as side-chain length. Ultimately, it is assumed that if the reactivity ratios for both the first and second sets of polymers are all approximately equal to unity, then copolymer systems in between should also have reactivity ratios of approximately one.

4.2a ^{13}C -NMR Analysis Technique

To obtain information about the reactivity ratios for copolymerization of n-alkyl acrylates, it was necessary to have a reliable method to determine the composition of the copolymers formed. Initial efforts showed that ^1H -NMR was not sensitive enough to distinguish the various protons in the long hydrocarbon side-chains of the poly (n-alkyl acrylates). Poly (n-alkyl acrylates) with side-chain lengths ranging from $n = 6$ to $n = 22$ were analyzed using ^{13}C -NMR. Homopolymers with side-chain lengths of 10 or more carbons produced identical spectra with respect to peak location. Poly (hexyl acrylate) had a slightly different spectra with peaks shifted from those observed with the other acrylates examined. This was a result of fewer carbon-carbon interactions in the side-chains. Figures 4.1 and 4.2 show examples of the two types of spectra; the one for poly (dodecyl acrylate) in Figure 4.1 is typical of the spectra for poly (n-alkyl acrylates) with side-chain lengths of 10 or more carbons and is slightly different from the spectra of poly (hexyl acrylate) in Figure 4.2. All peak characterization was performed similarly to previous studies in the literature. [6] As shown in the spectra for poly (dodecyl acrylate),

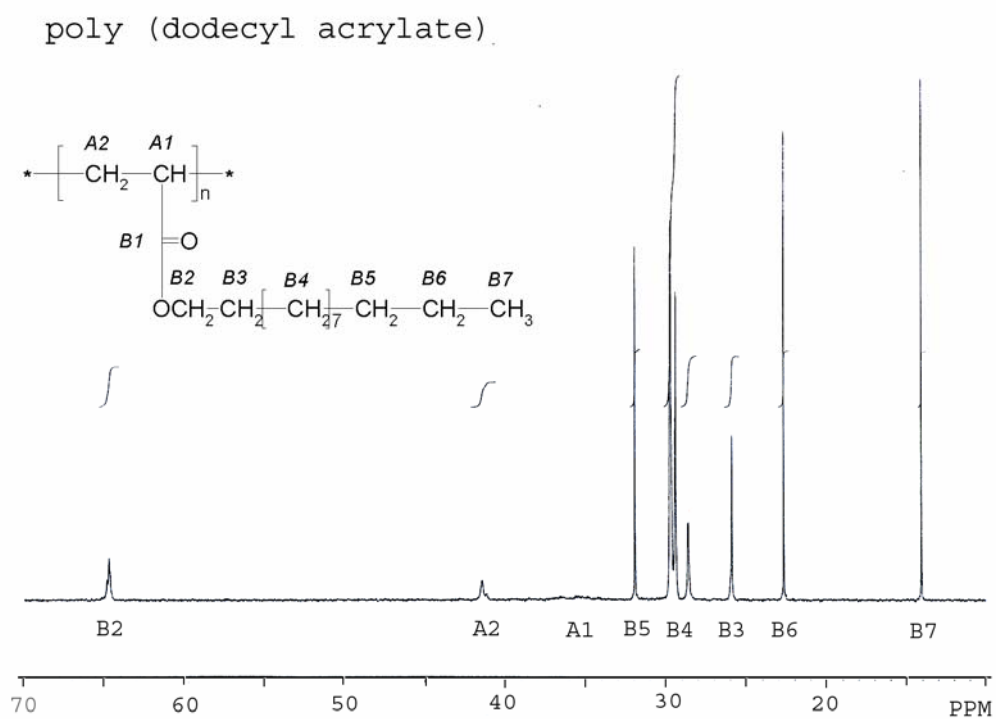


Figure 4.1 ^{13}C -NMR spectra for poly (dodecyl acrylate).

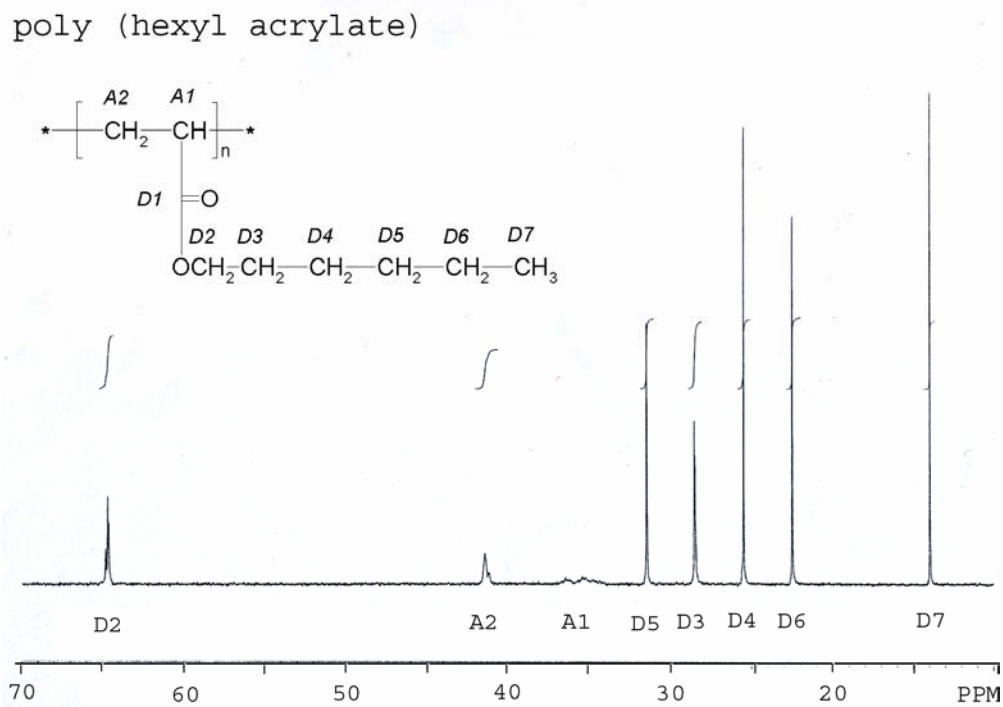


Figure 4.2 ^{13}C -NMR spectra for poly (decyl acrylate).

^{13}C -NMR is only sensitive enough to distinguish among the end carbons, producing a single peak for each carbon. The middle carbons in the side-chain are grouped into a common peak at 30 PPM. Since peak frequency is the same for the longer poly (n-alkyl acrylates) investigated here, the individual polymers may be characterized by the area under the common middle peak. The polymers were compared by normalizing the middle peak (B4) with the outermost, least restrictive peak (B7). The normalized peak areas are plotted as a function of side-chain lengths for the homopolymers in Figure 4.3

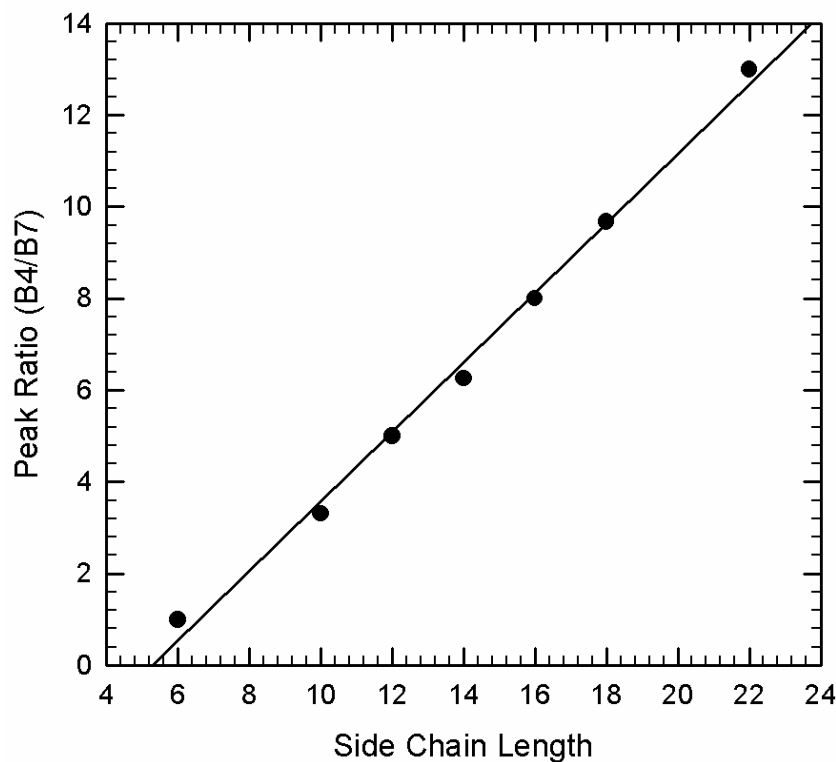


Figure 4.3 Calibration curve generated from ^{13}C -NMR data for poly (n-alkyl acrylates) of varied side chain lengths (n).

with the best linear fit to the data given by

$$P = 0.79(n) - 4.76 \quad (4.1)$$

where P = peak ratio B4/B7 and n = side-chain length. From the relationship of the peak ratios versus the side-chain length determined from the n-alkyl acrylate homopolymers, the compositions of alkyl acrylate copolymers were determined as follows. ^{13}C -NMR analyses were conducted on all the copolymers mentioned above. Copolymers of side-chain lengths greater than or equal to 10 carbons were analyzed similarly to the homopolymers; i.e., the central B4 peak was normalized by the tail carbon B7. From the measured peak ratio, P , the average number of carbons, $\bar{n}$, in the side-chains of copolymers can be computed from Equation 4.1. In turn, the values of $\bar{n}$ should be related to the number of carbons in the two comonomers by

$$\bar{n} = F_1 n_1 + F_2 n_2 \quad (4.2)$$

where F_i = the mole fraction of monomer i in the copolymer. Thus, F_i can be calculated from the ^{13}C -NMR spectra.

Copolymers composed of poly (hexyl acrylate) and poly (dodecyl acrylate) or poly (behenyl acrylate) generated ^{13}C -NMR spectra with peaks for both acrylates in the polymer as shown in Figure 4.4 for P(A6-co-A12) made from a 50/50 mol% monomer mixture. The composition of these copolymers was determined by calculating the fraction of the B5 peak over the sum of the B5 and D5 peaks in the spectra. The B5 and D5 peaks were chosen for comparison because they both represented a single carbon in the side-chain and were the most proximal carbons to the tail end of the chain with unique and distinguishable peaks in the spectra.

poly (hexyl-co-dodecyl acrylate)

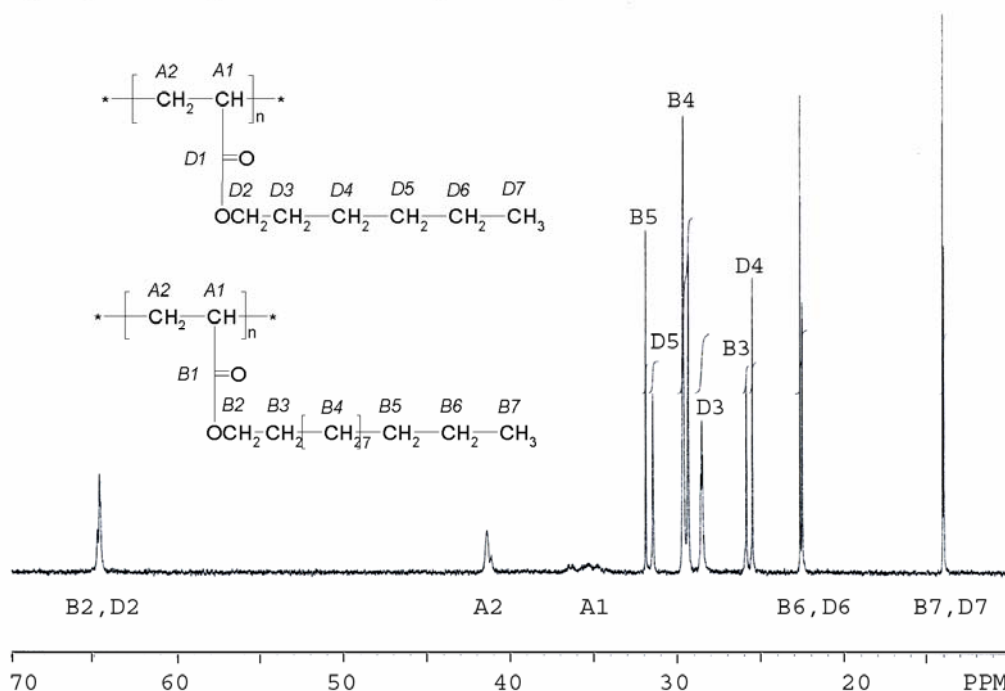


Figure 4.4 ^{13}C -NMR spectra for poly (hexyl-co-dodecyl acrylate).

4.2b Reactivity Ratio Calculations

The reactivity ratios (r_1 and r_2) for a given copolymer system are defined by the copolymer equation

$$F_1 = \frac{f_1(1 + [r_1 - 1]f_1)}{(r_1 + r_2 - 2)f_1^2 + 2(1 - r_2)f_1 + r_2} \quad (4.3)$$

the measured values of the copolymer composition, F_i , and the known monomer composition, f_i are plotted in Figure 4.5.[7] In addition, these data sets were regressed by Microsoft Excel with Equation 4.3 to obtain the best fit r_1 and r_2 for the data. Table

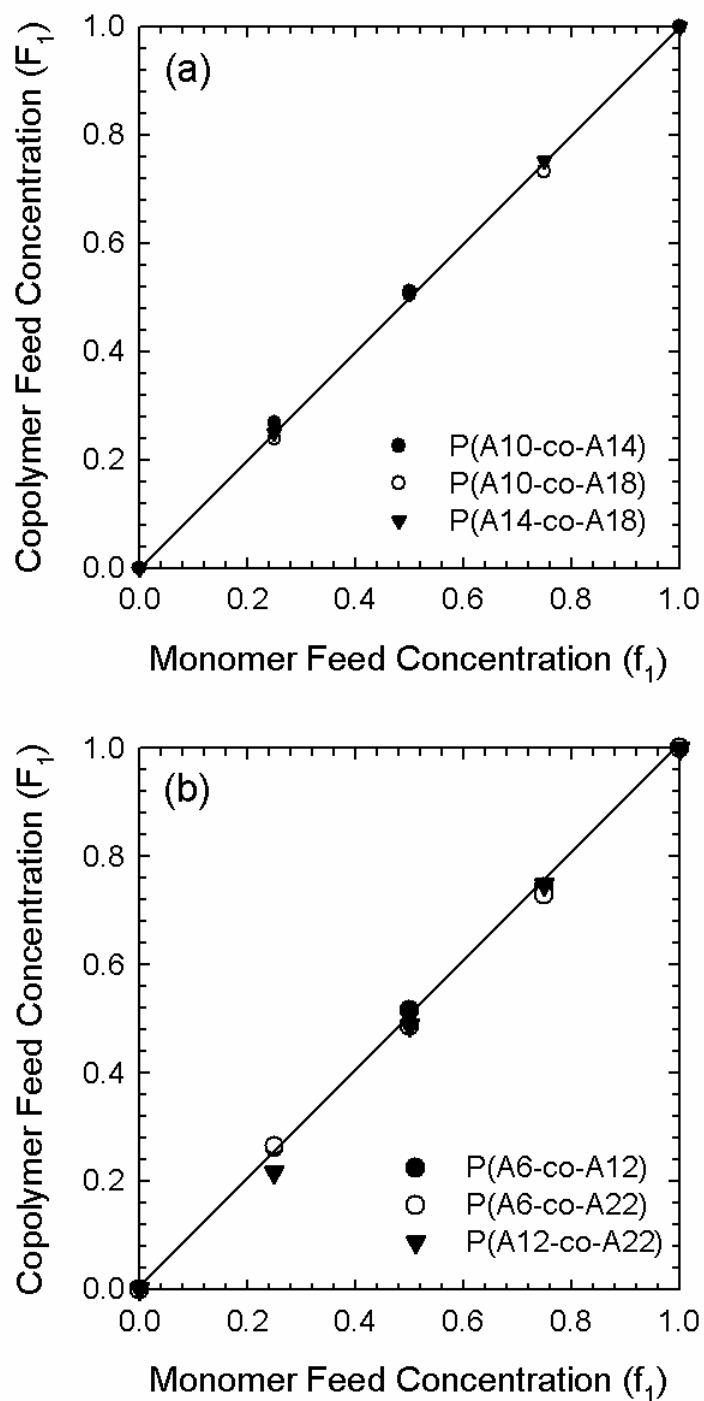


Figure 4.5 Dependence of copolymer composition on composition of reactant mixtures for (a) P(A10-co-A14), P(A10-co-A18), and P(A14-co-A18) and (b) P(A6-co-A12), P(A6-co-A22), and P(A12-co-A22) mixtures.

Table 4.1
Reactivity Ratio Values for Poly (n-alkyl acrylate) Copolymers

				$\frac{100}{n} \sum [(F_1)_{\text{exp}} - (F_1)_{\text{calc}}]^2$ *	
Copolymer	r₁	r₂	r₁ r₂	Best fit r₁ and r₂	r₁ = r₂ = 1
P(A10-co-A14)	0.9	0.8	0.7	0.45	1.20
P(A10-co-A18)	0.9	1.0	0.9	0.82	0.97
P(A14-co-A18)	1.0	1.0	1.0	0.20	0.29
P(A6-co-A12)	0.9	0.9	0.8	0.45	0.98
P(A6-co-A22)	0.9	0.9	0.8	0.70	0.90
P(A12-co-A22)	1.1	1.3	1.4	0.36	1.60

* The average root-mean square difference in Table 4.1 is calculated from the measured copolymer composition, $(F_1)_{\text{exp}}$, and the composition calculated from Equation 4.3, $(F_1)_{\text{calc}}$, using the respective reactivity ratios (r_1 and r_2); this was averaged over the number of data sets (n) used in the calculation of the reactivity ratios.

4.1 lists the reactivity ratios for each of the copolymers examined and the average root mean squared difference in F_i between the experimental data and the values from Equation 4.3 using both the best fit reactivity ratios listed in Table 4.1 and for the case when the reactivity ratios are assumed to be equal to one, i.e., $F_i = f_i$. The calculated reactivity ratios for the poly (n-alkyl acrylate) copolymers are all close to unity, within a range from 0.8 to 1.3 with the ratios for P(A14-co-A18) being equal to one. The average root-mean square differences between the measured mole percent of monomer 1 in the copolymer and copolymer composition calculated from Equation 4.3 with the best fit reactivity ratios are all less than 1%. A small mean difference is expected since the reactivity ratios were calculated to provide the best fit of the data. A similar mean deviation was also calculated from the data assuming the reactivity ratios are equal to one. These mean deviations are somewhat larger but still below the probable error of the

overall experimental technique. Though the polymerizations and ^{13}C -NMR characterizations were performed with the most careful of techniques, experimentally its difficult to be certain of an accuracy to within 2%. Therefore, even though the best fit reactivity ratios calculated for each of the copolymer systems are not always equal to one, the small error in assuming the ratios are equal to one suggests this is a fully adequate approximation. While only a limited set of copolymers were included in this study, the combinations used would suggest that this approximation should be valid for copolymers based on any pairs of n-alkyl acrylate monomers with n in the range from 6 to 22.

Though previous studies have assumed no composition drift occurs for copolymers of n-alkyl acrylates, this point has never been definitively investigated.[2, 4, 5] Previous work on n-alkyl acrylate – acrylic acid copolymers showed that for acrylates with side-chain lengths ranging from A14 through A22, the reactivity ratios remained constant, independent of side-chain length, supporting the observation shown here that acrylates with long alkyl side-chains are similar enough in structure not to exhibit significant composition drift when copolymerized together.[8] The n-alkyl acrylate-acrylic acid study did not include short side chains. Cases where there are large differences in side-chain lengths create the most probable region for copolymer composition drift. Jordan et al thoroughly examined copolymers of n-alkyl acrylates with $n = 1$ to $n = 4$ which form amorphous homopolymers and octadecyl acrylate which forms a semi-crystalline homopolymer, but performed all polymerizations to 100% conversion ignoring any possible drift.[2] Later work suggested the presence of limited composition drift with the methyl acrylate-octadecyl acrylate copolymer reporting reactivity ratios of 1.56 and 0.84, respectively; signifying an almost random distribution.[4]

4.3 Physical Properties as a Function of Conversion

The uniformity of composition of the copolymers was further tested by comparing the differential scanning calorimetry (DSC) and permeation properties for copolymers made at <10 and 100% conversion.

4.3a DSC Behavior

The DSC experiments were conducted by initially heating the polymer to some temperature well above the melting point (T_m) at 20°C/min, cooling to well below the T_m

Table 4.2
Melt Temperature and Heat of Fusion Data For Poly(n-alkyl acrylate) Copolymers
Polymerized to Different Conversions

Number of alkyl carbons, n		Mol % Monomer 1	T_m (°C)		ΔH_f (J/g)	
Monomer 1	Monomer 2		100%	10%	100%	10%
10	14	25	11.7	11.9	44.5	40.7
10	14	50	3.7	3.8	20.1	17.8
10	18	25	42.4	42.3	78.8	81.3
10	18	50	31.1	31.9	50.8	50.1
10	18	75	-1.1	-1.3	30.8	30.2
14	18	25	41.9	41.7	85.1	83.1
14	18	50	32.7	33.5	70.2	69.7
14	18	75	25.7	25.7	61.5	60.8
6	12	25	-7.6	-6.2	26.8	25.5
6	22	25	54.2	52.7	101.8	99.8
6	22	50	48.7	48.9	77.4	76.3
6	22	75	34.5	33.5	43.2	41.5
12	22	25	56.0	57.0	82.3	82.6
12	22	50	45.9	44.8	51.3	54.9
12	22	75	12.5	13.8	50.2	47.4

at a rate of 1°C/min, and a second heating at 10°C/min. The constant cooling rate insured a controlled thermal history for all polymers examined, while the second heating was

used to acquire the T_m and heat of fusion (ΔH_f) data. All of the copolymer systems are listed in Table 4.2 with the exception of P(A6-co-A12) with $F_1 = 0.50$ and 0.75 and P(A10-co-A14) with $F_1 = 0.75$. The T_m for these copolymers was too low to be analyzed by the DSC and, therefore, T_m and ΔH_f data could not be acquired for comparison.

The T_m data for the copolymers listed show very comparable values and trends for the <10 and 100% conversion copolymers. Generally, the difference in melting point between high and low conversion polymers is under 1°C and is 1.5°C in the most extreme case. The high and low conversion copolymers both show an increasing T_m with increasing concentration of the longer alkyl side chain.

To properly assess the heats of fusion by integration of the area under the melting peak, it is necessary to construct a rational baseline. Melting temperature is not affected by baseline construction since T_m is defined by the location of the peak of the curve. Figure 4.6 shows 4 examples of different types of DSC thermograms observed for these poly (n-alkyl acrylate) copolymers. Ideally, all scans should resemble schematic ‘a’ for P(A6-co-A22) with $F_1 = 0.25$. The baseline location for this heat flow curve is relatively straight forward since the pre- and post- melt curves are nearly straight and co-linear. In this case the reasonable baseline is the straight line connecting the post-melt and pre-melt line. For nearly all polymers, however, there is usually some measurable pre-melting present before the onset of the actual melt curve, and this complicates baseline construction because it is difficult to differentiate pre-melting from baseline non-linearity. Schematic ‘b’ shows P(A10-co-A18) with $F_1 = 0.25$ as an example of a typical

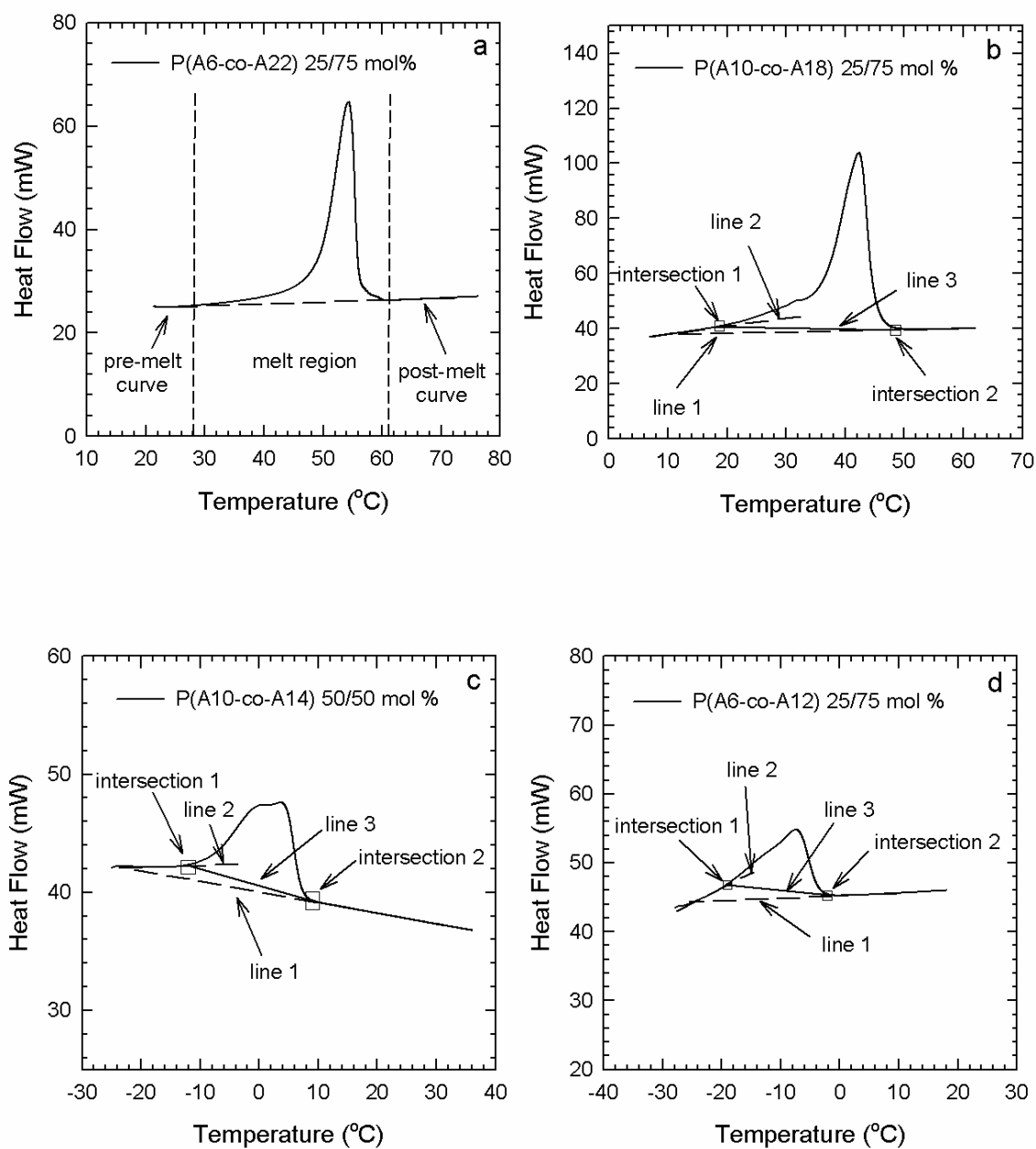


Figure 4.6 Four different examples of DSC spectra for poly (n-alkyl acrylate) copolymers.

thermogram for a poly (n-alkyl acrylate) homo or copolymer exhibiting pre-melting. In this case, an initial baseline is drawn by starting at the high temperature end of the post-melt line and using a ruler to extend the line through the melt region to the pre-melt curve, 'line 1'. A second line is then drawn by starting at the low temperature end of the pre-melt line and using a ruler to extend this line until it separates from the melt curve, 'line 2'. This point of separation is labeled 'intersection 1', while the location where 'line 1' separates from the curve is labeled 'intersection 2'. Finally, the baseline, 'line 3', is drawn by connecting 'intersection 1' with 'intersection 2'. This baseline method is generally accepted as the standard method for drawing baselines.[9] Schematic 'b' is representative of many longer side-chain length homo and copolymers where the pre-melt region is easily defined and usually represents less than <10% of the overall curve area. The pre-melt region, however, tends to become a little more ambiguous for shorter side-chain length polymers and especially copolymers where the melt curve tends to elongate. This is shown in schematics 'c' and 'd' for P(A10-co-A14) with $F_1 = 0.5$ and P(A6-co-A12) with $F_1 = 0.25$. The baseline in these examples is drawn with the same method used in schematic 'b'. In these cases, the 'pre-melt' regions are 26 and 30%, respectively, of the overall curve area. For many of these copolymers this 'pre-melt' region may become as large as 40% of the heat of fusion for the polymer. For these cases, this 'pre-melting' must be part of the actual melt curve, where the pre-melt and actual melt regions are indistinguishable. This may be due to the different side-chain lengths affecting the overall crystalline structure; this will be addressed further in future papers.

Ultimately, the method shown in schematic ‘a’ of Figure 4.6 was used to analyze the data shown in Table 4.2. Although this is not the standard method for analyzing polymers because it does not take pre-melting into account, it can be universally applied to all the poly (n-alkyl acrylate) copolymers examined here. It may be necessary in future work to subtract out the <10% or so of the heat of fusion attributed to pre-melting; however, in this situation, it did not seem necessary to do so. The values reported in Table 4.2 were all calculated by method ‘a’, and the difference in ΔH_f between the high and low conversion polymers is less than 4 J/g or about 12%. Such differences may easily be attributed to issues of baseline construction.

4.3b Permeation Behavior

Previous studies by Mogri and Paul thoroughly characterized the physical properties, particularly gas permeation, of poly (n-alkyl acrylate) homopolymers.[3, 10] Like most physical properties of poly (n-alkyl acrylates), the permeability of these materials exhibit a large ‘switch’ or ‘jump’ at the T_m of the polymer. This switch occurs because of the change from a semi-crystalline morphology below T_m to a completely amorphous state above T_m .

The magnitude of the jump is defined as the ratio of the permeability in the melt to the permeability in the semi-crystalline state; both values being extrapolated to the melting temperature. The magnitude of the jump depends both on the size of the penetrating gas molecules as well as the side-chain length of the polymer; large gas molecules exhibit larger jumps than smaller molecules. The side-chain length of the polymer contributes to the magnitude of the permeability jump because the number of carbon atoms ‘switching’ from the semi-crystalline to amorphous state increases with n.

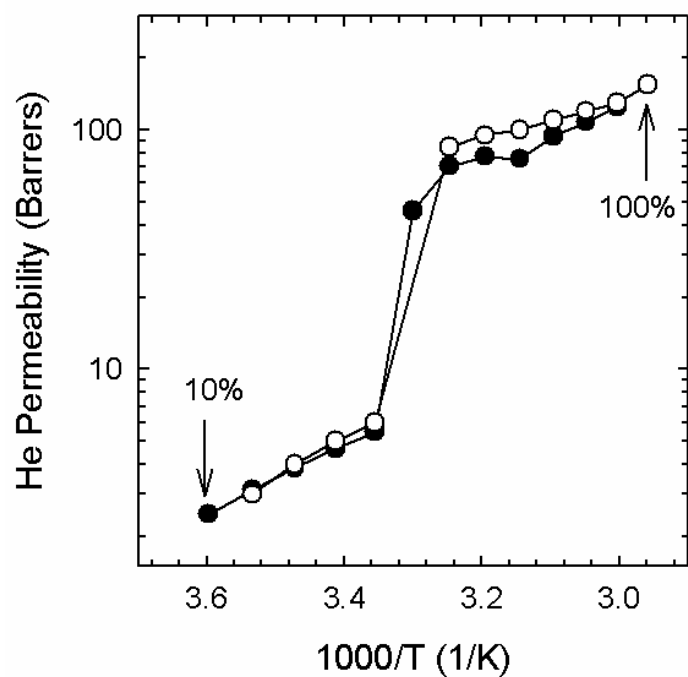


Figure 4.7 He Permeability measurements for P(A14-co-A18) 50/50 mol % copolymers synthesized to high (100%) and low (< 10%) conversion.

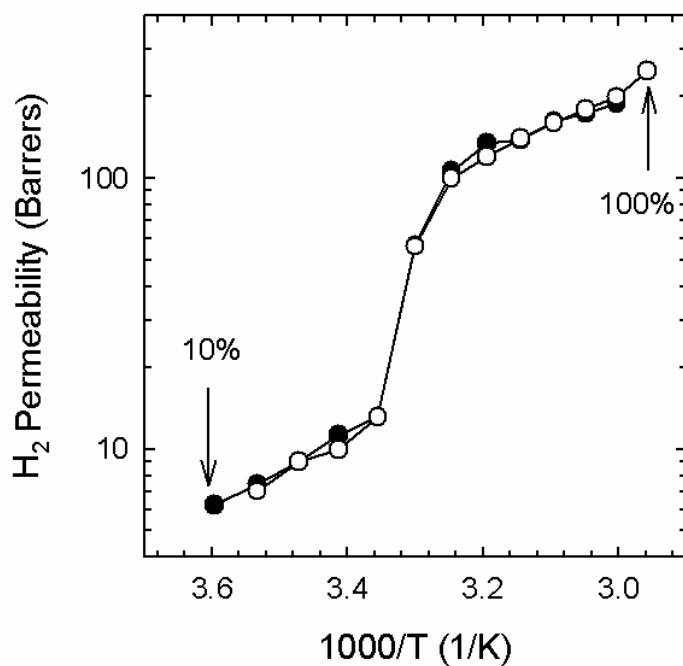


Figure 4.8 H₂ Permeability measurements for P(A14-co-A18) 50/50 mol % copolymers synthesized to high (100%) and low (< 10%) conversion.

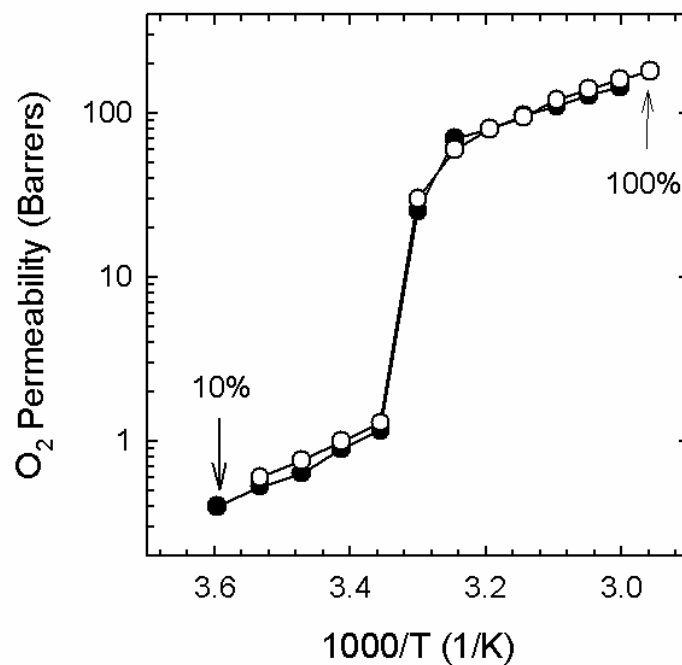


Figure 4.9 O₂ Permeability measurements for P(A14-co-A18) 50/50 mol % copolymers synthesized to high (100%) and low (< 10%) conversion.

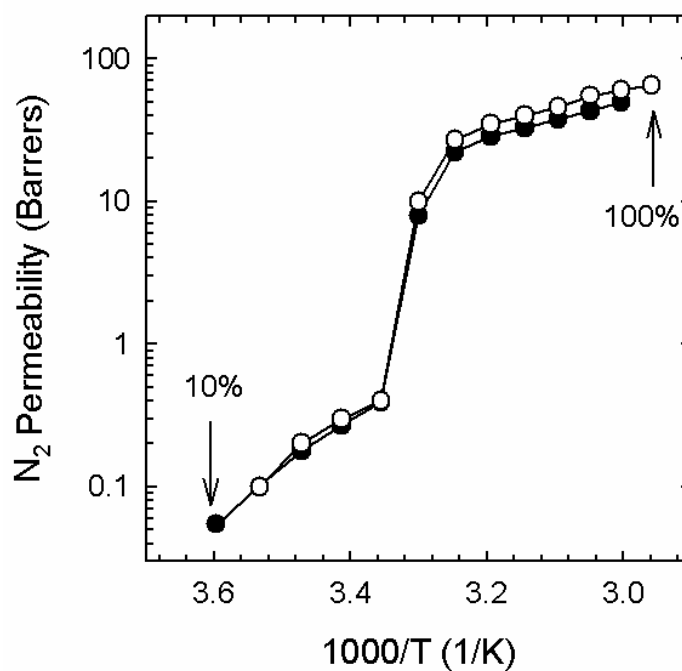


Figure 4.10 N₂ Permeability measurements for P(A14-co-A18) 50/50 mol % copolymers synthesized to high (100%) and low (< 10%) conversion.

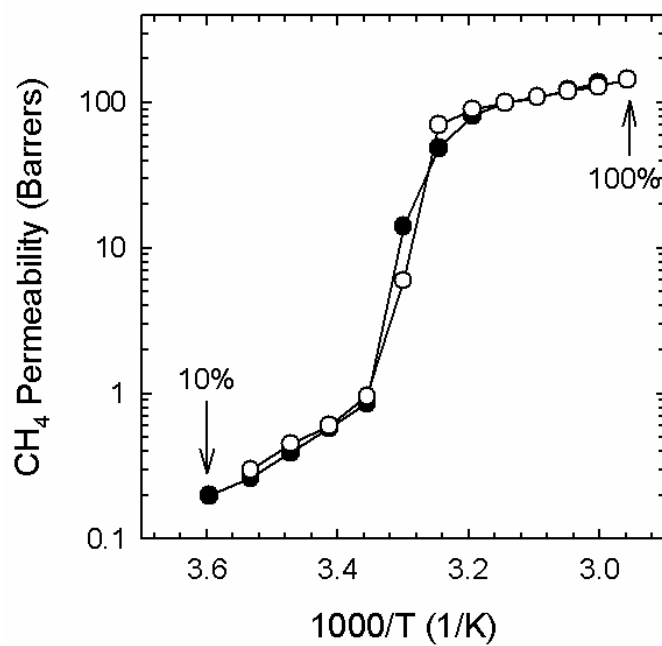


Figure 4.11 CH₄ Permeability measurements for P(A14-co-A18) 50/50 mol % copolymers synthesized to high (100%) and low (< 10%) conversion.

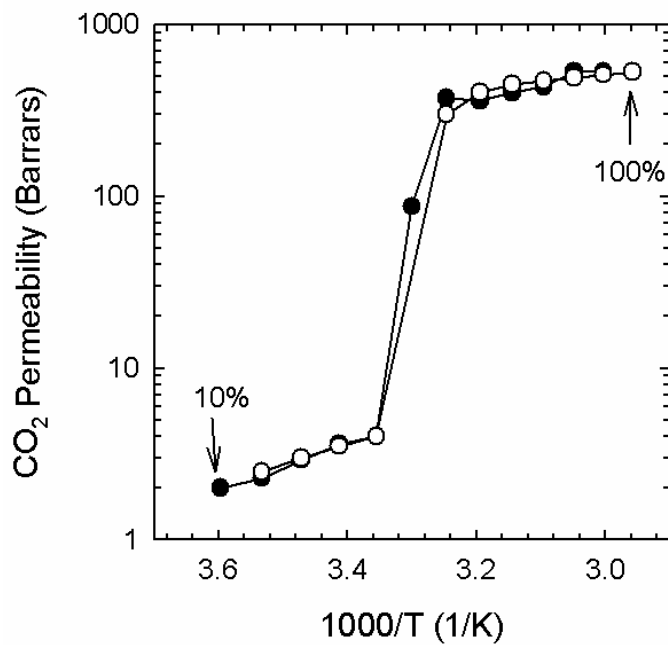


Figure 4.12 CO₂ Permeability measurements for P(A14-co-A18) 50/50 mol % copolymers synthesized to high (100%) and low (< 10%) conversion.

Chapter 7 will concentrate on the gas permeability behavior of various n-alkyl acrylate copolymers; however, the scope here is limited to a comparison of high and low conversion copolymers. The permeability of P(A14-co-A18) polymerized from a 50/50 mol% monomer mixture reacted to either <10 and 100% conversion are shown in Figures 4.7-4.12 for various gases as a function of temperature transversing the T_m . The permeation measurements of the copolymers are very comparable to Mogri and Paul's results for homopolymers of PA14 and PA18 where they reported jump ratios of 11 and 55 for N_2 in the polymers, respectively. The jump ratios for both the high and low conversion P(A14-co-A18) samples with N_2 are 33 and 33.2, respectively.[3, 10]

The jump ratio occurs for each gas is essentially the same for P(A14-co-A18) copolymers reacted to <10 and 100% conversion. It is evident in Figures 4.7-4.12 that the permeability data for both polymers are nearly interchangeable in both slope and jump. The plots are arranged in order of increasing jump magnitude, which correlates directly with increasing size of the penetrant gas molecule. The smallest molecule, helium, has the smallest permeation jump, while the largest molecule, CO_2 , has the largest jump. These trends occur for both the high and low conversion copolymers. Since there is no distinguishable difference in permeation behavior for the sample at any temperature or for any gas, it is, therefore, safe to use high conversion copolymers in future work without concern for compositional non-uniformity.

4.4 Conclusion

Copolymers composed of various pairs of n-alkyl acrylate monomers with side-chain lengths varying from 6 to 22 carbons were prepared to different conversions and molar concentrations to determine if there was any composition drift in the copolymers.

Using ^{13}C -NMR, the low yield copolymers were characterized and the reactivity ratios calculated. The experimental reactivity ratios were essentially equal to one meaning the copolymers formed have the same composition as the monomer mix from which it was made. This means no composition drift occurs during the polymerization which supports previous assumptions involving these copolymers. The physical properties of the high and low conversion copolymers were nearly identical, again indicating a uniform composition for these n-alkyl acrylate copolymers independent of conversion of the copolymerization reaction.

4.5 References

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Chapter 5

Thermal Properties of Poly (n-alkyl acrylates)

5.1 Introduction

The physical properties of poly (n-alkyl acrylates) have been of continuing interest since they were first investigated in the 1940's by Rehberg and Fisher.[1] Unlike conventional crystalline polymers where the backbone crystallizes, it is the long n-alkyl side-chains of these polymers that crystallize.[2-8] The melting - crystallization of the long side-chains, which occurs at a melting temperature (T_m) that can be controlled by side-chain length, causes significant changes in the physical properties of the polymer. It is, therefore, imperative to thoroughly understand the affects of copolymer combinations on its T_m and ΔH_f in order to relate them to other physical properties.

Two types of copolymers were evaluated here: those composed of two crystallizeable comonomers and those composed of one crystallizeable comonomer and one non-crystallizeable or borderline crystallizeable comonomer. All copolymers were synthesized and physical properties analyzed using techniques described in Chapter 3. The compositions of the copolymers were determined using the ^{13}C -NMR technique described in Chapter 4.

5.2 Homopolymers

The melt temperatures and heats of fusion of various poly (n-alkyl acrylate) were characterized using DSC and are shown in Figures 5.1a and b, respectively, as a function of the number of carbon atoms in the side chain, n ; also see Table 5.1. All data were

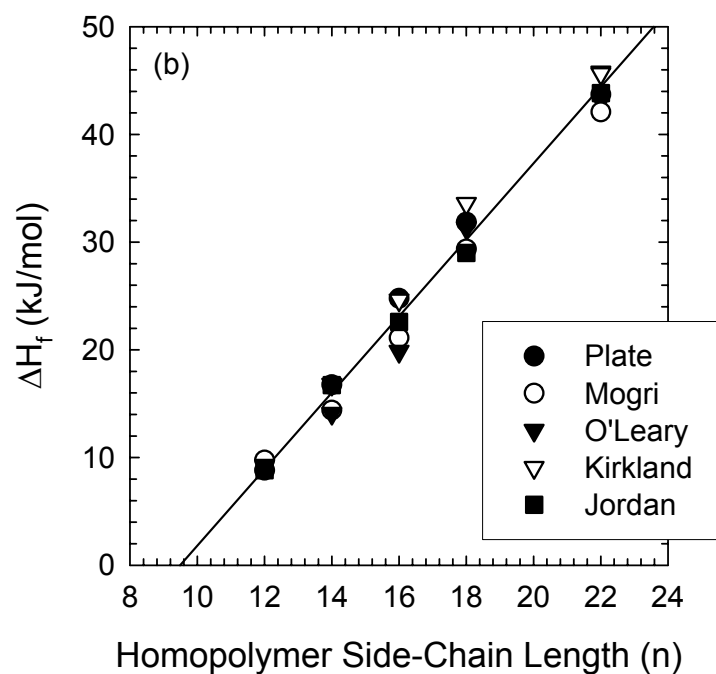
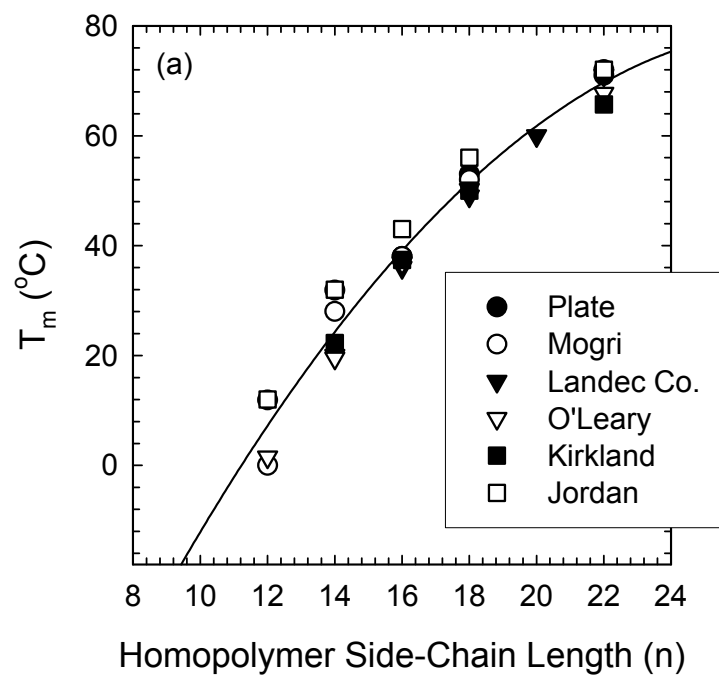


Figure 5.1 T_m (a) and ΔH_f (b) of n-alkyl acrylate homopolymers versus side-chain length (n). Data from this work as well as from the literature.[3, 5-7, 9-12]

Table 5.1
Melting Temperature, Heat of Fusion, and Molecular Weight Data Measured for
Poly (n-alkyl acrylate) Homopolymers

Homopolymer	Side-Chain Length (n)	T _m (°C)	ΔH _f (kJ/mol)	Polymer M _w	Repeat unit M _w (g/mol)
PA 6	6	-	-	105000	156
PA 10	10	-	-	112000	212
PA 12	12	1.5	9.3	165000	240
PA 14	14	19.5	14.1	245000	268
PA 16	16	36.4	23.1	161000	296
PA 18	18	50.2	31.3	106000	325
PA 22	22	67.7	45.8	(a)	381

^(a) M_w could not be measured since PA 22 is no soluble in THF

measured on the second heating. The heat of fusion, ΔH_f, is shown in units of kJ/mol of monomer repeat units because these units give a useful linear relationship with side-chain length; whereas, the heat of fusion per unit mass of polymer is clearly non-linear in n. The results from this study agree well with previous reports in the literature.[3, 5-7, 10-12] Slight deviations among the various data sets may be attributed to experimental methods of analysis including thermal history or baseline approximation.

Regression lines were drawn through the T_m and ΔH_f data in Figure 5.1 for subsequent use in comparing copolymers. The melt temperatures of the homopolymers increase with side-chain length following a second order polynomial trend. The heats of fusion increased linearly with side-chain length when plotted in units of kJ/mol.

$$\Delta H_f = 3.54n - 33.4 \quad (5.1)$$

The linear equation goes to ΔH_f = 0 at a value of n equal to 9.4. The minimum side-chain length needed for crystallization reported in the literature ranges from 8-9.2.[3, 6]

According to Equation 5.1, PA 10 should be a crystallizeable polymer, while all polymers with fewer carbons in their side chains should not. Due to equipment limitations, we are

not able to make measurements on PA 10 in the crystalline state. Since this value of n is near the border of crystallizeable - non-crystallizeable polymers, copolymers containing the alkyl acrylate with $n = 10$ have been included in both categories.

5.3 Crystalline / Crystalline Combination Copolymers

Thermal analyses of copolymers were performed in the same manner as for homopolymers. Table 5.2 lists the melt temperature, heat of fusion, and molecular weight of the copolymers examined in this study. An average side-chain length $\langle n \rangle$ was defined as follows

$$\langle n \rangle = (x_1)(n_1) + (1 - x_1)(n_2) \quad (5.2)$$

where x_1 is the mole fraction of monomer one having n_1 side-chain carbons and $x_2 = (1 - x_1)$ is the mole fraction of monomer 2 having n_2 side-chain carbons; this quantity is useful for analyzing the data.

In principle, crystallization of copolymers, whether the crystallizing units are in the main chains or side chains, can form two extremes of behavior. The most common is where the presence of the minor component causes melting point depression of the major component as described by simple theories.[13-16] Typically when both units are capable of crystallizing, the melting point shows depressions on both sides of the T_m -composition diagram resembling a eutectic while crystallinity may go to zero, or reach a minimum in the mid-composition range. The other is where the two units co-crystallize with the typical signature of a steady progression of the melting point and heat of fusion from that of the one homopolymer to that of the other as composition changes.[15-22]

As will be

Table 5.2
Melting Temperature, Heat of Fusion, and Molecular Weight Data Measured for
Poly (n-alkyl acrylate) Crystallizeable / Crystallizeable Copolymers

Copolymer	Mol % monomer 1^(a)	Average Side- Chain Length <n>	T_m (°C)	ΔH_f (kJ/mol)	Copolymer M_w
P(A10-co-A14)	75	11	-	-	215000
	50	12	3.7	4.8	141000
	25	13	11.9	9.5	191000
P(A10-co-A18)	75	12	-1.1	7.6	185000
	50	14	21.2	11.8	235000
	25	16	38.5	21.3	197000
P(A14-co-A18)	75	15	25.7	17.4	468000
	50	16	32.7	20.8	119000
	25	17	41.7	24.5	109000
P(A10-co-A22)	75	13	6.4	10.3	192000
	50	16	35.5	13.3	118000
	25	19	50.5	21.0	138000
P(A12-co-A22)	75	14.5	21.0	12.9	338000
	50	17	44.2	17.0	102000
	25	19.5	56.0	25.4	108000
P(A14-co-A22)	75	16.5	30.2	16.0	161000
	50	19	44.0	18.1	115000
	25	21.5	54.9	27.8	108000
P(A16-co-A22)	75	17.5	40.7	21.5	121000
	50	19	48.2	24.6	116000
	25	20.5	57.2	30.2	100000
P(A18-co-A22)	75	19	53.4	28.2	100000
	50	20	56.9	31.9	111000
	25	21	61.5	35.6	173000

^(a) Note that monomer 1 refers to the first monomer listed in the copolymer, i.e., for P(A14-co-A18), monomer 1 refers to A14

shown in the following, this is the more common situation in the copolymers described here. However, there are clearly intermediate situations that are somewhat more difficult to categorize. The following gives a rather detailed analysis of the thermal properties of the current copolymers as this is useful for interpreting their gas permeation characteristics.

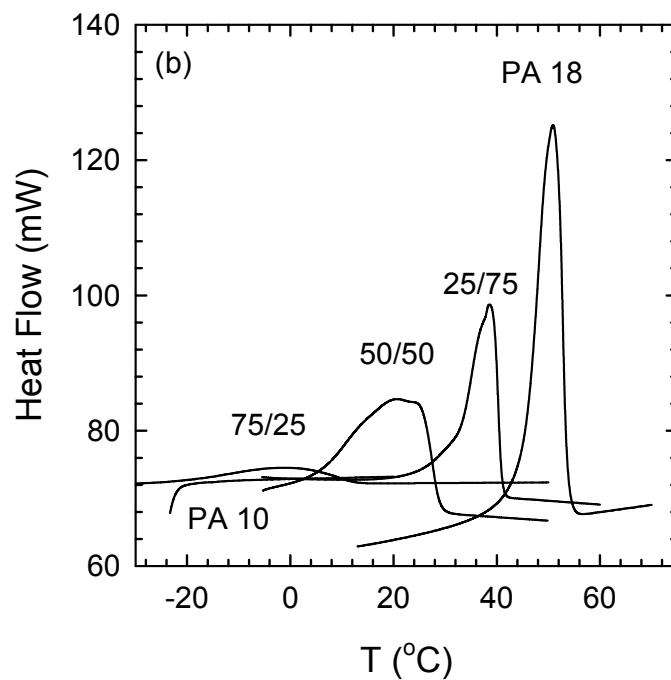
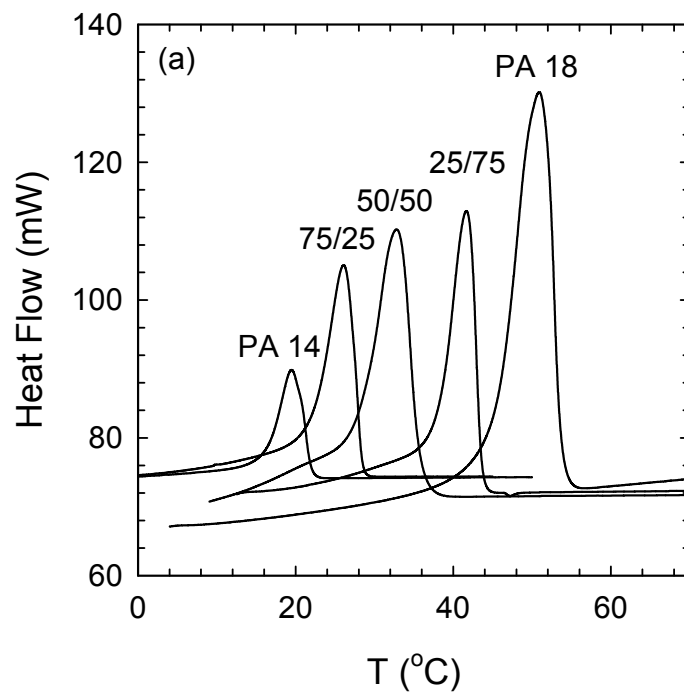


Figure 5.2 DSC endotherms for various compositions of P(A14-co-A18) (a) and P(A10-co-A18) (b) copolymers.

Sample DSC endotherms for two copolymers systems are shown in Figures 5.2a and b. The endotherms qualitatively reflect the distribution of crystallite sizes. In general, as the side-chain lengths for n-alkyl acrylate homopolymers increase, the more methylene units in each repeat unit are able to crystallize causing larger crystallites to form having a more uniform size distribution and a higher melting point. As the side chains become shorter, the number of carbons entering the lattice decreases causing a broader distribution of smaller crystallites and a correspondingly lower melting point. Figure 5.2a shows endotherms for copolymers of P(A14-co-A18) polymerized in molar increments of 25% ranging from 0-100. There is a steady progression of peak location, width, and height for the copolymers ranging from PA 14 to PA 18 consistent with the progression of a corresponding series of n-alkyl acrylate homopolymers. This indicates that the two types of units co-crystallize as expected from the literature.[5] Figure 5.2b shows endotherms for copolymers of P(A10-co-A18) in molar increments of 25% ranging from 0-100. As explained earlier PA 10 does not crystallize in the temperature range examined here. In principle, A10 units have perhaps only one carbon in its side chain for co-crystallizing with A18 units. There is also a distinct possibility that the A10 units do not co-crystallize with the A18 units at all, but rather act as a non-crystallizeable spacer that more appropriately might be thought of as a melting point depression phenomenon. This possibility will be addressed more thoroughly in Section 5.4. In any case, as the concentration of A10 in the copolymer increases, the crystallites become smaller and have a broader distribution of size of perfection reflected in the lowering and broadening of the endotherms with increasing A10 concentration.

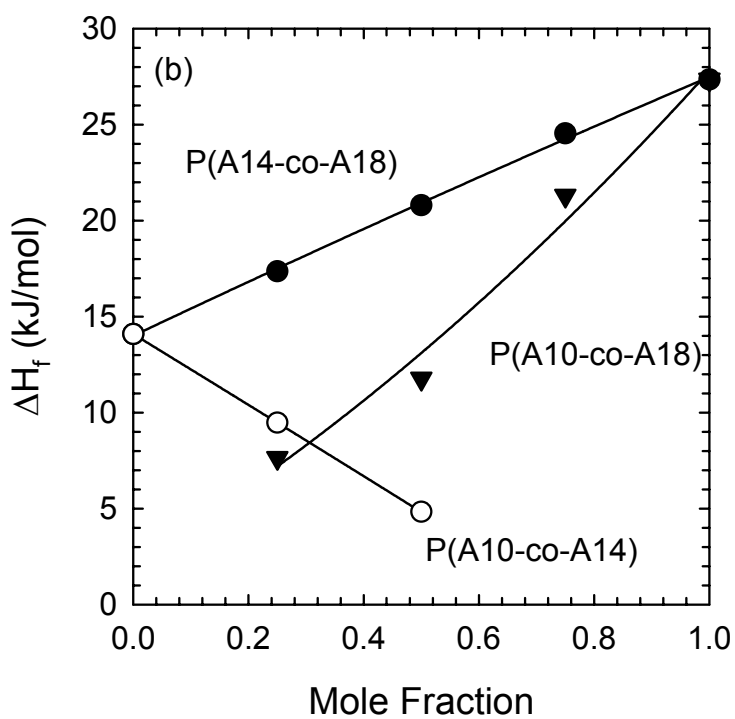
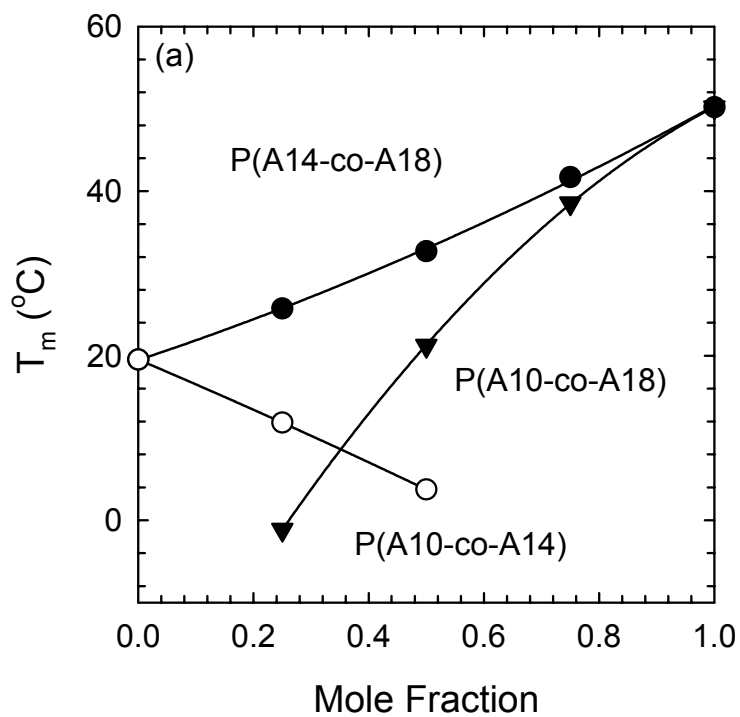


Figure 5.3 Dependence of T_m (a) and ΔH_f (b) on copolymer composition for materials based on monomers A10, A14, and A18.

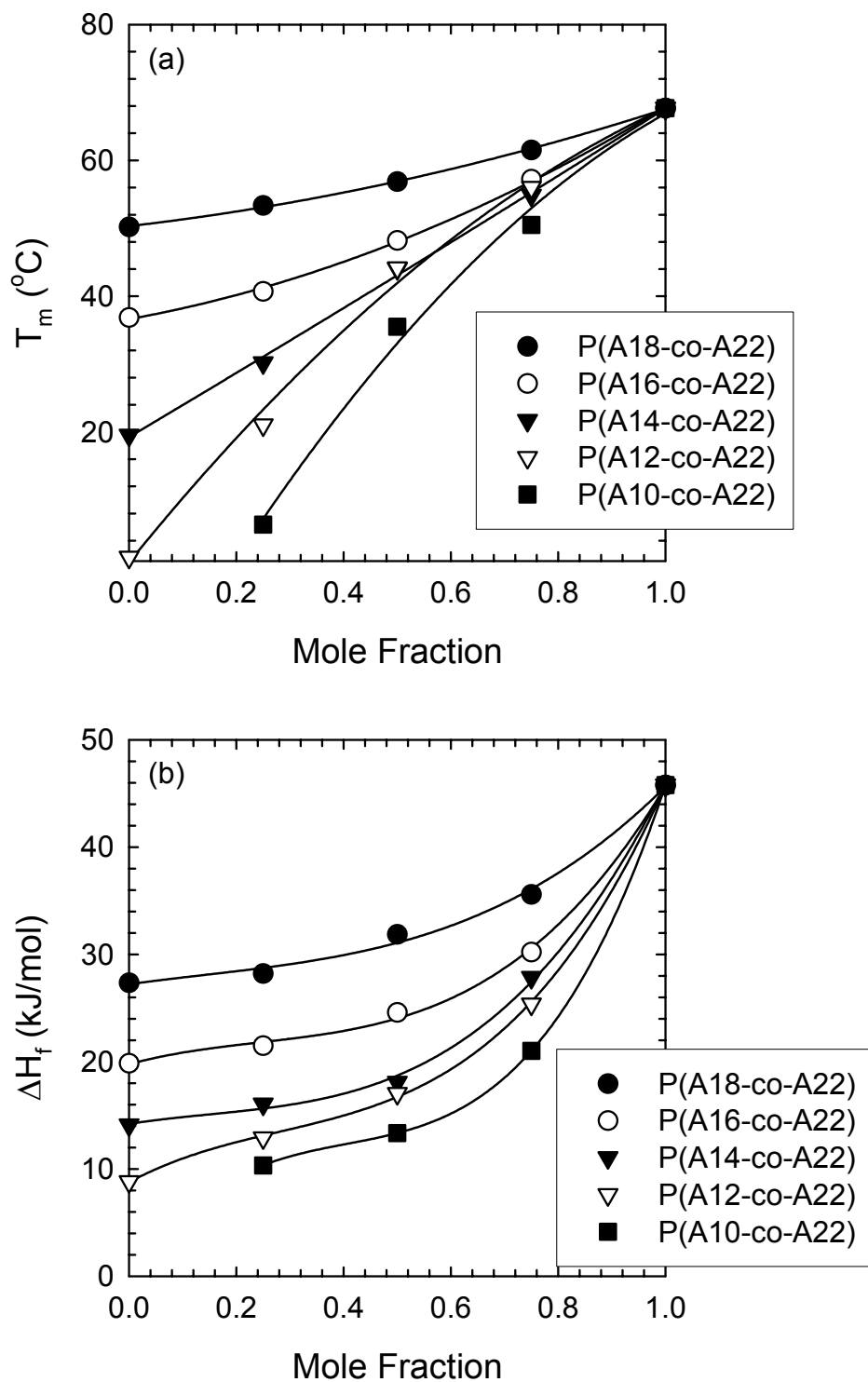


Figure 5.4 Dependence of T_m (a) and ΔH_f (b) on copolymer composition for materials based on monomer A22 and other n-alkyl acrylate monomers.

Figures 5.3 and 5.4 summarize the melting behavior of a number of copolymers of n-alkyl acrylates, where the n for each is always 10 or higher, by plotting T_m and ΔH_f versus the copolymer composition. For all cases when n is larger than 10 for each n-alkyl acrylate monomer, we see a characteristic signature of co-crystallization in that both T_m and ΔH_f are monotonic functions of composition. The copolymer melting point may lie slightly above or below the line connecting the homopolymer values when plotted versus mole fraction depending on the values of n_1 and n_2 . On the other hand, ΔH_f always lies below the line suggesting that copolymerization generally leads to less crystallinity than expected; however, the extent of this departure depends on n_1 and n_2 .

For copolymers containing A10, the T_m and ΔH_f versus composition relations are also monotonic but tend to show slightly different patterns particularly as the other monomer unit has shorter side chains. For example, if we extrapolate the homopolymer T_m data in Figure 1a to $n = 10$ we get $T_m \approx -12^\circ\text{C}$. The T_m data for A10 copolymers with A18 and A22 appear to be going to a much lower intercept, in the 100% A10 limit, than this. Copolymers of A10 and A14 show $\Delta H_f = 0$ even when there is 25 mol% A14 in the material, see Figure 5.3b. On the other hand, copolymers of A10 with A18 and A22 seem to be headed to a finite ΔH_f in the limit of 100% A10.

The copolymer thermal property relationships are more critically analyzed with respect to the behavior of homopolymers of similar side-chain lengths in Figures 5.5 – 5.7, using plots versus n for the homopolymer and $\langle n \rangle$ for the copolymers. Figure 5.5 compares T_m and ΔH_f results for P(A14-co-A18) copolymers with those of homopolymers. The open circles represent the homopolymer data while the closed circles correspond to the results for the copolymers. For this system, both T_m and ΔH_f

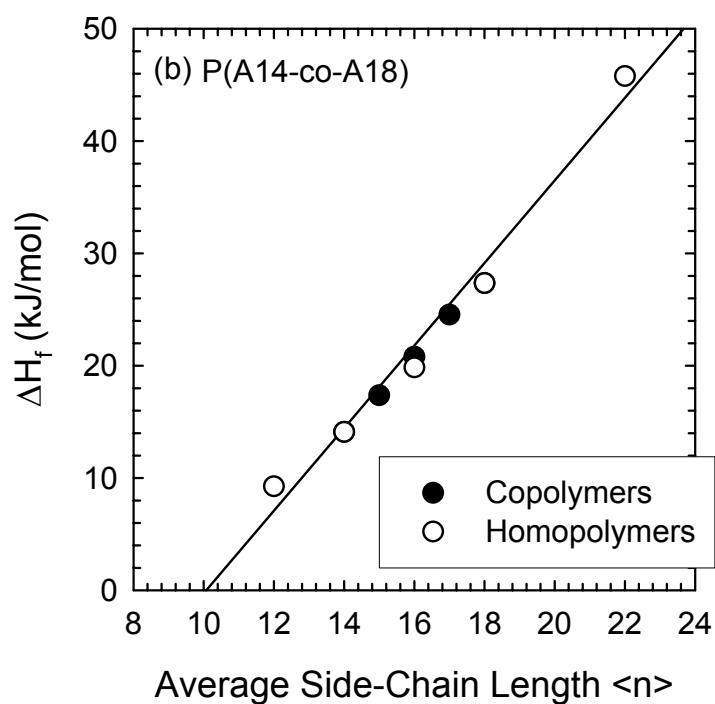
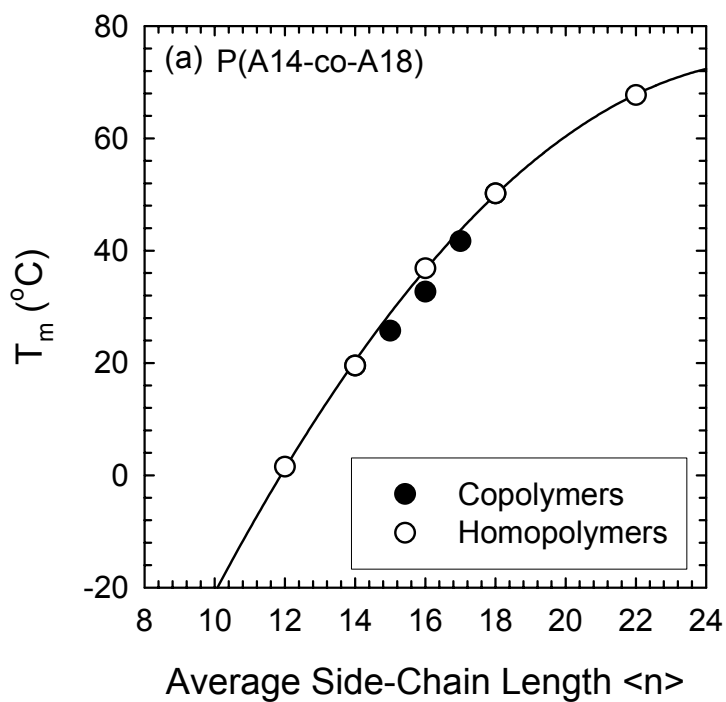


Figure 5.5 Homopolymer and P(A14-co-A18) copolymer comparisons for melting temperature (a) and heat of fusion (b) shown as a function of the average side-chain length of the copolymer or side-chain length of the homopolymer.

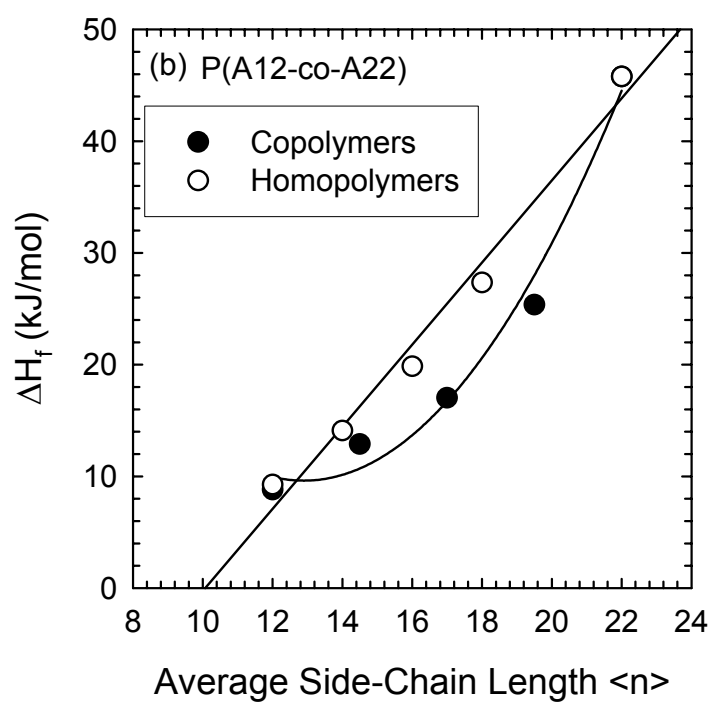
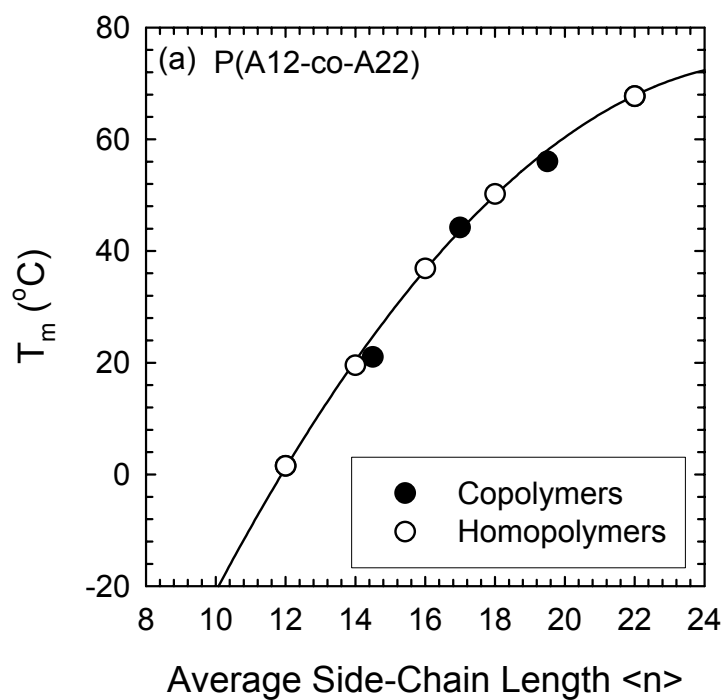


Figure 5.6 Homopolymer and P(A12-co-A22) copolymer comparisons for melting temperature (a) and heat of fusion (b) shown as a function of the average side-chain length of the copolymer or side-chain length of the homopolymer.

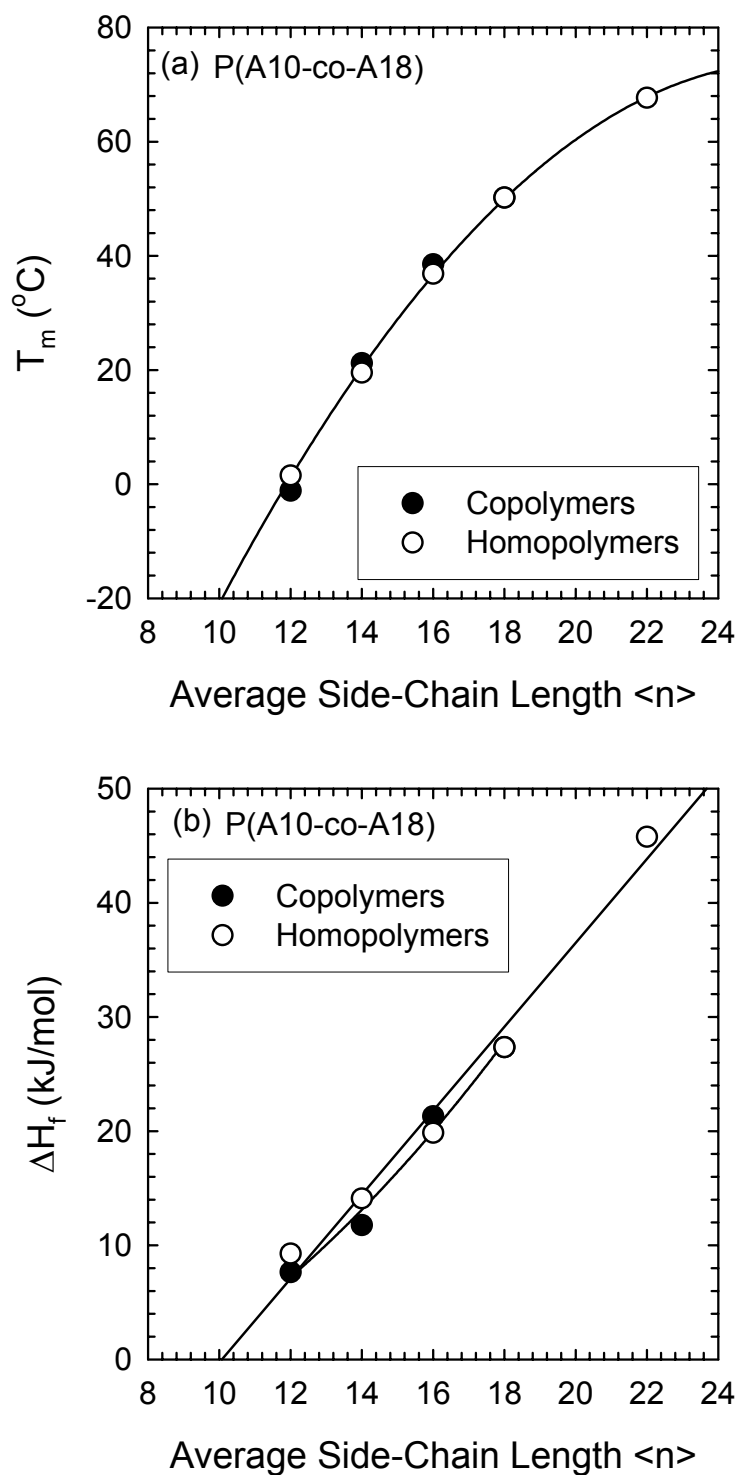


Figure 5.7 Homopolymer and (A10-co-A18) copolymer comparisons for melting temperature (a) and heat of fusion (b) shown as a function of the average side-chain length of the copolymer or the side-chain length of the homopolymer.

track closely along the trend established by the homopolymers. The side chains of the two monomers are of similar lengths, both crystallize, and their endotherms from Figure 5.2a progress in similar shape and size as those of the homopolymers. It is understandable that copolymers of these two similar monomers would thermally behave like a homopolymer with an equivalent side-chain length. Similar plots are shown in Figures 5.6 and 5.7 for P(A12-co-A22) and P(A10-co-A18). In both systems, the copolymer T_m results are also in close agreement with the homopolymer T_m results; however, the copolymer ΔH_f results are significantly lower than the line established for the homopolymers. Both monomer units in the P(A12-co-A22) system are readily crystallizeable while for the P(A10-co-A18) copolymers the A10 units are on the border of being crystallizeable. Both systems have monomers of significantly different side-chain lengths. Their endotherms also resemble those shown in Figure 5.2b with smaller, broader melting peaks than homopolymers or the P(A14-co-A18) copolymer system. These copolymers are able to co-crystallize because of the conformational freedom in the amorphous backbone allowing side-chains of different lengths to crystallize in the same lattice. In the case of P(A12-co-A22), both monomers contain reasonably long side-chains in addition to having a side-chain length difference of 10 carbons. This difference may impede and restrict the conformational freedom of the backbone inhibiting the overall crystallinity of the copolymer system as seen by the depressed heat of fusion relative to homopolymers. On the other hand, the P(A10-co-A18) system does not show the extent of the ΔH_f depression as the P(A12-co-A22) system despite the fact that the A10 units do not readily crystallize on their own.

5.4 Crystalline / non-Crystalline Copolymers

Several prior studies have reported on the thermal properties of poly (n-alkyl acrylate) copolymers having one crystallizeable and one non-crystallizeable comonomer.[2, 3, 23-27] Greenburg and Alfrey reported an early study on the melting temperatures of P(A1-co-A18) and P(A1-co-A16).[23] Jordan later performed a thorough investigation of copolymers of methyl, ethyl, butyl, and octyl acrylates with A18.[27] In his study, Jordan described the acrylates that were unable to crystallize as ‘spacers’ because they depress the T_m and ΔH_f by interrupting the ordered long side chains of A18 without crystallizing in the lattice.[27]

In main-chain copolymers, a non-crystallizeable comonomer concentration as low as 25 mol% may entirely prevent any crystallization.[3, 27] Side-chain copolymers, with flexible amorphous backbones and proximal side-chain carbons, can crystallize when the non-crystallizeable comonomer content is as high as 90 mol %.[3] Non-crystallizeable comonomers affect the copolymer in several ways. First, they reduce the overall concentration of crystallizeable side-chains, second, they interrupt and impede the crystallizeable side chains from forming perfect crystals and, third, they force the amorphous backbone to contort to accommodate the side chains crystallizing around the non-crystallizeable side-chains; this hinders the conformational freedom of the backbone.[3, 25-27] These effects reduce the overall crystallinity which, in turn, depresses the T_m and ΔH_f of the copolymers.

DSC thermograms for the copolymer systems P(A6-co-A22), P(A10-co-A14), and P(A10-co-A18) are shown in Figure 5.8 and 5.2b. All data were obtained on a

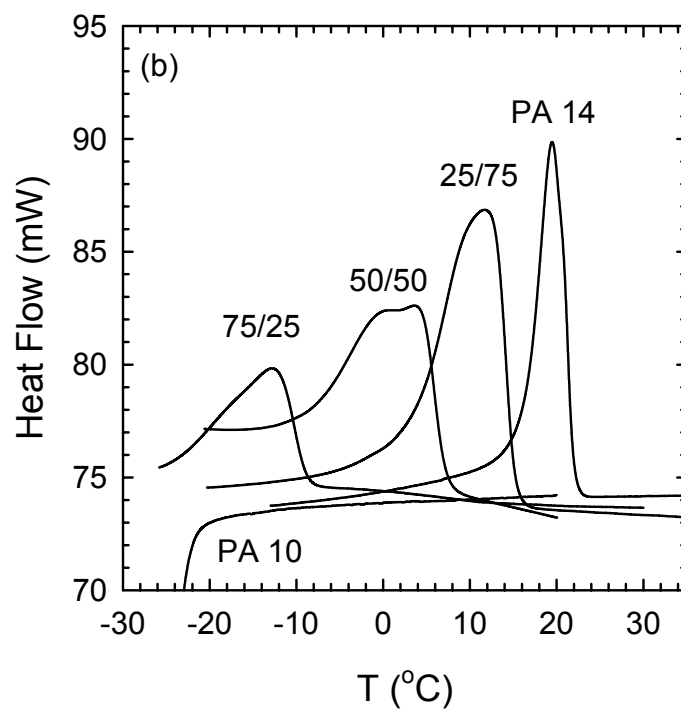
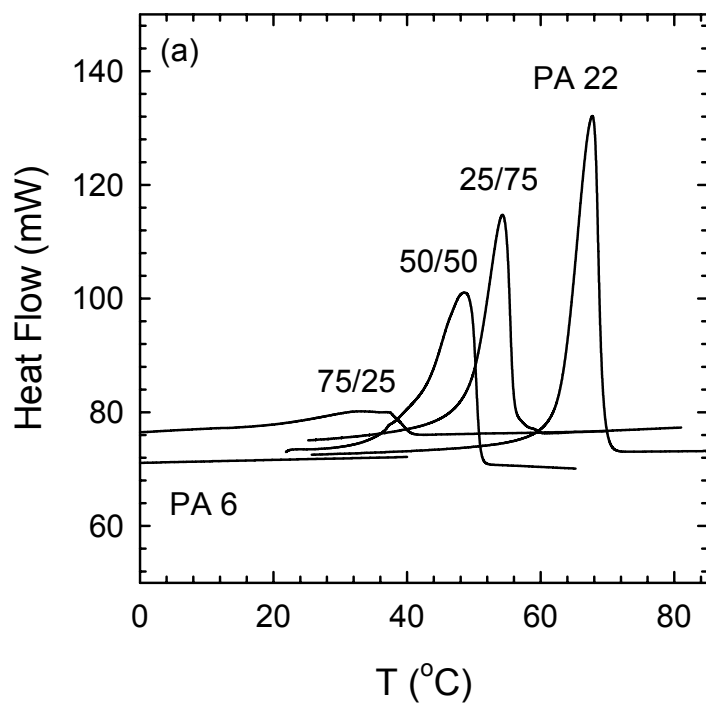


Figure 5.8 DSC scans for various compositions of P(A6-co-A22) (a) and P(A10-co-A14) (b) copolymers.

second heating. Just as with the crystalline / crystalline copolymers, these copolymers show a single melting peak that grows broader and smaller in area as the concentration of the shorter side-chain length comonomer increases resulting in an overall increase in smaller crystals. In the case of P(A6-co-A22), the A6 comonomer inhibits the crystallization of A22 units by physically separating them which causes the formation of small and more imperfect crystals. As discussed earlier, the crystal lattices of the P(A10-co-A14) and P(A10-co-A18) copolymer systems are ambiguous because A10 is technically crystallizeable at low enough temperatures; however, the ability for this comonomer to substantially crystallize into a lattice with other comonomers is

Table 5.3
Melting Temperature, Heat of Fusion, and Molecular Weight Data Measured for Poly (n-alkyl acrylate) Crystallizeable / non-Crystallizeable Copolymers

Copolymer	Mol % monomer 1^(a)	Average Side- Chain Length <n>	T_m (°C)	ΔH_f (kJ/mol)	Copolymer M_w
P(A10-co-A14)	75	11	-	-	215000
	50	12	3.7	4.8	141000
	25	13	11.9	9.5	191000
P(A10-co-A18)	75	12	-1.1	7.6	185000
	50	14	21.2	11.8	235000
	25	16	38.5	21.3	197000
P(A6-co-A22)	75	10	33.5	1.6	186000
	50	14	48.7	20.6	210000
	25	18	54.2	33.0	237000
P(A8-co-A22)	75	11.5	-	-	255000
	50	15	47.2	18.2	149000
	25	18.5	57.0	33.2	187000
P(A10-co-A22)	75	13	6.4	10.3	201000
	50	16	35.5	13.3	253000
	25	19	50.5	21.0	189000

^(a) Note that monomer 1 refers to the first monomer listed in the copolymer, i.e., for P(A10-co-A18), monomer 1 refers to A10

questionable. The thermograms in Figures 5.8b and 5.2b for P(A10-co-A14) and P(A10-co-A18) are similar to those for P(A6-co-A22) in Figure 5.8a. These figures suggest that both types of copolymer systems have reduced crystallinity and crystallite size with increasing composition of the smaller comonomer.

The data points and solid lines in Figures 5.9a and b represent the T_m and ΔH_f versus the mole fraction of A22 units in copolymers with A6 and A8, also see Table 5.3. The dashed lines represent the results for copolymers of A22 with other n-alkyl acrylates with n ranging from 10 to 18, i.e., crystalline / crystalline copolymers reported in Figures 5.4a and b. The values of T_m and ΔH_f for P(A6-co-A22) and P(A8-co-A22) copolymers are nearly the same at a given mole fraction of A22; the trend or progression with n for these copolymers is clearly different than that for the series where $n > 10$, i.e.; where co-crystallization occurs. The T_m curves for $n = 6$ and 8 are in about the same range as seen for copolymers of A22 with n-alkyl acrylate monomers having $n = 14 - 16$. On the other hand, the ΔH_f curves crosses the lines shown for the systems that co-crystallize and approaches zero at just over 20 mol % of A22. While the P(A6-co-A22) and P(A8-co-A22) copolymers show nearly identical T_m and ΔH_f trends, P(A10-co-A22) falls in the progression observed for crystallizeable-crystallizeable copolymers. The latter shows the largest depression in ΔH_f because of the differences in side-chain lengths between A10 and A22 entering the lattice. This indicates that when copolymerized with A22, A10 behaves like a crystallizeable polymer in terms of T_m and ΔH_f . This also appears to be the case in Figures 5.3a and b for P(A10-co-A14) and P(A10-co-A18). The difference in behavior between P(A10-co-A22) and P(A8-co-A22) is the most dramatic considering A8 and A10 only differ by two side-chain carbons.

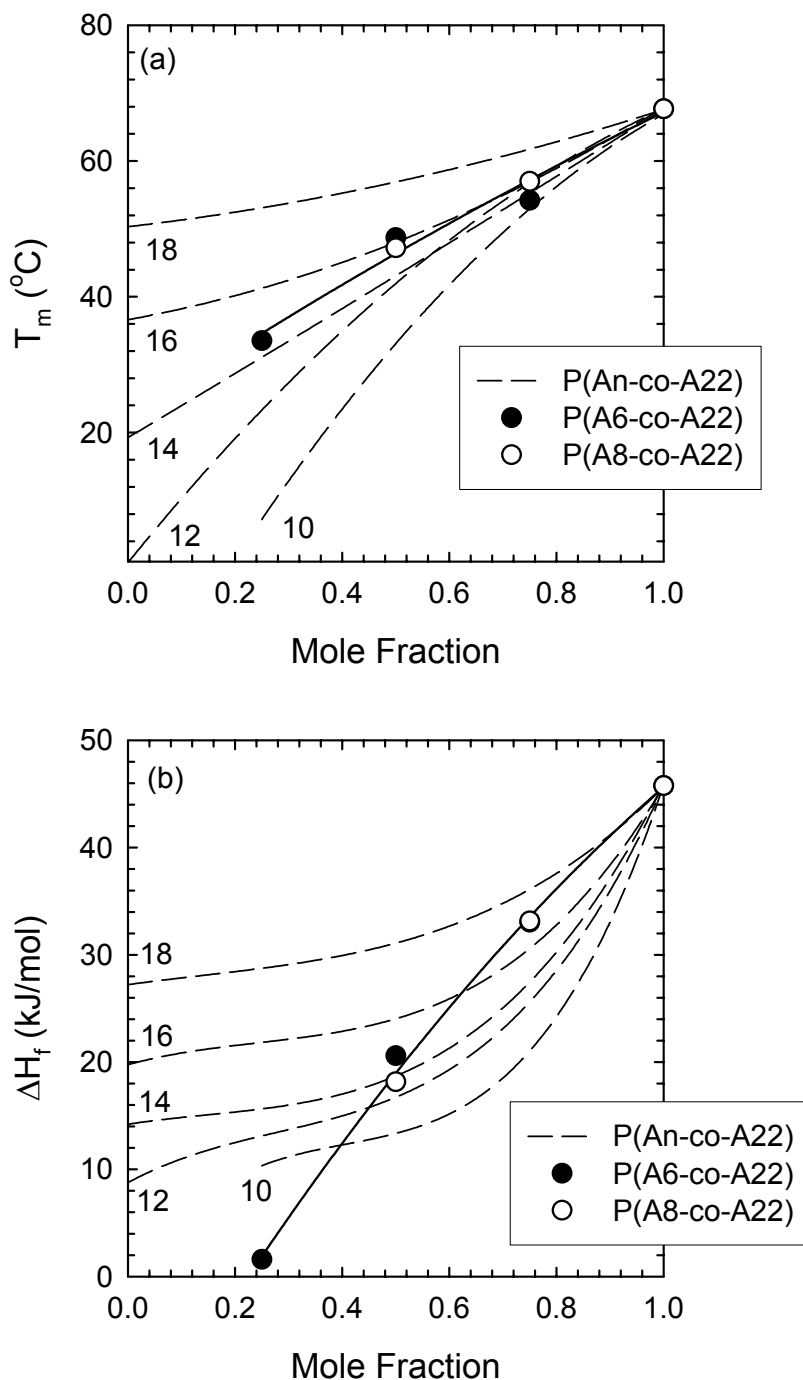


Figure 5.9 Dependence of T_m (a) and ΔH_f (b) on copolymer composition for materials based on A22 and other n-alkyl acrylate monomers. The data points shown are for copolymers containing spacers, P(A6-co-A22) and P(A8-co-A22), while the dashed lines represent copolymers containing two crystallizeable comonomers, P(An-co-A22).

The T_m and ΔH_f values for copolymers can be plotted as a function of the average number of crystallizeable side-chain carbons $\langle n_{cr} \rangle$, as shown in Figures 5.10a and b.

This quantity is defined as follows

$$\langle n_{cr} \rangle = (n_1 - 9.4)x_1 + (n_2 - 9.4)x_2 \quad (5.3)$$

where n_1 and n_2 are the number of carbon atoms in the longer and shorter side-chain lengths, respectively, and x_1 and x_2 are the mole fractions of these monomer units.

When comonomer 2 has fewer than 9.4 carbon atoms in its side chain, i.e., $(n_2 - 9.4) < 0$, this term in Equation 5.3 is set equal to zero, since a comonomer with fewer than 9.4 carbons cannot crystallize. This is a convenient way of comparing the T_m and ΔH_f of the different types of copolymers based on the average number of carbons participating in the crystal lattice rather than all the side-chain carbons. Although T_m and ΔH_f for all copolymer systems decrease as n_{cr} decreases, those for the A6 and A8 copolymer systems decrease less rapidly than the P(An-co-A22) systems in Figure 5.10a. This is the difference between systems that co-crystallize versus systems that experience melting point depression or the ‘spacer’ effect in the terminology of Jordan.[27] The comonomers capable of co-crystallizing with A22 units have a greater effect on the T_m than the ‘melting point’ depression caused by either A6 or A8. The A6 and A8 spacers do not have as great an effect on T_m since they simply impede crystal formation rather than actually altering the basic nature of the crystal itself. These effects are also apparent in Figure 5.10b where it is shown that ΔH_f of all copolymers decrease with decreasing number of crystallizing side-chain carbons; reducing the composition of A22 decreases the number of carbons capable of crystallizing. The ΔH_f of P(A6-co-A22) and P(A8-co-

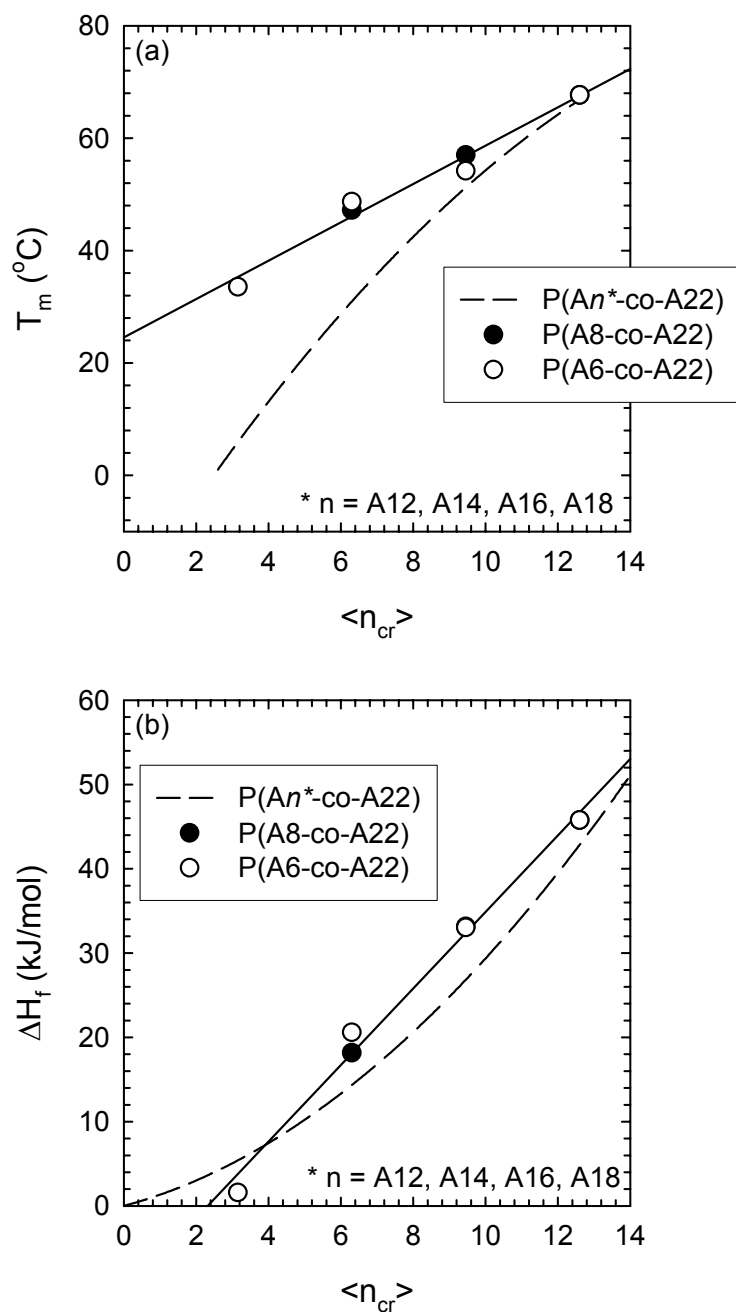


Figure 5.10 Dependence of T_m (a) and ΔH_f (b) on number of crystallizeable side-chain carbons, $\langle n_{cr} \rangle$, for materials based on monomer A22 and other n-alkyl acrylate monomers. The data points are for copolymers containing spacers, P(A6-co-A22) and P(A8-co-A22), while the dashed lines represent copolymers with two crystallizeable comonomers, P(An-co-A22). The average number of crystallizeable carbons $\langle n_{cr} \rangle$, was calculated with Equation 5.3.

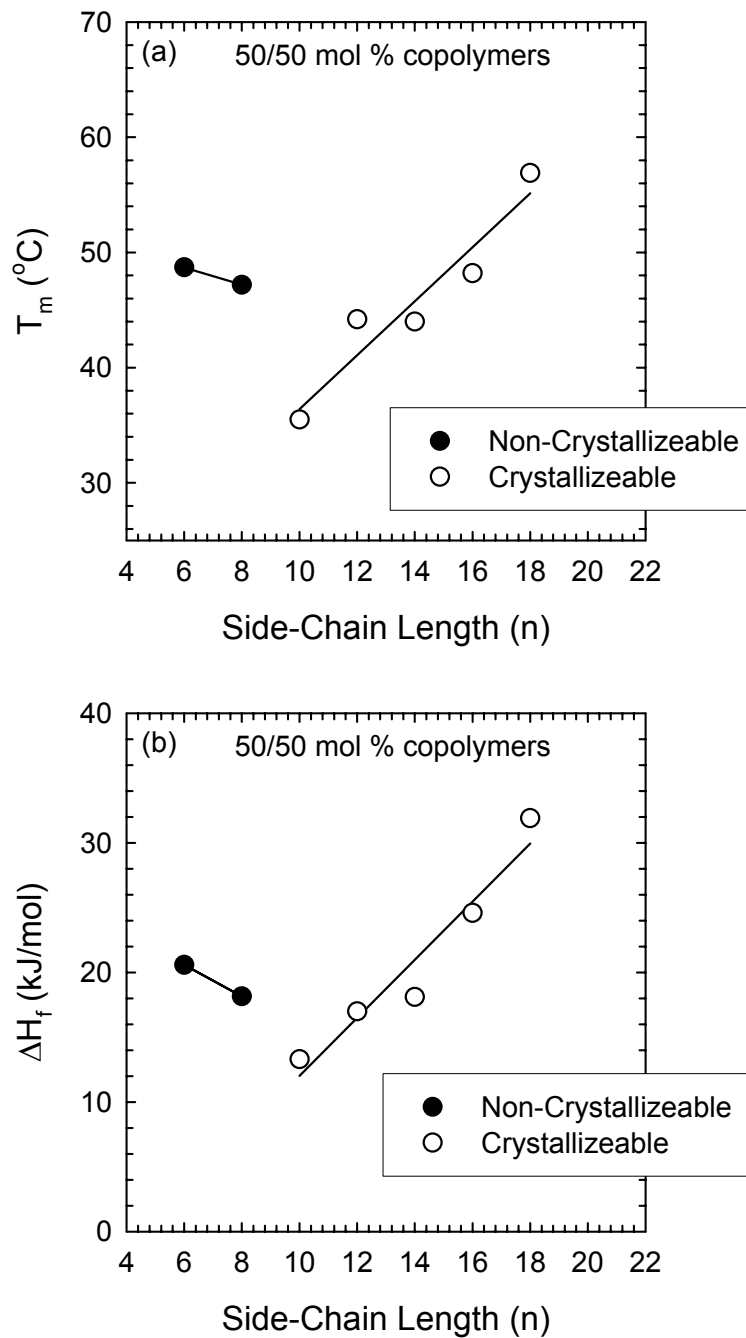


Figure 5.11 Dependence of T_m (a) and ΔH_f (b) of copolymers of P(An-co-A22) with 50/50 mol% based on the side-chain length of the comonomer with the shorter alkyl unit. The solid points are for copolymers containing spacers, P(A6-co-A22) and P(A8-co-A22), while the open points are for copolymers with two crystallizeable comonomers, P(An-co-A22), i.e., where $n > 10$.

A22) more or less decrease linearly with n_{cr} and goes to zero at n_{cr} just over 2. Note that the side-chain carbons of A22 are the only units that can form crystals. For the copolymers of A22 with n-alkyl acrylates with $n = 10$ to 18, ΔH_f shows a non-linear relationship that goes to zero at $n_{cr} = 0$. For $n_{cr} < 4$, ΔH_f for the co-crystallizeable copolymers are lower than for those where co-crystallization does not occur. These trends reflect both the reduction in concentration of crystallizeable side chains as well as the physical differences between altering the nature of the crystal by co-crystallization versus restricting the ability of the copolymer to crystallize by reducing the conformational freedom of the backbone and thereby forcing the A22 side chains to form less perfect crystals. These trends are similar to observations reported by Jordan for the P(A8-co-A18) system.[27]

The consequences of these differences are shown by plots of T_m and ΔH_f for 50/50 mol % copolymers of A22 versus the number of carbons in the side chain of the shorter comonomer shown in Figure 5.11. Copolymers of A22 with the non-crystallizeable comonomers A6 and A8 are represented by the solid points while those with the crystallizeable comonomers, A10, A12, A14, A16, and A18, are represented by open points. The T_m and ΔH_f curves show exactly opposite trends with n for the two types of systems with a minimum in appearing at $n \approx 10$ for both T_m and ΔH_f .

5.5 Conclusions

The thermal properties of poly (n-alkyl acrylate) homopolymer and copolymer systems were evaluated in terms of their average side-chain lengths. The thermal properties, T_m and ΔH_f of the homopolymers show a direct correlation with side-chain

length. Copolymers containing two crystallizeable comonomers exhibit isomorphic behavior with similar relationships between T_m and ΔH_f and the average side-chain length as seen for the homopolymers. They did, however, exhibit some depression in ΔH_f relative to that of the homopolymers which increases as the difference in the number of carbons in the side chains of the two monomers increases. This qualitatively measures the reduction in crystallite size for the copolymers as a function of composition.

The non-crystallizeable comonomers affect the copolymer by interrupting and impeding the crystallizeable side chains from forming perfect crystals and impinging order on the amorphous backbone. The formation of smaller and less perfect crystals causes a 'depression' in the melting temperature. Unlike copolymers with two crystallizeable comonomers that enter the lattice and alter the basic nature of the crystal, non-crystallizeable comonomers only impede crystal formation; therefore, the co-crystallizing side chains can affect the T_m and ΔH_f more than the latter.

PA 10 is an unusual polymer in that its side chains are on the border of being crystallizeable; therefore, the thermal properties of several copolymers containing A10 were evaluated and compared to other copolymers. It was determined that the T_m and ΔH_f for copolymers of P(A6-co-A22) and P(A8-co-A22) exhibit melting point depression caused by the non-crystallizeable side chains limiting crystal formation, while the A10 in P(A10-co-A22) behaved like a crystallizeable comonomer which enters the lattice and alters the nature of the crystal.

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Chapter 6

Structural Properties of Poly (n-alkyl acrylates)

6.1 Introduction

Chapter 5 described the basic relationships among copolymer side-chain length, composition, and crystallite distribution using thermal properties measured with DSC. This chapter is dedicated to describing similar relationships for crystalline ($T < T_m$) and amorphous ($T > T_m$) poly (n-alkyl acrylates) using small angle X-ray scattering (SAXS) to measure d-spacings. Simple models are developed for the side-chain packing of the homopolymers and compared to copolymers with two crystallizeable comonomers as well as a single crystallizeable comonomer.

6.2 Homopolymers

Small angle X-ray scattering (SAXS) was performed on the homopolymers and copolymers in both the amorphous ($T > T_m$) and crystalline ($T < T_m$) states. Figure 6.1 shows typical plots of scattering intensity versus the scattering angle, 2θ , for both states. Wide angle X-ray diffraction has revealed that poly (n-alkyl acrylates) are paraffin-like in their ability to form hexagonal crystal lattice structures. SAXS provides insight about how the thicknesses of these crystals change with n ; unfortunately, this technique provides no information about the lateral dimensions of the crystal which would be needed to determine the aspect ratio that affects permeation in the solid form. Hsieh and Morawetz reported a SAXS study using crystalline fibers of PA 16 and PA 18. They found three scattering peaks; two diffuse peaks at 16 and 47 Å as well as a weak, well-

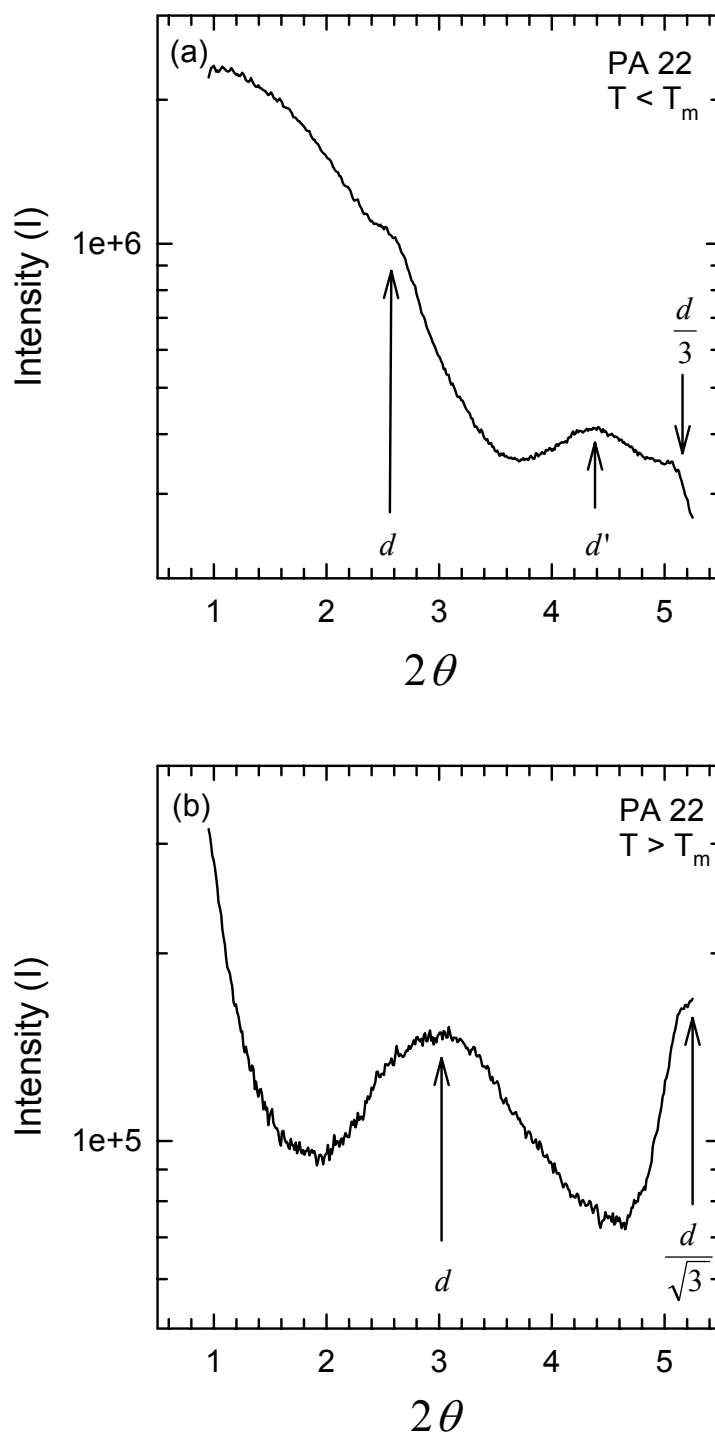


Figure 6.1 Typical SAXS intensity versus 2θ plots for poly (n-alkyl acrylate) homopolymers. Data shown for crystalline PA 22 ($T < T_m$) (a) and molten PA 22 ($T > T_m$) (b).

defined, sharp peak at 28 Å for PA 18.[1, 2] The crystalline side-chains pack similar to a lamellar system where the multiple scattering peak spacings progress in the following ratios 1:1/2:1/3:1/4:...[3-5] In some systems, some of the peaks may be missing owing to a minimum in the form factor scattering.[3]. It is proposed that the similarly shaped 16 and 47 Å peaks characterize the same lamellar-like packing formation where the 16 Å peak appears to be a third order reflection of the 47 Å peak; the second order peak appears to be missing. The peak at 28 Å is much weaker and shaped differently and does not appear to be a multiple of the other two peaks, therefore, it is interpreted as independent of the other peaks. These two types of reflections suggest two different packing structures: an end-to-end packing of the side chains (see Figure 6.2a) which gives rise to the 16 and 47 Å peaks and an inter-digitating structure (see Figure 6.2b) that leads to the 28 Å peak.[1, 2] Later work by Plate showed that poly (n-alkyl acrylates) exhibit an increase in d-spacings for both packing structures as the side chains become longer.[6] Figure 6.1a shows the SAXS intensity versus 2θ for crystalline PA 22 examined over a d-spacing region of 16-100 Å. The crystalline sample of PA 22 illustrates the two types of packing reported in the literature. In this figure, the peaks labeled d and $\frac{d}{3}$ correspond to the 47 and 16 Å peaks Hsieh observed for PA 18 while d' corresponds to the equivalent peak for PA 22 as the 28 Å peak for PA 18. It should also be noted that the 16 Å peak observed for PA 18 is at approximately 17.5 Å for PA 22 and can, therefore, be seen in Figure 6.1a; this was the only polymer whose $\frac{d}{3}$ peak was within the experimental range analyzed here. The relationship between the spacings for

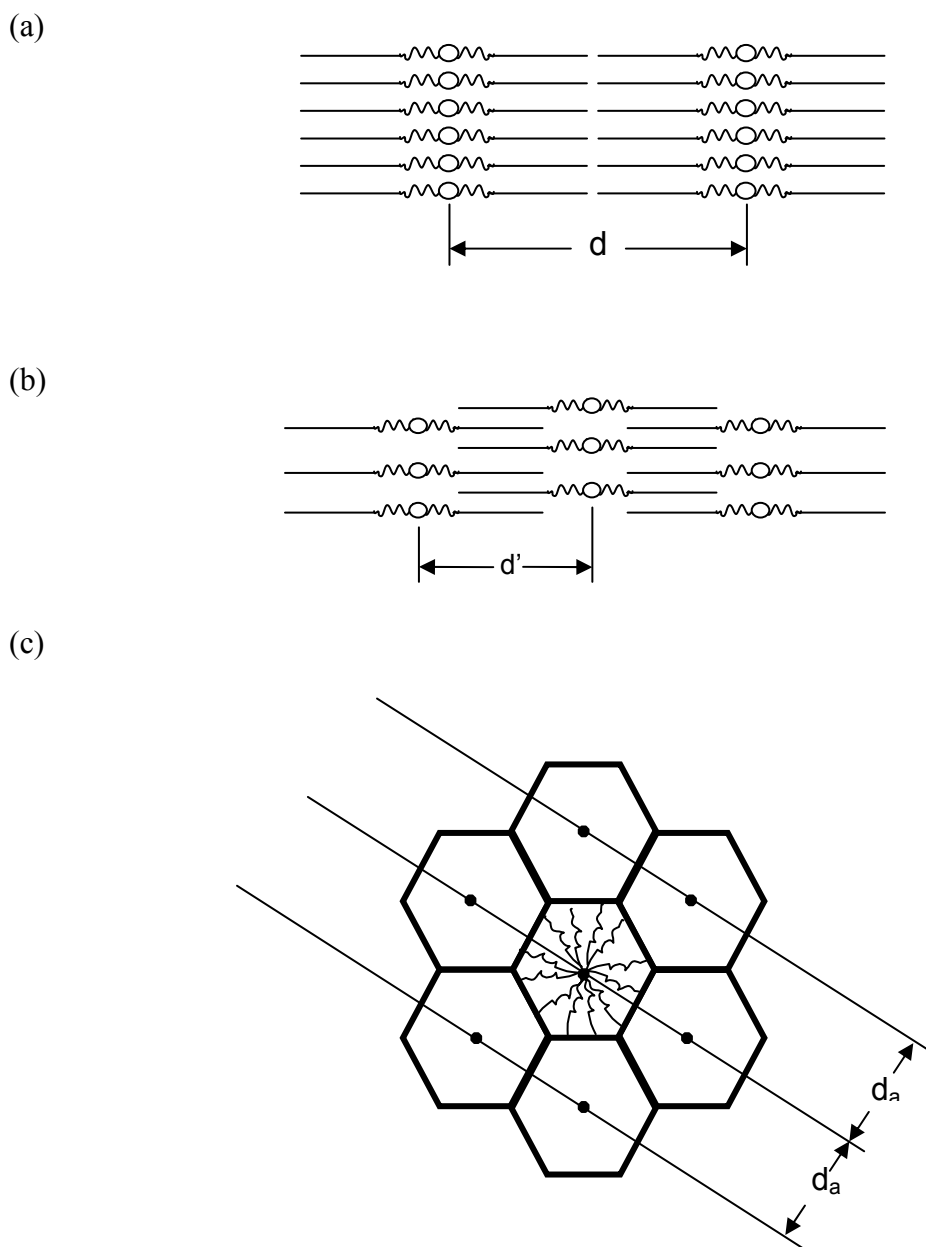


Figure 6.2 Schematics of end-to-end (a) and interdigitating (b) side-chain packing proposed by Hsieh and Morawetz for crystalline poly (n-alkyl acrylates) where the open circles represent the axes of the polymer main chains with side chains extending out in an all trans conformation in the crystalline region (solid lines); amorphous portion is represented by wavy lines.[2] Proposed hexagonal packing lattice (c) for amorphous n-alkyl acrylate polymers where the open circles represent the polymer main chain axes and curvy lines represent the side chains.

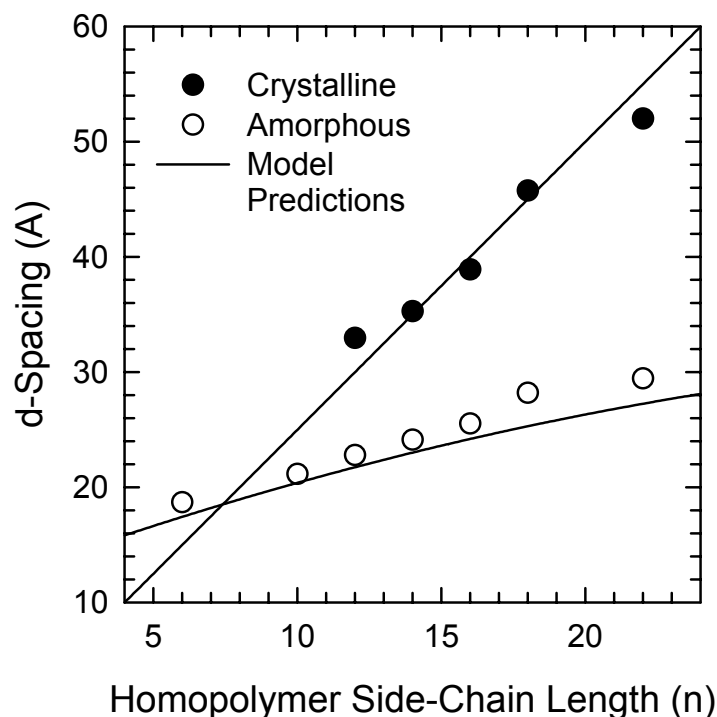


Figure 6.3 d-spacing (Å) values for crystalline and amorphous n-alkyl acrylate homopolymers as a function of side-chain length measured by SAXS (points) and calculated from model predictions (lines). According to the interpretation given in the text, the crystalline values correspond to d as defined in Figures 8a and 9a while the amorphous values correspond to d as defined in Figures 8b and 9b.

the d peak and the homopolymer side-chain length, n , is shown in Figure 6.3; these results are similar to those reported in the literature.[1, 2, 6] As the side chains become longer, this d-spacing increases, indicating the crystallites become thicker with increasing n . The linear relationship between heat of fusion and the crystalline d-spacing in Figure 6.4a clearly illustrates a direct correlation between the two as might be expected.

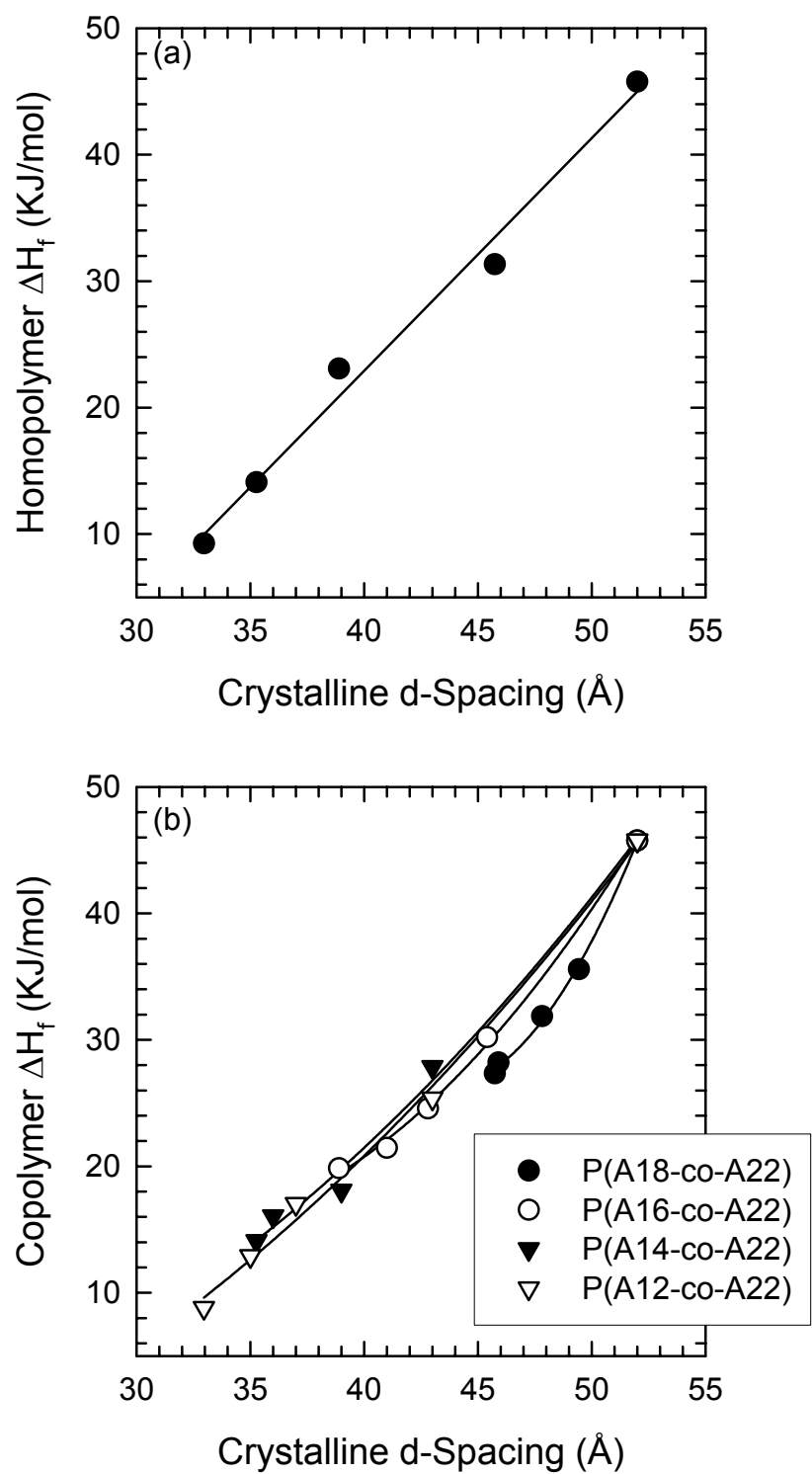


Figure 6.4 Relationships between ΔH_f and crystalline d-spacing for homopolymers (a) and copolymers (b) of A 22.

A simple mathematical model was developed to describe the d-spacings for the crystalline polymers with side-chains arranged in an end-to-end packing structure. It was assumed, as shown in Figure 6.2a, that the d-spacing of the polymers is the distance between the main chains, shown as open circles in the schematic. Assuming an all trans conformation between the crystallizeable carbons ($n - 9.4$), each methylene pair in the crystalline region contributes two times the projected carbon-carbon bond length, i.e., $2(1.25\text{\AA})$, to the thickness of the crystal. Note that in this packing arrangement, increasing n by one adds two methylene units to the crystals. An additional length, β_0 , is added to account for the amorphous regions, i.e.,

$$d = (2)(1.25)(n - 9.4) + \beta_0 \quad (6.1)$$

This latter parameter was estimated using a least squares fit of the experimental SAXS data to Equation 6.3 where the slope of d versus n was constrained to be $2.5\text{ \AA}$ as explained above. The result is

$$d = 2.5(n - 9.4) + 23.5 \quad (6.2)$$

which provides a reasonably good fit to the experimental results as seen in Figure 6.3.

SAXS experiments were also performed on amorphous poly (n -alkyl acrylates) in order to explore any local order in the molten state. An earlier study of amorphous acrylates by Plate reported only a single broad, weak peak in the amorphous state.[6] Figure 6.1b shows the scattering intensity for amorphous PA 22 plotted as a function of 2θ , where there is a peak at $2\theta \sim 3^\circ$ that shifts to smaller angles for poly (n -alkyl acrylates) with smaller values of n . The open circles in Figure 6.3 show the d-spacings computed from the amorphous scattering peaks from the various homopolymers at $T > T_m$. This d-spacing increases with n but with a different slope than the peaks observed at

$T < T_m$. Molten PA 22 shows another peak on the edge of the range that can be measured by this instrument. This peak is never visible for lower values of n . We propose the following interpretation of these results and suggest a simple model that seems to confirm the picture.

Figure 6.2a asserts that the axes of the main chains (circles in Figure 6.2) are forced to line up with a close lateral spacing because of the constraints imposed by forming side-chain crystals. However, the latter constraints are removed upon melting; hence, the main chain axes move apart while the side chains take on random conformations radiating away from the backbone in all directions to fill up the space between the main chains. We assume there is some local short order in which the main chain axes arrange for short distances in a more or less hexagonal close packing arrangement as suggested in 6.2c. We further assume that the SAXS scattering reflects the d-spacings between main chains as schematically illustrated in Figure 6.2c. The scattering should, thus, be similar to that from hexagonally close-packed cylinders where the scattering peak position ratios progress as $1 : \frac{1}{\sqrt{3}} : \frac{1}{2} : \frac{1}{\sqrt{7}} : \dots$ [3-5] Thus, in Figure 6.1b, we label the two peaks mentioned as d_a and $\frac{d_a}{\sqrt{3}}$. We assume that the peak in the center of the figure represents the first peak. The second peak at the far right of the figure, on the edge of the experimental data, labeled $\frac{d_a}{\sqrt{3}}$, represents the second order of the first peak, d_a . Thus, we focus on the peak labeled d_a for the following analysis. In summary, we assume that both peaks measure the same repeat hexagonal ordering of the main chain axes and their spacings are simply ratios of each other.

The following provides a simple way to estimate the characteristic d-spacing defined by the model in Figure 6.2c. The cross-sectional area, A , for each main chain is that of the hexagon shown there which, by simple geometry, is related to d_a as follows

$$A = \frac{2}{\sqrt{3}} d_a^2 \quad (6.3)$$

Consequently, the volume occupied by each n-alkyl acrylate repeat unit, V , can be obtained by assuming the main chain carbons are in an approximate trans-conformation, i.e., axial length = $2(1.25 \text{ \AA})$, such that

$$V = (2)(1.25 \text{ \AA}) \left(\frac{2}{\sqrt{3}} d_a^2 \right) \quad (6.4)$$

The mass within this volume, in atomic mass units, can be computed by multiplying the molecular weight of each repeat unit by three which is the number of main chains within each hexagonal unit, i.e.,

$$\text{mass} = 3 * (72 + 14n) \quad (6.5)$$

The ratio of the mass to the volume of a repeat unit, after appropriate unit conversions, can be set equal to the approximate density of 0.88 g/cm^3 for amorphous poly (n-alkyl acrylates) to obtain the following relationship between d_a (in $\text{\AA}$) and n is

$$d_a = \sqrt{\frac{(72 + 14n)}{0.51}} \quad (6.6)$$

The calculated d-spacings from the model are shown as the solid line in Figure 6.3. The model line has essentially the same slope as the open points but lies slightly below the experimental observations. The latter disparity may be attributed to the assumption that the backbone carbon-carbon bonds are in an all trans conformation whereas many of these bonds will be in gauche positions. These kinks and coils in the

main chains create would cause the measured d-spacing to be slightly greater than predicted by the model.

6.3 Crystalline / Crystalline Combination Copolymers

The characteristic dimensions of the crystalline and amorphous phases of copolymers of various n-alkyl acrylate monomers as measured by SAXS are shown in Figure 6.3. The d-spacing values are plotted in terms of mole fraction of the A22 comonomer for comparisons with the results in Figure 5.4. Plotting the data versus weight fraction does not change the overall trends. As the concentration of A22 in the copolymer increases, so does the d-spacing of the copolymers indicating a larger crystallite as expected for a longer average side-chain length. The crystalline phase d-spacings observed for copolymers shown in Figure 6.5a reveal similar trends as seen in Figure 5.4b for the heat of fusion for these copolymers. The greater the difference in the number of carbons in the side chains of the two monomers in a copolymer, the greater the depression in crystal size and in heat of fusion. A plot of ΔH_f versus d-spacing for the crystalline copolymers, see Figure 6.4b, shows a similar correlation between crystal size and ΔH_f as observed for homopolymers, see Figure 6.4a.

The amorphous copolymer d-spacings, see Figure 6.5b, increase more or less linearly with mole fraction of A22. This is a logical progression for amorphous copolymers that is in accord with the model derived earlier, i.e., the d-spacing values should increase with the average side-chain length of the copolymer.

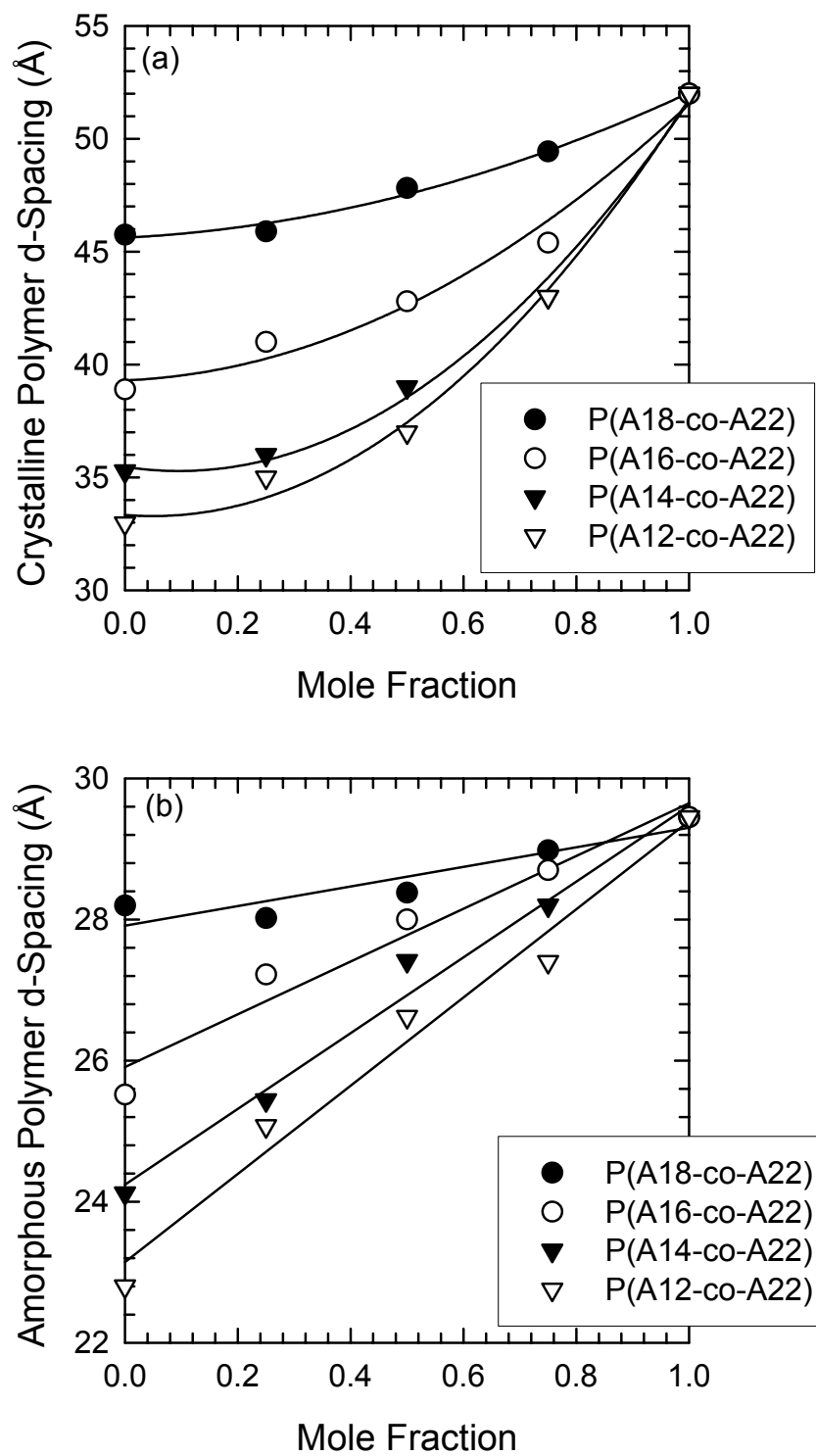


Figure 6.5 Relationship between d-spacings and copolymer composition in the crystalline (a) and amorphous (b) states for various copolymers based on A22.

6.4 Crystalline / non-Crystalline Copolymers

Two prior studies of small angle X-ray scattering (SAXS) have been reported for poly (n-alkyl acrylate) copolymers containing one long, crystallizeable comonomer and one non-crystallizeable comonomer.[1, 2, 6] Although each investigated a single copolymer system, both reported similar findings involving the packing structure of the side chains. Unlike the crystallizeable-crystallizeable copolymers examined earlier, the copolymers with only one crystallizeable comonomer described here had relatively low melting temperatures which limited the number of samples that could be examined in the crystalline state by SAXS. Therefore, the SAXS results reported provide only a small glimpse into the structural behavior of these polymers adding to the initial studies by Plate and Hsieh; however, it is not thorough enough to draw any definitive conclusions or to attempt to model their behavior.[1, 6]

Hsieh and Morawetz studied P(A1-co-A16) copolymers and observed both interdigitating and end-to-end side-chain packing forms, illustrated in Figures 6.2a and b, as well as a strong increase in d-spacing values as the concentration of A1 increased, although the average side-chain length and amount of crystallinity decreased in the copolymers.[1] Plate and Shibaev reported on poly(isopropyl acrylate-co-A18) copolymers and observed only one packing form along with a very slight increase in d-spacing values as the concentration of isopropyl acrylate increased, even though the average side-chain length and amount of crystallinity decreased.[6] They determined that the d-spacing values measured for their homopolymers were for the interdigitating packing form. Using the notation from Figure 6.1, Plate and Shibaev determined that the

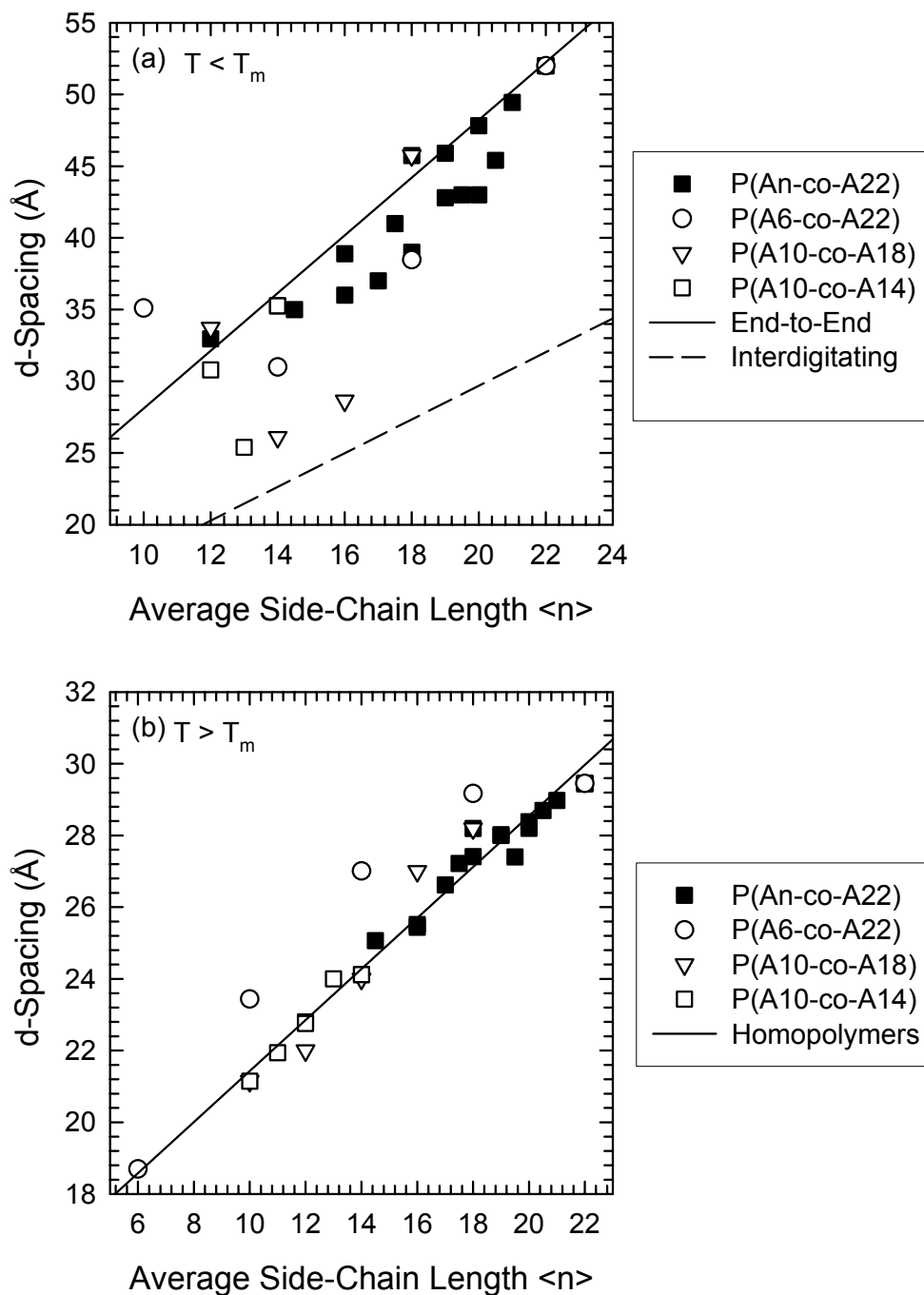


Figure 6.6 Small angle X-ray d-spacings for homopolymers (lines) and poly (n-alkyl acrylate) copolymers (points) measured in the semi-crystalline (a) and amorphous (b) states. For the crystalline polymers, the upper homopolymer line reflects d-spacing for end-to-end crystal packing while the lower line represents the d-spacings for interdigitating crystal packing as reported by Plate.[6]

d peak disappeared while the d' peak increased in intensity.[6] They attributed this shift in packing to the distortions in the amorphous backbone caused by the longer side chains crystallizing around the non-crystallizable comonomers.[6]

We examined the small angle spacings for P(A6-co-A22), P(A10-co-A14), and P(A10-co-A18) copolymer systems in both the crystalline and amorphous states; the spacings are shown in Table 6.1. Figure 6.6a shows the d-spacings for crystalline polymers and copolymers versus their average side-chain length. The d-spacings for the end-to-end packing form, d peak, and the interdigitating packing form, d' peak, for n-alkyl acrylate homopolymers are shown as the solid and dashed lines, respectively. The

Table 6.1
d-spacings for amorphous and crystalline copolymers

Copolymer	Mol % monomer 1	Average Side-Chain Length <n>	Molten d-Spacing (Å)	Crystalline d-Spacing (Å)
P(A6-co-A22)	100	6	18.7	-
	75	10	23.4	33.0
	50	14	27.0	34.0
	25	18	29.2	36.5
	0	22	29.5	52.0
P(A10-co-A14)	100	10	21.2	-
	75	11	21.9	-
	50	12	22.8	24.6
	25	13	23.6	30.7
	0	14	24.1	35.3
P(A10-co-A18)	100	10	21.2	-
	75	12	22.0	28.3
	50	14	24.0	29.1
	25	16	27.0	33.6
	0	18	28.2	45.7

solid line for end-to-end packing was taken from Figure 6.3 earlier. Since the interdigitating, or d' peaks were often too weak to measure, Plate's data were used to compute the dashed line; this has a slope of 1.2 which is approximately half the slope of the line for the end-to-end form.[6] This correlates well with the simple model derived previously, where the end-to-end packing formation had a slope of 2(1.25), and an interdigitating packing formation would be expected to have half that slope. In Figure 6.6a, the crystallizeable-crystallizeable copolymers from Figure 6.4a are represented by the darkened square data points in Figure 6.6a while the P(6-co-A22), P(A10-co-A14), and P(A10-co-A18) d-spacing values are the open data points. In general, most of the d-spacing values for our copolymers lie slightly below those for the homopolymers packed end-to-end. This is logical since copolymers form slightly smaller and less perfect crystals than homopolymers. It also shows that, in general, both types of copolymers are predominantly end-to-end packed and their d-spacing values increase with the average side-chain length, or crystallinity, of the copolymer. P(A6-co-A22) with the composition 75/25 mol % has a larger d-spacing value than the homopolymers while P(A10-co-A14) with 75/25 mol % and P(A10-co-A18) with 75/25 and 50/50 mol % copolymers are closer to the interdigitating packing line. This scattering at lower average side-chain lengths may be similar to the observations Hsieh and Plate observed for their copolymers in this local region; however, when considering the broader range of average side-chain lengths, these points are contrary to the majority of the d-spacing values observed.[1]

SAXS experiments were also performed on the three sets of copolymers in the amorphous state with the results shown in Figure 6.6b. Homopolymer d-spacing values

from Figure 6.3 are plotted as the solid line while copolymers composed of two crystallizeable comonomers from Figure 6.4b are plotted as the solid square data points. The d-spacings for the P(A6-co-A22), P(A10-co-A14), and P(A10-co-A18) copolymers are plotted as the open data points in this figure. As suggested by the model proposed for amorphous copolymers earlier, the d-spacing values of these amorphous copolymers increase with the volume or mass of carbons in the copolymers. Interestingly, the d-spacings for the P(A6-co-A22) copolymers lies somewhat over the experimental line in the amorphous state ($T > T_m$) while the crystalline d-spacings ($T < T_m$) lie below the expected line.

6.5 Conclusions

Small angle X-ray scattering measurements were made on homopolymer and copolymer systems of varying side-chain length. The d-spacing values of the crystalline homopolymers were measured and described by a simple model. A simple molecule packing model was proposed that reasonably predicts the amorphous d-spacing as a function of side-chain length. Crystalline copolymers showed nearly identical trends of d-spacing with composition as seen for the heat of fusion. Amorphous copolymers show a nearly linear relationship between d-spacing and composition as expected from the simple packing model.

Limited small angle X-ray scattering (SAXS) experiments were used to measure the crystalline and amorphous d-spacings for three different copolymer systems. It was determined that the crystalline side chains of the P(A6-co-A22), P(A10-co-A14), and P(A10-co-A18) copolymers have a predominantly end-to-end packing form with slightly

smaller d-spacing values attributed to the formation of smaller and less perfect crystals.

As was expected, the amorphous copolymers have d-spacings similar to those the homopolymers and copolymers with two crystallizeable comonomers.

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Chapter 7

Gas Permeation Properties of Poly (n-alkyl acrylates)

7.1 Introduction

Chapters 5 and 6 established the thermal and structural relationships among various copolymers systems. It was determined that as the side-chain length increases, so do the melting temperatures, heats of fusion, and crystallite sizes. This chapter focuses on establishing trends for the gas permeability through copolymer systems and relating them to the thermal and structural properties established previously. Like the previous chapters, this chapter begins with a thorough look at the permeability properties of homopolymers and expands to copolymers with a heavy emphasis on copolymers that contain one crystallizeable comonomer and one borderline comonomer.

All synthesis and analytical characterization was performed as described in Chapter 3. Permeability experiments were performed on samples with a controlled thermal history having been cooled at a rate of 1°C/min. All polymers were characterized with six penetrant gases, though most of the figures shown in the chapter involve the permeability of O₂ and CO₂ gas through poly (n-alkyl acrylate) membranes. Additional figures for the gas permeability of He, H₂, CH₄, and N₂ gases may be found in Appendix B.

7.2 Homopolymers

The permeability of small molecules, like gases, in poly (n-alkyl acrylates) exhibit a large ‘jump’ or ‘switch effect’ over the range of the side-chain crystalline melting.[1-6]

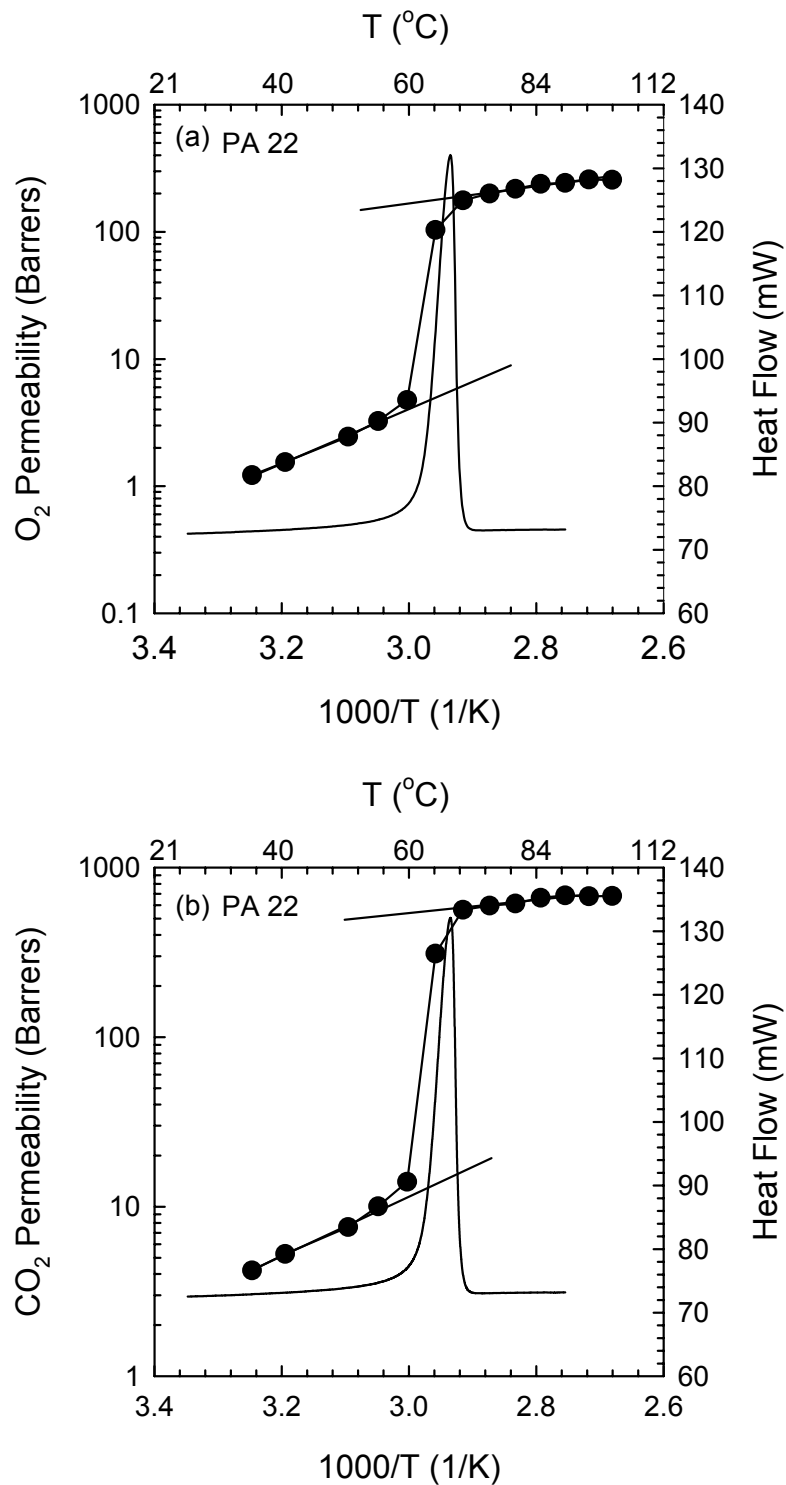


Figure 7.1 Gas permeability coefficients as a function of temperature (on Arrhenius coordinates) through the melting temperature region for PA 22 for O₂ (a) and CO₂ (b) with DSC scans superimposed.

Figure 7.1 illustrates the typical permeation jump for O₂ and CO₂ for the PA 22 homopolymer; for convenient reference, the DSC thermogram is superimposed on this plot. The permeability of these semi-crystalline polymers has been analyzed in terms of a modified two-phase model proposed by Michaels and Bixler.[7] The model suggests that the permeability of a penetrant in a semi-crystalline polymer, P_c , is related to that of the completely amorphous polymer, P_a , by

$$P_c = \frac{P_a(1 - \phi)}{\tau\beta} \quad (7.1)$$

where ϕ is the volume fraction of the crystal phase, τ is a tortuosity factor and β accounts for immobilization of amorphous chain segments by the presence of the crystals. It is assumed that the crystallites are impermeable to the penetrants and, therefore, all gas transport occurs through the molten polymer. Thus, the crystals create a tortuous path for diffusion. The permeability ‘jump’ occurs as the crystallites melt at the polymer’s T_m causing a significant increase in gas transport through the membrane. This correlation between the melting endotherm and the permeation jump is clearly shown in Figure 7.1 where the onset and end of the melting peak marks the beginning and end of the permeation jump. Above and below the T_m , permeability increases with temperature in an Arrhenius fashion. At the onset of melting, the permeability jumps nearly two orders of magnitude and returns to an Arrhenius temperature relationship at the end of the melting. The activation energies above and below T_m are different since apparently β (see Equation 7.1) depends on temperature. This phenomenon, as reported previously, is reversible and reproducible given the same thermal history.[1-3] Although the permeability for six penetrant gases was measured, only O₂ and CO₂ are shown here as

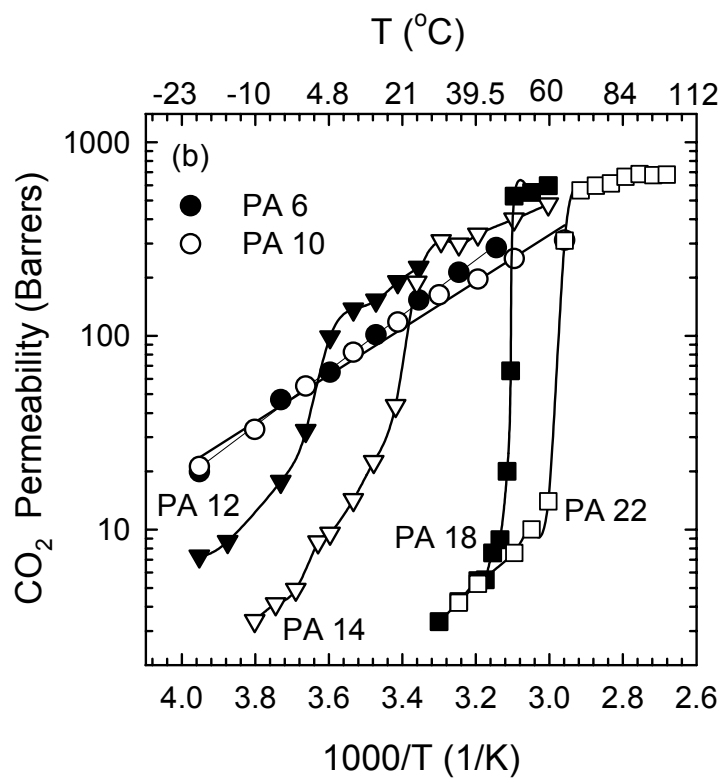
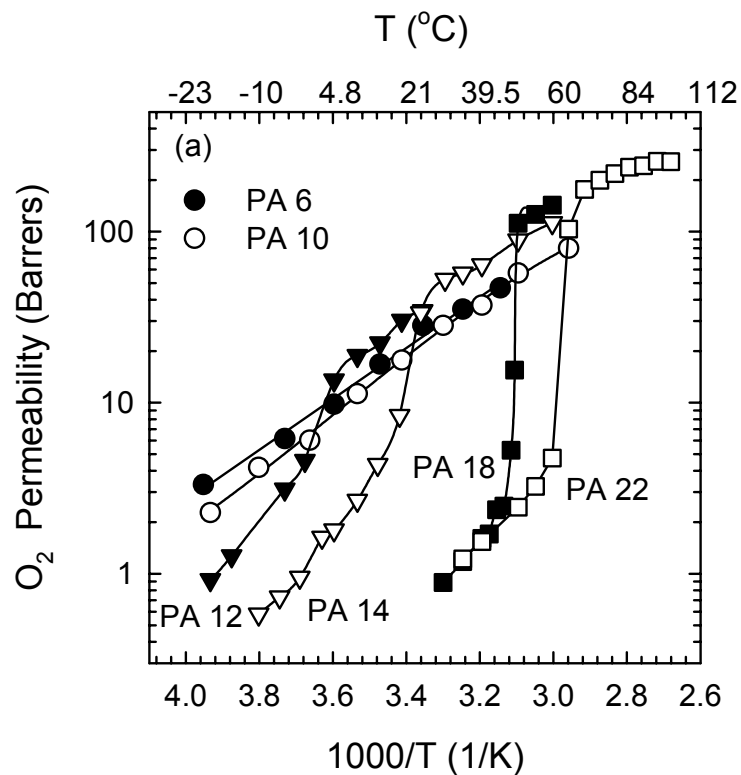


Figure 7.2 Permeability of O_2 (a) and CO_2 (b) for homopolymers with side-chain lengths ranging from 6 to 22 carbons as a function of temperature on Arrhenius coordinates.

examples because they are the most important for modified atmosphere packaging applications. Similar permeation experiments were conducted on n-alkyl acrylate homopolymers with side-chain lengths from 6 to 22 with the results shown in Figure 7.2. Various parameters were extracted for comparison and analysis of the permeability data listed in Table 7.1. The quantities E_a and E_c are the activation energies for permeation in the amorphous and semi-crystalline states, respectively, while P_{35}^+ and P_{35}^- correspond to the molten and crystalline polymers, respectively, extrapolated to 35°C. Extrapolation to 35°C is an arbitrary temperature for calculating the jump ratio (P_{35}^+ / P_{35}^-); however, as explained by Mogri and Paul, the temperature used greatly affects the magnitude of the jump ratios, or calculated jump heights, since the activity energies of the molten and crystalline polymers are not the same.[3] We selected 35°C because it is near the median melting temperature for the homopolymers and most of the copolymer systems examined later. As the side-chain lengths of the polymers increase, E_a decreases while the permeability and jump ratios increase. These trends were described in detail by Mogri and Paul for similar n-alkyl acrylate homopolymers.[1, 3]

As may be seen in Figure 7.2, as the side-chain lengths decrease for these homopolymers, the widths of the permeation jumps increase and gradually flatten out into the simple Arrhenius curves observed for the completely amorphous PA 6 and PA 10. These trends are similar to the endotherm trends for these polymers. As previously discussed, n-alkyl acrylate homopolymers with longer side chains are more crystalline than those with shorter side chains. This fact is responsible for the increase in permeation jump ratio with increasing n illustrated in Figure 7.3. A polymer with a long side-chain

has a greater change in morphology at T_m than one with a shorter side chain and, therefore, the permeability jump is greater. It should be emphasized that a significant

Table 7.1
Activation Energy and Permeability Data Extrapolated to 35°C for Various Gases
Through Poly (n-alkyl acrylate) Homopolymers

Homopolymer		Gas					
		He	H ₂	O ₂	N ₂	CH ₄	CO ₂
PA 6	E _a	6.5	6.7	6.8	8.0	6.8	6.5
	E _c	-	-	-	-	-	-
	P ₃₅ ⁺	49.8	89.8	35.3	13.9	31.6	211.7
	P ₃₅ ⁻	-	-	-	-	-	-
	P ₃₅ ⁺ /P ₃₅ ⁻	-	-	-	-	-	-
PA 10	E _a	7.9	6.8	7.4	7.9	6.7	5.5
	E _c	-	-	-	-	-	-
	P ₃₅ ⁺	43.4	71.6	31.1	12.0	35.0	167.3
	P ₃₅ ⁻	-	-	-	-	-	-
	P ₃₅ ⁺ /P ₃₅ ⁻	-	-	-	-	-	-
PA 12	E _a	7.6	8.9	7.1	8.7	6.4	5.7
	E _c	7.4	8.3	11.0	5.7	6.5	4.5
	P ₃₅ ⁺	6.4	125.9	52.0	22.5	49.0	304.9
	P ₃₅ ⁻	30.3	53.0	42.0	2.6	8.0	35.7
	P ₃₅ ⁺ /P ₃₅ ⁻	2.1	2.4	1.2	8.6	6.1	8.5
PA 14	E _a	5.8	5.7	5.7	6.5	6.5	3.7
	E _c	6.1	6.4	8.7	10.1	8.6	6.6
	P ₃₅ ⁺	52.5	96.2	56.5	22.4	65.7	303.1
	P ₃₅ ⁻	12.2	14.9	6.6	2.2	4.5	21.3
	P ₃₅ ⁺ /P ₃₅ ⁻	4.3	6.5	8.5	10.3	14.8	14.3
PA 18	E _a	5.4	6.3	5.2	6.3	5.6	2.7
	E _c	6.6	7.1	10.6	12.9	13.1	9.2
	P ₃₅ ⁺	68.9	113.4	75.0	29.1	90.9	428.2
	P ₃₅ ⁻	8.1	8.2	1.2	0.3	0.6	4.3
	P ₃₅ ⁺ /P ₃₅ ⁻	8.5	13.8	62.7	95.8	143.9	100.1
PA 22	E _a	4.2	2.0	4.0	4.0	4.0	1.5
	E _c	6.8	7.4	9.7	11.6	11.5	8.5
	P ₃₅ ⁺	107.2	256.7	92.1	34.9	65.0	461.2
	P ₃₅ ⁻	8.7	9.0	1.0	0.3	0.5	4.2
	P ₃₅ ⁺ /P ₃₅ ⁻	12.4	28.5	92.1	120.7	123.0	110.3

Note that the activation energies E_a and E_c have units of kcal/mol while the permeability coefficients P₃₅⁺ and P₃₅⁻ have units of Barrers, i.e.: 10⁻¹⁰ cm³(STP)-cm/(cm³-cmHg-sec).

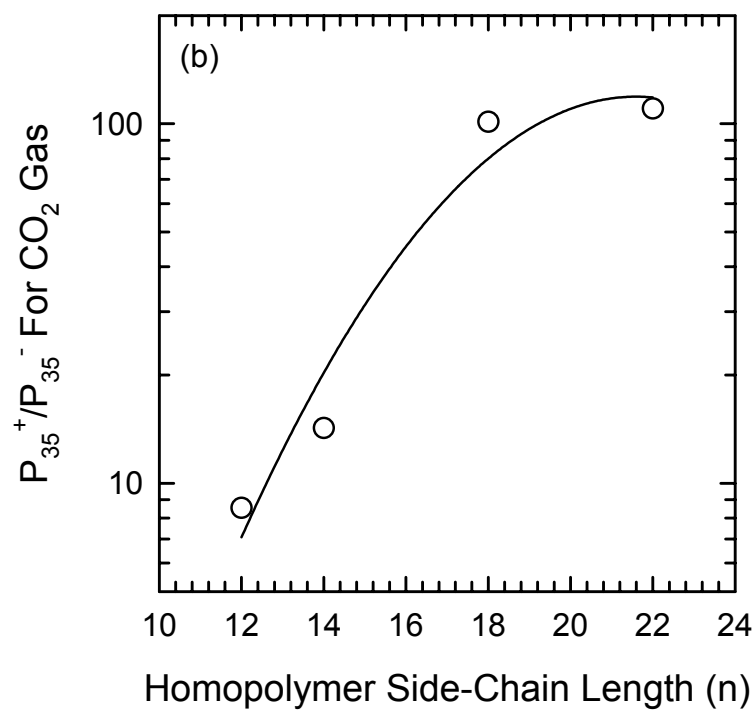
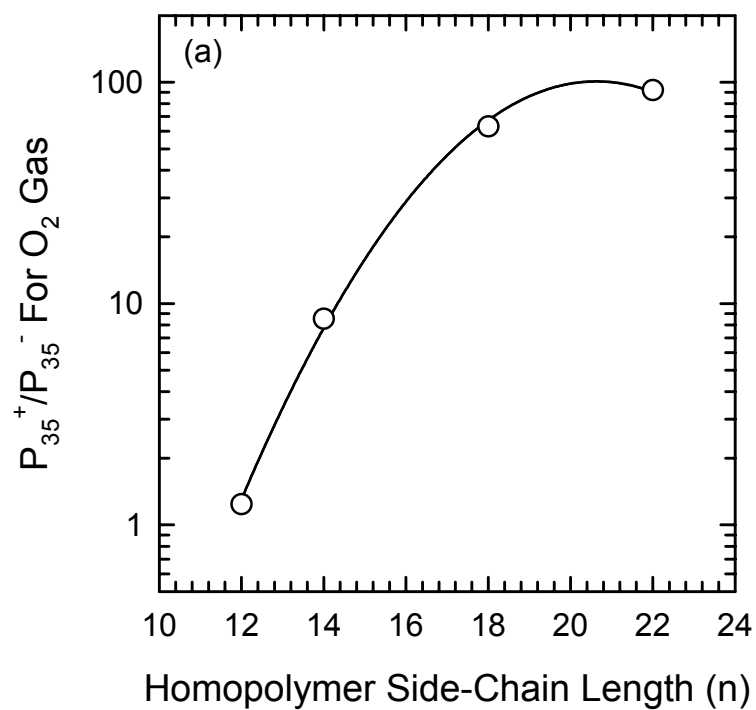


Figure 7.3 Homopolymer permeation jump ratios calculated at 35°C for O_2 (a) and CO_2 (b) gases as a function of side-chain length.

part of the jump arises because gas molecules must follow a tortuous path around the crystallites; this tortuosity disappears upon melting. The length of this pathway is affected by the lateral dimensions of the crystallites as well as their amount, i.e., crystallinity. At this point there is no simple way to quantify the dimensions, or aspect ratio, of the crystallites, the number of crystallites, or their arrangement with respect to one another. While speaking simply of crystallinity clearly does not tell the whole story, it seems that the crystallite dimensions and crystallinity must be closely coupled since the permeability jump seems to be described, at least to a first approximation, by just crystallinity. If the simple geometrical change in the tortuous path on melting were the only factor at play, then, as shown by Mogri and Paul, the permeation jump ratios should be the same for all gases which is not the case.[1-3] It seems that the crystallites tend to restrict mobility in the amorphous phase which Michaels and Bixler recognized in main-chain crystalline polymers and modified the conventional two phase model for permeation by including a ‘chain immobilization factor’, i.e. β , in the equation for permeability in semi-crystalline polymer.[7] In terms of this model, the permeation jump is given by

$$\frac{P_a}{P_c} = \frac{P_{35}^+}{P_{35}^-} = \frac{\tau\beta}{(1-\phi)} \quad (7.2)$$

The tortuosity factor should be the same for all penetrants since it is simply a geometric term; however, since the β term reflects changes in segmental dynamics, it may not be the same for all penetrants. Temperature dependence of β would explain why the activation energies for permeation can be larger below T_m than above. The loss of the chain immobilization effect on melting, accounts for much of the permeation jump seen in side-chain crystalline polymers and all of the penetrant size dependence. The

increased crystallinity increases the impedance of this conformational freedom in the amorphous backbone. While Figure 7.3 only shows data for O₂ and CO₂, Figure 7.4 contains data for all the penetrant gases examined in this study. This figure shows the effects of penetrant size on the permeation jump for the various homopolymers. As discussed by Mogri and Paul, the magnitude of the permeation jumps for these semi-crystalline polymers varies greatly with penetrant gas size.[1-3] In general, an increase in gas molecular size results in an increase in the jump ratio as shown in Figure 7.4. This strong dependence on penetrant size seems to be unique to side-chain crystalline polymers; the jump in permeability on melting main-chain polymers seems to have a much weaker dependence on penetrant size.[1, 3]

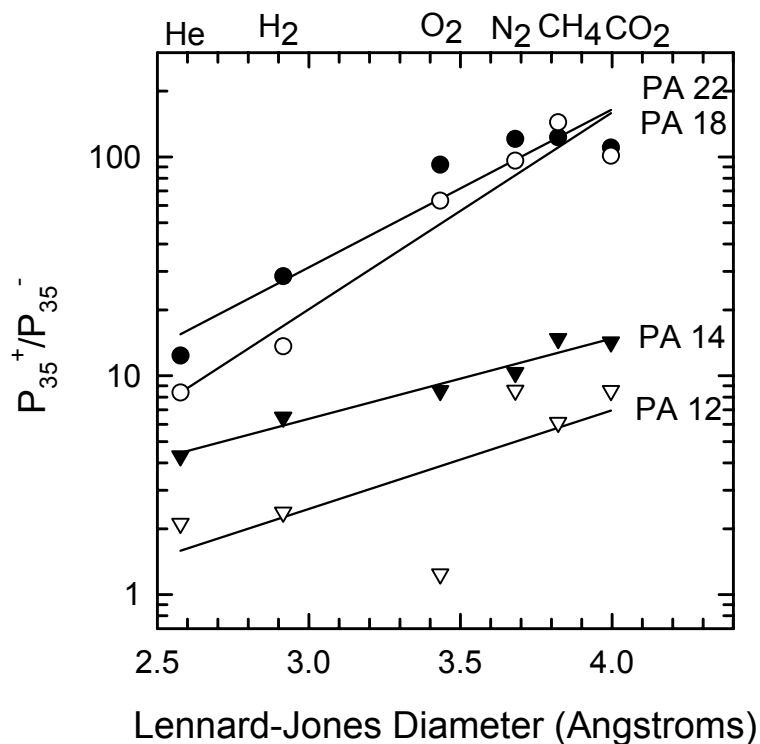


Figure 7.4 Permeation jump ratios for various gases calculated at 35°C for homopolymers with various side-chain lengths as a function of penetrant size.

Another informative way to examine the change in permeability on melting is to plot P_{35}^{+} and P_{35}^{-} versus n , see Figure 7.5 for O_2 and CO_2 data represented in this way. For both gases, the amorphous permeability increases with n . This increase is due to the increased free volume associated with the additional methylene units of the longer side chains [3]. This increase, however, is significantly less than the decrease in permeability of the crystalline polymer with increasing n . The crystalline phase permeability is clearly the dominant contributor to the overall permeation jump ratio for these gases shown in Figure 7.3. The crystals act as impenetrable barriers to gas forcing all transport to occur through the amorphous polymer. The increase in side-chain length of the polymers leads to an increase in crystallinity and a decrease in permeability.

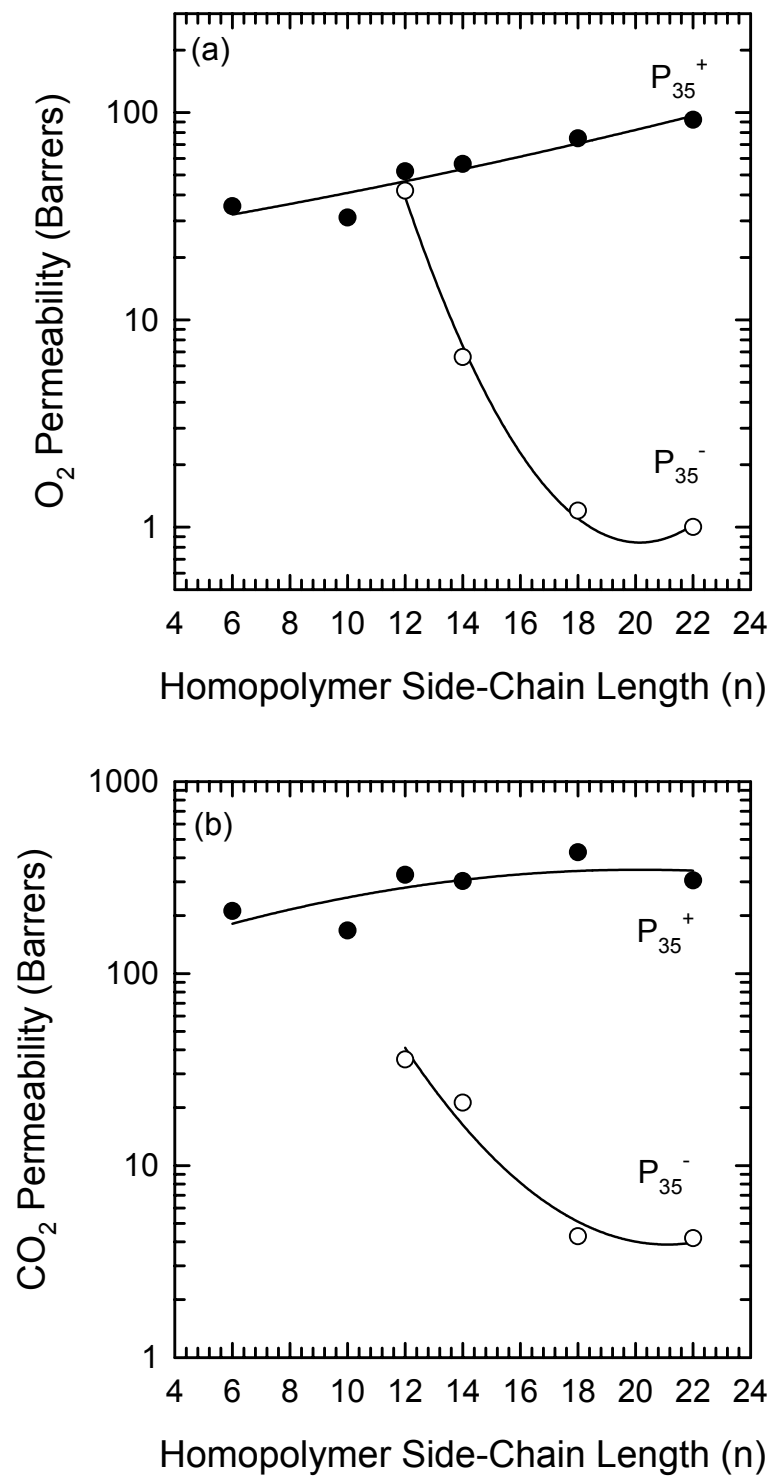
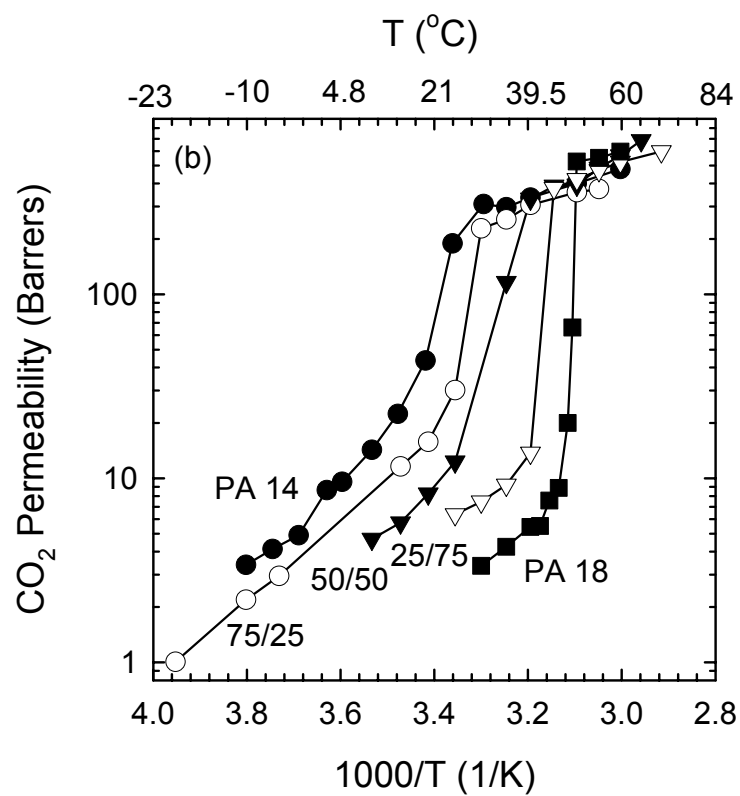
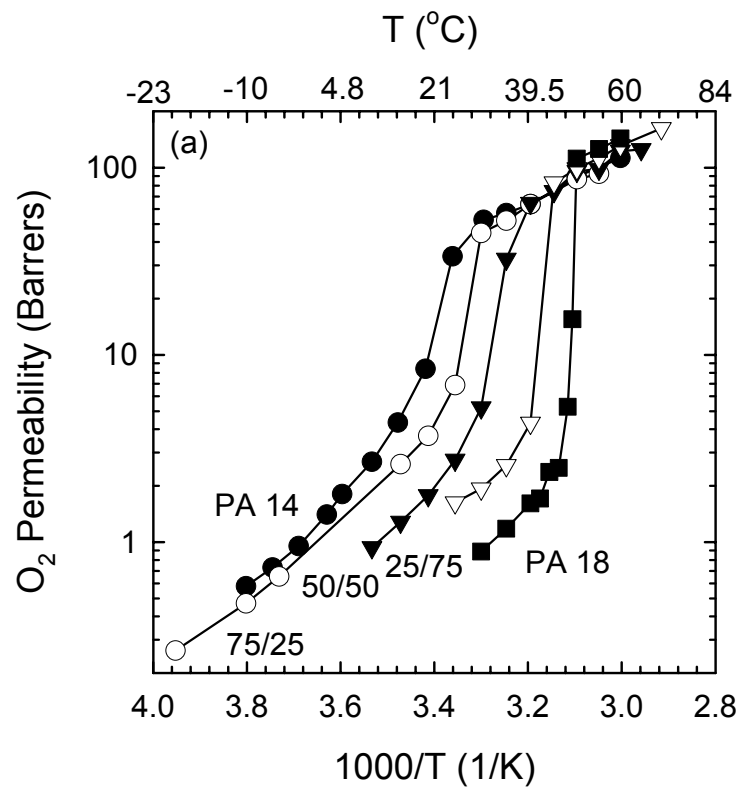


Figure 7.5 Permeability of amorphous (P_{35}^+) and crystalline (P_{35}^-) homopolymers extrapolated to 35°C for O_2 (a) and CO_2 (b) gas as a function of side-chain length.

7.3 Crystalline / Crystalline Combination Copolymers

Figure 7.6 shows the permeability for O₂ and CO₂ in P(A14-co-A18) and P(A12-co-A22) copolymers. These plots for copolymers are similar in nature to the plots for homopolymers shown in Figure 7.2. For the P(A14-co-A18) system, as the average side-chain lengths increases, the magnitude of the permeation jumps increases while the widths of the jumps decreases. Just as the P(A14-co-A18) endotherms of Figure 5.2a resemble the endotherms of n-acrylate homopolymers, the permeability trends for this system resemble the homopolymer permeability trends of Figure 7.2. The P(A12-co-A22) permeation trends follow an analogous pattern; in some cases the behavior is a little more complex than seen for the P(A14-co-A18) system. Some of these difference may stem from the larger difference in the length of the side-chain in the comonomers, i.e., $\Delta n = 10$ versus 6. The copolymer containing 75 mol% of A12, in particular, does not show a well-defined Arrhenious relationship in the crystalline phase before the jump begins; for this composition, the permeability jump is rather broad. PA 12 and the copolymer containing 50 mol% A12 do not seem to show such an increase in permeability in the vicinity of T_m . The endotherms for this system were similar to those of P(A10-co-A18) in Figure 5.2b where the distribution of crystallite sizes in the copolymer broaden with increasing concentration of the shorter monomer. This broadening and reduction of the height of the melting peak appears to be the cause for the gradual increase in permeability as the T_m is approached in the case of the 75 mol% A12 copolymer.

It turns out that DSC thermograms are a simple and effective way to predict the breadth of the permeation jump as the copolymer melting point is traversed, as shown in Figures 7.7 and 7.8 for P(A14-co-A18) and P(A12-co-A22) copolymers. Figures 7.7a



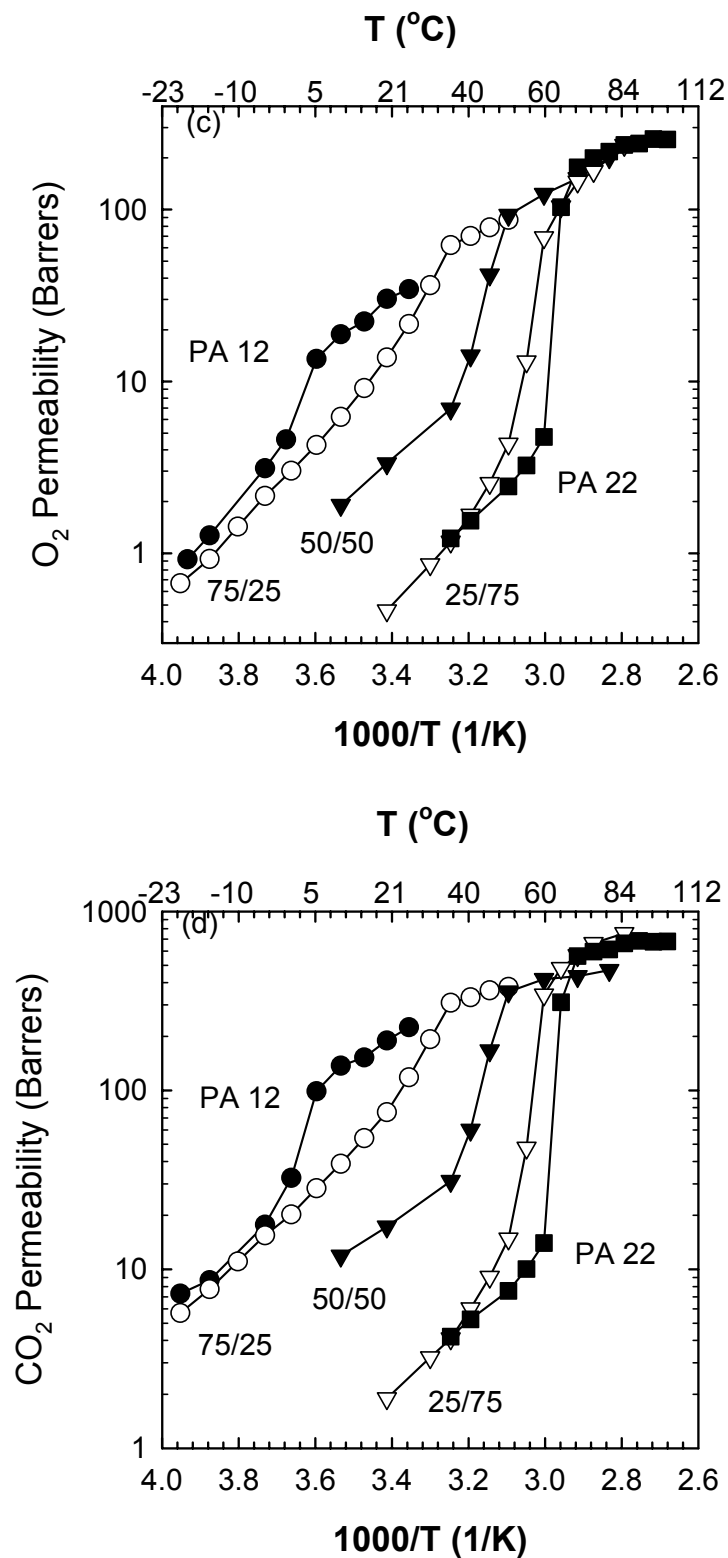


Figure 7.6 Permeability of various P(A14-co-A18) copolymers to O_2 (a) and CO_2 (b) and P(A12-co-A22) copolymers to O_2 (c) and CO_2 (d) as a function of temperature on Arrhenius coordinates.

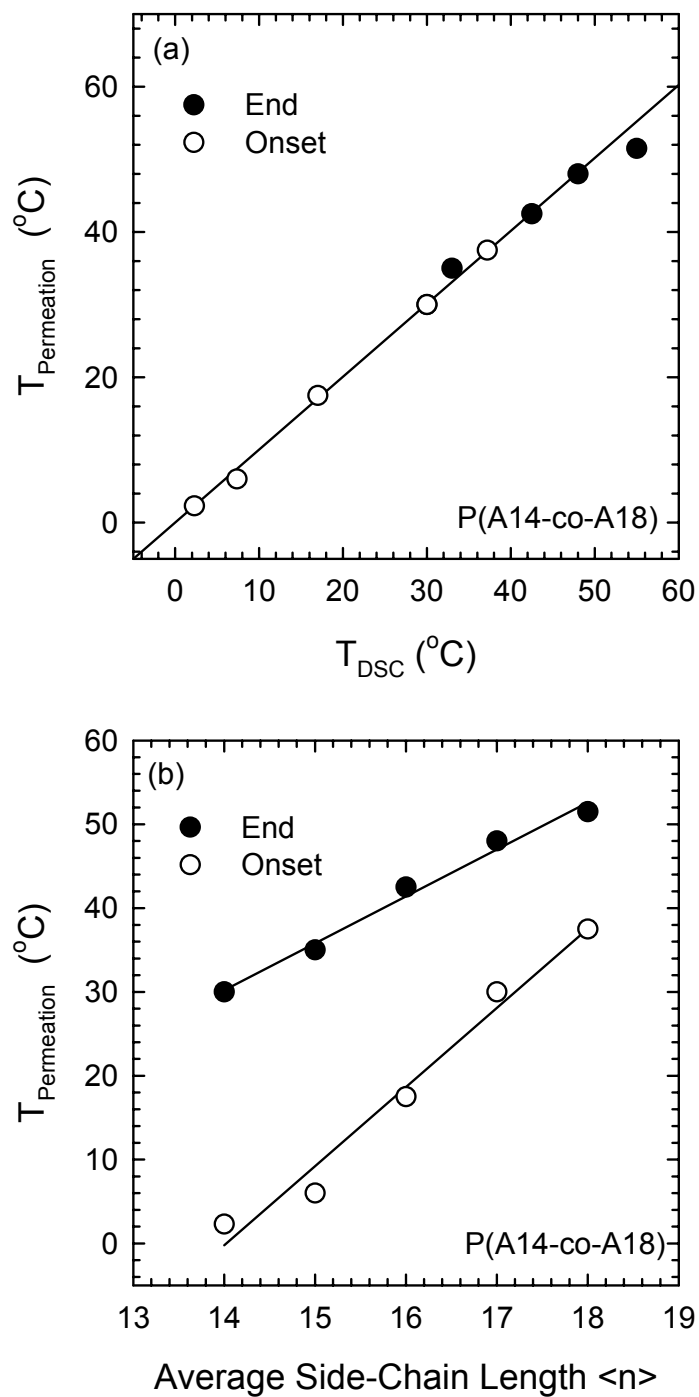


Figure 7.7 Relationship between onset and end temperatures for melting of P(A14-co-A18) copolymers as measured by permeability jumps and DSC endotherms (a).

Correlation between the onset and end temperatures for copolymers of P(A14-co-A18) with the average side-chain length (b).

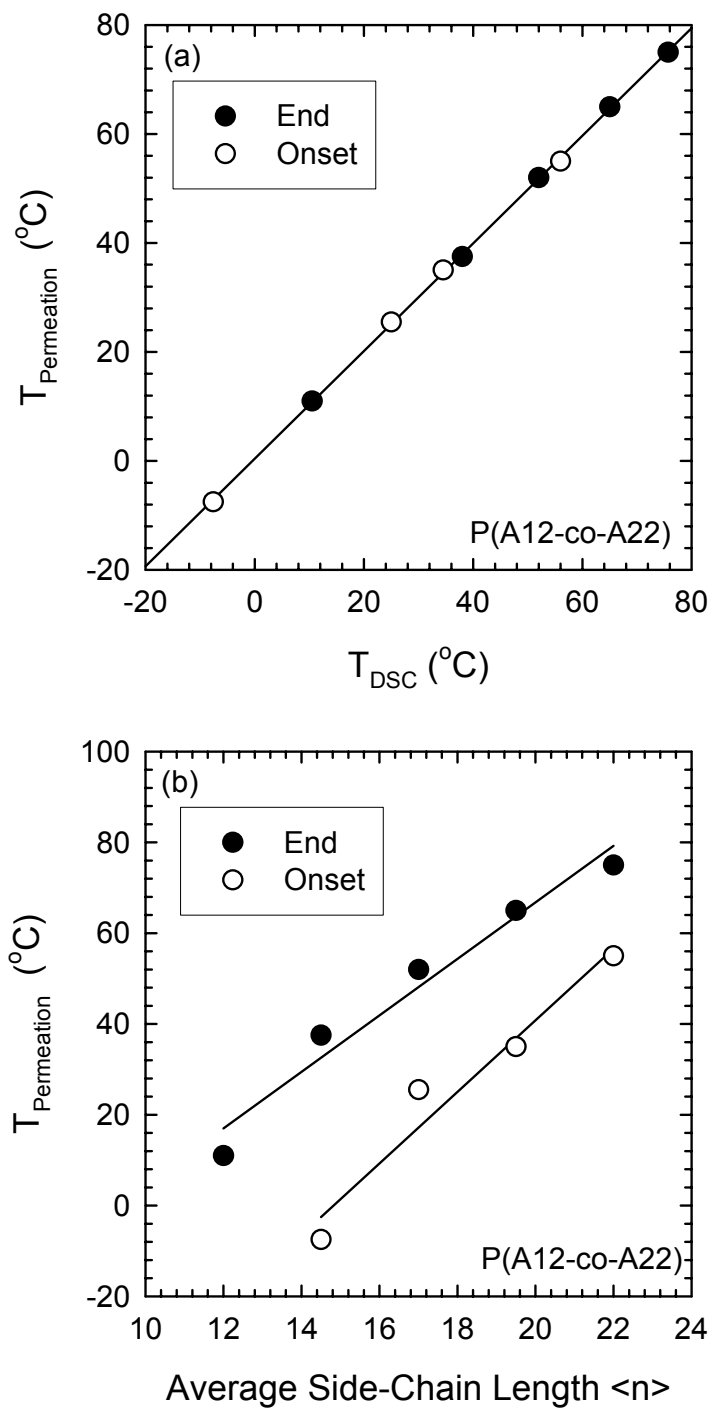


Figure 7.8 Relationship between onset and end temperatures for melting of P(A12-co-A22) copolymers as measured by permeability jumps and DSC endotherms (a).

Correlation between the onset and end temperatures for copolymers of P(A12-co-A22) with the average side-chain length (b).

and 7.8a compare the onset and end temperature of the melting peak, as measured using DSC, to the onset and end temperatures of the permeability jump. The baseline construction method used to define the DSC onset and end points was described in a previous publication.[4]. The onset and end of the permeability jump was defined by drawing the Arrhenious relationship for the pre- and post-melt permeabilities and locating the temperature where the permeability deviates from the Arrhenius line. As seen in Figures 7.7a and 7.8a, there is an excellent linear relationship between the temperatures obtained by DSC and by permeation experiments. This is a significant relationship in that simple DSC experiments can be used to judge the breadth of the permeability jump since changes in copolymer crystallinity define the permeation response. The onset and end temperatures measured by permeability are plotted as a function of average side chain length in Figures 7.7b and 7.8b. The difference between the onset and end temperatures for both copolymer systems tend to narrow as the side-chain length increases. This is consistent with the thermal and structural analysis of these systems; as the side-chain length increases, the melting peak becomes more narrow which suggests a more narrow size distribution of crystallites.

Table 7.2 summarizes the activation energies for the molten and crystalline copolymers, their absolute permeability to six gases and the permeability jump ratios evaluated at 35°C. Figure 7.9 shows how the jump ratio for P(A14-co-A18) and P(A12-co-A22) copolymers depend on copolymer composition and penetrant size. As in Figure 7.4 for homopolymers, the copolymer systems show larger permeation jumps with larger penetrant diameter and longer average side-chain length. Surprisingly, however, for both systems, the homopolymer with the shorter side chains tend to show a larger permeability

Table 7.2
Activation Energy and Permeability Data Extrapolated to 35°C for Various Gases
Through Poly (n-alkyl acrylate) Crystallizeable / Crystallizeable Copolymers

Copolymer	Mol % monomer 1	Gas						
			He	H ₂	O ₂	N ₂	CH ₄	CO ₂
P(A10-co-A14)	75	E _a	3.5	2.2	4.2	4.9	5.8	3.1
		E _c	-	-	-	-	-	-
		P ₃₅ ⁺	57.8	86.7	49.5	17.2	29.5	212.0
		P ₃₅ ⁻	-	-	-	-	-	-
		P ₃₅ ⁺ /P ₃₅ ⁻	-	-	-	-	-	-
P(A10-co-A14)	50	E _a	6.6	4.4	7.9	8.1	8.6	5.4
		E _c	-	-	-	-	-	-
		P ₃₅ ⁺	0.0	68.5	60.9	19.5	50.2	302.2
		P ₃₅ ⁻	-	-	-	-	-	-
		P ₃₅ ⁺ /P ₃₅ ⁻	-	-	-	-	-	-
P(A10-co-A14)	25	E _a	5.1	3.6	4.8	7.9	8.3	2.7
		E _c	4.4	3.0	6.5	6.5	4.8	5.1
		P ₃₅ ⁺	58.1	91.5	66.8	21.7	56.5	308.0
		P ₃₅ ⁻	18.2	14.6	11.7	2.3	2.6	44.9
		P ₃₅ ⁺ /P ₃₅ ⁻	3.2	6.2	5.7	9.6	21.6	6.9
P(A10-co-A18)	75	E _a	8.3	8.1	6.6	9.4	6.6	4.9
		E _c	-	-	-	-	-	-
		P ₃₅ ⁺	46.5	45.7	41.4	48.8	41.5	225.0
		P ₃₅ ⁻	-	-	-	-	-	-
		P ₃₅ ⁺ /P ₃₅ ⁻	-	-	-	-	-	-
P(A10-co-A18)	50	E _a	7.6	8.2	6.4	9.0	10.4	7.3
		E _c	-	-	-	-	-	-
		P ₃₅ ⁺	56.8	98.6	49.6	21.2	58.3	291.1
		P ₃₅ ⁻	-	-	-	-	-	-
		P ₃₅ ⁺ /P ₃₅ ⁻	-	-	-	-	-	-
P(A10-co-A18)	25	E _a	6.4	6.1	5.4	6.5	6.0	3.4
		E _c	7.3	8.5	9.4	11.5	10.6	7.8
		P ₃₅ ⁺	78.1	137.3	55.4	17.7	42.4	272.0
		P ₃₅ ⁻	18.7	23.1	4.9	1.6	3.6	21.6
		P ₃₅ ⁺ /P ₃₅ ⁻	4.2	6.0	11.3	11.0	11.7	12.6

Table 4 (cont'd)

P(A14-co-A18)	75	E_a	6.3	6.4	5.9	6.2	5.7	3.7
		E_c	7.6	7.4	9.9	13.1	12.7	10.3
		P_{35}^+	67.8	67.1	53.5	22.5	55.7	265.1
		P_{35}^-	12.4	12.0	8.0	3.1	8.1	37.5
		P_{35}^+/P_{35}^-	5.5	5.6	6.7	7.2	6.9	7.1
P(A14-co-A18)	50	E_a	6.6	6.0	4.5	5.6	7.2	6.0
		E_c	5.2	7.6	10.5	11.6	11.1	9.3
		P_{35}^+	54.0	194.0	66.8	19.4	51.0	273.4
		P_{35}^-	8.4	13.1	4.3	1.2	2.5	17.5
		P_{35}^+/P_{35}^-	6.4	7.2	15.7	16.7	20.5	15.6
P(A14-co-A18)	25	E_a	6.3	6.0	5.5	7.1	4.7	4.0
		E_c	7.3	8.1	4.6	6.9	12.3	4.7
		P_{35}^+	69.6	105.8	65.4	20.9	66.1	311.2
		P_{35}^-	9.1	9.0	2.2	0.6	1.3	8.8
		P_{35}^+/P_{35}^-	7.6	11.7	30.4	34.8	49.4	34.6
P(A12-co-A22)	75	E_a	4.3	4.1	4.5	3.8	5.6	2.7
		E_c	7.7	8.6	10.1	9.9	10.3	6.4
		P_{35}^+	118.2	204.0	62.3	22.4	43.6	267.2
		P_{35}^-	48.3	76.5	23.3	5.3	20.1	50.1
		P_{35}^+/P_{35}^-	2.5	2.7	2.7	4.2	2.2	5.2
P(A12-co-A22)	50	E_a	3.1	3.3	5.8	4.7	6.0	1.3
		E_c	7.8	8.3	8.9	9.1	12.3	6.7
		P_{35}^+	127.5	188.7	59.8	21.1	33.2	352.9
		P_{35}^-	31.5	38.4	7.0	2.1	5.0	30.9
		P_{35}^+/P_{35}^-	4.0	4.9	8.6	9.9	6.6	11.4
P(A12-co-A22)	25	E_a	6.2	8.1	4.0	4.0	4.0	1.5
		E_c	7.8	8.1	11.0	9.2	12.3	9.2
		P_{35}^+	85.3	89.7	77.1	19.6	37.5	375.4
		P_{35}^-	7.3	7.3	1.2	0.3	0.8	4.1
		P_{35}^+/P_{35}^-	11.7	12.3	66.5	57.2	46.8	91.5

Note that monomer 1 refers to the first monomer listed in the copolymer, i.e., for P(A14-co-A18), monomer 1 refers to A14.
All units are the same as in Table 3.

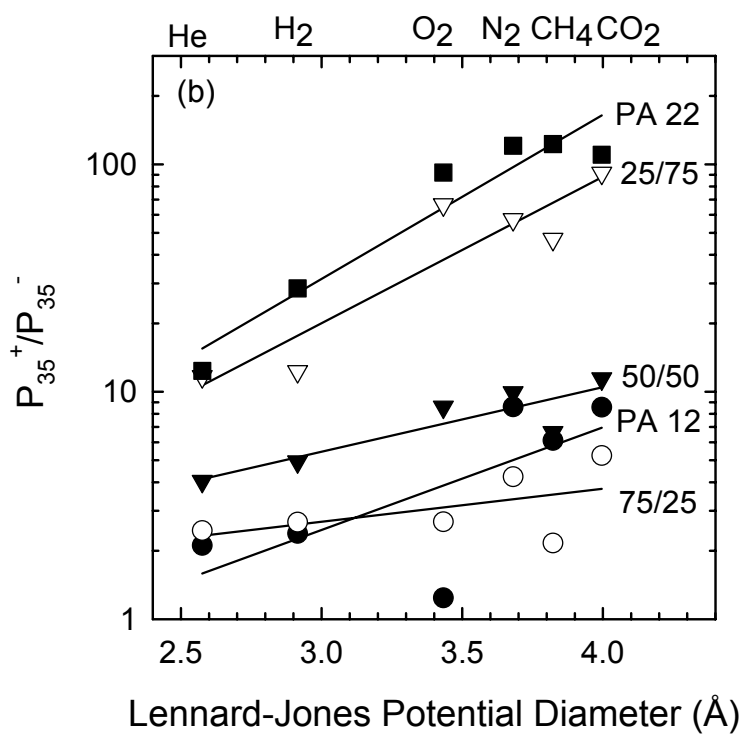
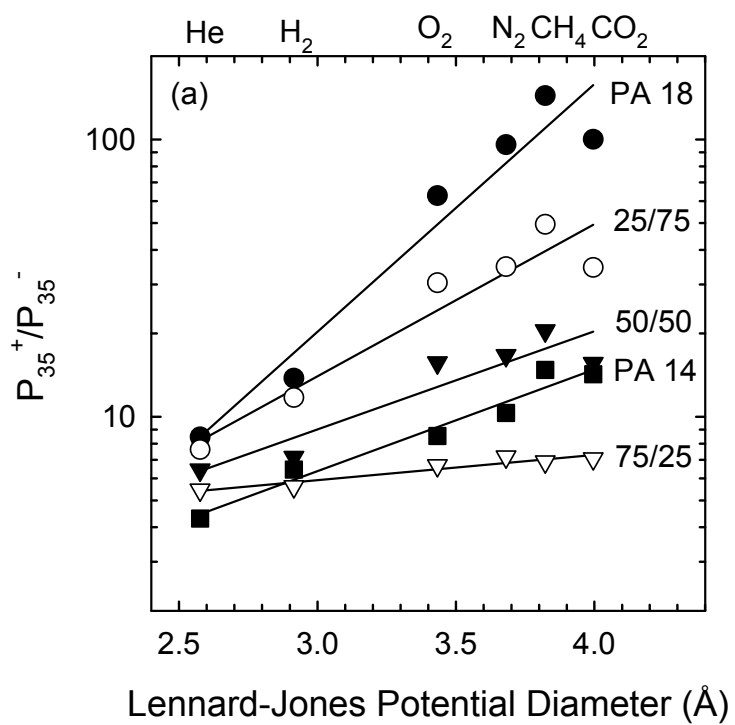


Figure 7.9 Permeation jump ratios calculated at 35°C for P(A14-co-A18) (a) and P(A12-co-A22) (b) copolymers as a function of side-chain length of the penetrant molecule.

jump ratio than the does the copolymer containing 25 mol% of the monomer with the longer side-chain for gases with diameters larger than hydrogen. Figures 7.10 and 7.11 show the jump ratios at 35°C for O₂ and CO₂ as a function of the average side-chain length of the copolymers. The points represent the copolymer data while the line represents the homopolymer data from Figure 7.3. In all cases shown, the copolymers exhibit somewhat smaller permeation jumps than do the homopolymers. The P(A14-co-A18) copolymer system shows smaller differences between the homopolymers and copolymers than does the P(A12-co-A22) system. The ΔH_f from thermal analysis and d-spacings from the SAXS analysis reveal a progressively lower value for the copolymer than the homopolymer at the same values of n with homopolymer-copolymer differences increasing as the differences in length of the side chains in the two copolymers becomes larger. This trend translates to the permeability jump ratio. Thus, even though there is evidence that the copolymers co-crystallize, it appears that there is some loss in crystallinity, and perhaps crystallite aspect ratio, compared to homopolymers of the same n and the effect grows as Δn becomes larger.

Figures 7.12 and 7.13 compare P_{35}^+ and P_{35}^- for O₂ and CO₂ for copolymer systems to that of homopolymers. This type of plot is useful since it shows that the permeability jump ratio is primarily a result of the reduction in permeability in the crystalline data as n increases, and to a lesser extent, the increase in permeability in the molten state as n increases. In addition, these plots provide a means of comparing the crystalline effects on permeability jump for homopolymers and copolymers. In Figure 7.12, the permeability through molten P(A14-co-A18) copolymers, shown as solid circles, clearly align with the molten homopolymer data, taken from Figure 7.5 and

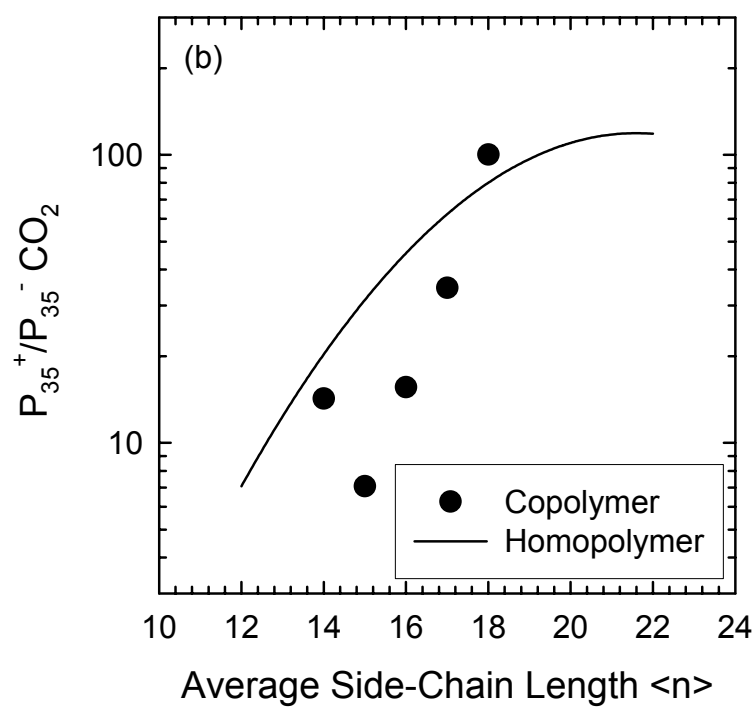
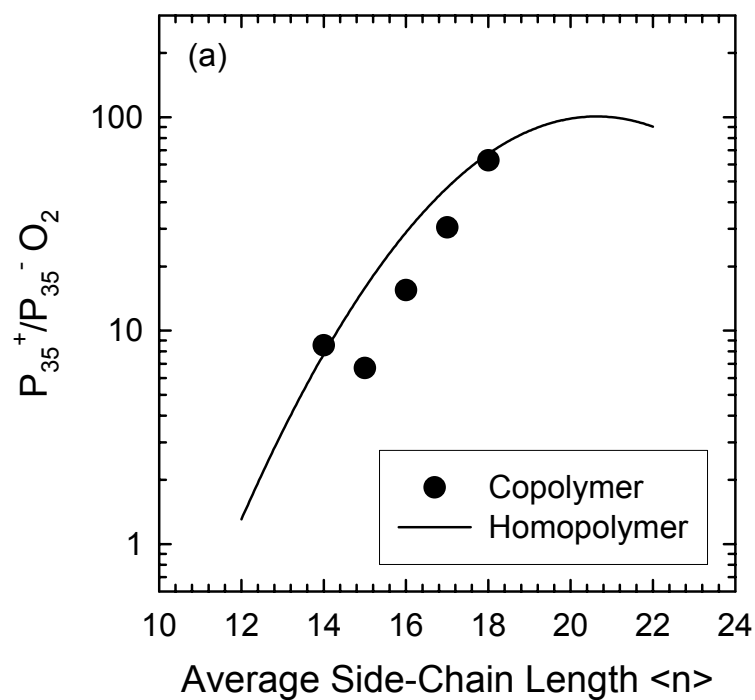


Figure 7.10 Comparison of permeation jump ratios for homopolymers (lines) with P(A14-co-A18) copolymers (points) calculated at 35°C for O₂ (a) and CO₂ (b) gases.

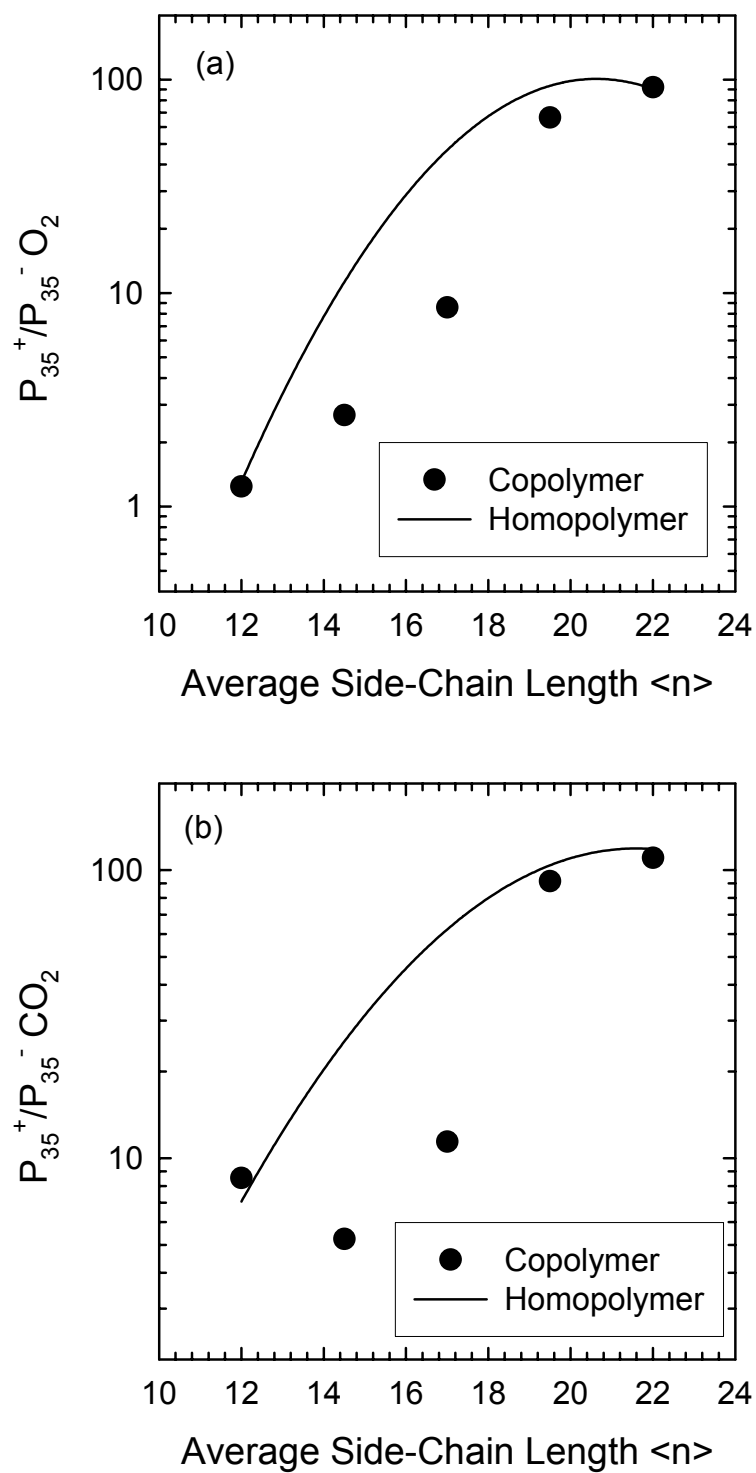


Figure 7.11 Comparison of permeation jump ratios for homopolymers (lines) with P(A12-co-A22) copolymers (points) calculated at 35°C for O_2 (a) and CO_2 (b) gases.

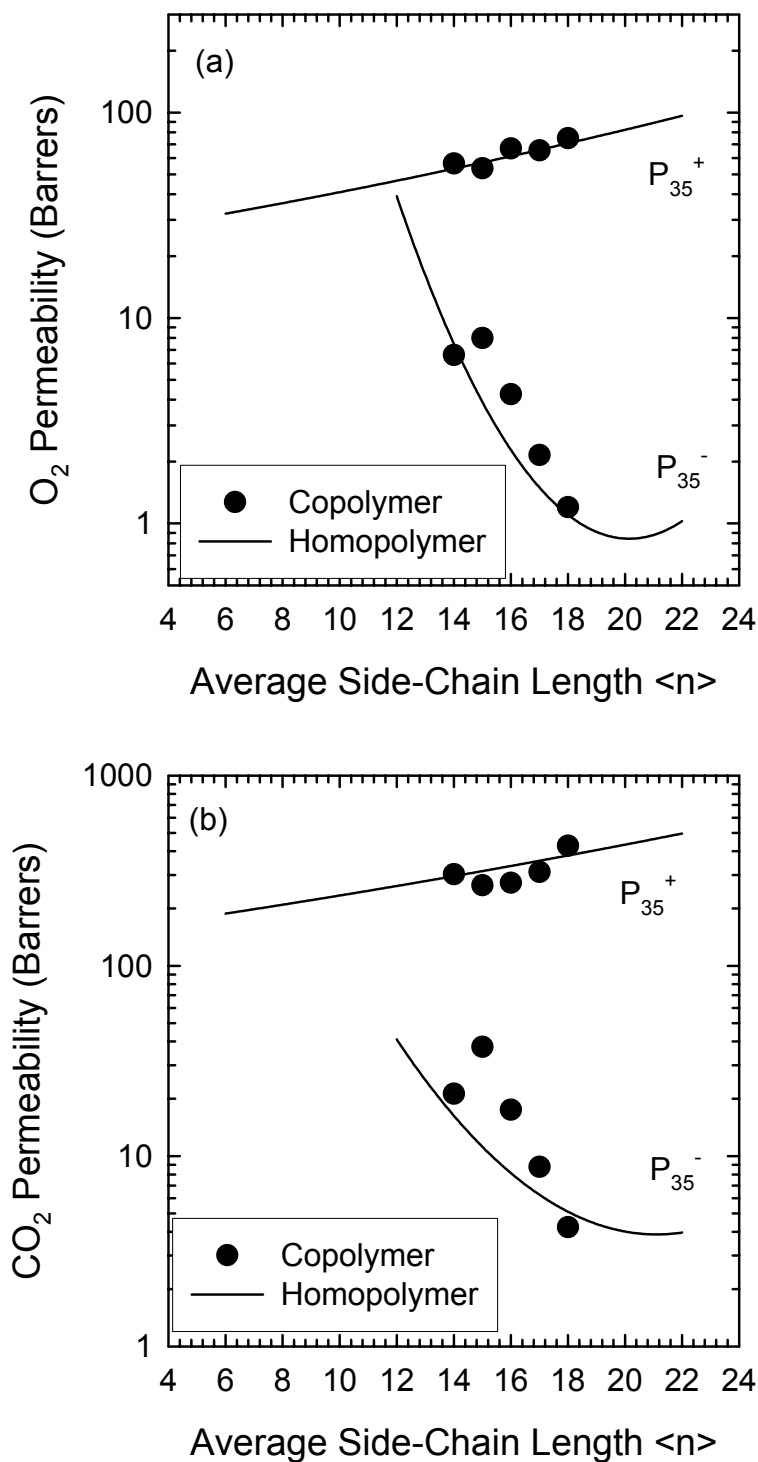


Figure 7.12 Comparison of the O_2 (a) and CO_2 (b) permeability of amorphous (P_{35}^+) and crystalline (P_{35}^-) homopolymers (lines) and P(A14-co-A18) copolymers (points) calculated at 35°C as a function of side-chain length.

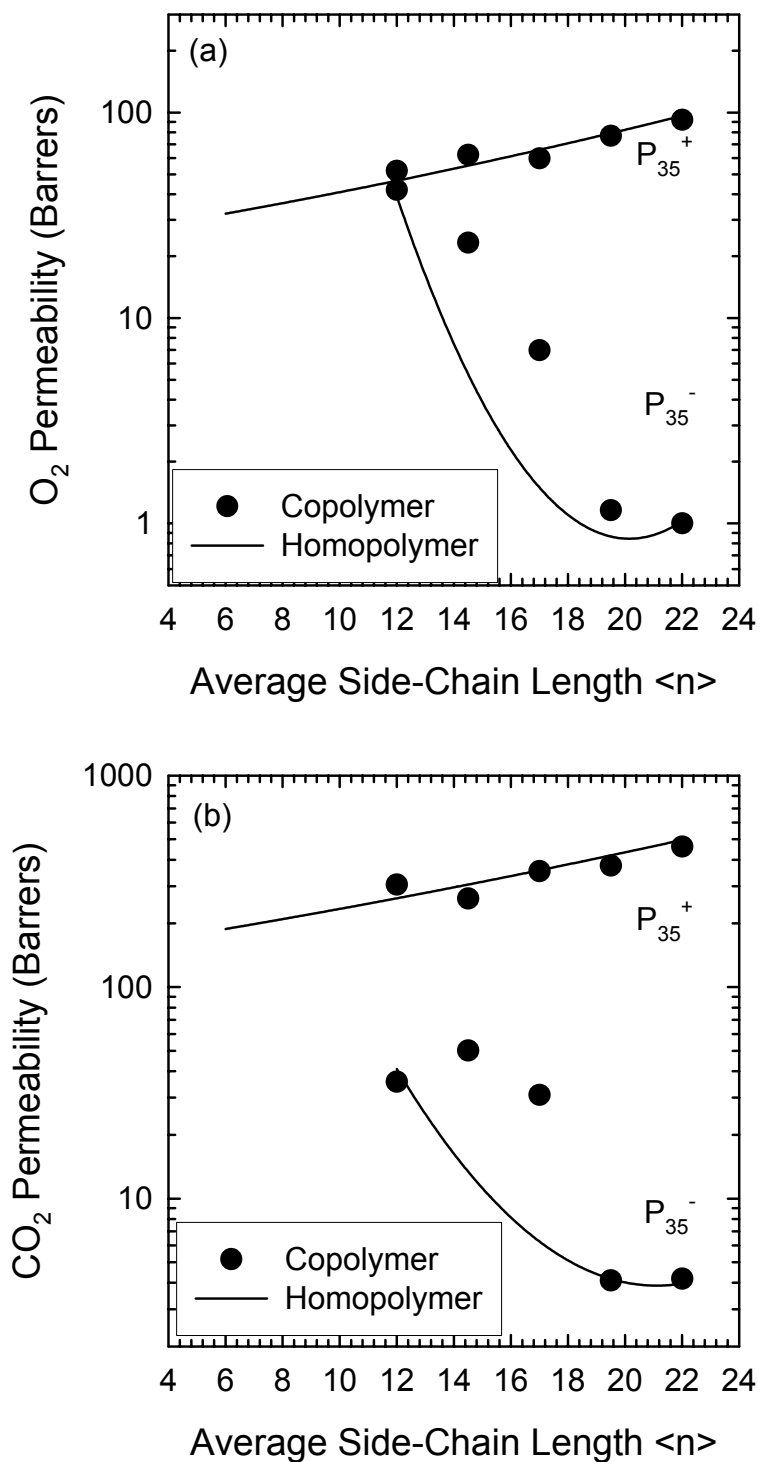


Figure 7.13 Comparison of the O_2 (a) and CO_2 (b) permeability of amorphous (P_{35}^+) and crystalline (P_{35}^-) homopolymers (lines) and P(A12-co-A22) copolymers (points) calculated at 35°C as a function of side-chain length.

shown as the lines in Figure 7.12. This is also true for the permeability of molten P(A12-co-A22) copolymers shown in Figure 7.13. The permeability of crystalline P(A14-co-A18) copolymers deviates slightly from the homopolymer data; the maximum deviation occurs at 75 mol% A14 copolymer. At this composition, there is an increased permeability through the crystalline polymer compared to PA 14. This increased permeability, caused by a decrease in crystallinity is reflected in the diminished jump ratio of Figures 7.9a and 7.10. P(A12-co-A22) copolymers, in Figure 7.13, show an even greater deviation from the crystalline homopolymer data. This is due to the decrease in crystallinity of this copolymer system, as seen from the ΔH_f and d-spacing results in Figures 5.4b and 6.5a. This increase in crystalline permeability is the compelling factor in the decreased jump ratio for this system shown in Figures 7.9b and 7.11.

7.4 Crystalline / non-Crystalline Copolymers

Figures 7.14 through 7.19 show the permeability of O₂ and CO₂ through copolymers of P(A6-co-A22), P(A10-co-A14), and P(A10-co-A18). The permeabilities of the copolymers before and after the melting range are described by the expected Arrhenius-temperature relationships. The copolymers that crystallize show the permeation ‘jump’ on melting as observed previously.

Figures 7.14a and b shows the permeability for O₂ and CO₂ through copolymers of P(A6-co-A22). PA 6 is a purely amorphous polymer and, therefore, exhibits a classic Arrhenius-temperature relationship. As the A22 monomer is copolymerized with A6, the copolymers develop the typical well-defined jump in permeability at the melting temperature. Figures 7.15a and b compare the DSC thermograms for P(A6-co-A22)

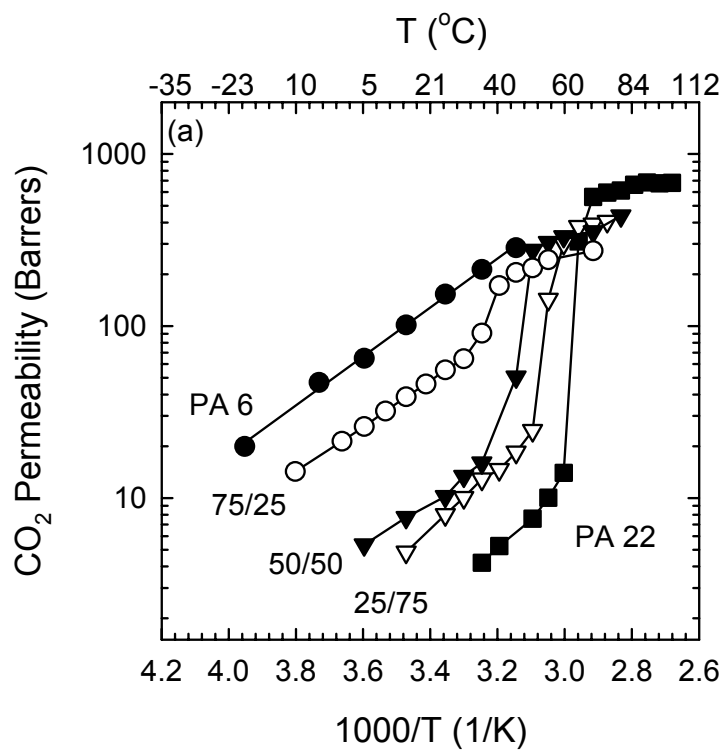
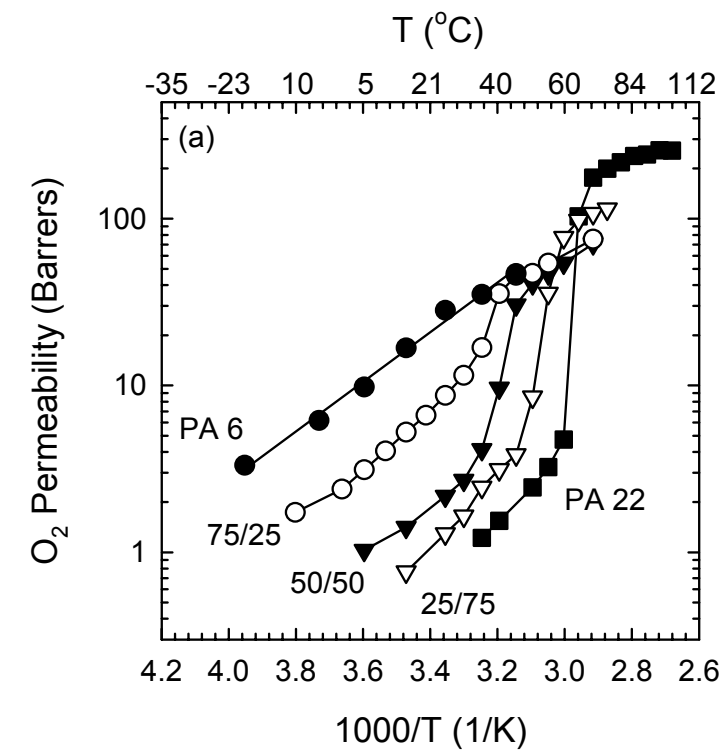


Figure 7.14 Permeability of O_2 (a) and CO_2 (b) in P(6-co-A22) copolymers as a function of temperature plotted on Arrhenius coordinates.

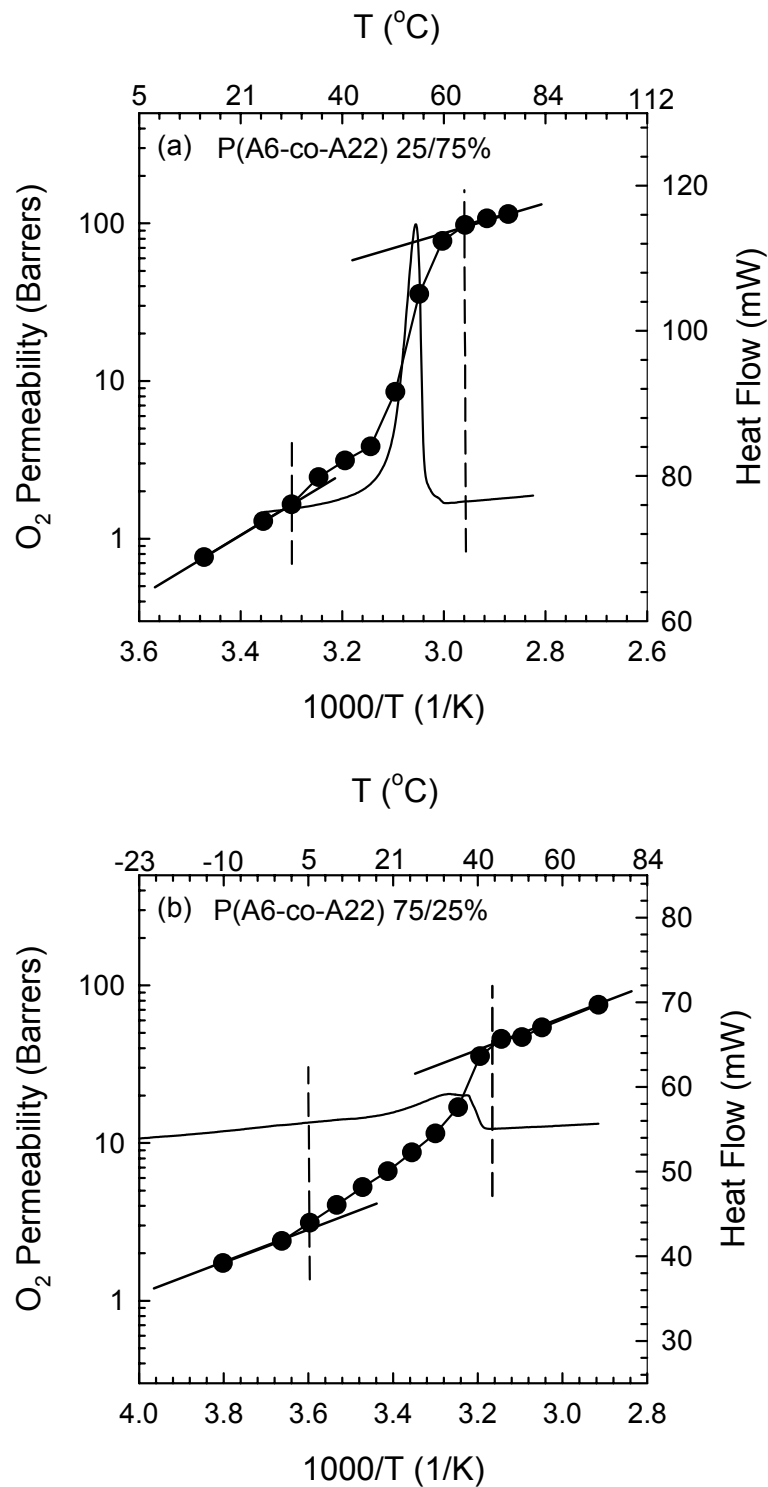


Figure 7.15 Permeability of O_2 in (a) P(A6-co-A22) with 25/75% and (b) with 75/25% as a function of temperature on Arrhenius coordinates with DSC thermograms superimposed on the same temperature scale. The onset and end temperature of the melting peak and permeation jumps are marked with dashed lines.

copolymers with 25/75 and 75/25 mol%, respectively, to their permeabilities. As marked by dashed lines in the figures, the melting onset and end temperatures for the thermograms and permeability jumps are approximately the same; this was also the case for PA 22 and P(A6-co-A22) with 50/50 mol%. These figures emphasize the existence of well-defined permeation jumps for all the P(A6-co-A22) copolymers, including the 75/25% composition that has a limited melting peak.

The permeability coefficients for O₂ and CO₂ through the P(A10-co-A14) copolymers are shown in Figures 7.16a and b. PA10 does not have a T_m in the experimental temperature range and is, therefore, considered amorphous like PA6. As the concentration of A10 increases in the copolymers, the permeation jumps increase in breadth and decrease in height until they eventually take on the Arrhenius-temperature relationship. Unlike the P(A6-co-A22) copolymers and crystallizeable / crystallizeable copolymers, only two compositions, PA14 and P(A10-co-A14) with 25/75%, appear to have strong and well-defined permeation jumps in Figure 7.16. The permeability of O₂ gas through the P(A10-co-A14) 25/75, 50/50, and 75/25% compositions is more closely examined in Figure 7.17. Figure 7.17a is similar to Figure 7.15a for P(A6-co-A22) 25/75%; the melting onset and end temperatures for the permeation jumps and DSC thermograms correlate well with each other. The permeation jump is well defined and easily contrasted from the pre-and post - T_m permeability behavior. Figure 7.17b compares the O₂ permeability response and the DSC endotherm for the P(A10-co-A14) 50/50% composition; it is clear that a permeability jump is present and that the melt onset and end temperatures align, although they are not as well defined as for the 25/75%

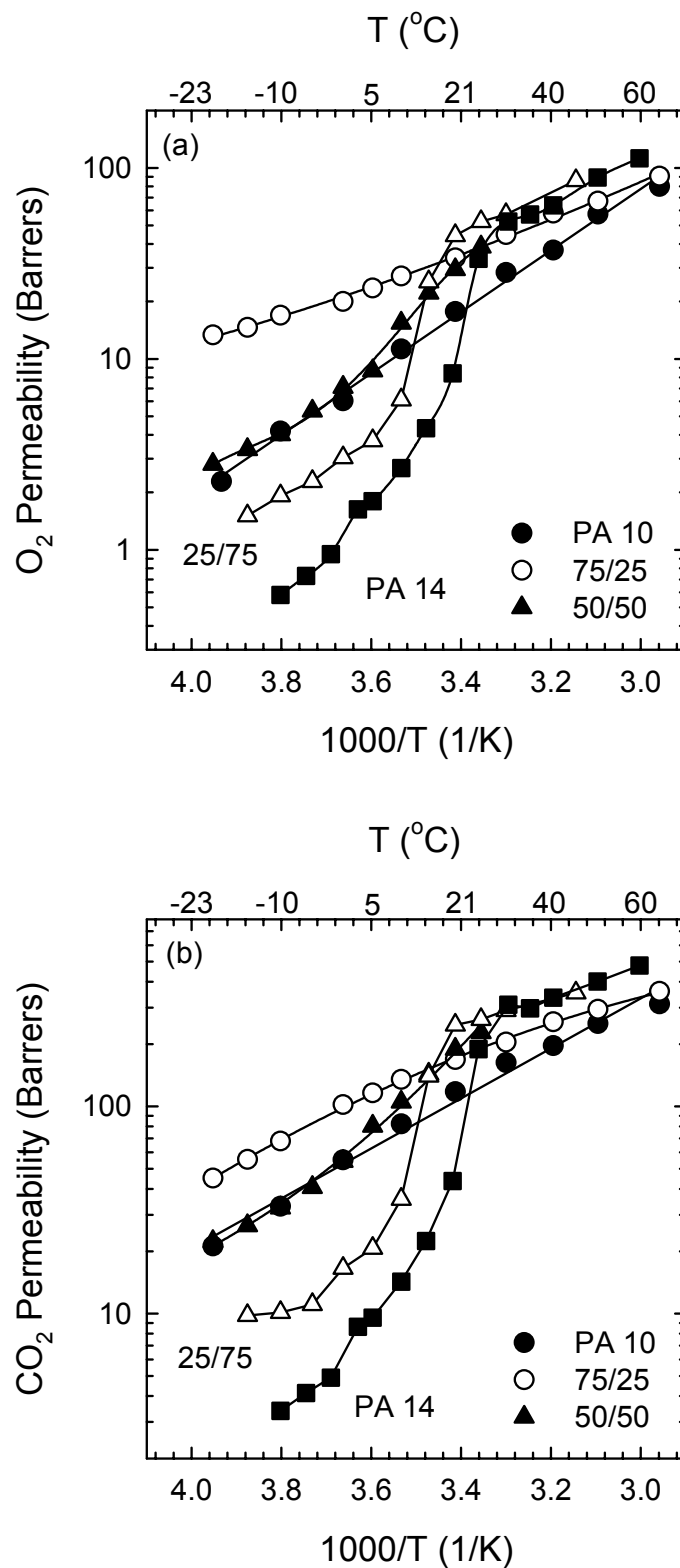
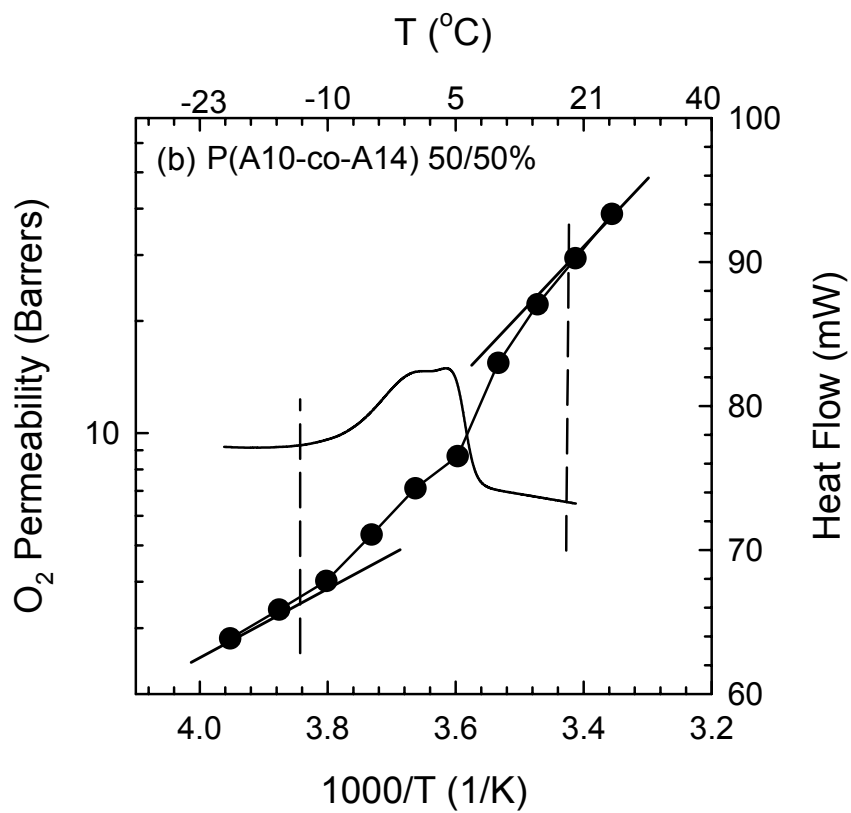
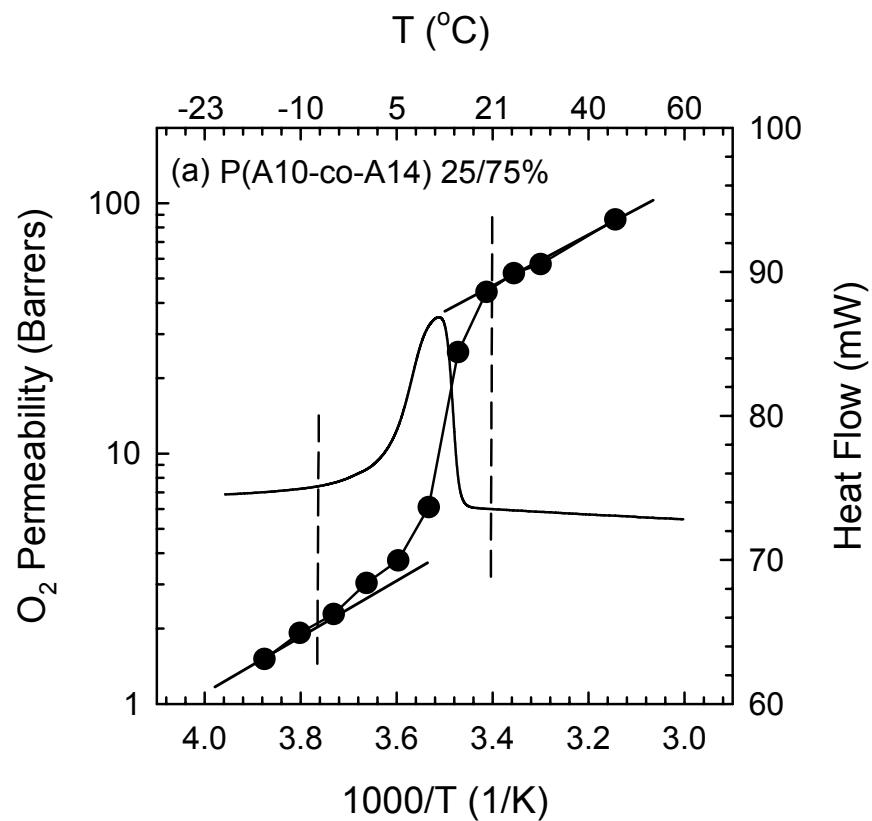


Figure 7.16 Permeability of O_2 (a) and CO_2 (b) in P(10-co-A14) copolymers as a function of temperature on Arrhenius coordinates.



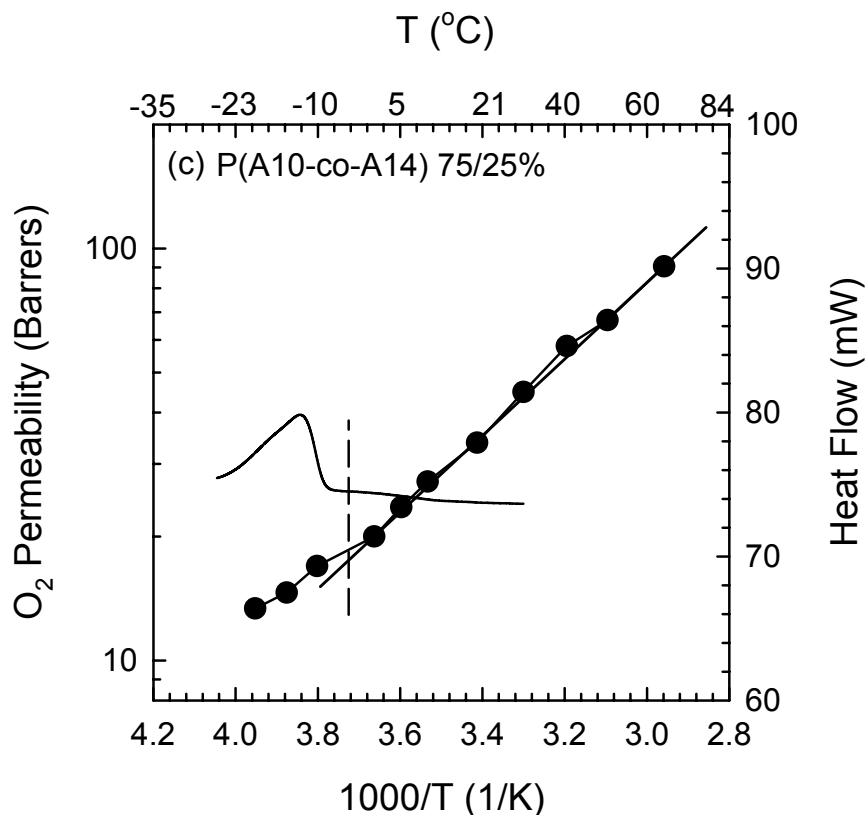


Figure 7.17 Permeability of O₂ in (a) P(A10-co-A14) with 25/75%, (b) 50/50%, and (c) 75/25% as a function of temperature on Arrhenius coordinates with DSC thermograms superimposed on the same temperature scale. The onset and end temperature of the melting peak and permeation jumps are marked with dashed lines.

composition in Figure 7.17a. The P(A10-co-A14) 75/25% copolymer appears to have no measurable permeability jump when compared to the other copolymers in Figure 7.16; however, when examined in detail as in Figure 7.17c, a slight change in slope of the permeability is present at the onset and end of the melting curve. The melting peak for the copolymer is in the lowest measurable temperature range for both experiments and only the end melting temperature can be accurately assessed; however, it is presumed that the permeation data would show a similar subtle change in slope at the onset temperature for melting.

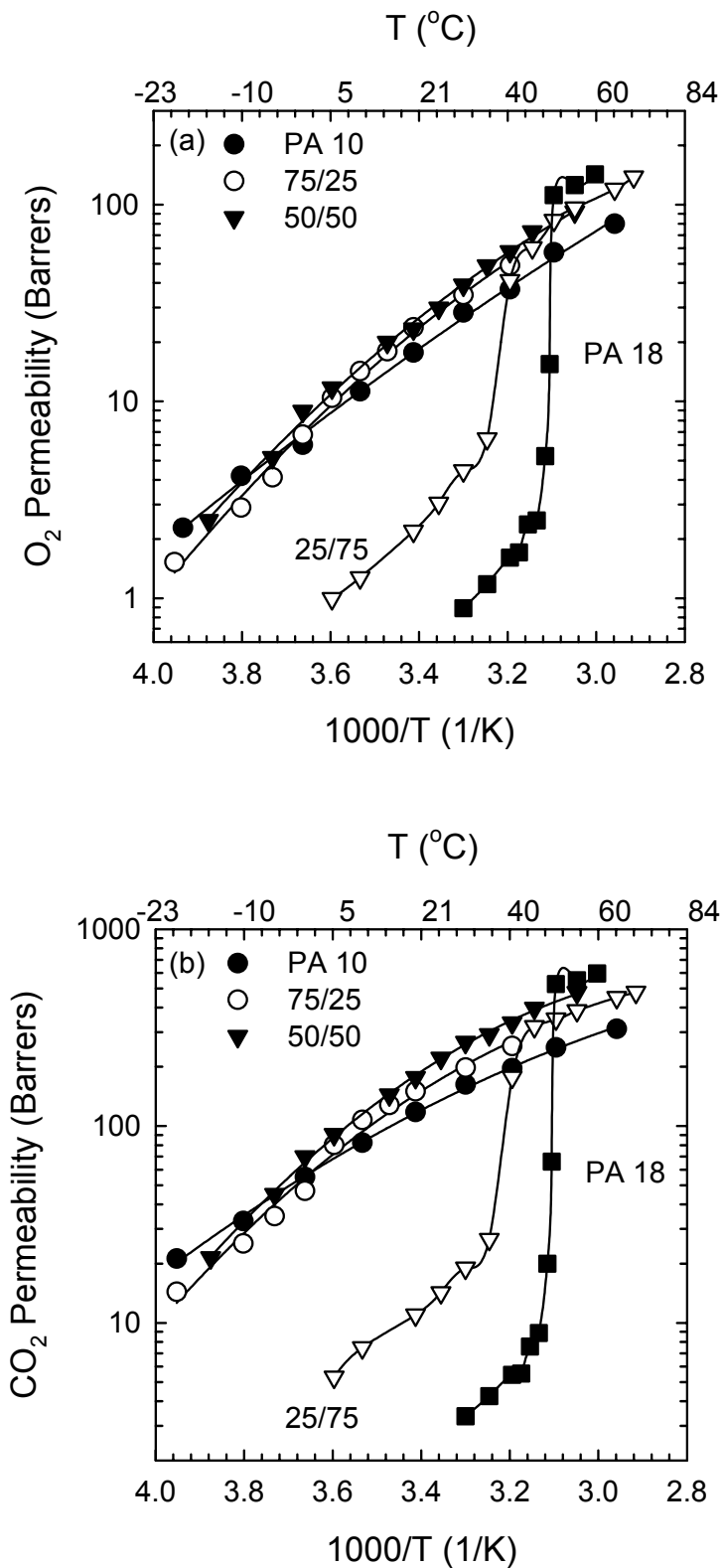
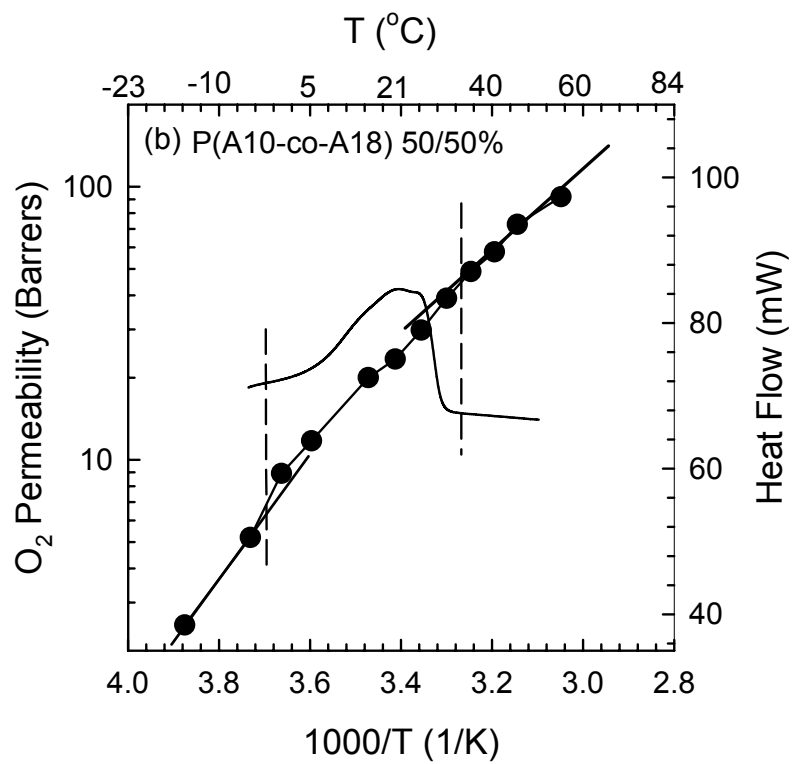
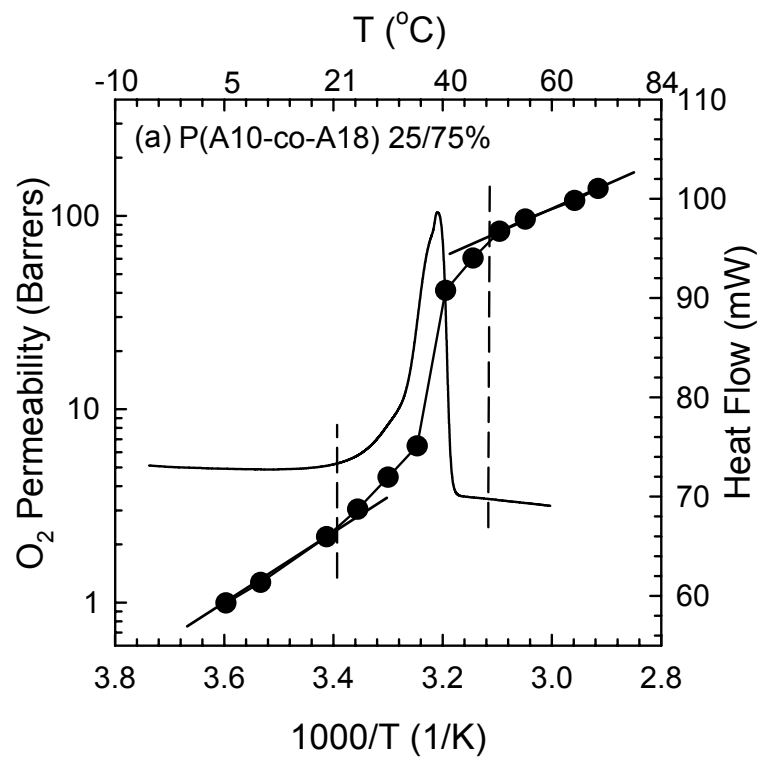


Figure 7.18 Permeability in O_2 (a) and CO_2 (b) for P(10-co-A18) copolymers as a function of temperature on Arrhenius coordinates.



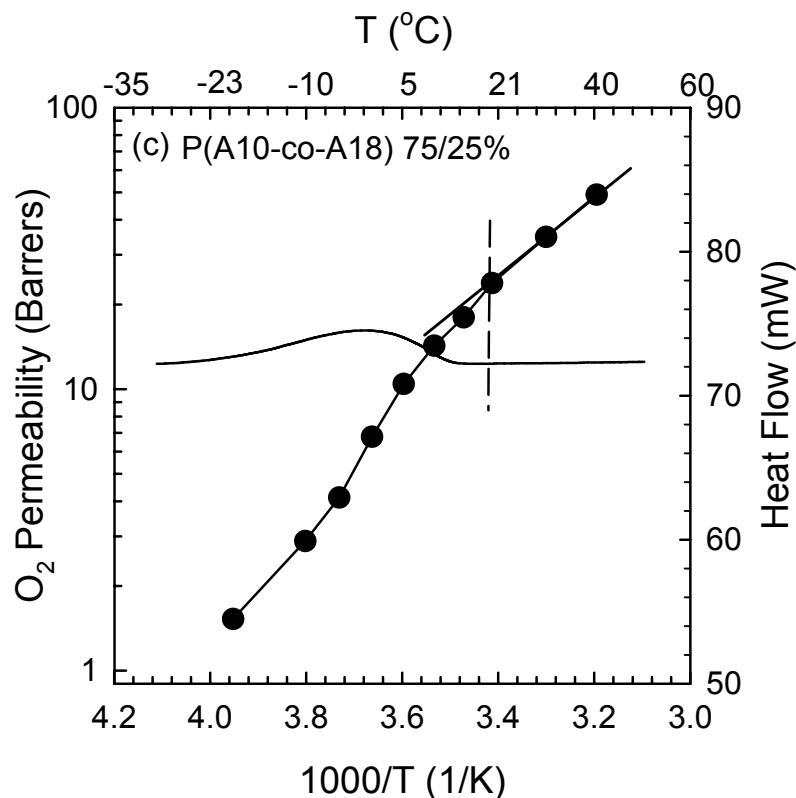


Figure 7.19 Permeability in O₂ gas for (a) P(A10-co-A18) with 25/75%, (b) 50/50%, and (c) 75/25% as a function of temperature on Arrhenius coordinates with DSC thermograms superimposed in the same temperature scale. The onset and end temperature of the melting peak and permeation jumps are marked with dashed lines.

Permeation data for O₂ and CO₂ are shown in Figure 7.18 for copolymers of P(A10-co-A18); like the P(A10-co-A14) system in Figure 7.16, only the PA18 and P(A10-co-A18) 25/75% composition show significant, well-defined permeation jumps. Figures 7.19a through 7.19c evaluate the permeation jumps more closely for comparisons with their melting endotherms. P(A10-co-A18) with 25/75% , see Figure 7.19a, shows the typical well-defined permeation jump; the melting onset and end temperatures of the endotherm match those of the permeation jump. The 50/50 and 75/25% compositions, Figures 7.19b and 7.19c, have much less defined permeation jumps; however, like

P(A10-co-A14) 75/25%, subtle, broad jumps do exist and correlate with the DSC endotherms. This broadening of the permeation jump in these copolymers may ultimately be a solution to modified atmospheric packaging needs by providing a broad, rapid change in permeability with temperature.

The onset and end temperatures as defined above are plotted in Figures 7.20a, 7.21a, and 7.22a for P(A6-co-A22), P(A10-co-A14), and P(A10-co-A18) copolymer systems, respectively. As expected, linear relationships exist between the onset and end temperatures obtained from the DSC thermograms and from the permeation jump. The onset and end temperature for the permeation jumps are plotted versus the average side-chain length of the copolymers, $\langle n \rangle$, in Figures 7.20b, 7.21b, and 7.22b as open and closed data points, respectively. Similar data for P(A14-co-A18), taken from Figure 7.8b, are superimposed onto the plots as dashed lines to compare the crystallizeable-crystallizeable copolymers to the crystallizeable / non-crystallizeable copolymers. The P(A6-co-A22) system is very different from P(A14-co-A18) while the data for P(A10-co-A14) and P(A10-co-A18) copolymers, see Figures 7.21b and 7.22b, align well with the corresponding results for the P(A14-co-A18) system.

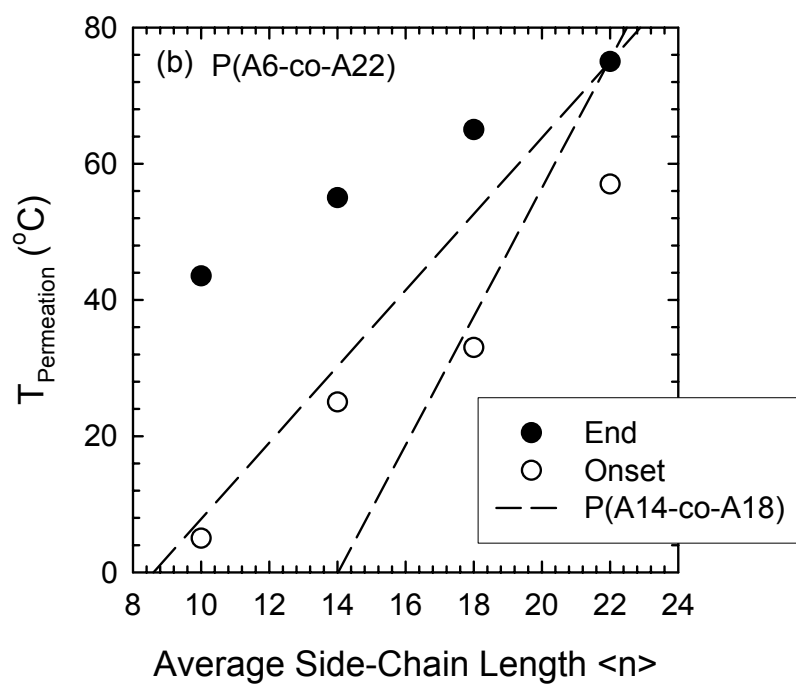
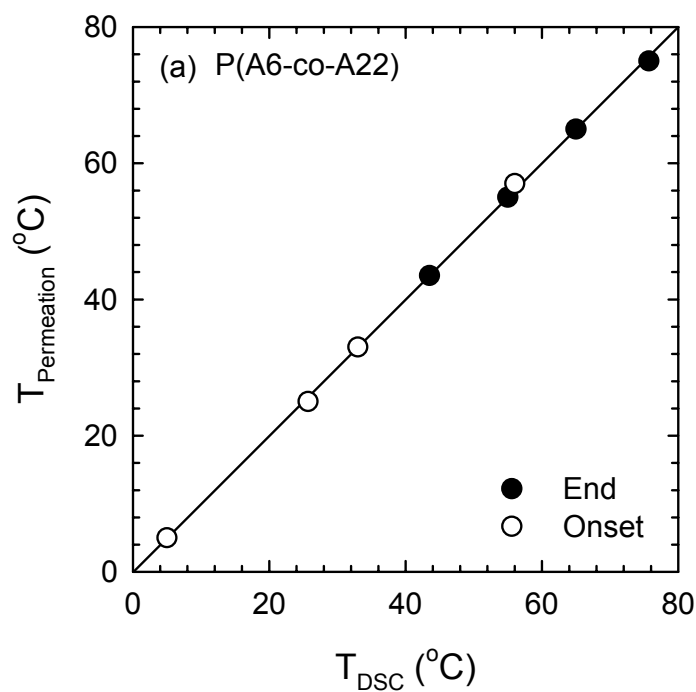


Figure 7.20 Relationship between onset and end temperatures for melting of P(A6-co-A22) copolymers as measured by permeability jumps and DSC endotherms (a). Correlation between the onset and end temperatures for copolymers of P(A6-co-A22) (points) and P(A14-co-A18) (dashed-lines) with the average side-chain length (b).

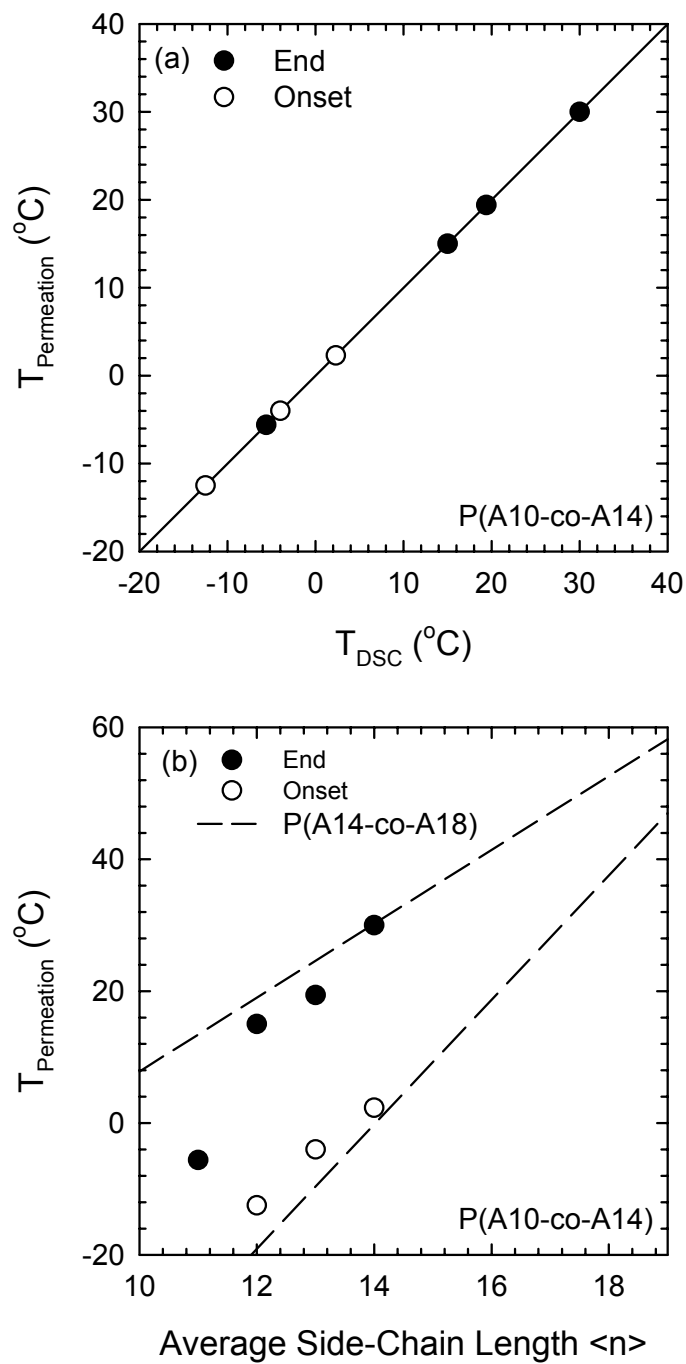


Figure 7.21 Relationship between onset and end temperatures for melting of P(A10-co-A14) copolymers as measured by permeability jumps and DSC endotherms (a). Correlation between the onset and end temperatures for copolymers of P(A10-co-A14) (points) and P(A14-co-A18) (dashed-lines) with the average side-chain length (b).

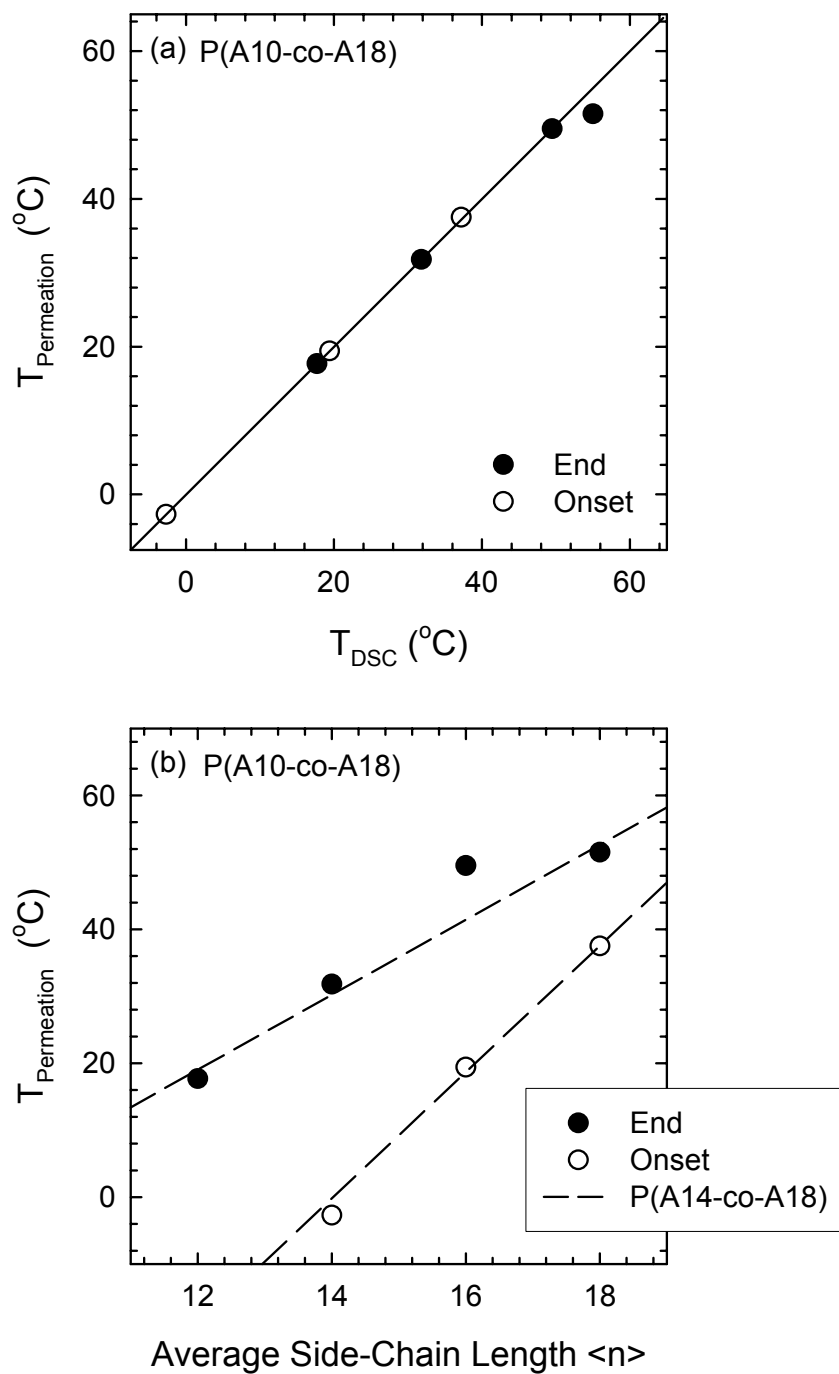


Figure 7.22 Relationship between onset and end temperatures for melting of P(A10-co-A18) copolymers as measured by permeability jumps and DSC endotherms (a). Correlation between the onset and end temperatures for copolymers of P(A10-co-A18) (points) and P(A14-co-A18) (dashed-lines) with the average side-chain length (b).

The magnitudes of the permeation jumps for each copolymer system were calculated using Equation 7.2 and are listed in Table 7.3. Permeation jump ratios for each of the copolymers are plotted versus the penetrant gas diameter in Figures 7.23-7.25 and average side-chain length of the copolymer in Figures 7.26-7.28. The amorphous and crystalline permeabilities are also shown in terms of the average side-chain length in Figures 7.29-7.31. As mentioned previously, the Arrhenius activation energies in the semi-crystalline state (E_c) are greater than those in the amorphous phases (E_a); therefore, the reference temperature to which the permeability data are extrapolated affects the jump ratio calculations.[1-3] Hence, we have used a median temperature for all the copolymers in a system as the reference temperature: 35°C for P(A6-co-A22) and P(A10-co-A18), and 10°C for P(A10-co-A14). Using a 35°C reference temperature for P(A10-co-A14) results in slightly skewed jump ratios because the melting temperatures of the copolymers in this system are significantly lower than 35°C. Although the onset and end melting temperatures could be observed for most of the copolymers in Figures 7.17 and 7.19, the permeability-temperature relationship could not always be measured adequately well enough below T_m to draw a conclusive Arrhenius-temperature line. Therefore, permeability jump ratios were not calculated for all the compositions.

Table 7.3
Activation Energy and Permeability Data Extrapolated to 35°C or 10 °C for
Various Gases Through Poly (n-alkyl acrylate) Crystallizeable / non-Crystallizeable
Copolymers

Copolymer	Mol % monomer 1		Gas					
			He	H ₂	O ₂	N ₂	CH ₄	CO ₂
P(A6-co-A22)	75	E _a	2.7	2.9	3.7	1.7	3.5	1.6
		E _c	6.1	9.3	10.5	12.3	12.6	4.8
		P ₃₅ ⁺	106.6	154.3	57.1	33.0	63.7	301.7
		P ₃₅ ⁻	13.9	13.9	2.4	0.7	1.7	12.9
		P ₃₅ ⁺ /P ₃₅ ⁻	7.7	11.1	23.4	44.9	37.2	23.4
P(A6-co-A22)	50	E _a	2.9	2.9	6.2	2.3	3.9	3.1
		E _c	6.0	6.0	6.2	7.9	9.8	5.9
		P ₃₅ ⁺	110.1	162.9	25.4	24.3	32.5	224.0
		P ₃₅ ⁻	15.2	17.2	3.0	1.1	2.2	15.0
		P ₃₅ ⁺ /P ₃₅ ⁻	7.3	9.5	8.5	22.6	15.1	15.0
P(A6-co-A22)	25	E _a	5.7	3.1	5.1	5.7	5.5	2.4
		E _c	7.6	7.1	5.5	9.2	7.3	5.8
		P ₃₅ ⁺	87.1	184.0	32.1	11.5	29.9	186.3
		P ₃₅ ⁻	45.1	55.5	8.0	4.9	11.6	72.5
		P ₃₅ ⁺ /P ₃₅ ⁻	1.9	3.3	4.0	2.4	2.6	2.6
P(A10-co-A14)	75	E _a	3.5	2.2	4.2	4.9	5.8	3.1
		E _c	-	-	-	-	-	-
		P ₁₀ ⁺	35.1	63.3	26.9	8.5	12.8	136.5
		P ₁₀ ⁻	-	-	-	-	-	-
		P ₁₀ ⁺ /P ₁₀ ⁻	-	-	-	-	-	-
P(A10-co-A14)	50	E _a	6.6	4.4	7.9	8.1	8.6	5.4
		E _c	-	-	-	-	-	-
		P ₁₀ ⁺	21.8	36.2	19.2	6.1	14.6	139.3
		P ₁₀ ⁻	-	-	-	-	-	-
		P ₁₀ ⁺ /P ₁₀ ⁻	-	-	-	-	-	-
P(A10-co-A14)	25	E _a	5.1	3.6	4.8	7.9	8.3	2.7
		E _c	4.4	3.0	6.5	6.5	4.8	5.1
		P ₁₀ ⁺	28.0	54.3	33.5	6.9	17.2	209.0
		P ₁₀ ⁻	9.7	9.5	4.6	0.9	1.3	21.5
		P ₁₀ ⁺ /P ₁₀ ⁻	2.9	5.7	7.4	7.8	13.1	9.7

P(A10-co-A14)	0	E_a	5.8	5.7	5.7	6.5	6.5	3.7
		E_c	6.1	6.4	8.7	10.1	8.6	6.6
		P_{10}^+	22.7	42.3	24.8	8.7	25.8	177.5
		P_{10}^-	5.0	5.9	1.9	0.5	1.3	8.3
		P_{10}^+/P_{10}^-	4.5	7.1	13.2	17.3	20.1	21.5
P(A10-co-A18)	75	E_a	8.3	8.1	6.6	9.4	6.6	4.9
		E_c	-	-	-	-	-	-
		P_{35}^+	46.5	45.7	41.4	48.8	41.5	225.0
		P_{35}^-	-	-	-	-	-	-
		P_{35}^+/P_{35}^-	-	-	-	-	-	-
P(A10-co-A18)	50	E_a	7.6	8.2	6.4	9.0	10.4	7.3
		E_c	-	-	-	-	-	-
		P_{35}^+	56.8	98.6	49.6	21.2	58.3	291.1
		P_{35}^-	-	-	-	-	-	-
		P_{35}^+/P_{35}^-	-	-	-	-	-	-
P(A10-co-A18)	25	E_a	6.4	6.1	5.4	6.5	6.0	3.4
		E_c	7.3	8.5	9.4	11.5	10.6	7.8
		P_{35}^+	78.1	137.3	55.4	17.7	42.4	272.0
		P_{35}^-	18.7	23.1	4.9	1.6	3.6	21.6
		P_{35}^+/P_{35}^-	4.2	6.0	11.3	11.0	11.7	12.6

Note that monomer 1 refers to the first monomer listed in the copolymer, i.e., for P(A14-co-A18), monomer 1 refers to A14.
All units are the same as in Table 3.

Figures 7.23-7.25 show that the jump ratios for all the copolymer systems increase with increasing average side-chain length as well as penetrant diameter. The results are similar to those for the homopolymers (Figure 7.2) and copolymer systems (Figure 7.9) reported previously. As explained, the permeation jump is partially the result of melting the impermeable crystallites thus eliminating the tortuous path for the molecules as well as the effects included in the chain immobilization factor, β , which accounts for how the segmental dynamics of the polymer are affected by the crystallites.[7-9] This factor becomes greater the larger the penetrant molecule.

The jump ratios for the copolymer systems are also plotted against their average side-chain lengths in Figure 7.26-7.28. The copolymers are plotted as data points while homopolymers, taken from Figure 7.3, are superimposed as lines in the plots. These are similar to Figures 7.10 and 7.11 for P(14-co-A18) and P(A12-co-A22). Although both P(A10-co-A14) and P(A10-co-A18) have only two permeation jumps each and therefore limited data points each to compare to the homopolymer data, in general, they like those for P(A6-co-A22), are lower than the homopolymer jump ratios.

Figures 7.29-7.31 compare the permeabilities for O₂ and CO₂ gas at 35°C, P_{35}^{+} and P_{35}^{-} , for the copolymer systems. Results for P(A6-co-A22), P(A10-co-A14), and P(A10-co-A18) are plotted as data points while the homopolymers, taken from Figure 7.5, are superimposed as lines. Both the amorphous and crystalline phase permeability data for all three systems align fairly well with the homopolymer data. The increasing amorphous phase permeabilities reflect the increased free volume associated with additional carbons in the longer side-chain lengths while the decrease in crystalline state permeability with average side-chain length is associated with the an increased tortuosity

due to the larger and more perfect crystals; the crystalline state permeabilities are the dominant factor in the permeation jump ratios.[1, 3]

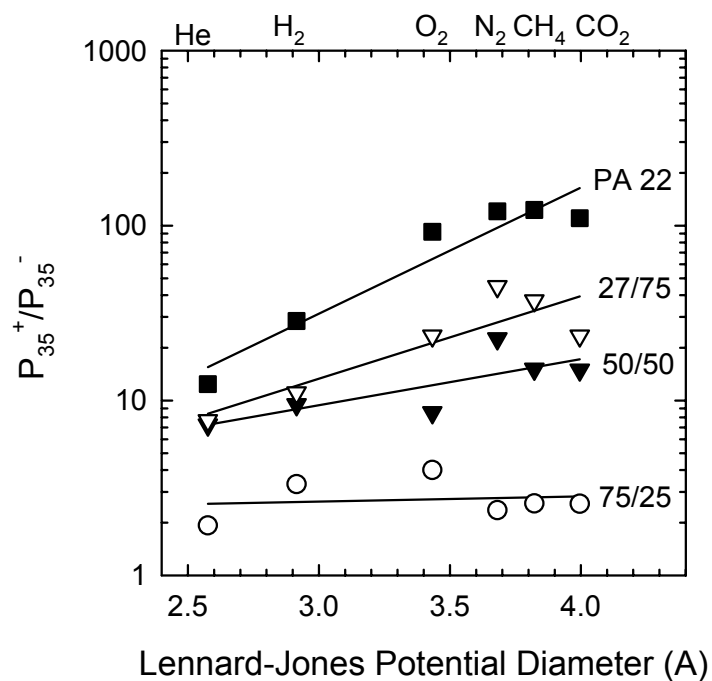


Figure 7.23 Permeation jump ratios calculated at 35°C for P(A6-co-A22) copolymers shown as a function of the penetrant molecule diameter.

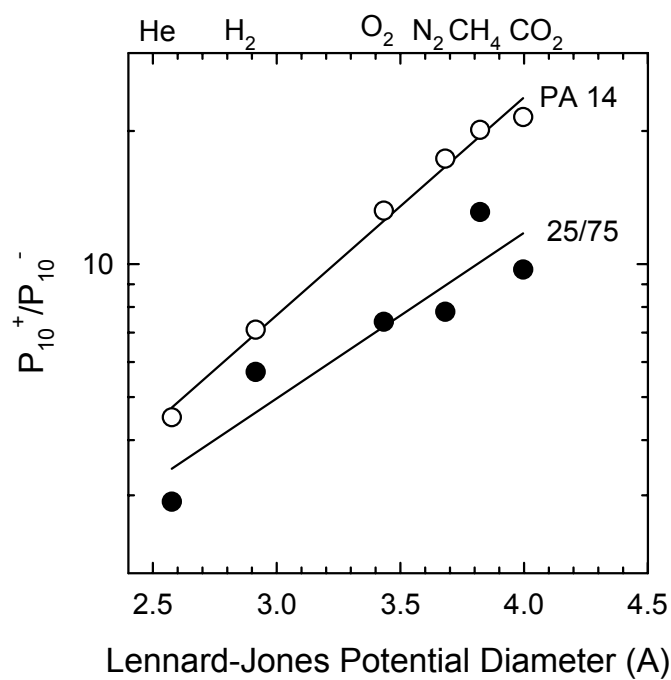


Figure 7.24 Permeation jump ratios calculated at 10°C for P(A10-co-A14) copolymers shown as a function of the penetrant molecule diameter.

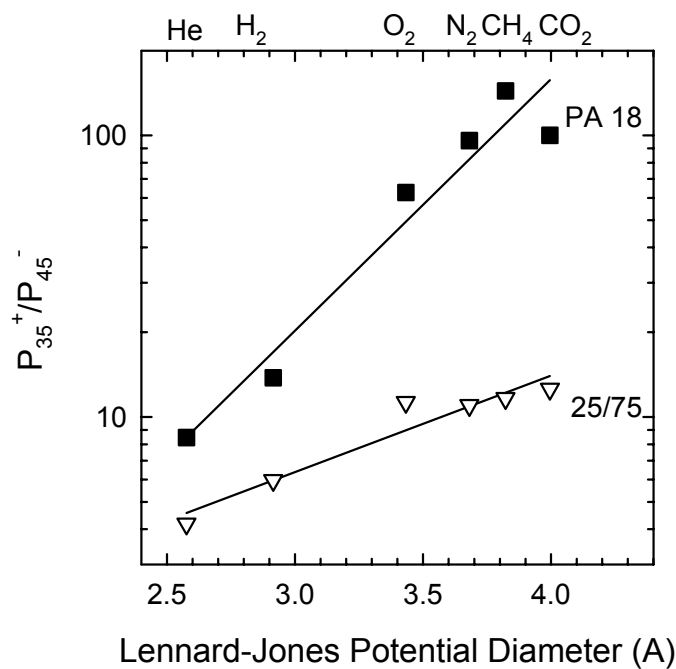


Figure 7.25 Permeation jump ratios calculated at 35°C for P(A10-co-A18) copolymers shown as a function of the penetrant molecule diameter.

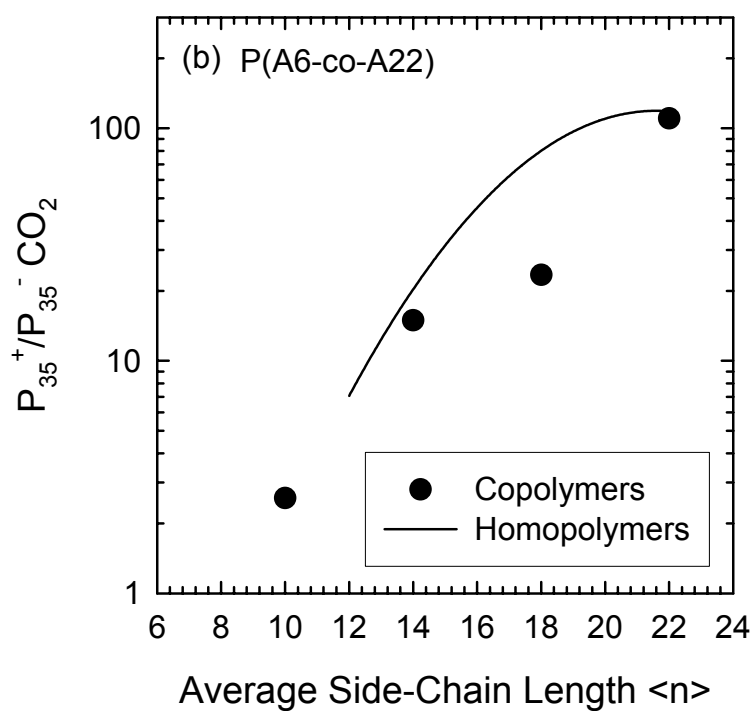
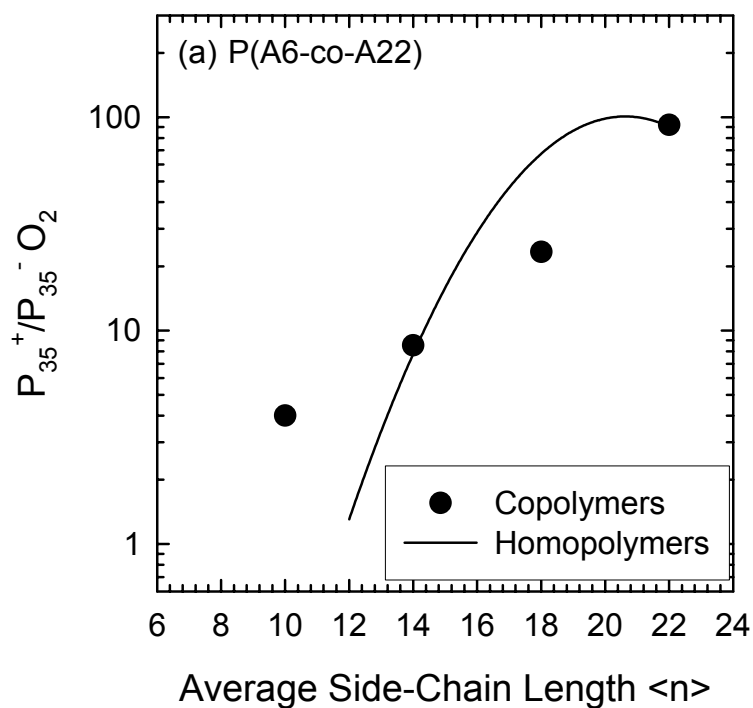


Figure 7.26 Comparison of permeation jump ratios for homopolymers (lines) with P(A6-co-A22) copolymers (points) calculated at 35°C for O₂ (a) and CO₂ (b) gases.

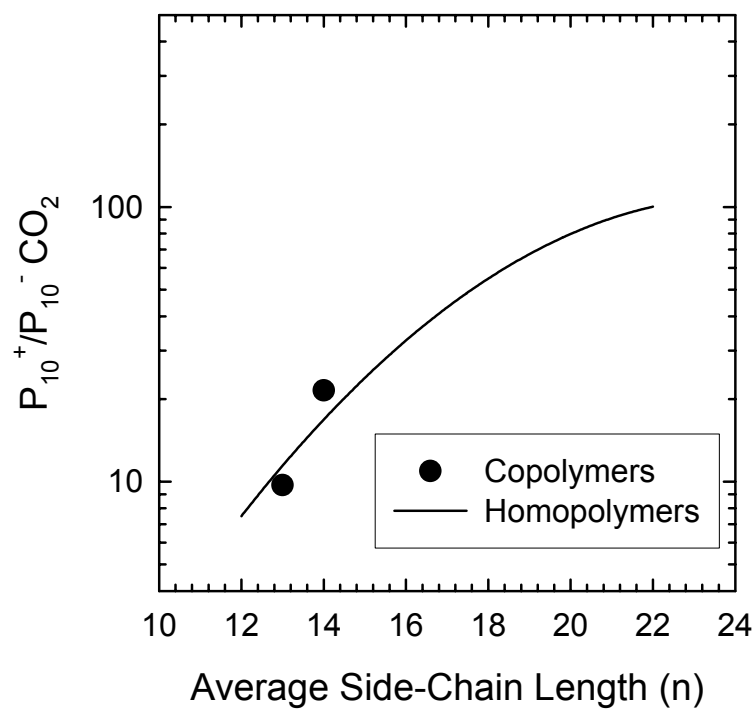
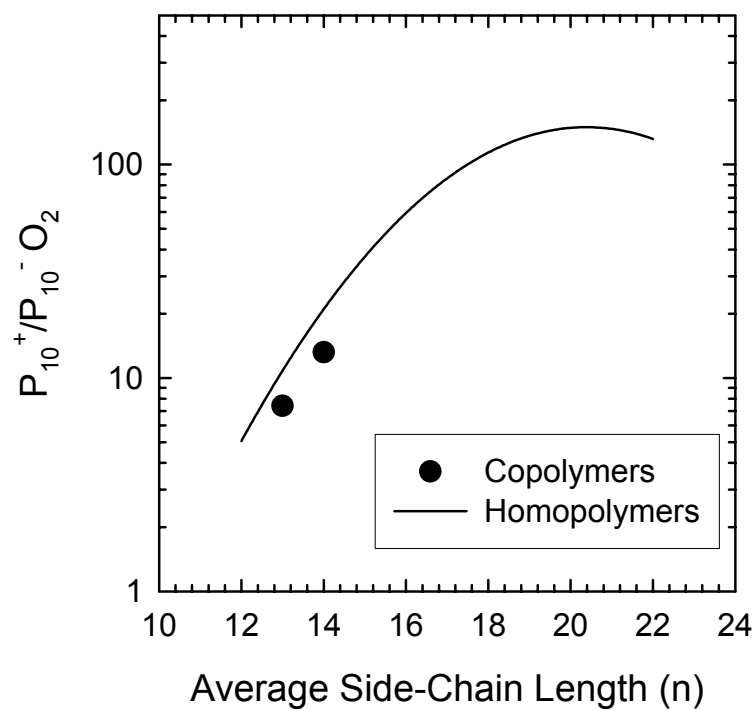


Figure 7.27 Comparison of permeation jump ratios for homopolymers (lines) with P(A10-co-A14) copolymers (points) calculated at 10°C for O₂ (a) and CO₂ (b) gases.

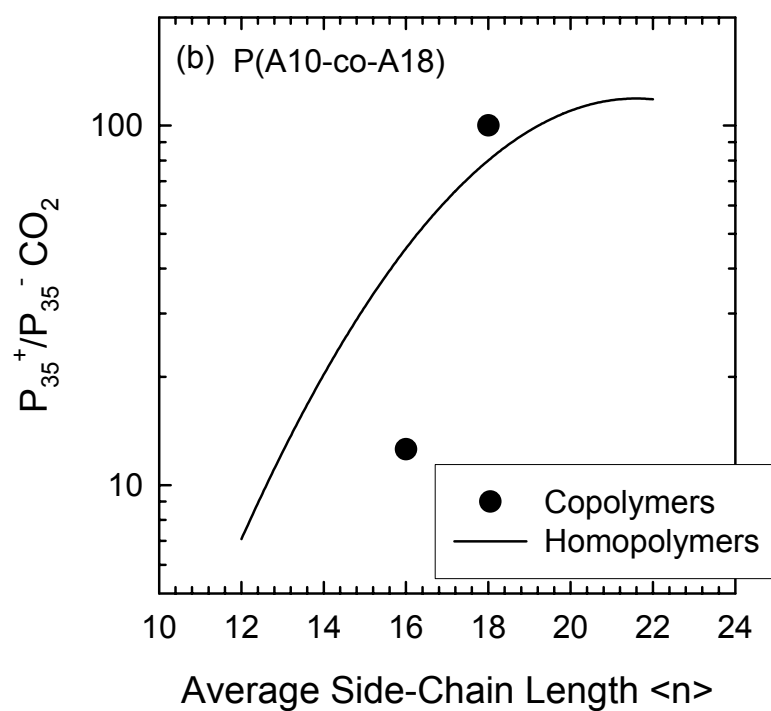
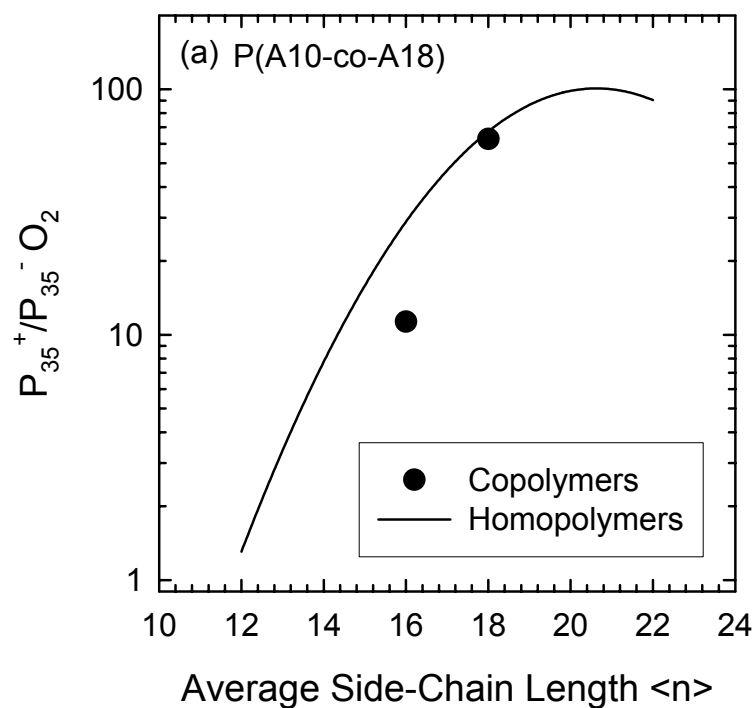


Figure 7.28 Comparison of permeation jump ratios for homopolymers (lines) with P(A10-co-A18) copolymers (points) calculated at 35°C for O₂ (a) and CO₂ (b) gases.

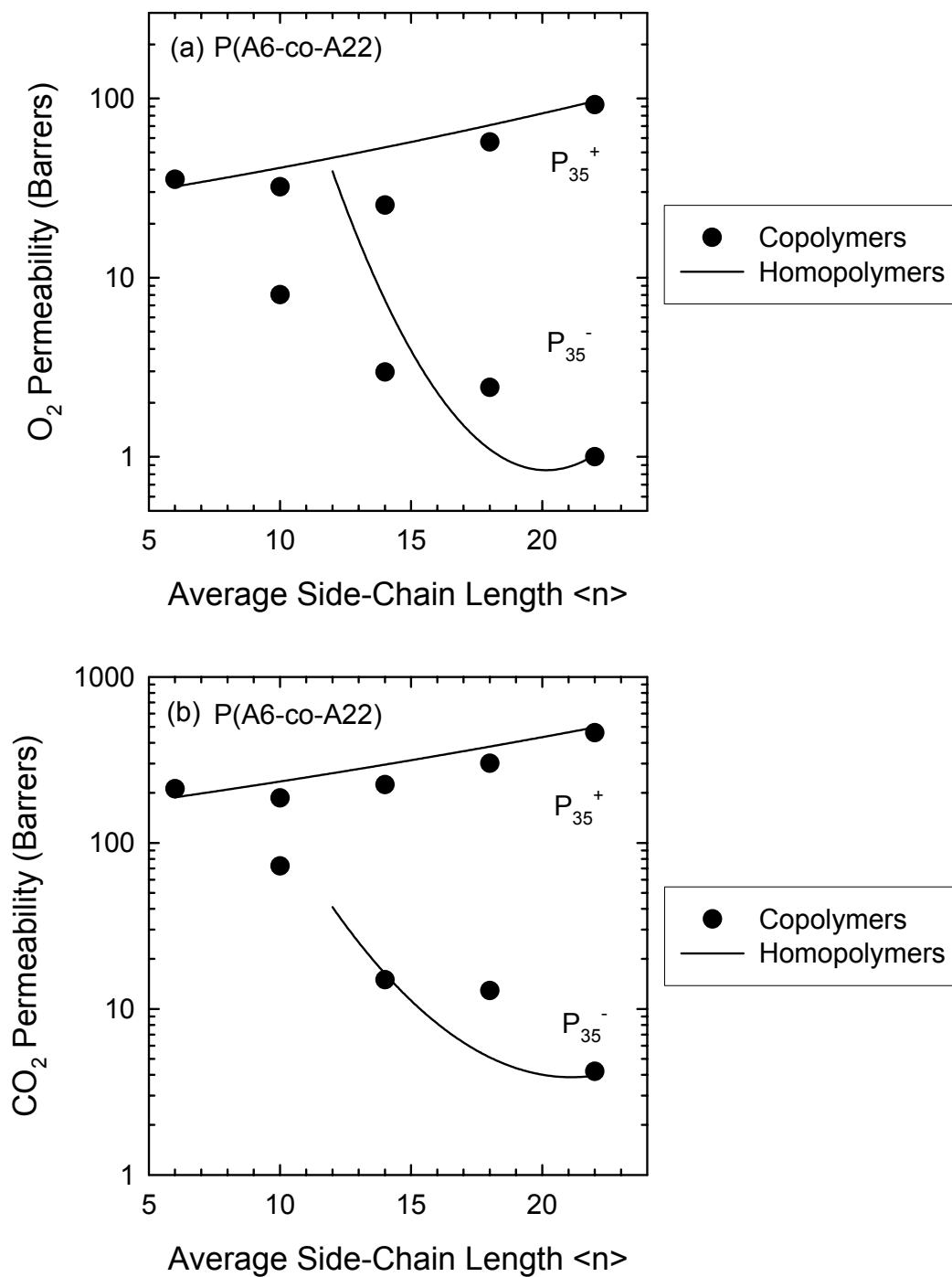


Figure 7.29 Comparison of the O_2 (a) and CO_2 (b) permeability of amorphous (P_{35}^+) and crystalline (P_{35}^-) homopolymers (lines) and $P(A6-co-A22)$ copolymers (points) calculated at $35^\circ C$ as a function of side-chain length.

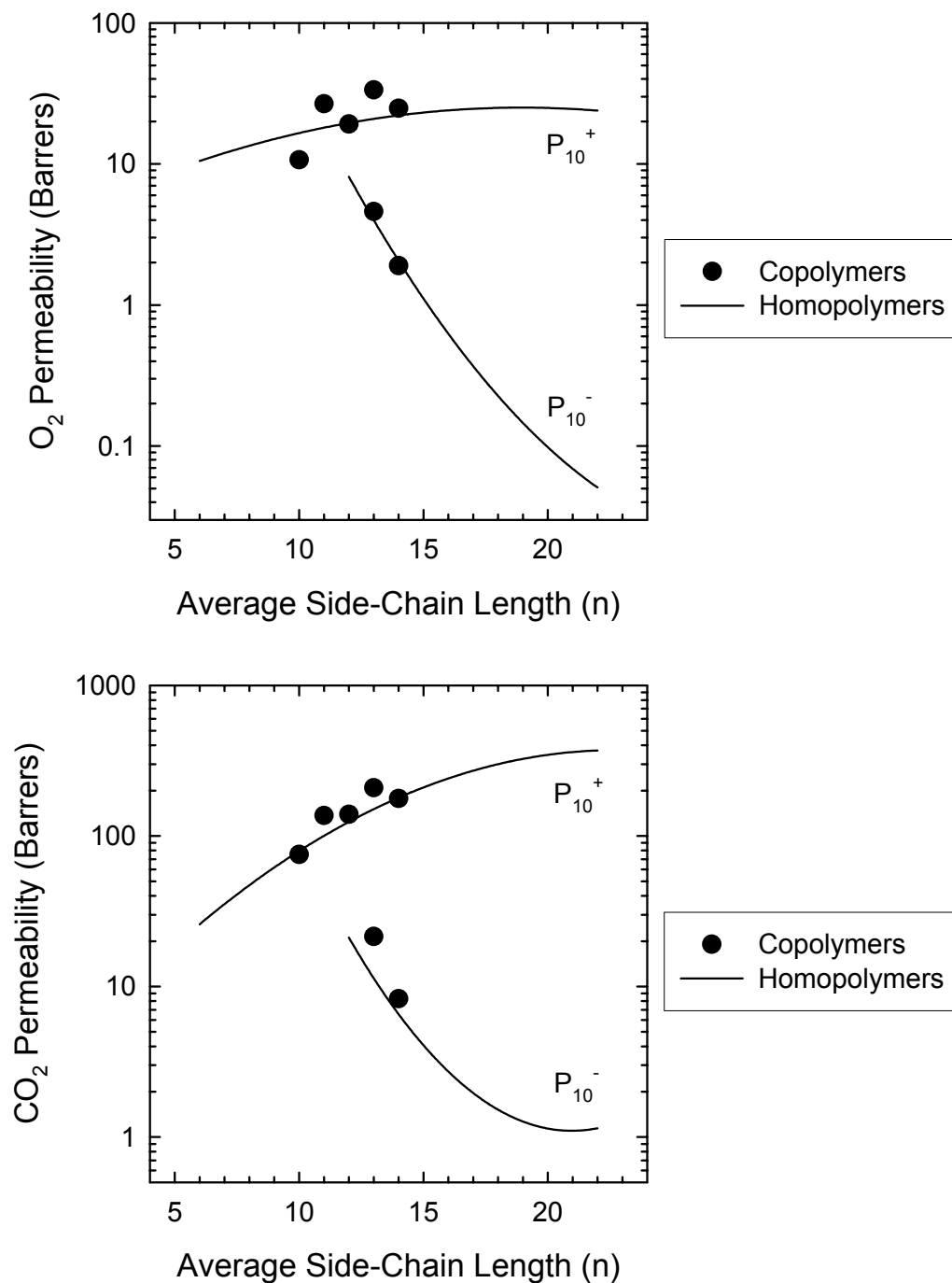


Figure 7.30 Comparison of the O₂ (a) and CO₂ (b) permeability of amorphous (P₁₀⁺) and crystalline (P₁₀⁻) homopolymers (lines) and P(A10-co-A14) copolymers (points) calculated at 10°C as a function of side-chain length.

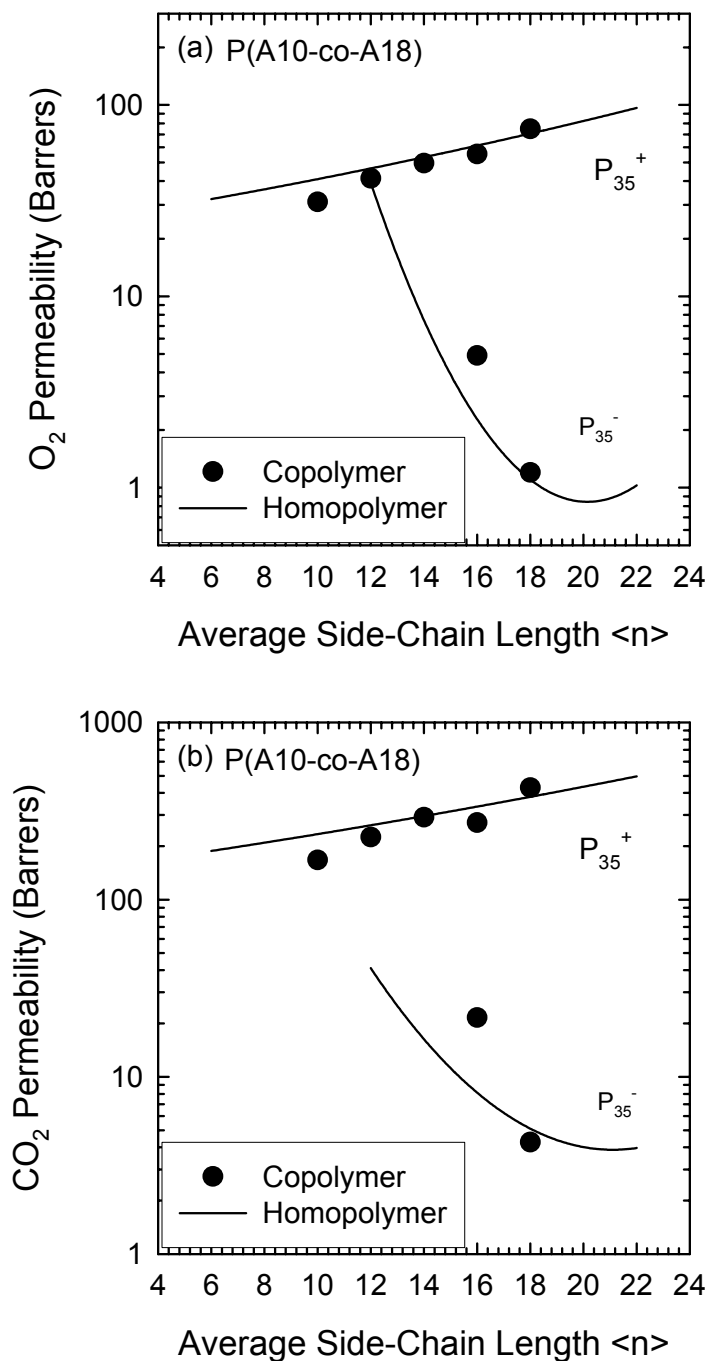


Figure 7.31 Comparison of the O_2 (a) and CO_2 (b) permeability of amorphous (P_{35}^+) and crystalline (P_{35}^-) homopolymers (lines) and P(A10-co-A18) copolymers (points) calculated at 35°C as a function of side-chain length.

7.5 Conclusions

Permeability measurements for the homopolymers and copolymers exhibit the signature ‘jump’ in permeability as the T_m is transversed. Comparison between DSC thermograms and gas permeability of the polymers over a range of temperatures revealed a correlation between both the width of the permeability jumps and endotherms as well as the permeation jump ratio and crystallinity for most of the copolymers. The permeation jumps for the P(A10-co-A14) and P(A10-co-A18) copolymer systems, however, are much smaller and less defined with increasing A10 concentration. While the onset and end melting temperatures and permeation jump temperatures aligned well, as observed with other systems, the onset and end of the permeation jump were very subtle due to its broadening. Ultimately, DSC endotherms are an excellent tool for predicting the permeation jump behavior of n-alkyl acrylate homopolymers and copolymers comprised of two crystallizeable monomers.

7.6 References

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Chapter 8

Conclusions and Recommendations

The thermal, structural, and gas permeability properties for poly (n-alkyl acrylate) copolymers containing either two crystallizeable comonomers or one crystallizeable and one non-crystallizeable (or borderline crystallizeable) comonomer have been studied. A brief summary of the conclusions given at the end of Chapters 5, 6, and 7 as well as recommendations for future work are given below.

8.1 Conclusions

8.1a Thermal Properties

The thermal properties of the homopolymers show a direct correlation between the T_m and ΔH_f and side-chain length. The crystalline-crystalline copolymers exhibit isomorphic behavior with similar relationships between T_m and ΔH_f and the average side-chain length as seen for the homopolymers. The copolymers, however, did exhibit some depression in ΔH_f relative to that of the homopolymers which increases as the difference in the number of carbons in the side chains of the two monomers increases. This qualitatively measures the reduction in crystallite size for the copolymers as a function of composition. The non-crystallizeable comonomers affect the copolymer by interrupting and impeding the crystallizeable side chains from forming perfect crystals and impinging order on the amorphous backbone. The formation of smaller and less perfect crystals causes a 'depression' in the melting temperature. Unlike copolymers with two crystallizeable comonomers that enter the lattice, altering the basic nature of the crystal,

non-crystallizeable comonomers only impede crystal formation; therefore, the co-crystallizing side chains affect the T_m and ΔH_f more than the latter. PA 10 is an unusual polymer in that its side chains are on the border of being crystallizeable; therefore, the thermal properties of several copolymers containing A10 were evaluated and compared to other copolymers. It was determined that the T_m and ΔH_f for copolymers of P(A6-co-A22) and P(A8-co-A22) exhibit melting depression caused by the non-crystallizeable side chains limiting crystal formation, while the A10 in P(A10-co-A22) behaves like a crystallizeable comonomer entering the lattice and, thereby, altering the nature of the crystal.

8.1b Structural Properties

Small angle X-ray scattering measurements were made on homopolymer and copolymer systems of varying side-chain length. The d-spacing values of the crystalline homopolymers were measured and described by a simple model. A simple molecular packing model was proposed that reasonably well predicts the amorphous d-spacing as a function of side-chain length. In the crystallite, the copolymer side chains pack in a predominantly end-to-end form with slightly smaller d-spacing values than for homopolymers, which is attributed to the slightly smaller crystals. Crystalline copolymers showed nearly identical trends of d-spacing with composition as seen for the heat of fusion. Amorphous copolymers show a nearly linear relationship between d-spacing and composition as expected from the simple packing model.

8.1c Gas Permeability Properties

Permeability measurements for the homopolymers and copolymers exhibit the signature ‘jump’ in permeability as the T_m is transversed. Comparison between DSC thermograms and gas permeability of the polymers over a range of temperatures revealed a correlation between both the width of the permeability jumps and endotherms as well as the permeation jump ratio and crystallinity. Ultimately, DSC endotherms are an excellent tool for predicting the permeation jump behavior of n-alkyl acrylate homopolymers and copolymers. The heat of fusion provides qualitative insight about the magnitude of the permeability jump in all systems except for those with A10. The permeation jumps for the P(A10-co-A14) and P(A10-co-A18) copolymer systems became smaller and less well defined as the concentration of A10 increased. While the onset and end melting temperatures and permeation jump temperatures aligned, as observed with other systems, the onset and end of the permeation jump were very subtle. This broadening of the permeation jump may ultimately be a solution to modified atmospheric packaging needs by providing a broad, rapid change in permeability with temperature.

8.2 Recommendations for Future Work

8.2a Mathematical Modeling

Poly (n-alkyl acrylates) are attractive polymers for potential uses in modified atmospheric packaging because of their unconventionally strong permeation–temperature relationships at T_m . Future work with these polymers should focus on modeling the temperature responsive membranes for uses in modified atmospheric packaging. Paul

and Clark recently published a paper addressing many of the mathematical requirements for such a model. This should be used as a platform for future modeling work.

8.2b Physical Blends of Copolymers

Currently, our labs have been focusing on both physical blends and copolymers of n-alkyl acrylates to achieve a polymer with a desirable permeability-temperature relationship for packaging. The permeability studies on different blends have thus far been limited by the miscibility of amorphous and crystalline polymers. Physical blends of these copolymers, especially those with one crystallizeable and one non-crystallizeable comonomer, with homopolymers may be more miscible and potentially increase the number of blend combinations evaluated. The subtle permeation jumps in copolymers with A10 were also promising for potential packaging applications. These copolymers maintain the jump in permeability; however, it is broadened over the melting temperature range which may more closely reflect the changes in respiration rates of produce with temperature. Laminates of the polymers and copolymers may also prove valuable for engineering a permeability jump, or series of jumps, capable of modifying atmospheric packaging over a broader temperature range.

8.2c Further Structural Analysis

Further SAXS analysis of the copolymer systems, especially systems containing only one crystallizeable comonomer and those with A10, may garnish further understanding of the structural relationships among side chains. As discussed in Chapter 6, and shown in Figure 6.6, the d-spacings for the copolymers tend to scatter from those

for the homopolymers in smallest side-chain length region. This is also the region that previous authors reported increases in d-spacings with reductions in average side-chain length. An in-depth study of these copolymers may help to further clarify the issue. We were also unable to develop simple mathematical models or relationships to quantify these systems due to limited data; therefore, future SAXS analysis may prove very useful.

As discussed in Chapter 7, there is currently no method for quantifying the side-chain crystallite dimensions, or aspect ratio, or their arrangement with respect to one another. If determined by some other structural analysis method, this information would prove useful for measuring the lateral dimension of the crystallite which determines the length of the tortuous pathway a gas molecule must take in order to diffuse around the crystallites. Currently tortuous pathway and crystallinity are coupled. Uncoupling the effects of the two may provide more insight into the difference between copolymer with A10 having broadened, smaller jumps and those for all other systems examined here.

8.2d Effects of Thermal History

Mogri and Paul performed several studies on the effects of thermal history on poly (n-alkyl acrylate) homopolymers.¹⁻³ They determined that varying the cooling rate of the polymer membranes alter the degree of crystallinity and subsequently the magnitude of the permeation jump ratio. All copolymer samples examined in this study were given a constant thermal history of 1°C/min. Future work with these polymers should include samples with varied thermal histories in order to determine their effect on the copolymer jump ratios. Permeation studies should also be performed on copolymer systems at different stages of being heated and cooled. This would provide insight into

membrane performance in a non-controlled setting, similar to conditions experienced while shipping produce.

8.2e Water Vapor and Ethylene Gas Studies

Permeability of water vapor and ethylene are extremely important variables in shipping produce. While the appropriate relative humidity inside a package of produce helps increase shelf life and limit growth of bacteria, ethylene gas is a natural ripening agent in nearly all produce; both factors also temperature dependant.⁴⁻⁸ Both of these penetrants must be considered when evaluating a membrane for modified atmospheric packaging purposes. Mogri and Paul performed a water vapor permeability study on PA18 demonstrating that water vapor behaves like other gas penetrants that they studied similar studies should be performed on copolymer systems.^{2,9} We also performed limited studies on the permeability of ethylene gas through copolymer membranes. In general, they also behaved similar to the other penetrants examined having a larger jump ratio for the increased gas diameter. A more complete study should be performed.

8.3 References

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Appendix A

Permeation DAQ System

This appendix contains the Labview code written to acquire pressure data as a function of time. Written with the help of Pavlos Tsiartas and Elizabeth Collister, the code records the data for up to 9 permeation cells at a given time. The change in downstream pressure is measured with a transducer powered by a PDR (power display readout). The PDR was previously connected to chart recorders with electrical wire that carried the pressure in terms of voltage. This system operates in a similar manner, recording the change in voltage, or pressure, with time electronically. The purpose of this Appendix is to explain the system to future users.

Both the inner loop for a single pressure input as well as the user interface are shown. The program contains an outer loop, not shown, that has an overall timer running for the entire system. Nine smaller loops (Figure A.1) are located within the outer loop that collects the pressure data from each permeation cell. The user interface is shown in Figure A.2. When the 'start / stop recording' button is selected for each permeation cell, the inner loop 'turns on' and the pressure data enters the loop while a second timer is started. The time collected is the difference between the overall timer and inner loop timer in seconds. The time and voltage input enter a 'Labview sub.vi', written by National Instruments, which simultaneously displays and records the voltage-temperature data. When the 'start / stop recording' button is pushed again, the small loop will stop and the computer will stop collecting the data for that permeation cell. A time delay, visible in Figure A.2, is used to control the number of data points collected. It is

normally set to 300 for a sample that runs up to 6 hours, to collect one data point every 300 milliseconds. This may be changed based on the duration of experiment and number of data points desired. The results are saved under a file name selected by the user and typed into the 'c:\test.txt' slot for each cell.

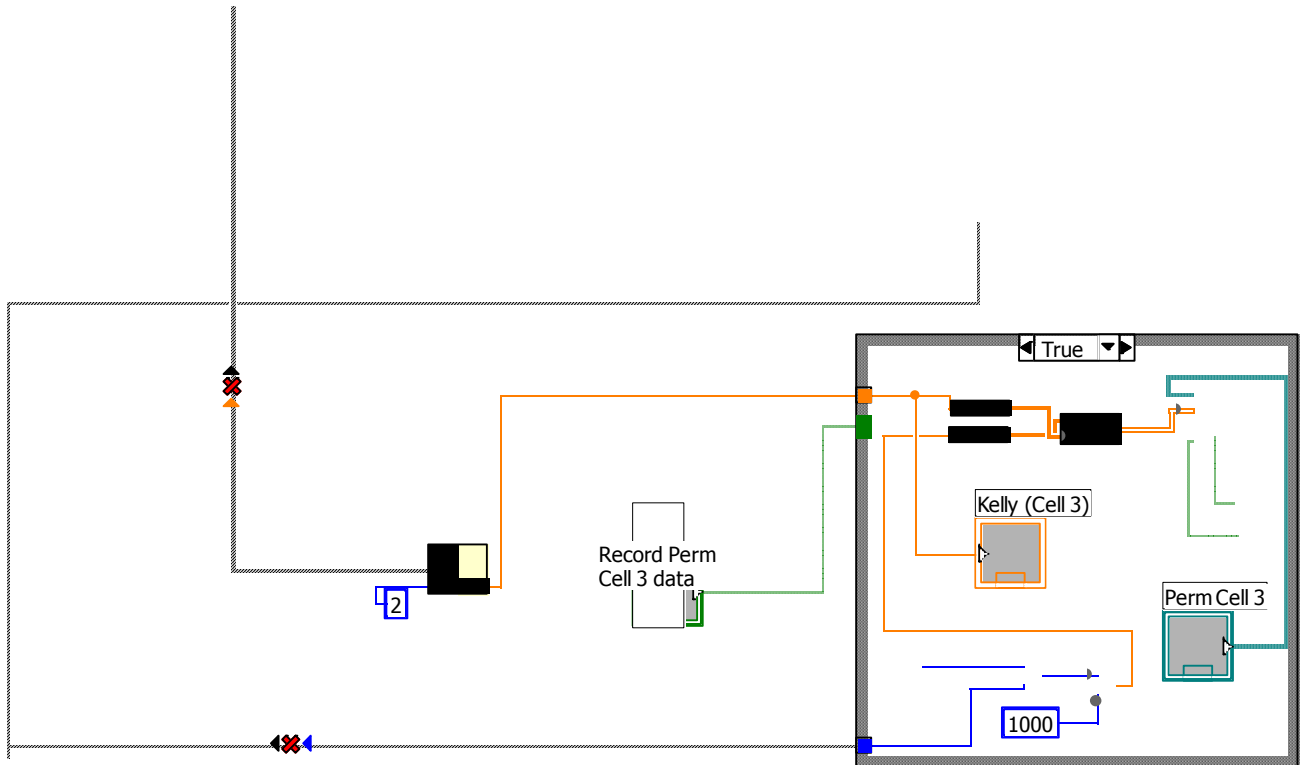


Figure A.1 Inner loop that records and exports voltage data for a single permeation cell.



Figure A.2 The user interface for the permeation DAQ program.

Appendix B

Additional Permeability Plots for Poly (n-alkyl acrylates)

This appendix contains permeability plots not shown previously in the thesis text. As explained in Chapter 7, the permeabilities of six gases, He, H₂, O₂, CO₂, CH₄, and N₂, were measured for every sample in this study; however, it focuses on the permeability of O₂ and CO₂ through various polymer membranes because they are the most important for modified atmospheric packaging purposes. The permeability properties of P(A10-co-A22) 50/50 mol % was also measured; however, because only one composition was analyzed, it was omitted from Chapter 7. Finally, as mentioned in the future recommendations section of Chapter 8, the permeability of ethylene gas was measured on a limited number of membranes. These results will also be shown here. The figures are ordered by polymer composition, starting with homopolymers and progressing to copolymers.

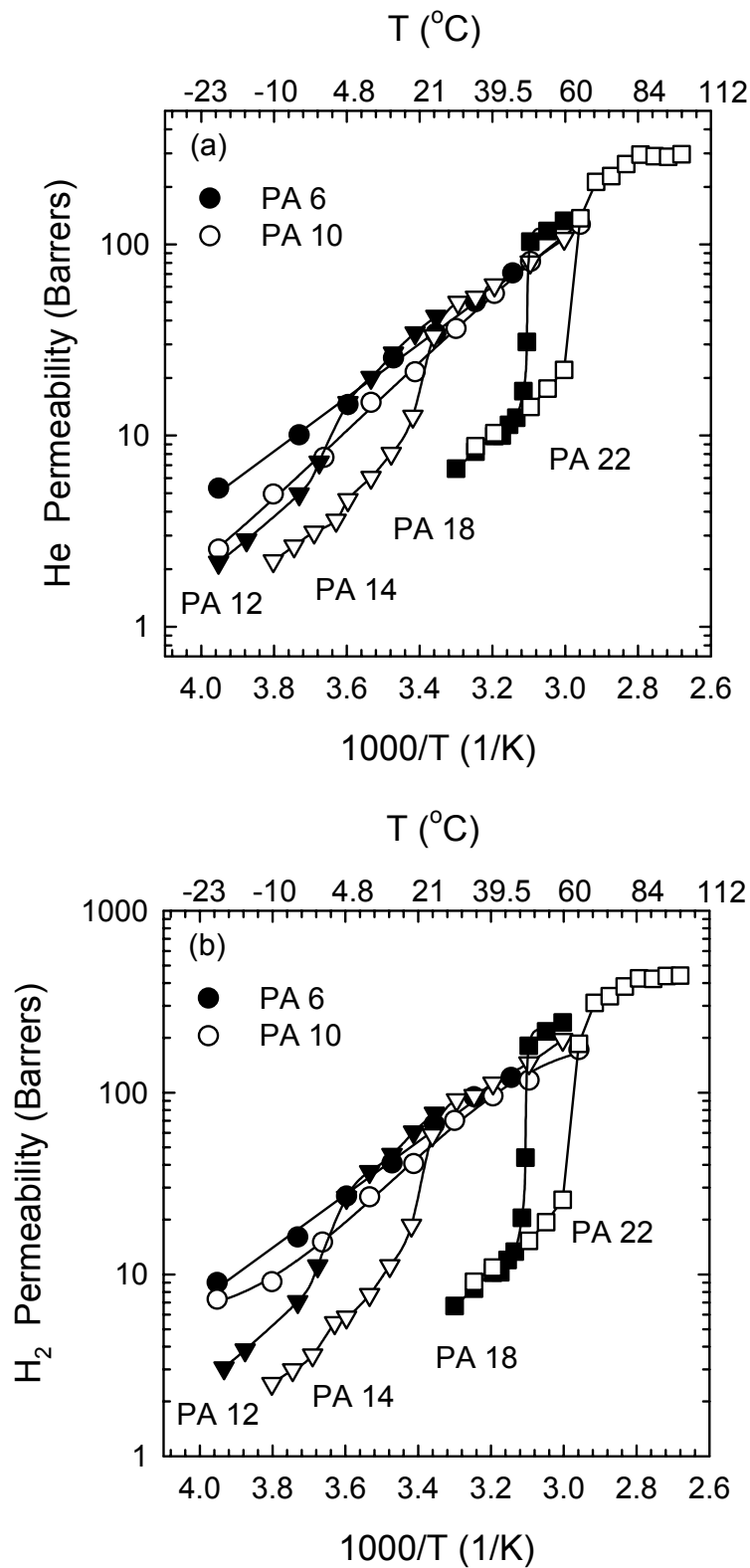


Figure B.1 Permeability of He (a) and H_2 (b) for homopolymers with side-chain lengths ranging from 6 to 22 carbons as a function of temperature on Arrhenius coordinates.

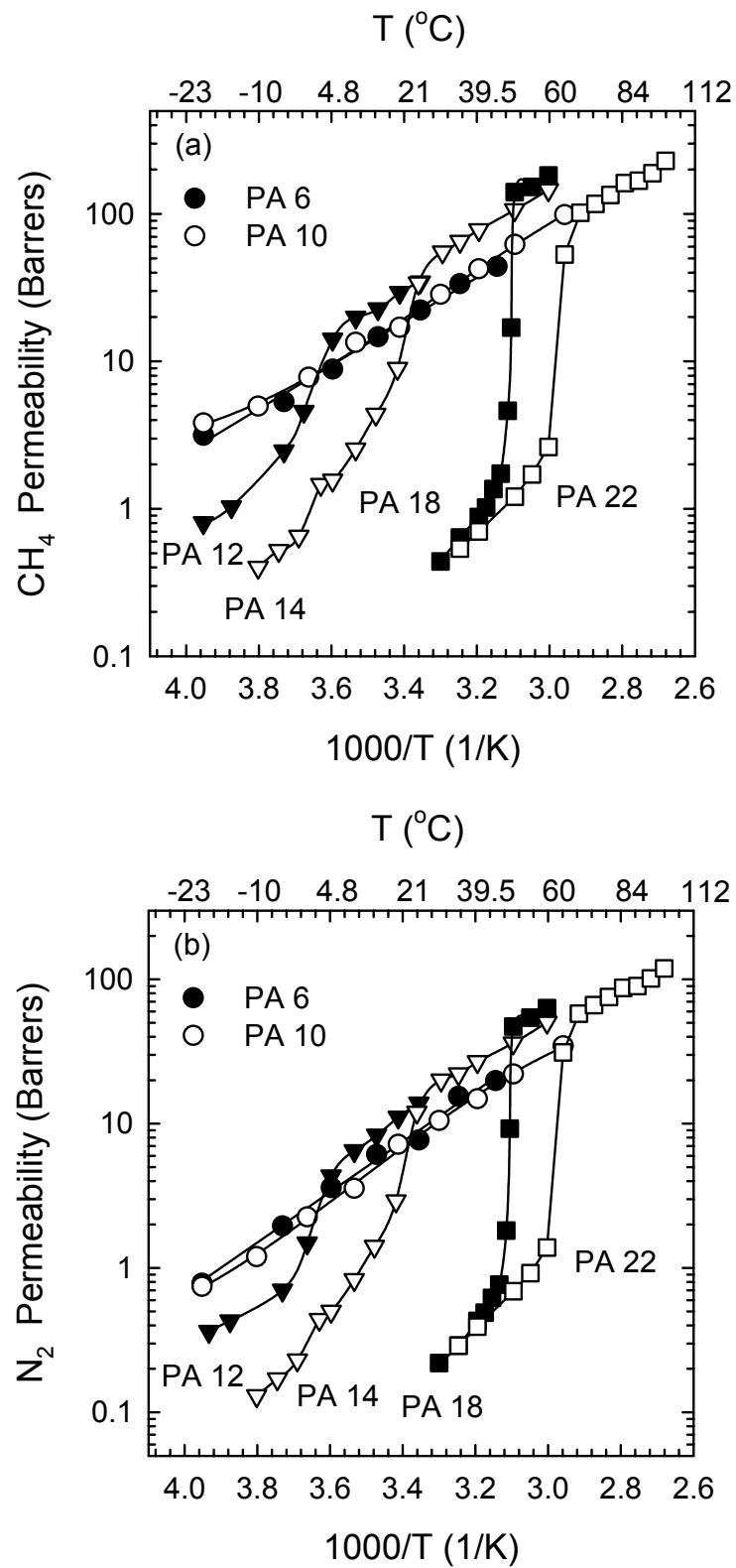


Figure B.2 Permeability of CH_4 (a) and N_2 (b) for homopolymers with side-chain lengths ranging from 6 to 22 carbons as a function of temperature on Arrhenius coordinates.

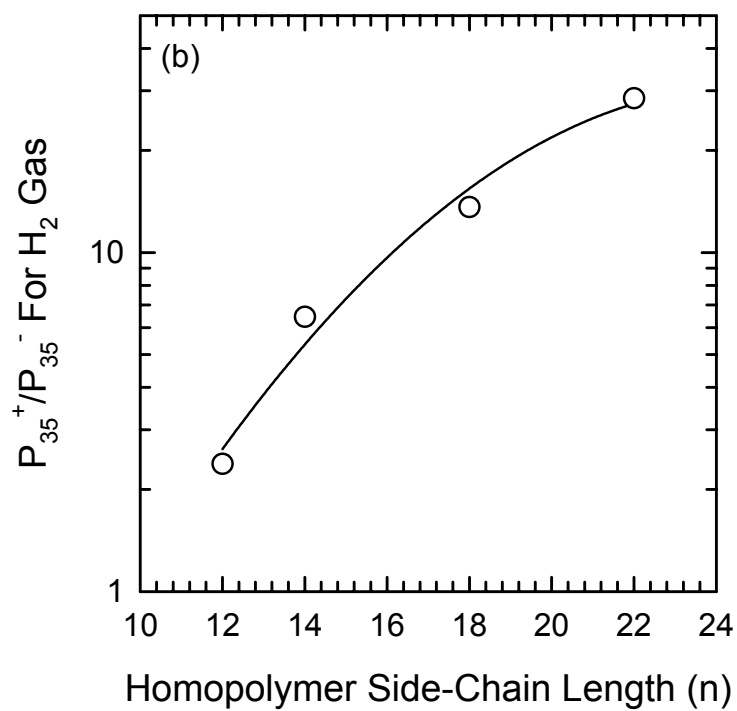
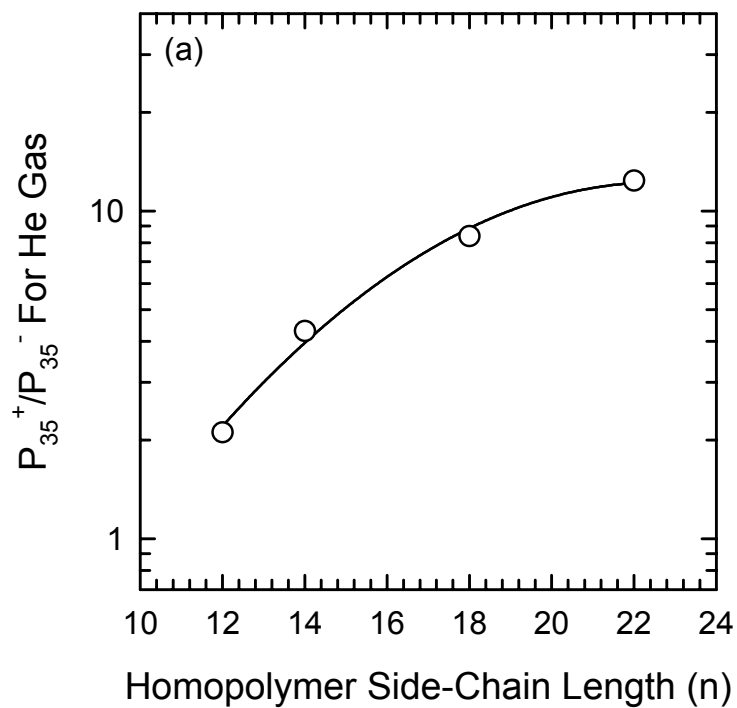


Figure B.3 Homopolymer permeation jump ratios calculated at 35°C for He (a) and H_2 (b) gases as a function of side-chain length.

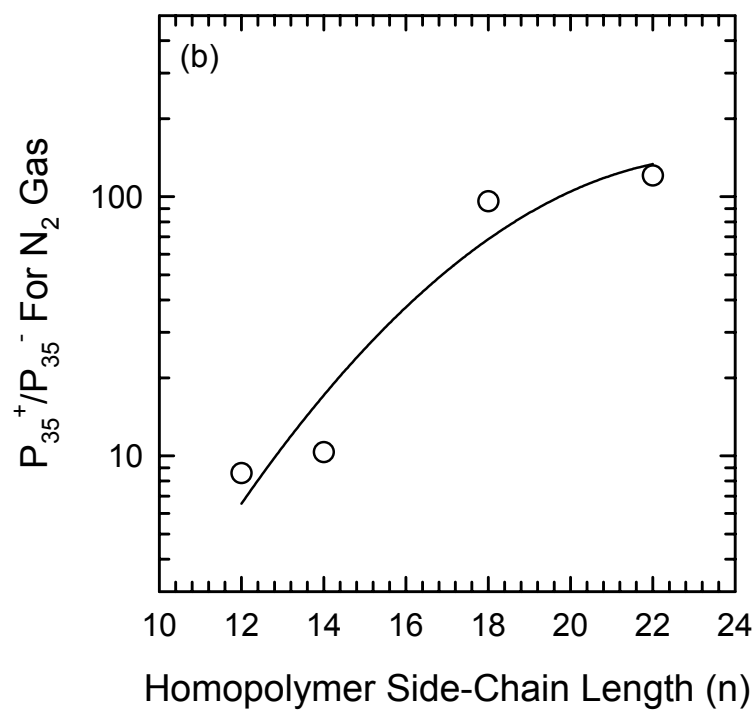
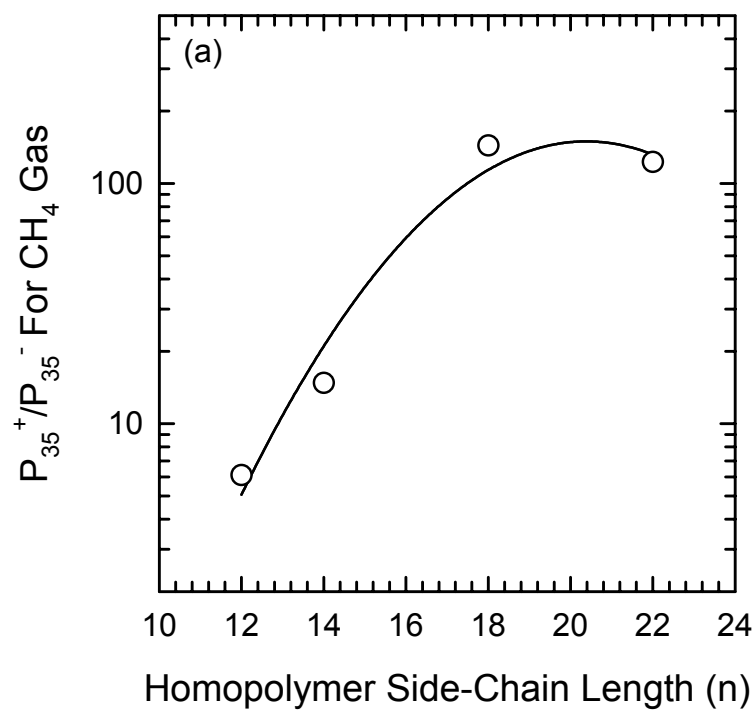


Figure B.4 Homopolymer permeation jump ratios calculated at 35°C for CH_4 (a) and N_2 (b) gases as a function of side-chain length.

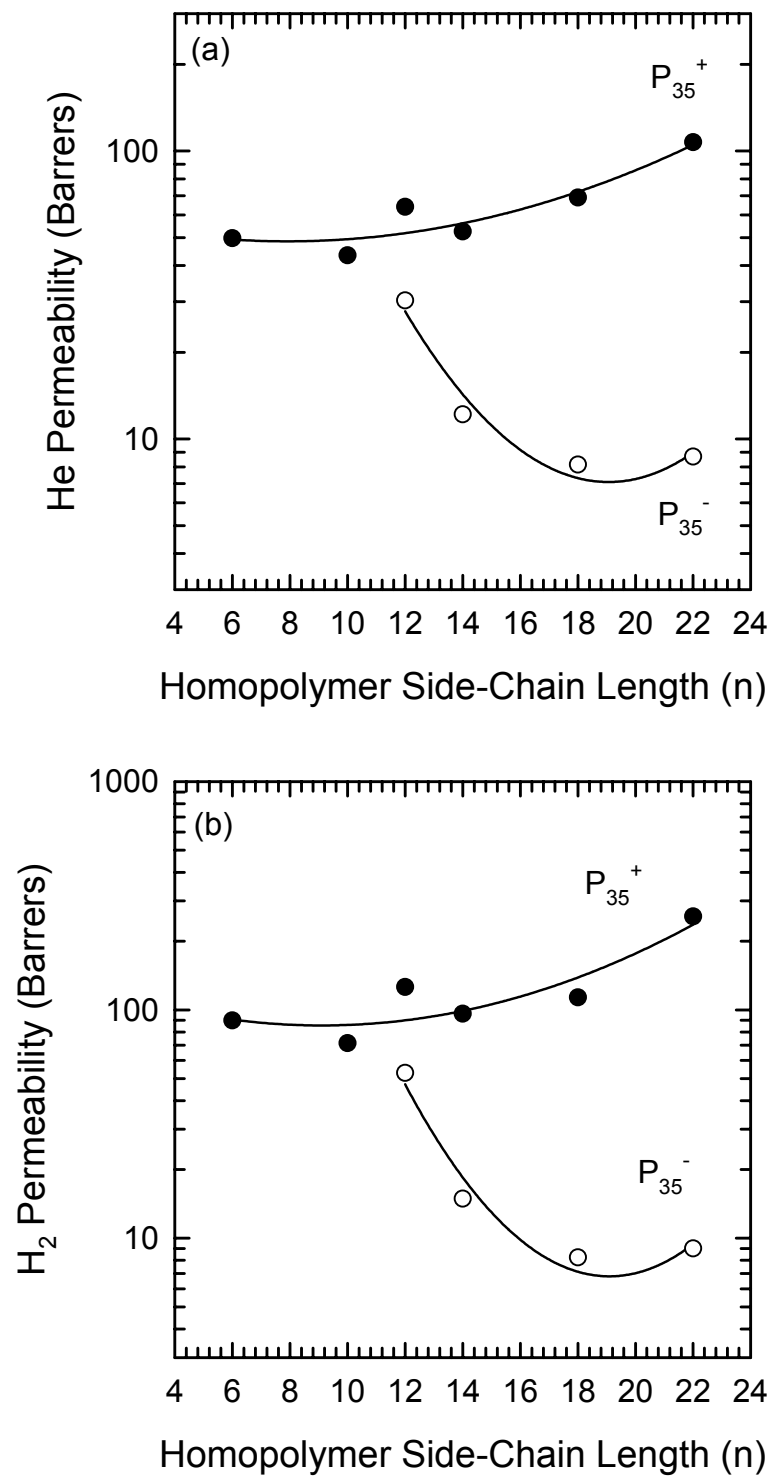


Figure B.5 Permeability of amorphous (P_{35}^{+}) and crystalline (P_{35}^{-}) homopolymers extrapolated to 35°C for He (a) and H_2 (b) gas as a function of side-chain length.

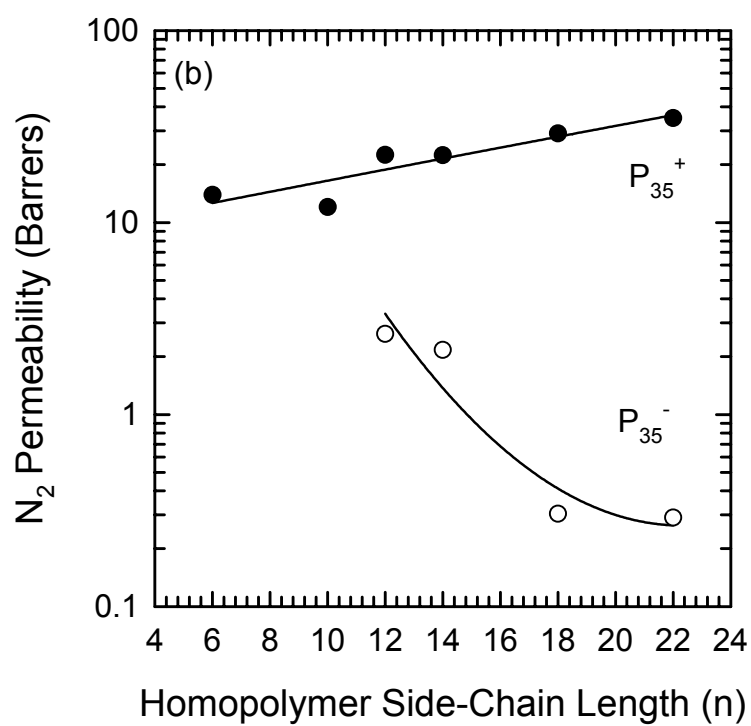
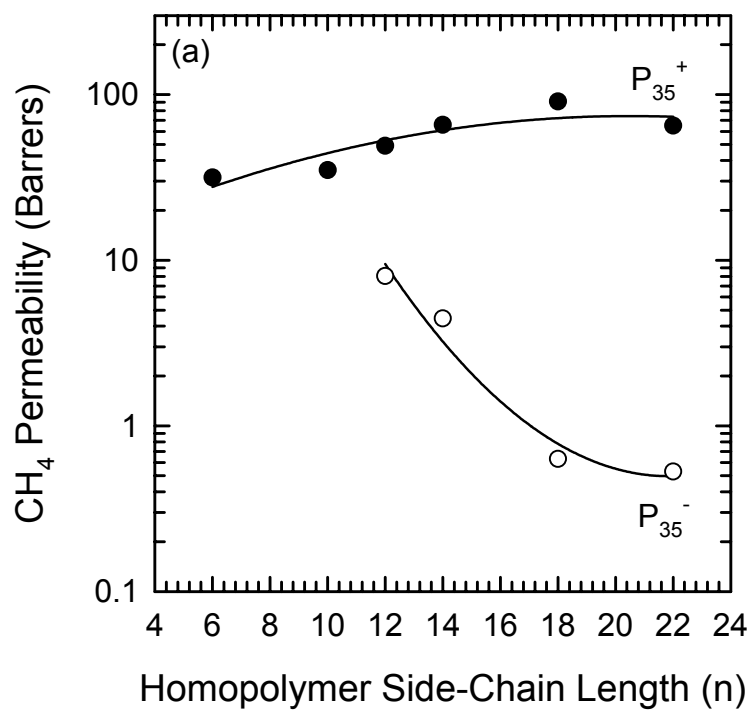


Figure B.6 Permeability of amorphous (P_{35}^{+}) and crystalline (P_{35}^{-}) homopolymers extrapolated to 35°C for CH₄ (a) and N₂ (b) gas as a function of side-chain length.

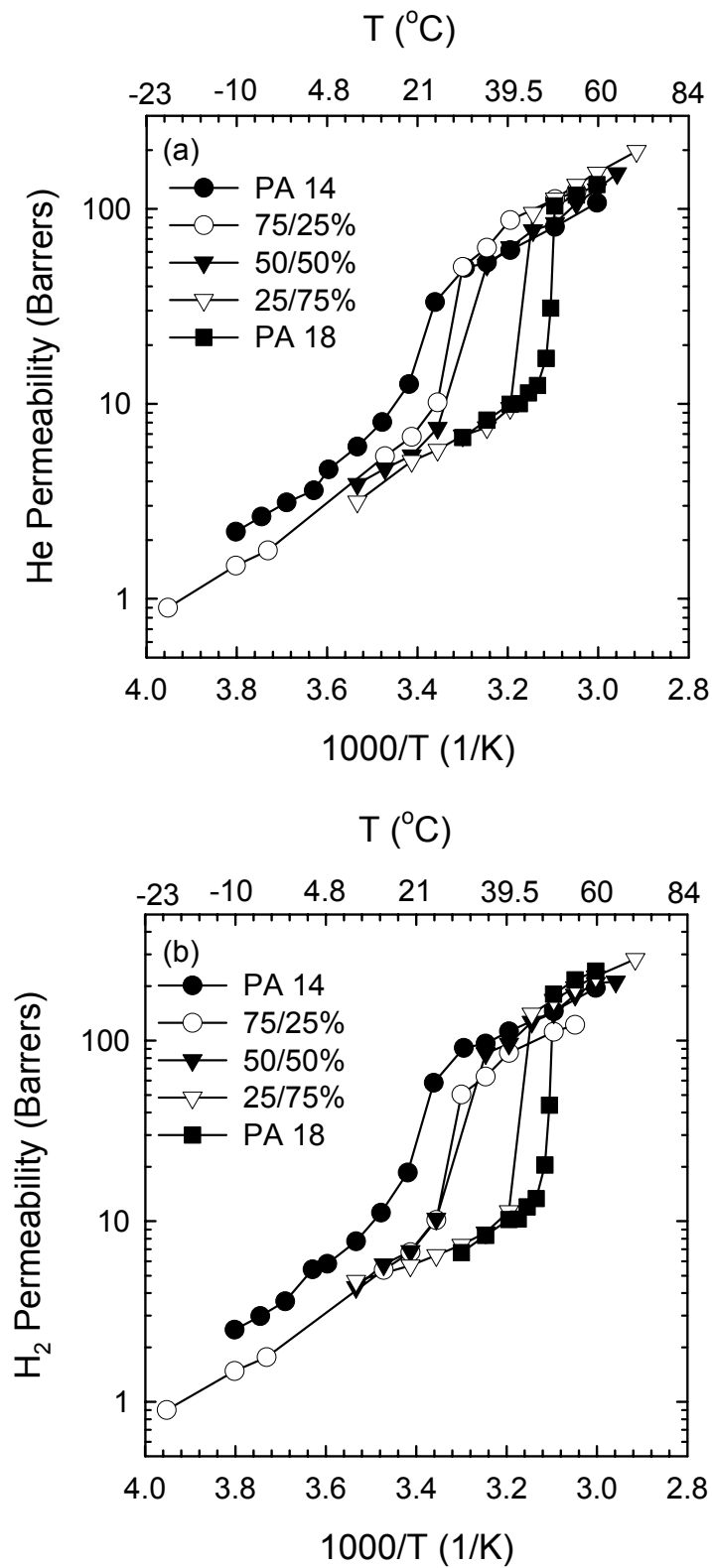


Figure B.7 Permeability of various P(A14-co-A18) copolymers to He (a) and H_2 (b) as a function of temperature on Arrhenius coordinates.

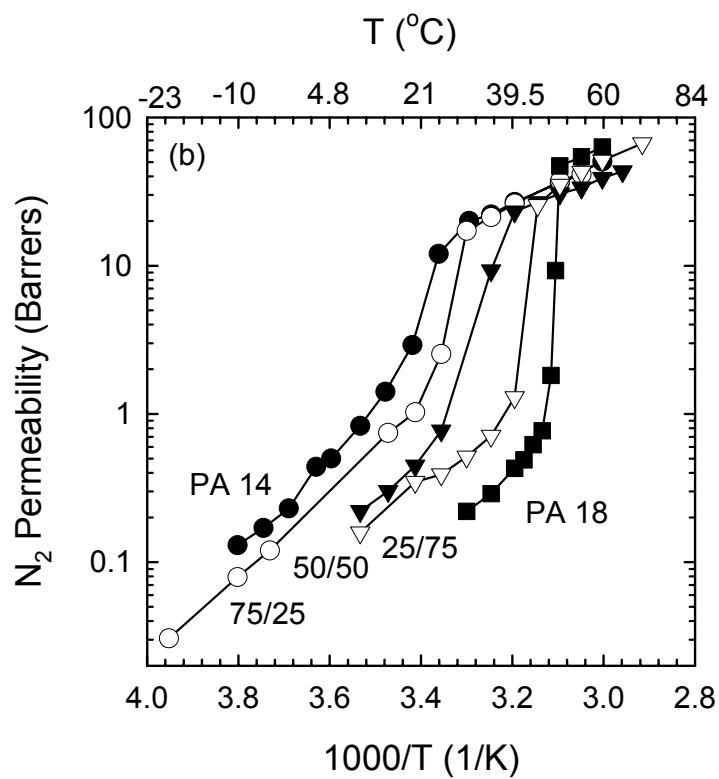
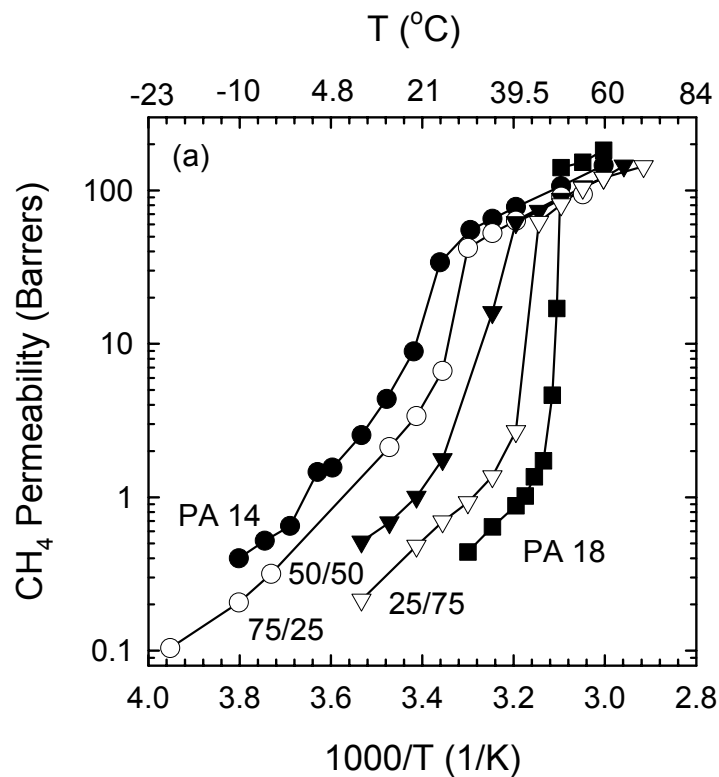


Figure B.8 Permeability of various P(A14-co-A18) copolymers to CH₄ (a) and N₂ (b) as a function of temperature on Arrhenius coordinates.

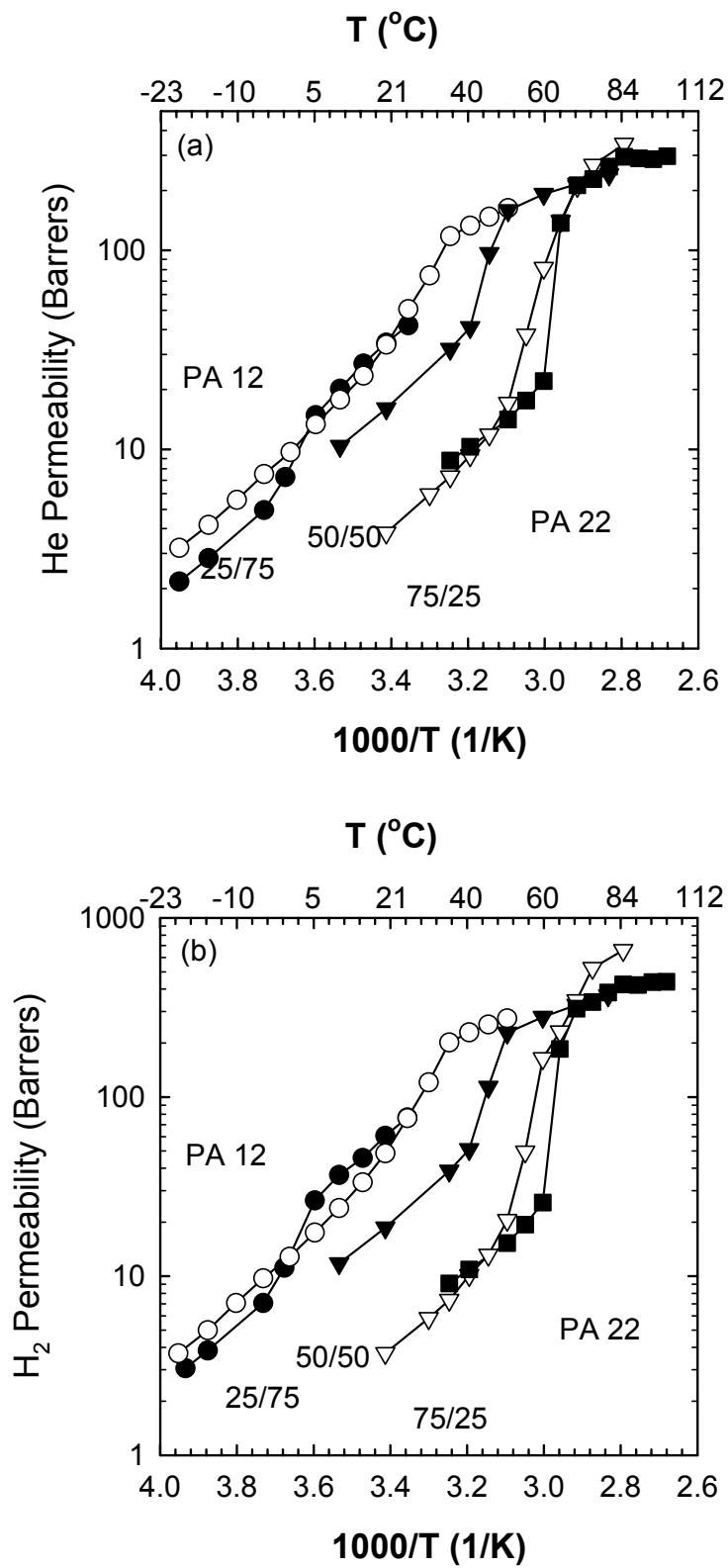


Figure B.9 Permeability of various P(A12-co-A22) copolymers to O_2 He (a) and H_2 (b) as a function of temperature on Arrhenius coordinates.

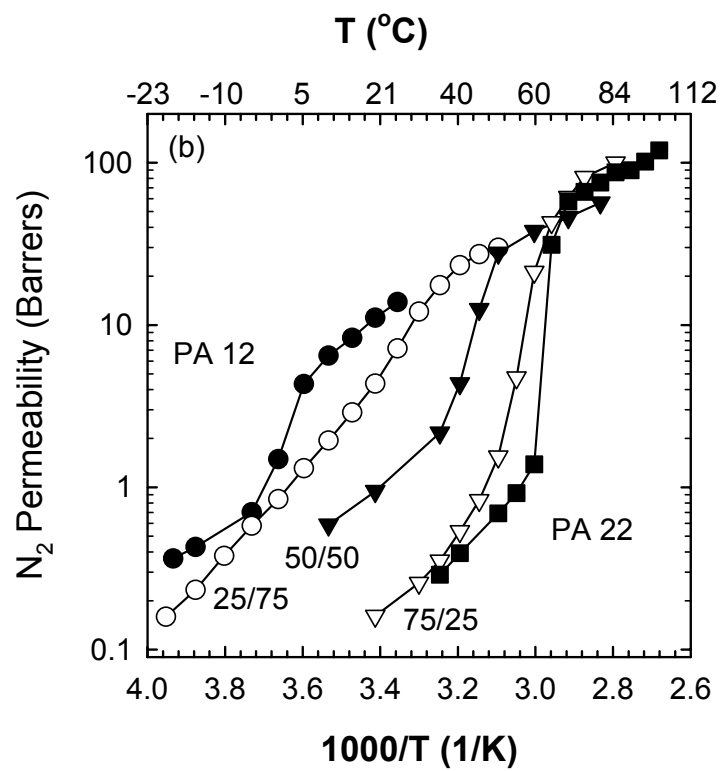
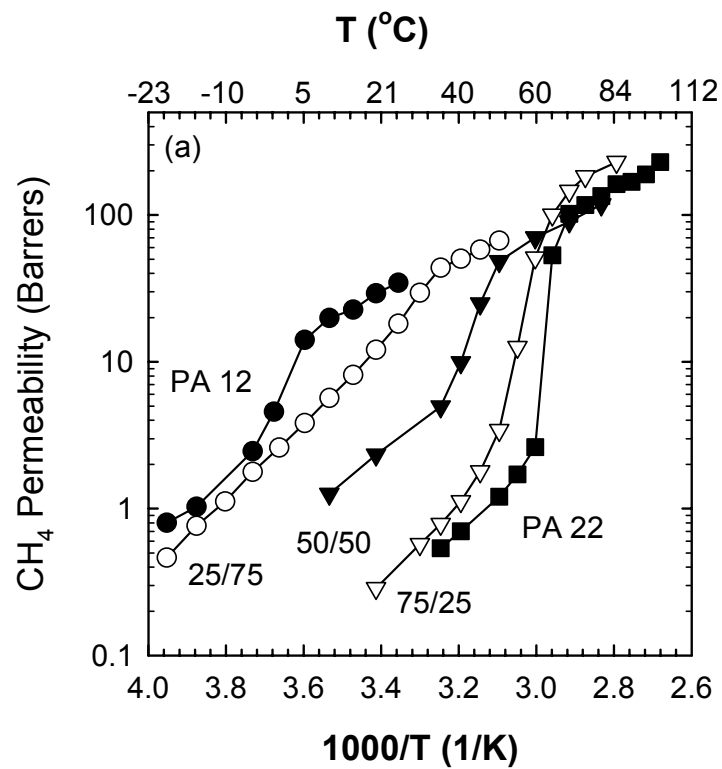


Figure B.10 Permeability of various P(A12-co-A22) copolymers to CH₄ (a) and N₂ (b) as a function of temperature on Arrhenius coordinates.

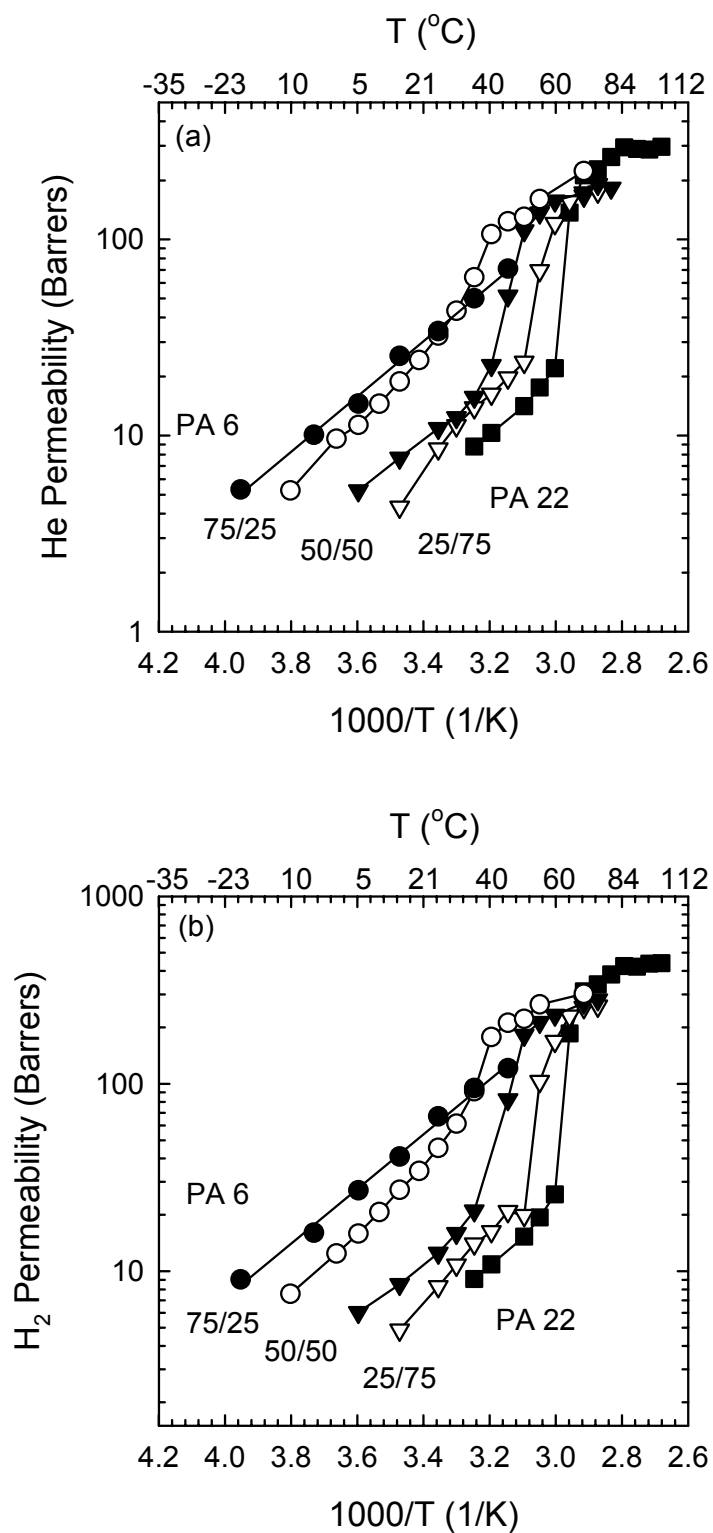


Figure B.11 Permeability of various P(A6-co-A22) copolymers to He (a) and H_2 (b) as a function of temperature on Arrhenius coordinates.

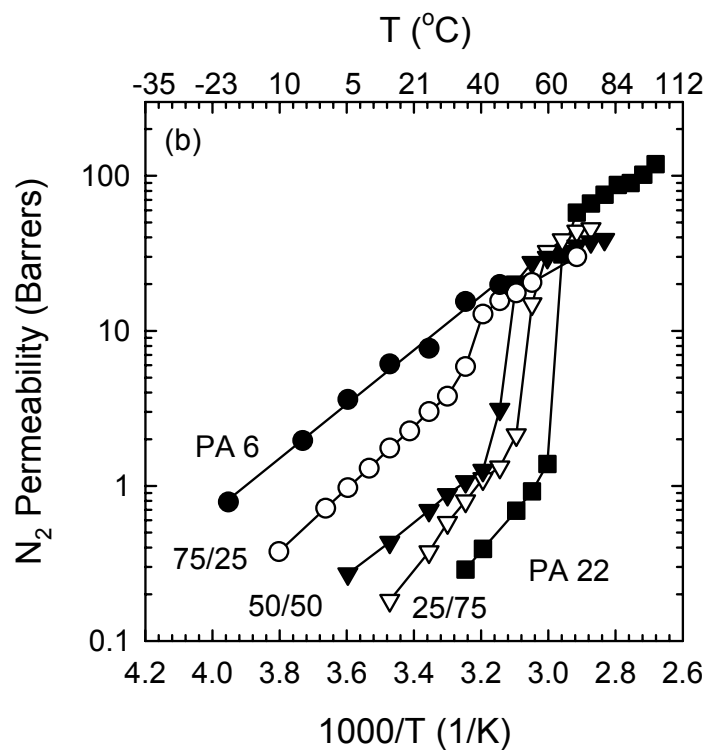
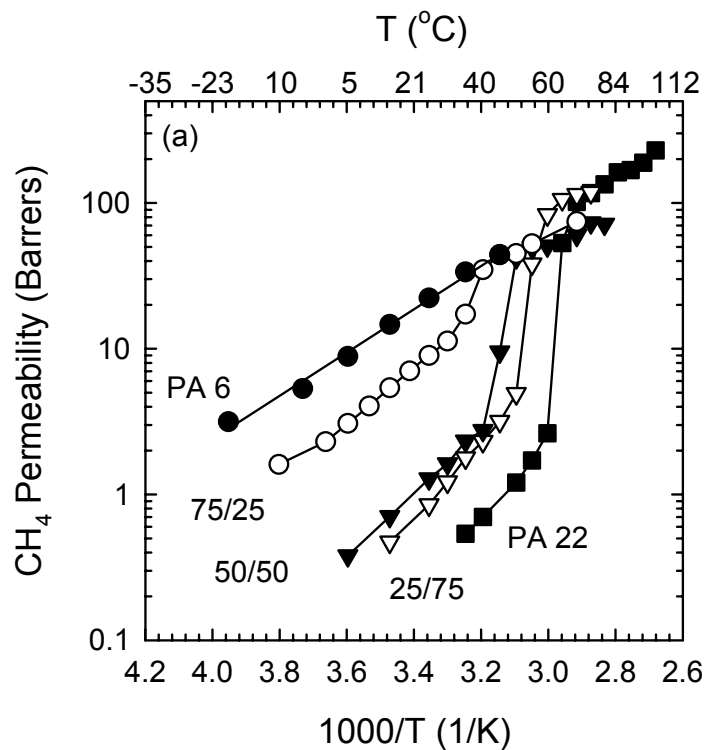


Figure B Permeability of various P(A6-co-A22) copolymers to CH₄ (a) and N₂ (b) as a function of temperature on Arrhenius coordinates.

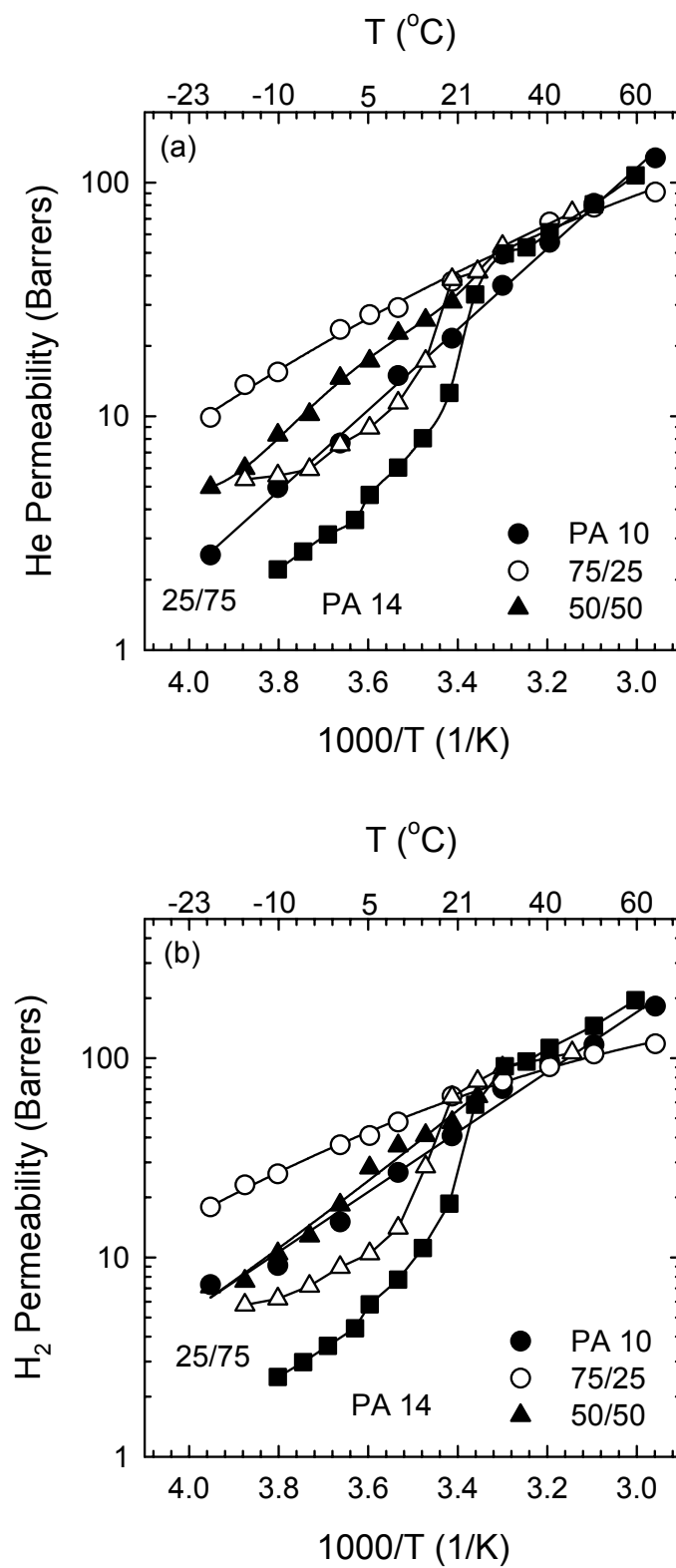


Figure B.13 Permeability of various P(A10-co-A14) copolymers to He (a) and H₂ (b) as a function of temperature on Arrhenius coordinates.

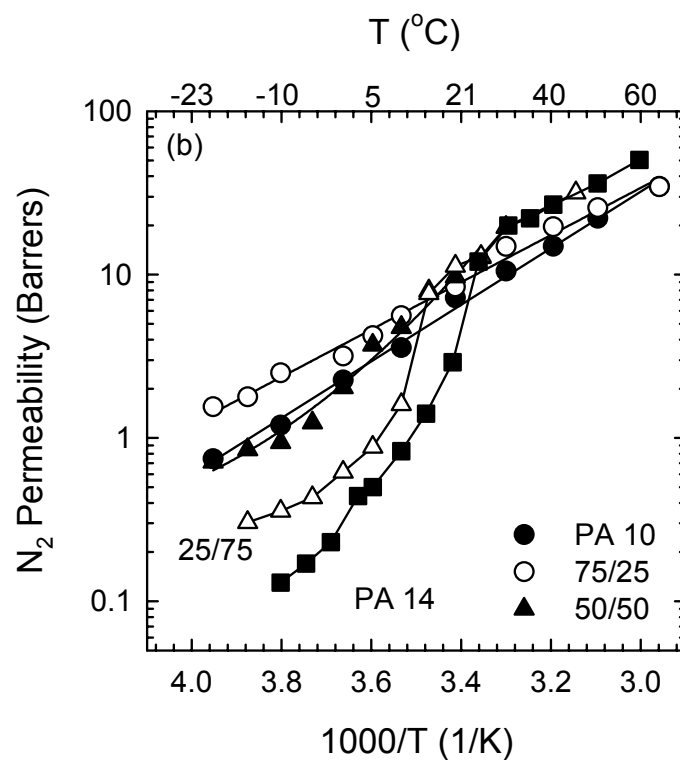
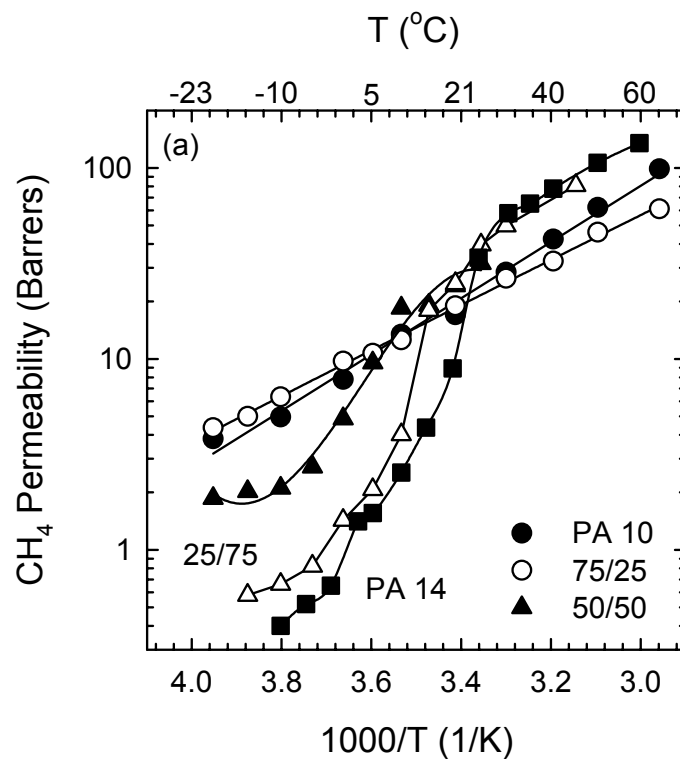


Figure B.14 Permeability of various P(A10-co-A14) copolymers to CH₄ (a) and N₂ (b) as a function of temperature on Arrhenius coordinates.

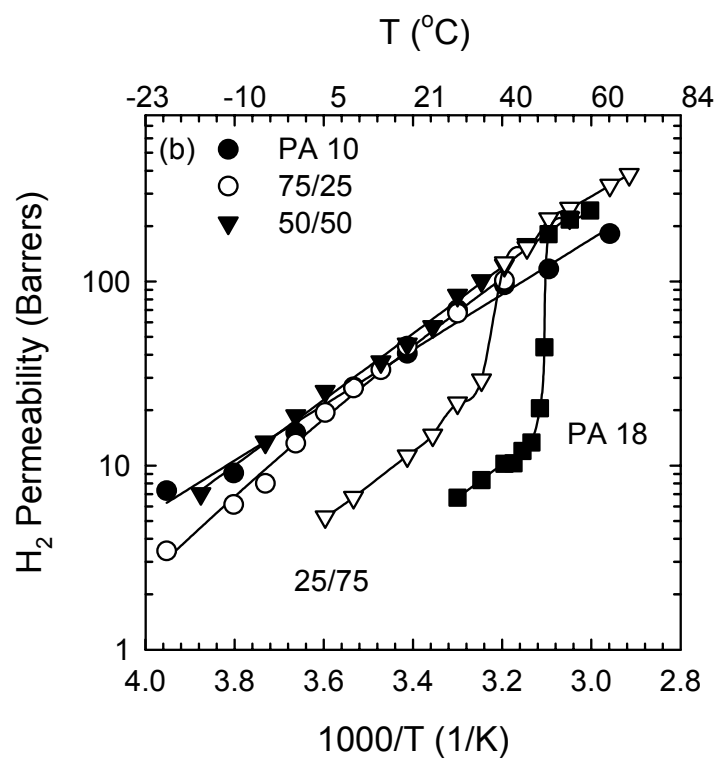
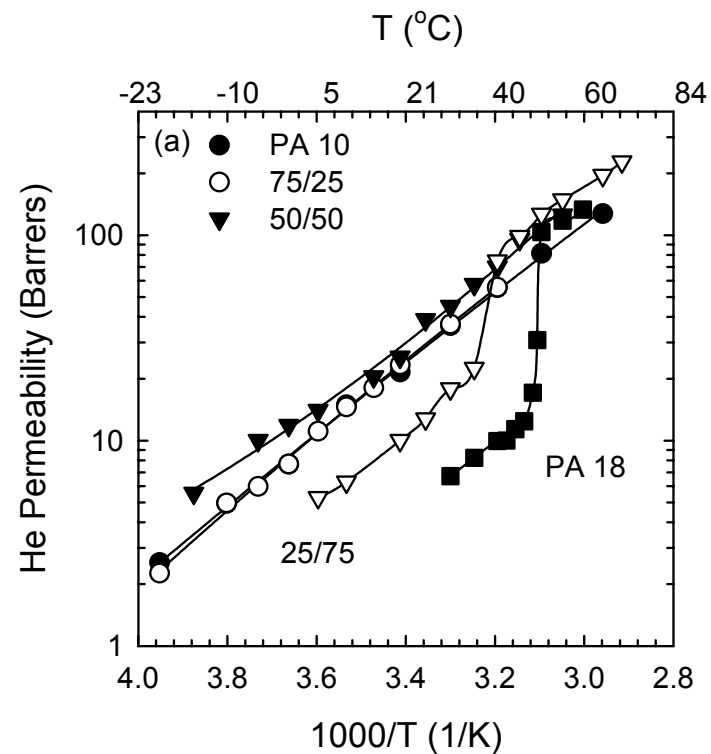


Figure B.15 Permeability of various P(A10-co-A18) copolymers to He (a) and H_2 (b) as a function of temperature on Arrhenius coordinates.

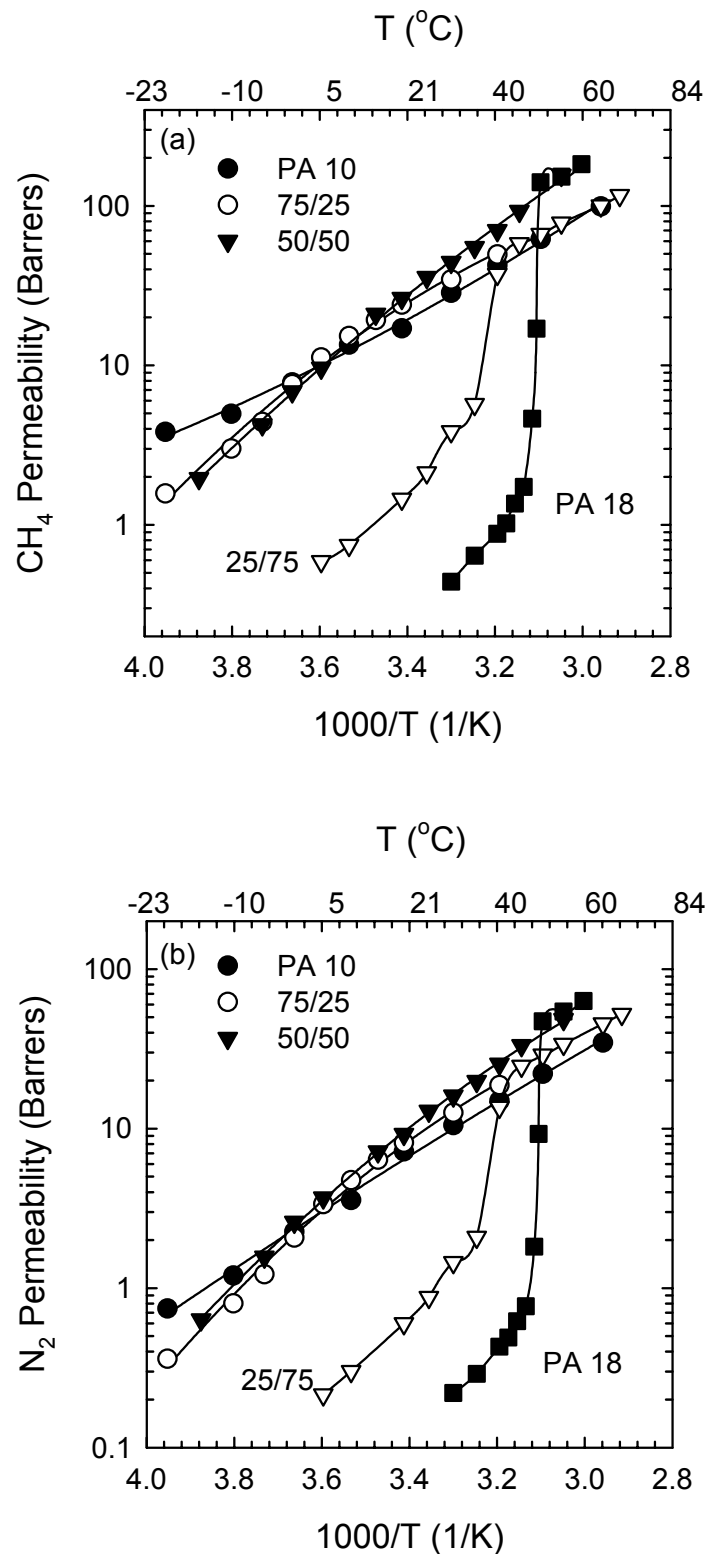


Figure B.16 Permeability of various P(A10-co-A18) copolymers to CH_4 (a) and N_2 (b) as a function of temperature on Arrhenius coordinates.

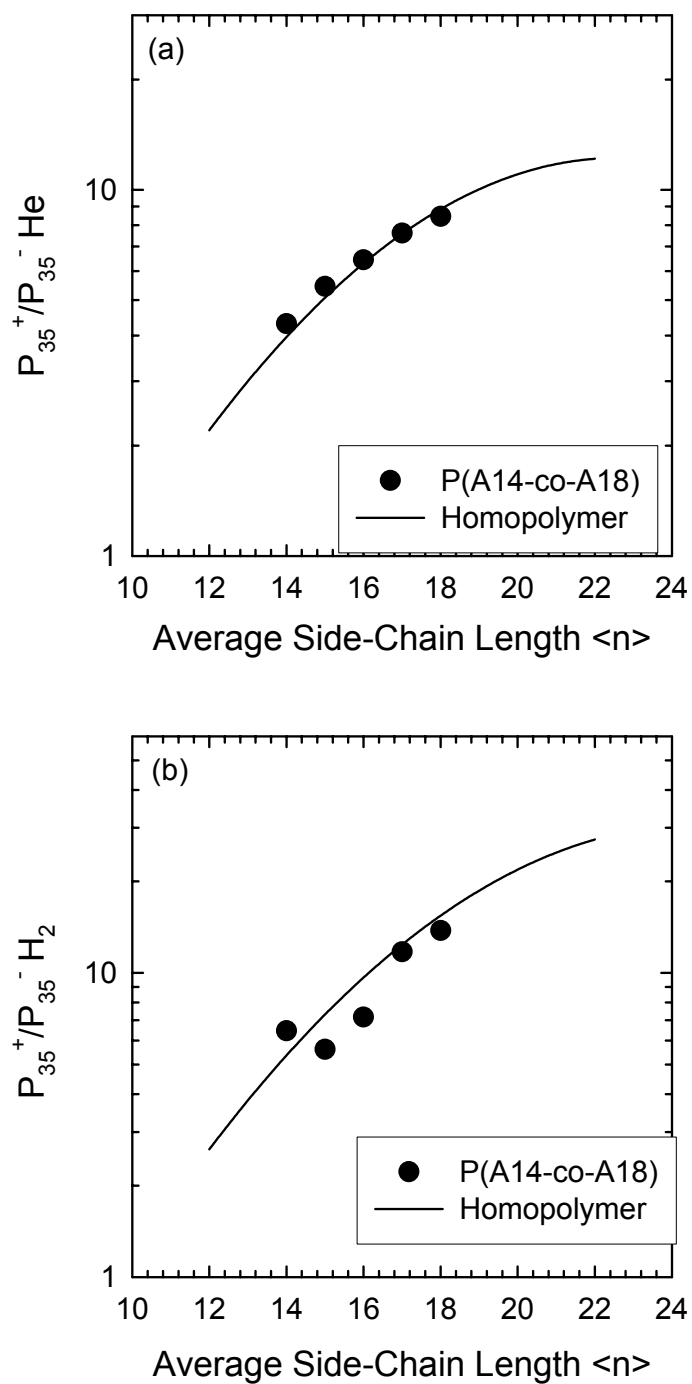


Figure B.17 Comparison of permeation jump ratios for homopolymers (lines) with P(A14-co-A18) copolymers (points) calculated at 35°C for He (a) and H₂ (b) gases.

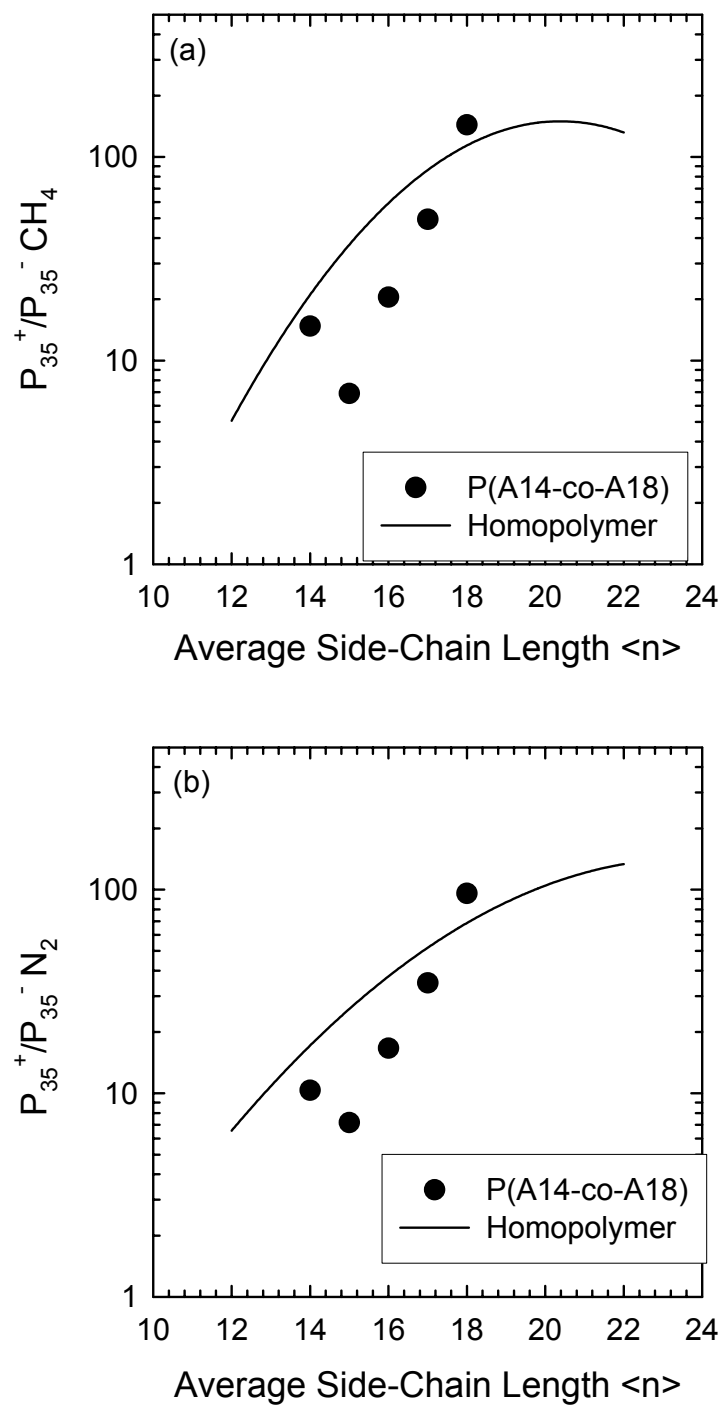


Figure B.18 Comparison of permeation jump ratios for homopolymers (lines) with P(A14-co-A18) copolymers (points) calculated at 35°C for CH_4 (a) and N_2 (b) gases.

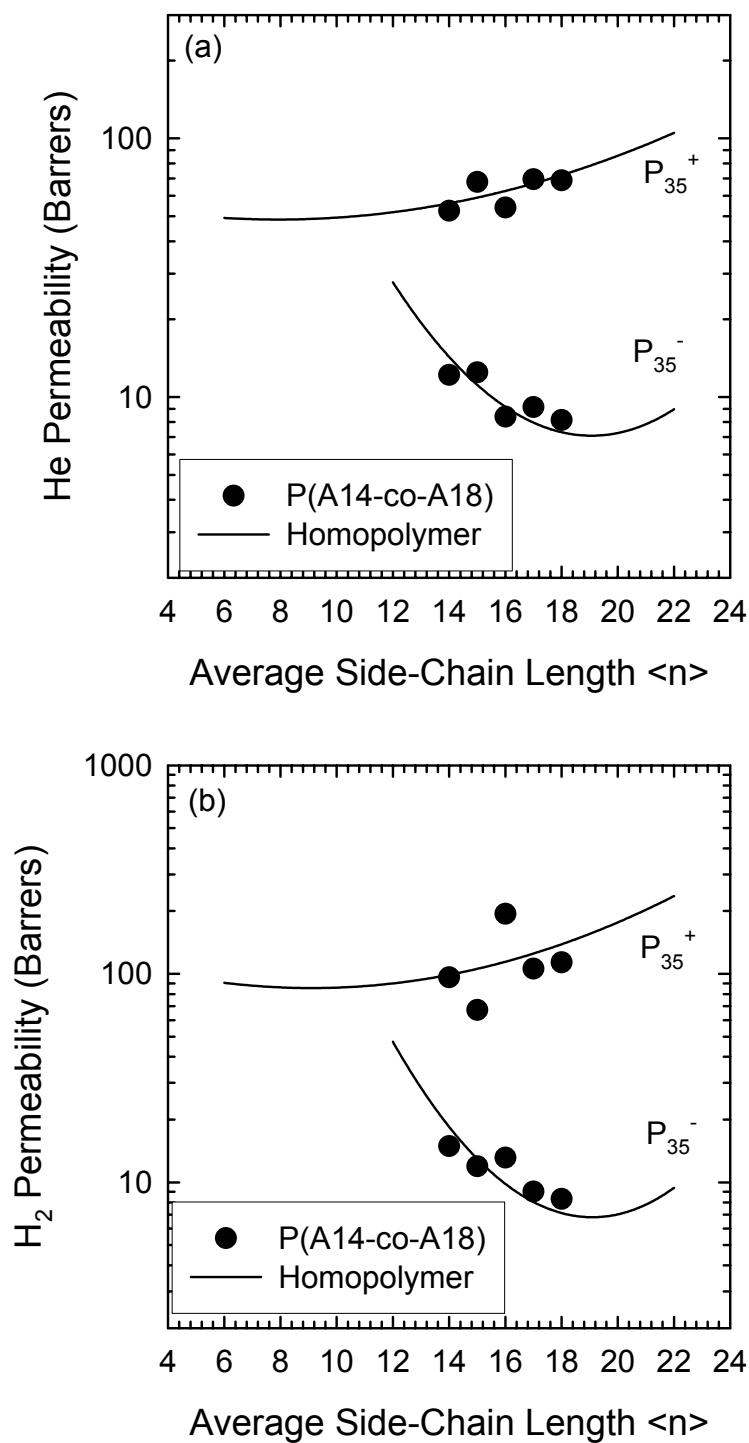


Figure B.19 Comparison of the He (a) and H₂ (b) permeability of amorphous (P_{35}^+) and crystalline (P_{35}^-) homopolymers (lines) and P(A14-co-A18) copolymers (points) calculated at 35°C as a function of side-chain length.

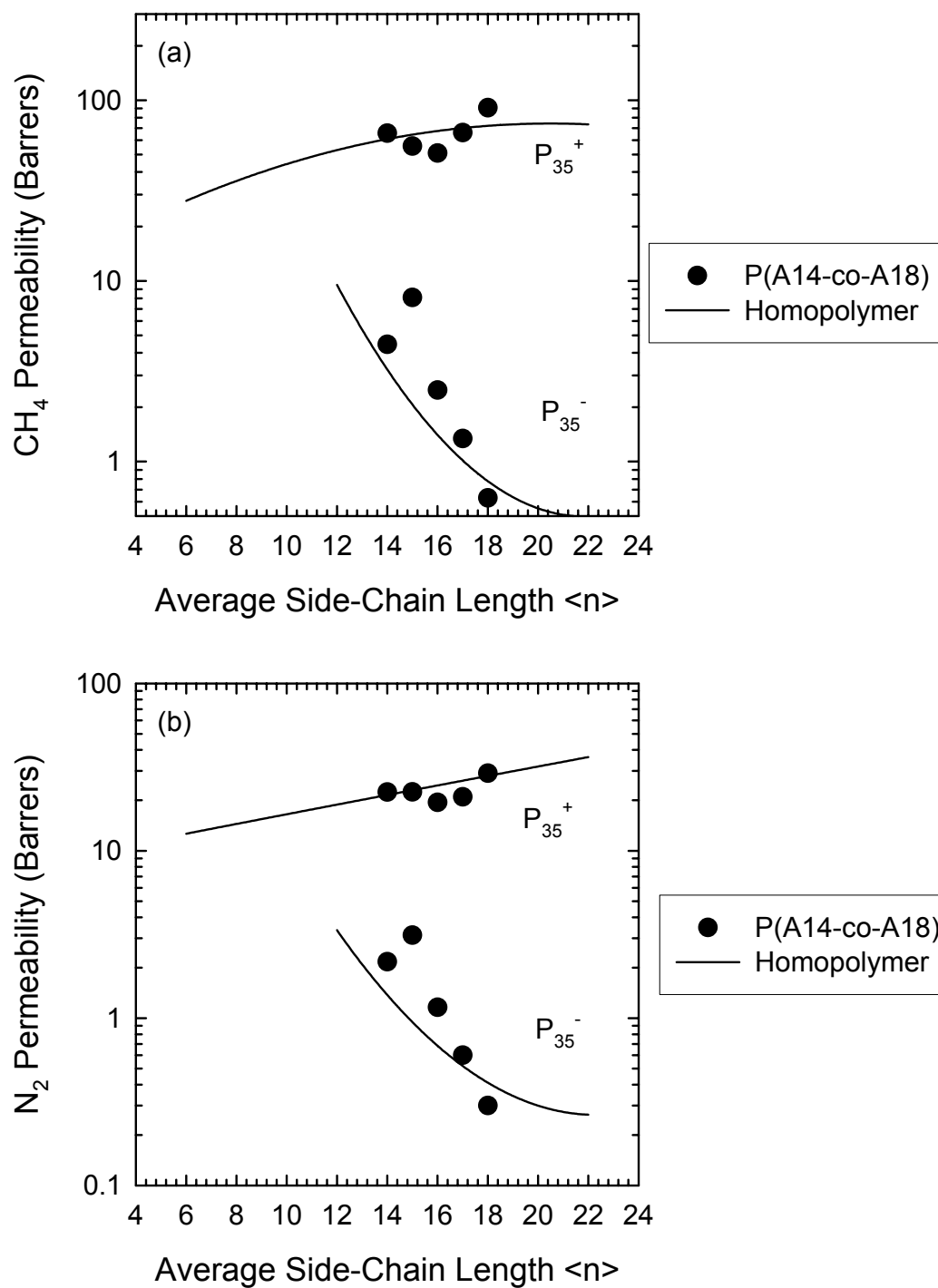


Figure B.20 Comparison of the CH_4 (a) and N_2 (b) permeability of amorphous (P_{35}^+) and crystalline (P_{35}^-) homopolymers (lines) and P(A14-co-A18) copolymers (points) calculated at 35°C as a function of side-chain length.

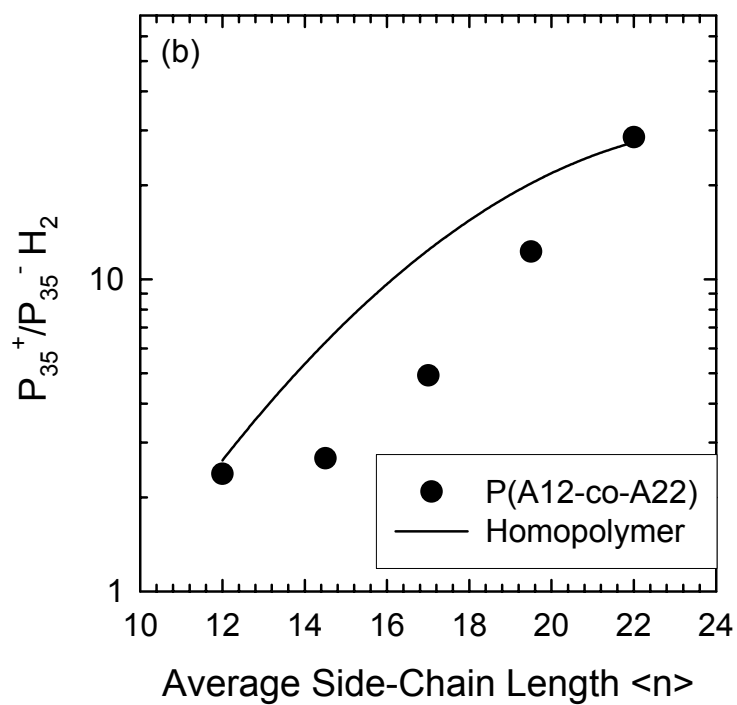
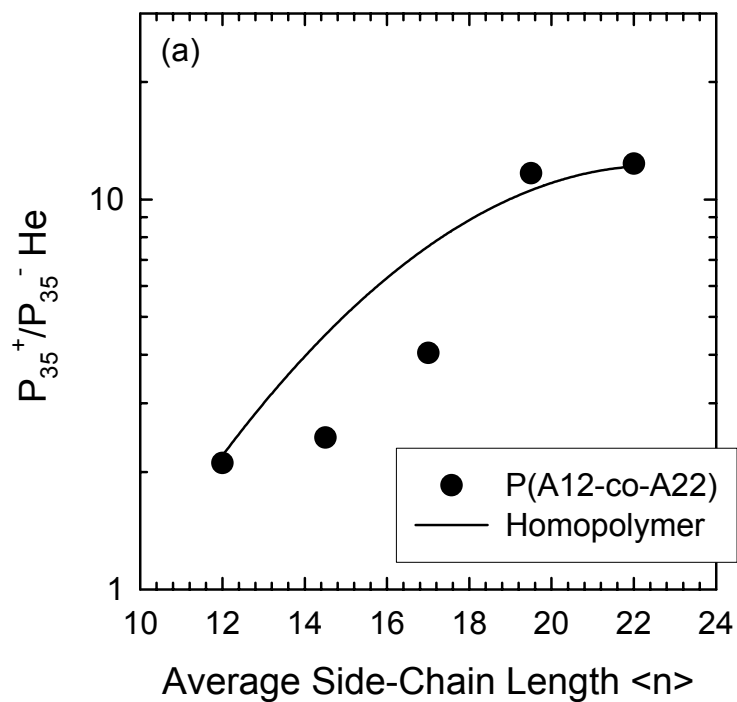


Figure B.21 Comparison of permeation jump ratios for homopolymers (lines) with P(A12-co-A22) copolymers (points) calculated at 35°C for He (a) and H₂ (b) gases.

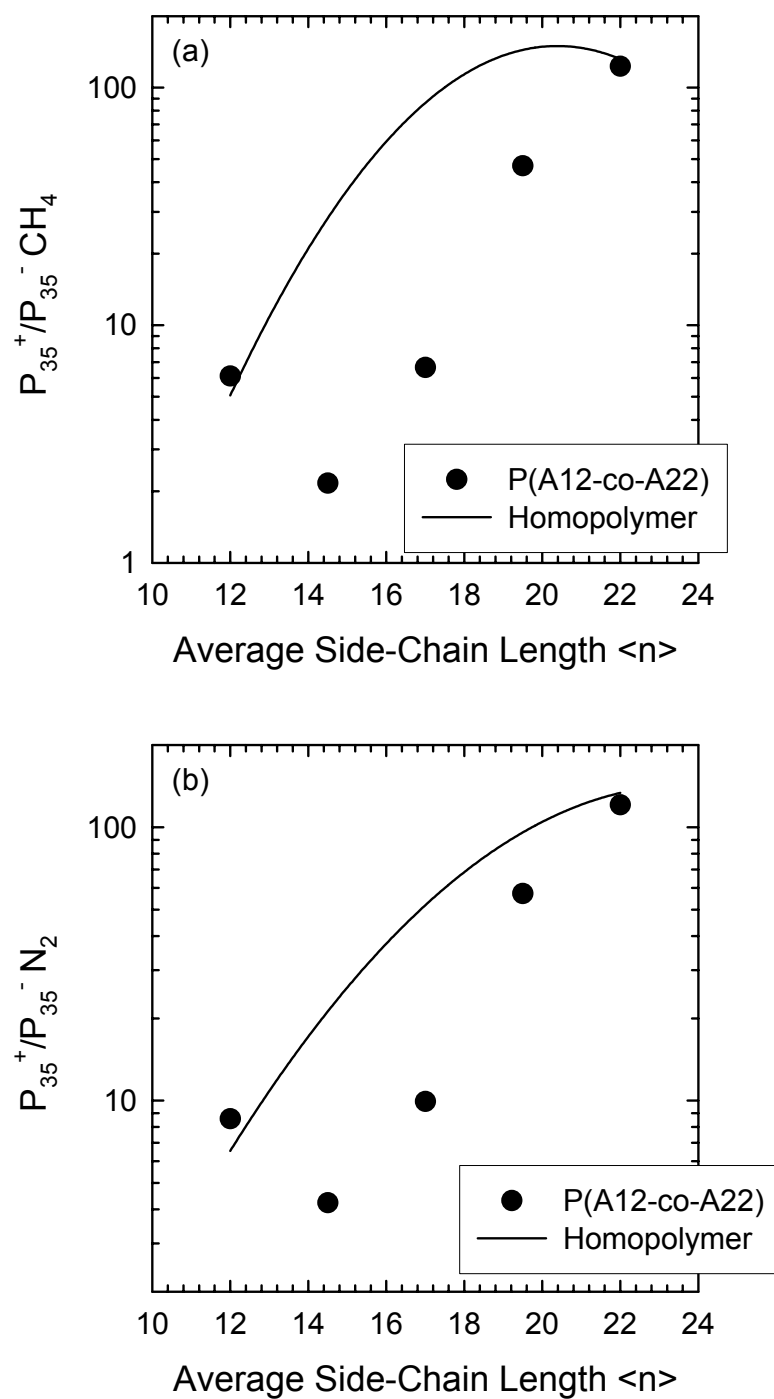


Figure B.22 Comparison of permeation jump ratios for homopolymers (lines) with P(A12-co-A22) copolymers (points) calculated at 35°C for CH_4 (a) and N_2 (b) gases.

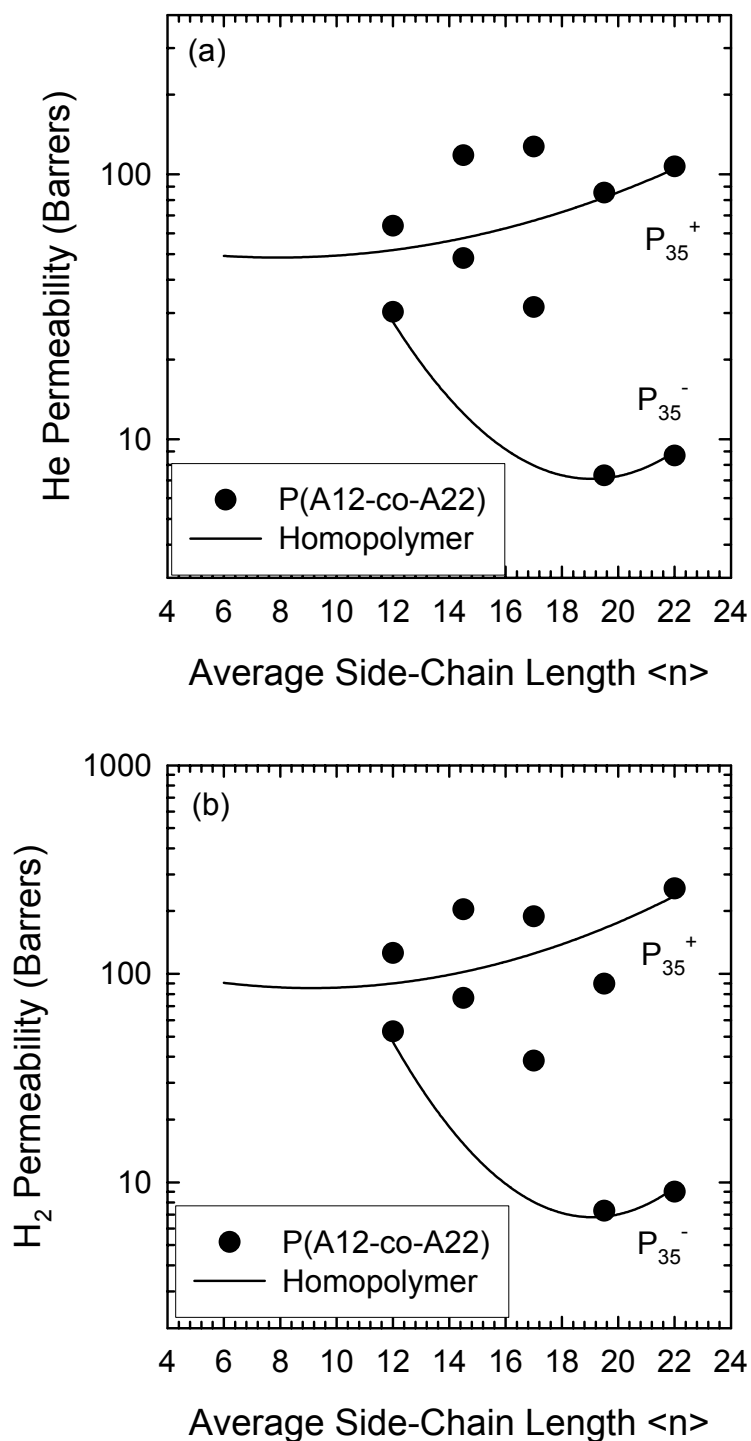


Figure B.23 Comparison of the He (a) and H₂ (b) permeability of amorphous (P_{35}^+) and crystalline (P_{35}^-) homopolymers (lines) and P(A12-co-A22) copolymers (points) calculated at 35°C as a function of side-chain length.

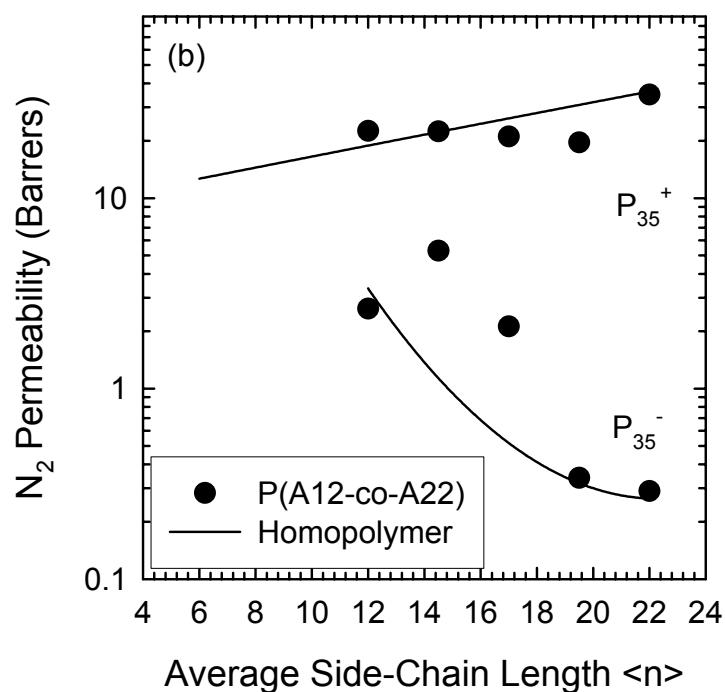
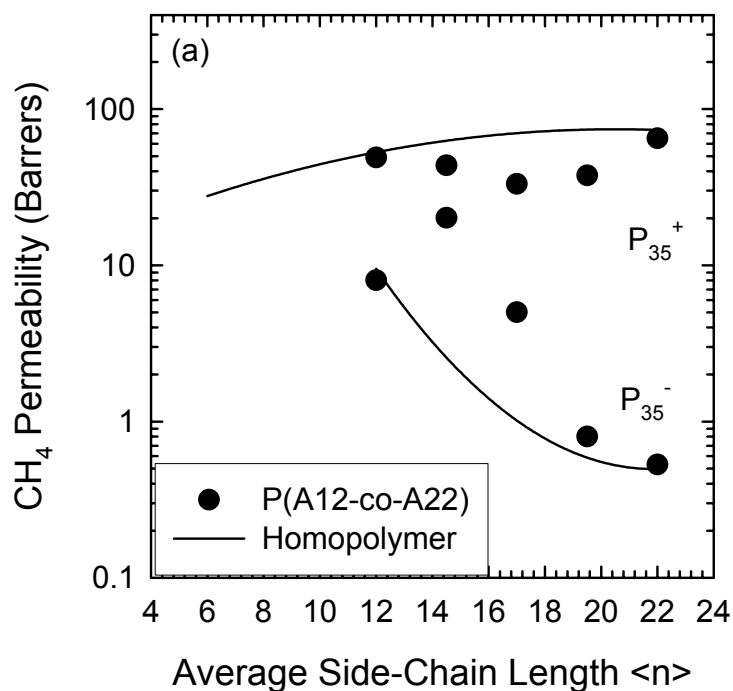


Figure B.24 Comparison of the CH₄ (a) and N₂ (b) permeability of amorphous (P_{35}^{+}) and crystalline (P_{35}^{-}) homopolymers (lines) and P(A12-co-A22) copolymers (points) calculated at 35°C as a function of side-chain length.

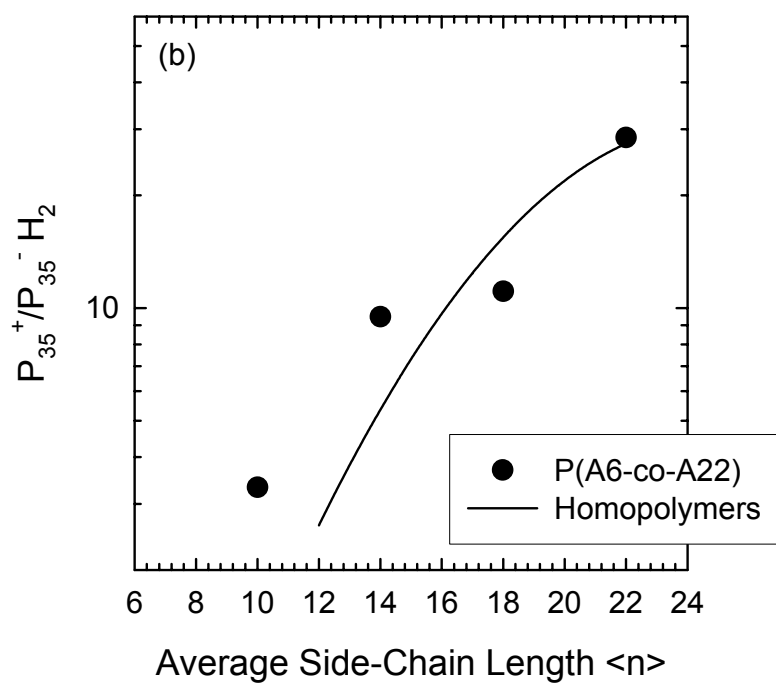
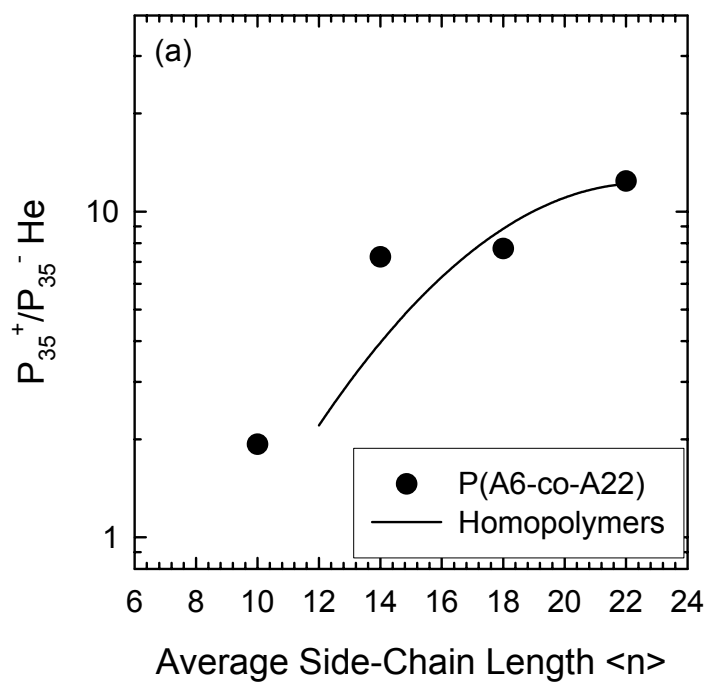


Figure B.25 Comparison of permeation jump ratios for homopolymers (lines) with P(A6-co-A22) copolymers (points) calculated at 35°C for He (a) and H₂ (b) gases.

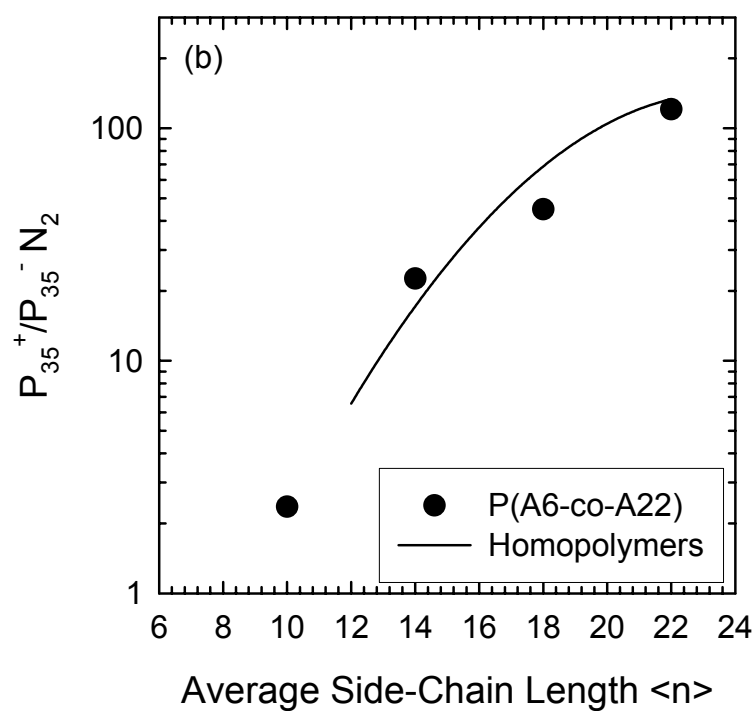
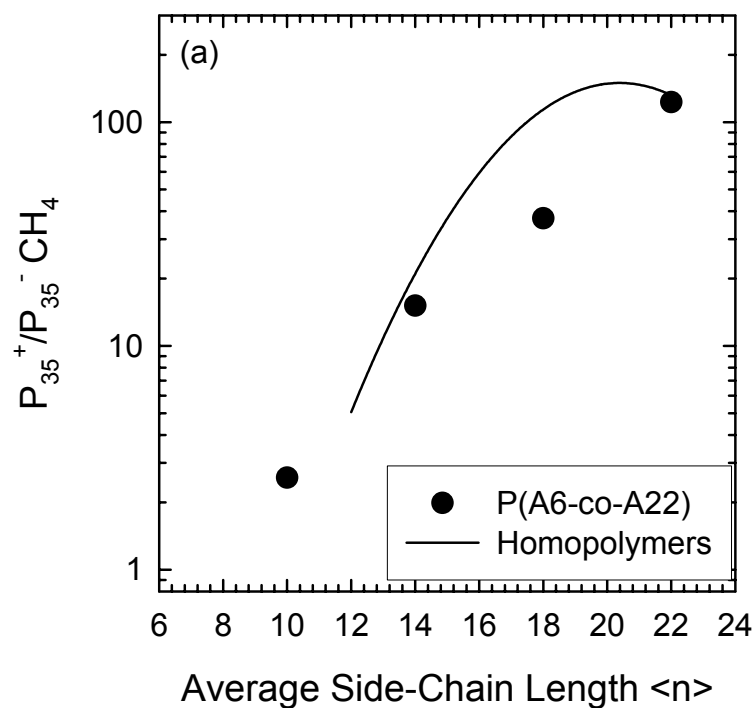


Figure B.26 Comparison of permeation jump ratios for homopolymers (lines) with P(A6-co-A22) copolymers (points) calculated at 35°C for CH₄ (a) and N₂ (b) gases.

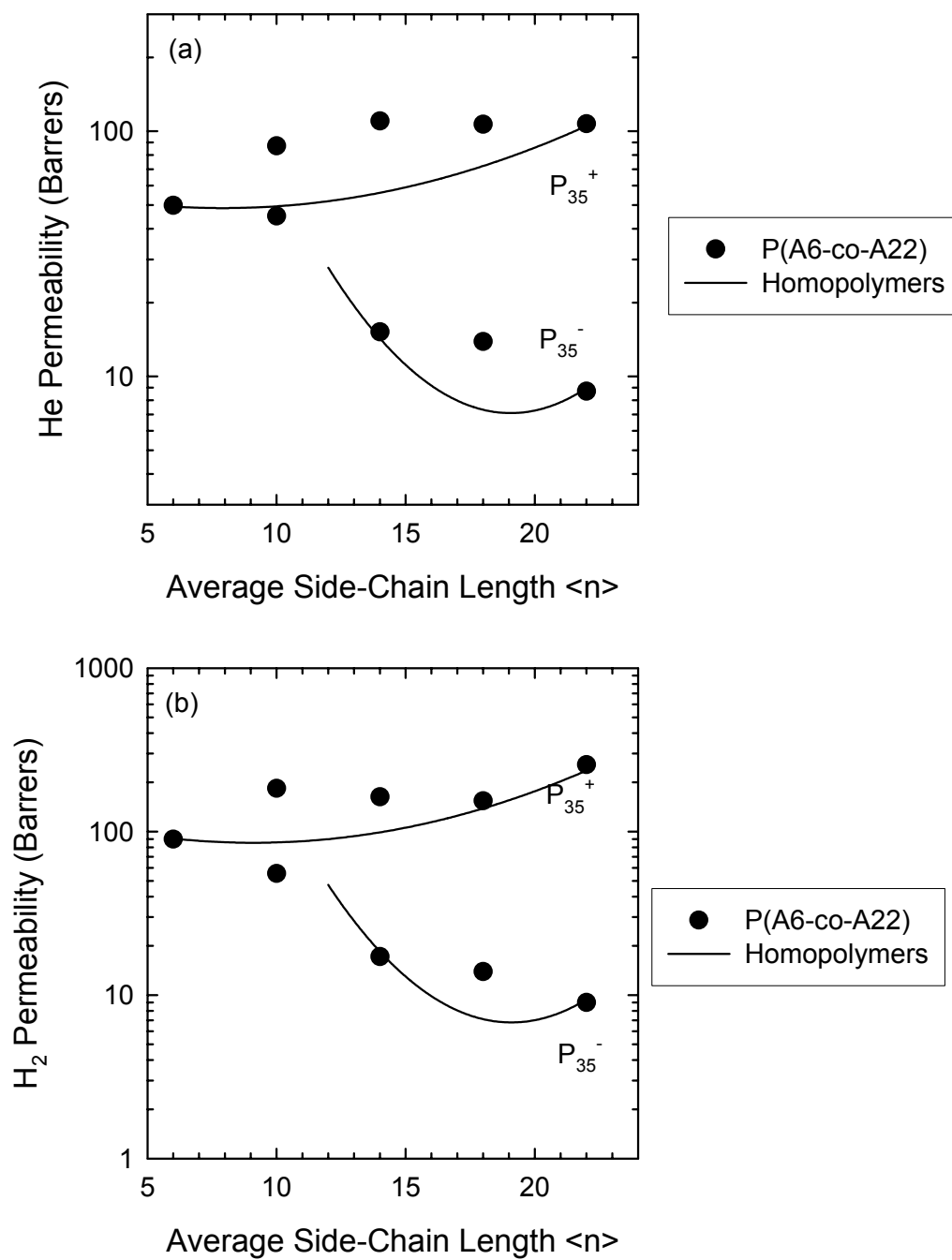


Figure B.27 Comparison of the He (a) and H_2 (b) permeability of amorphous (P_{35}^{+}) and crystalline (P_{35}^{-}) homopolymers (lines) and P(A6-co-A22) copolymers (points) calculated at 35°C as a function of side-chain length.

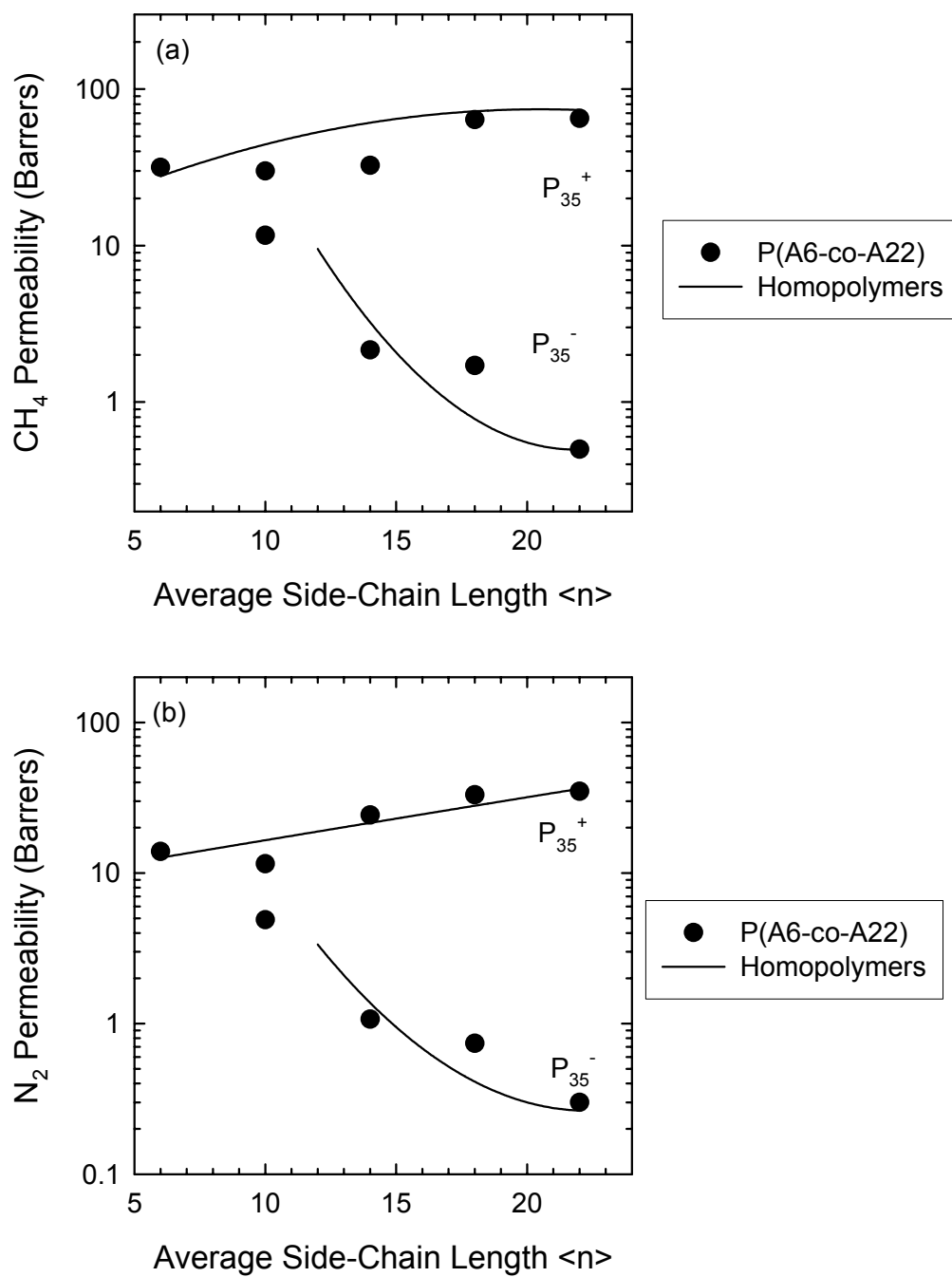


Figure B.28 Comparison of the CH_4 (a) and N_2 (b) permeability of amorphous (P_{35}^+) and crystalline (P_{35}^-) homopolymers (lines) and P(A6-co-A22) copolymers (points) calculated at 35°C as a function of side-chain length.

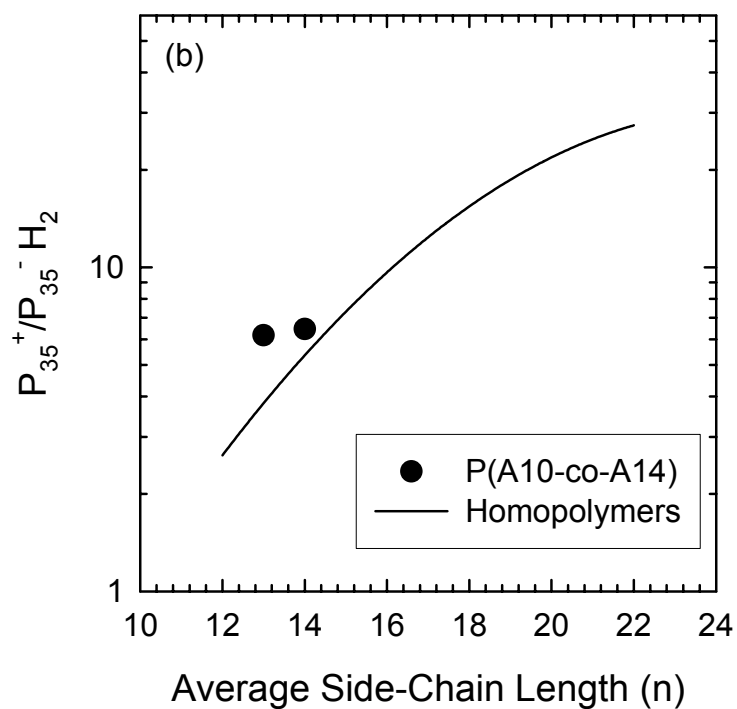
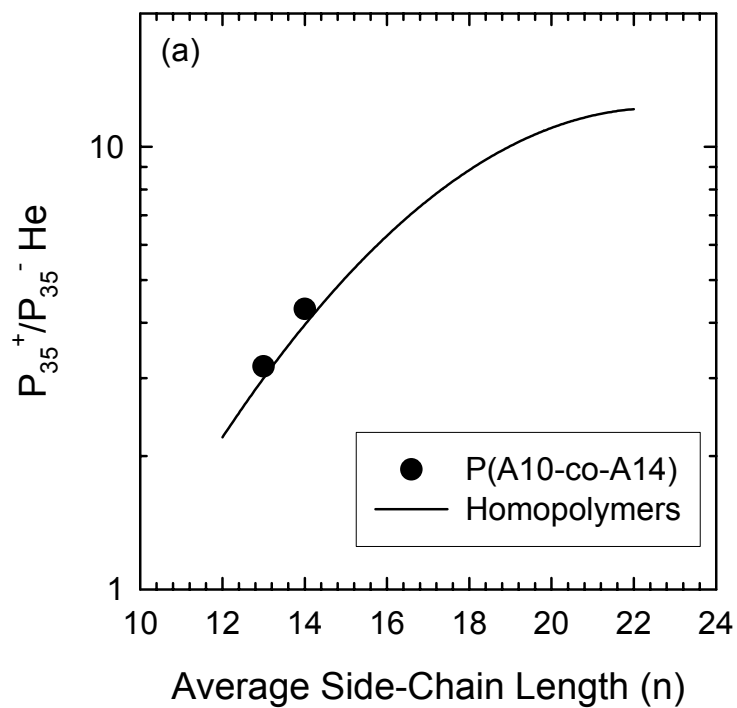


Figure B.29 Comparison of permeation jump ratios for homopolymers (lines) with P(A10-co-A14) copolymers (points) calculated at 35°C for He (a) and H₂ (b) gases.

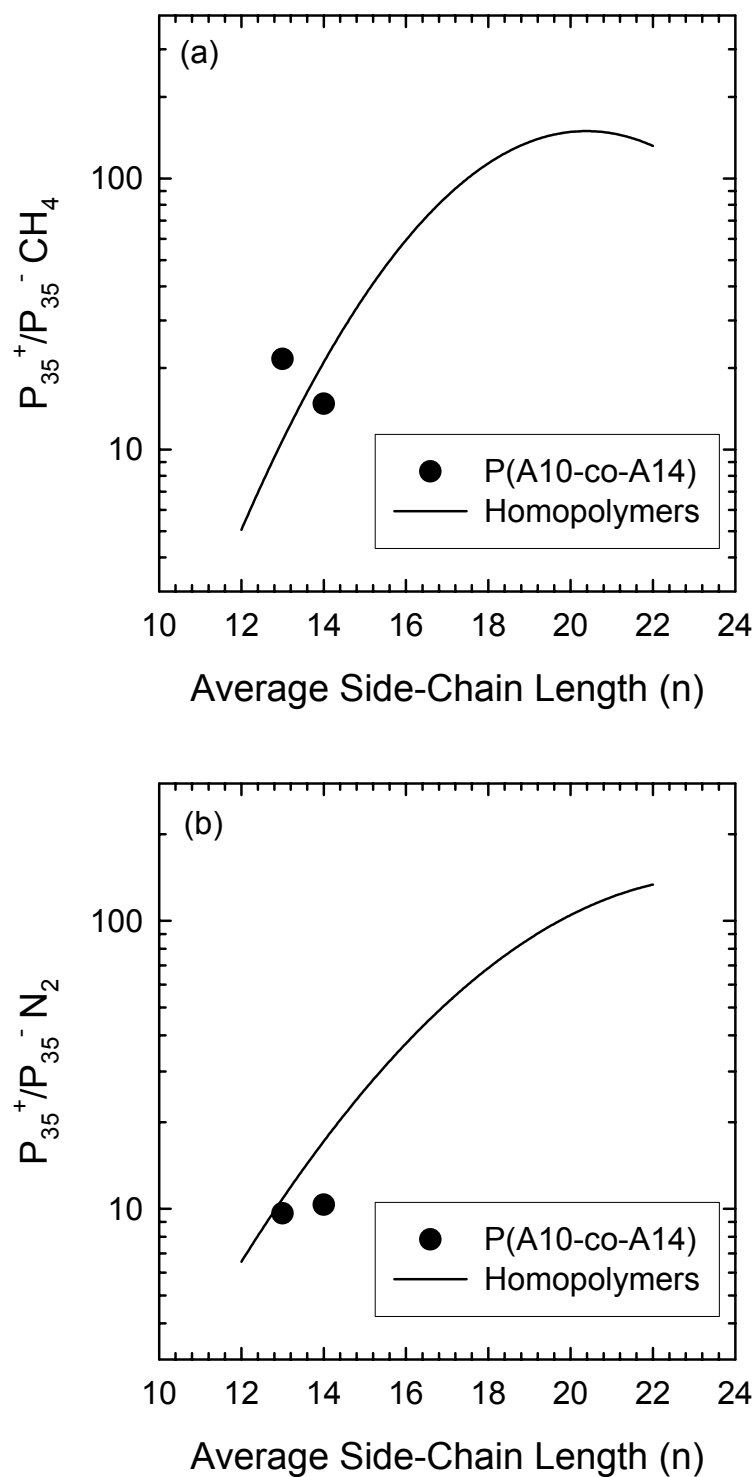


Figure B.30 Comparison of permeation jump ratios for homopolymers (lines) with P(A10-co-A14) copolymers (points) calculated at 35°C for CH_4 (a) and N_2 (b) gases.

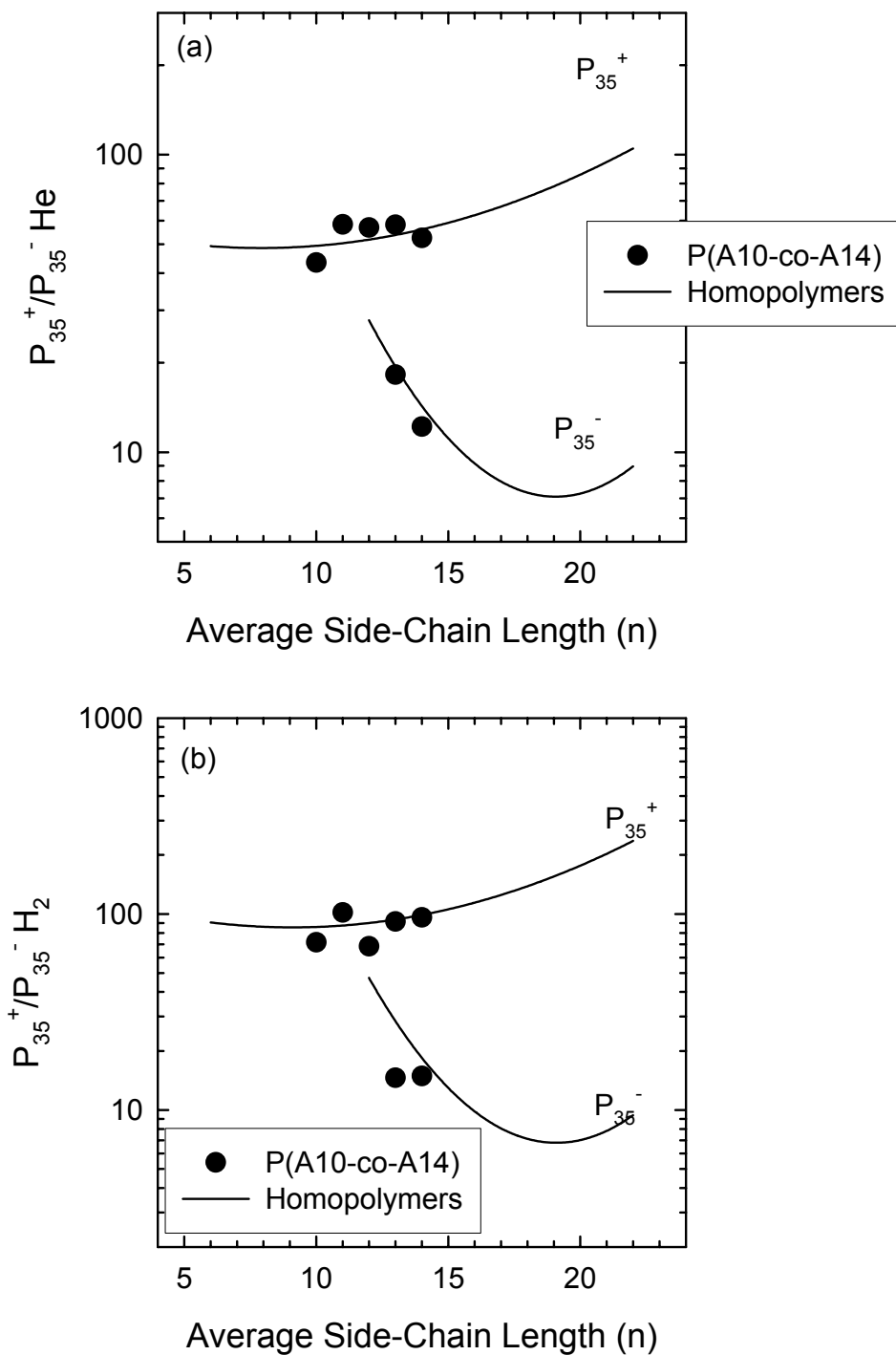


Figure B.31 Comparison of the He (a) and H_2 (b) permeability of amorphous (P_{35}^+) and crystalline (P_{35}^-) homopolymers (lines) and P(A10-co-A14) copolymers (points) calculated at 35°C as a function of side-chain length.

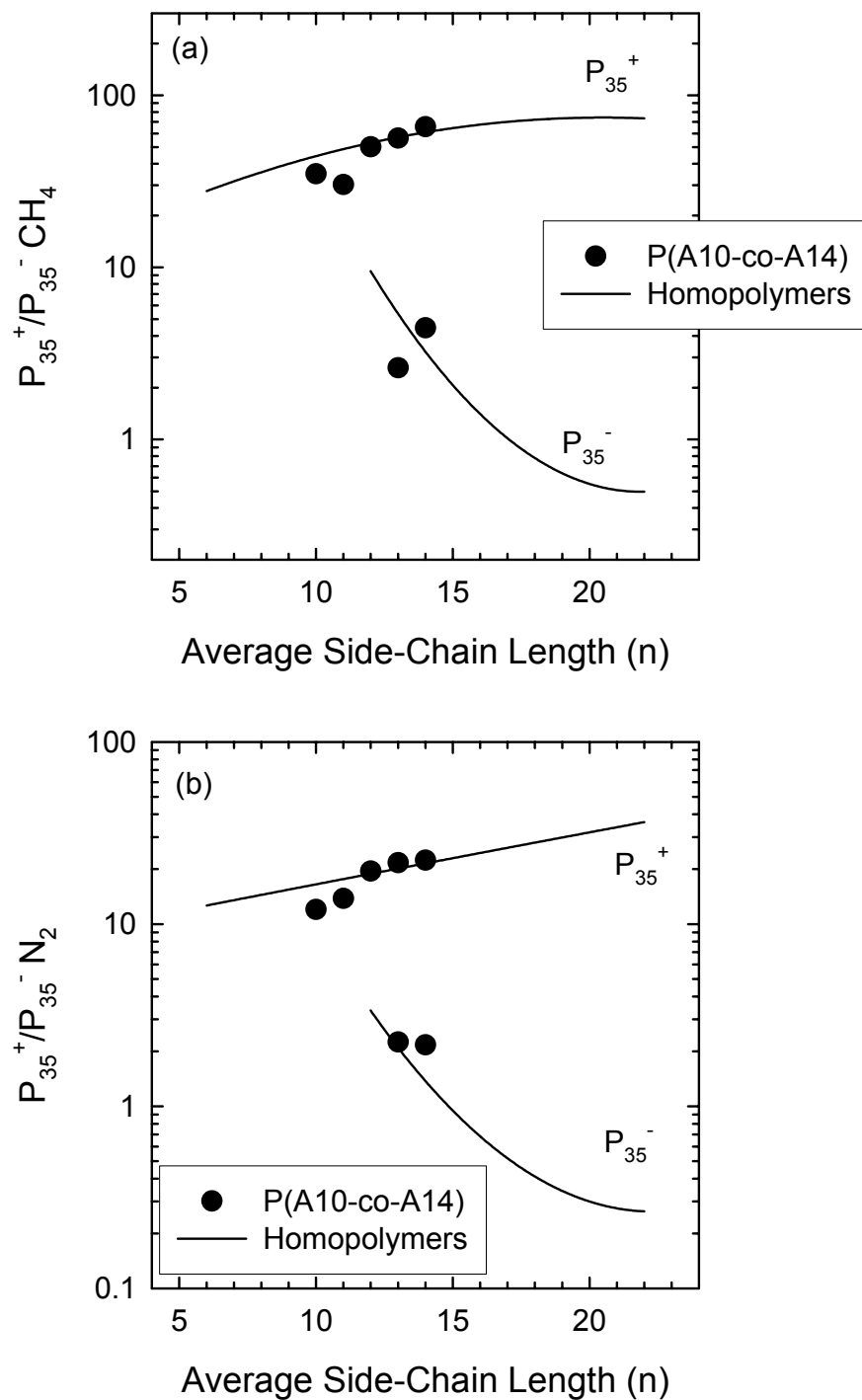


Figure B.32 Comparison of the CH_4 (a) and N_2 (b) permeability of amorphous (P_{35}^+) and crystalline (P_{35}^-) homopolymers (lines) and P(A10-co-A14) copolymers (points) calculated at 35°C as a function of side-chain length.

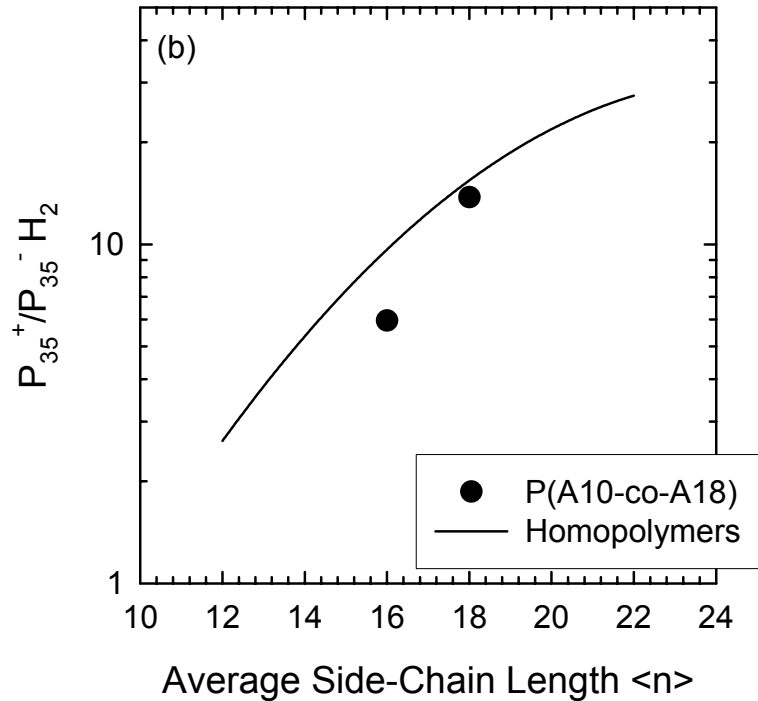
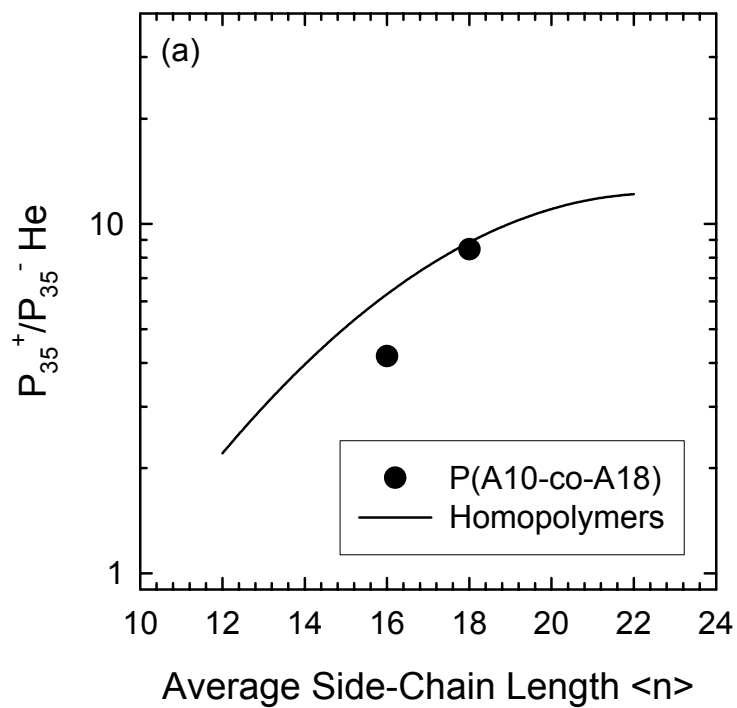


Figure B.33 Comparison of permeation jump ratios for homopolymers (lines) with P(A10-co-A18) copolymers (points) calculated at 35°C for He (a) and H₂ (b) gases.

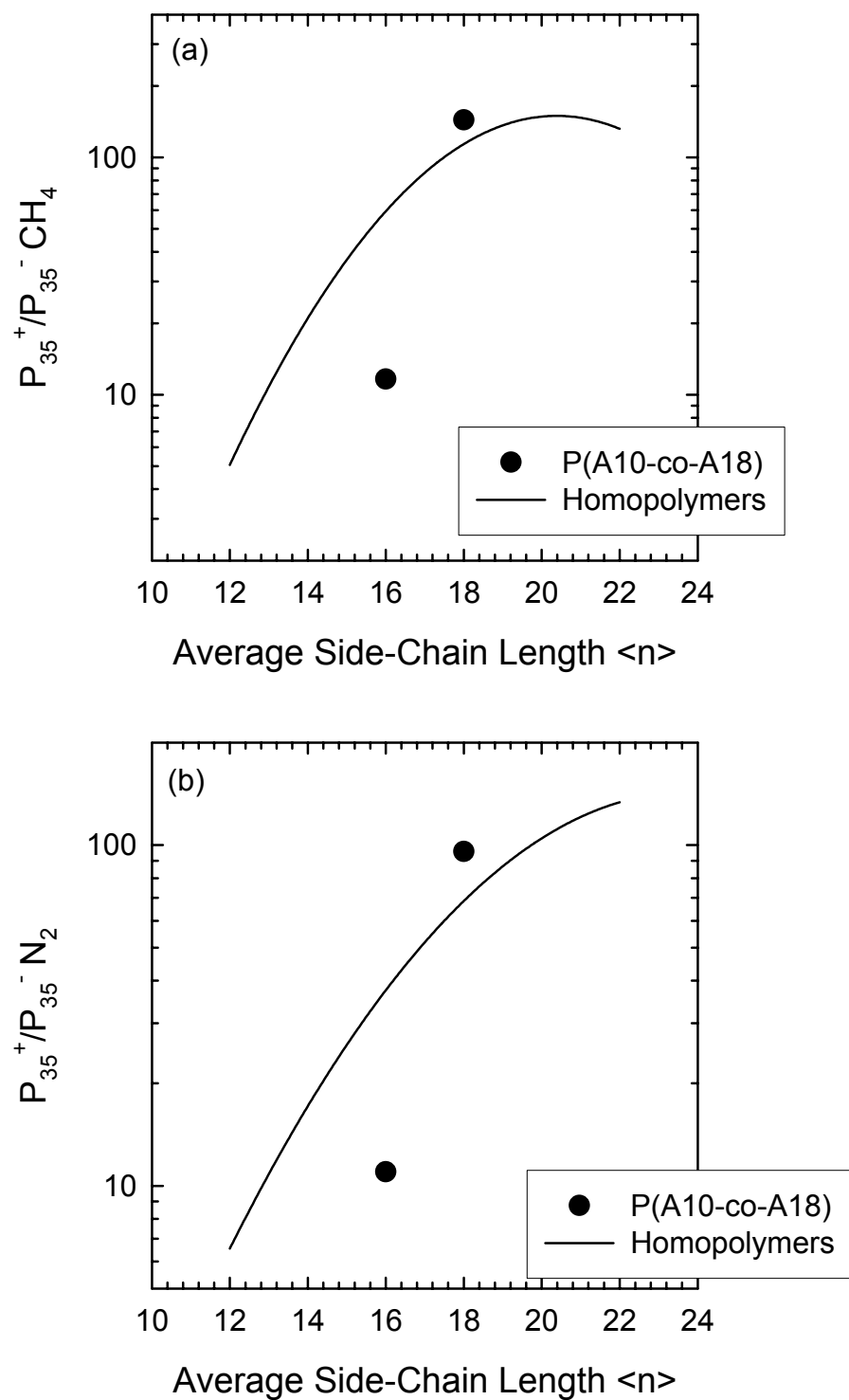


Figure B.34 Comparison of permeation jump ratios for homopolymers (lines) with P(A10-co-A18) copolymers (points) calculated at 35°C for CH_4 (a) and N_2 (b) gases.

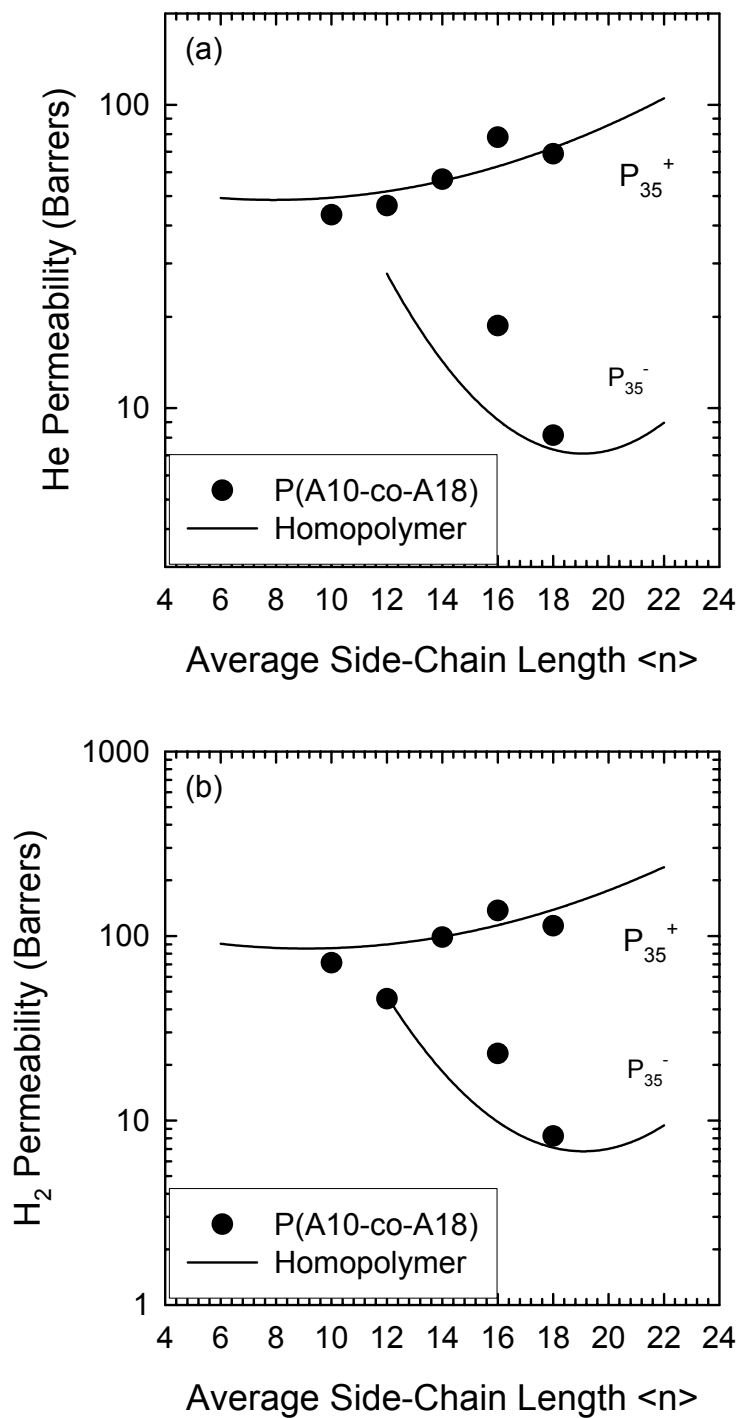


Figure B.35 Comparison of the He (a) and H_2 (b) permeability of amorphous (P_{35}^{+}) and crystalline (P_{35}^{-}) homopolymers (lines) and P(A10-co-A18) copolymers (points) calculated at 35°C as a function of side-chain length.

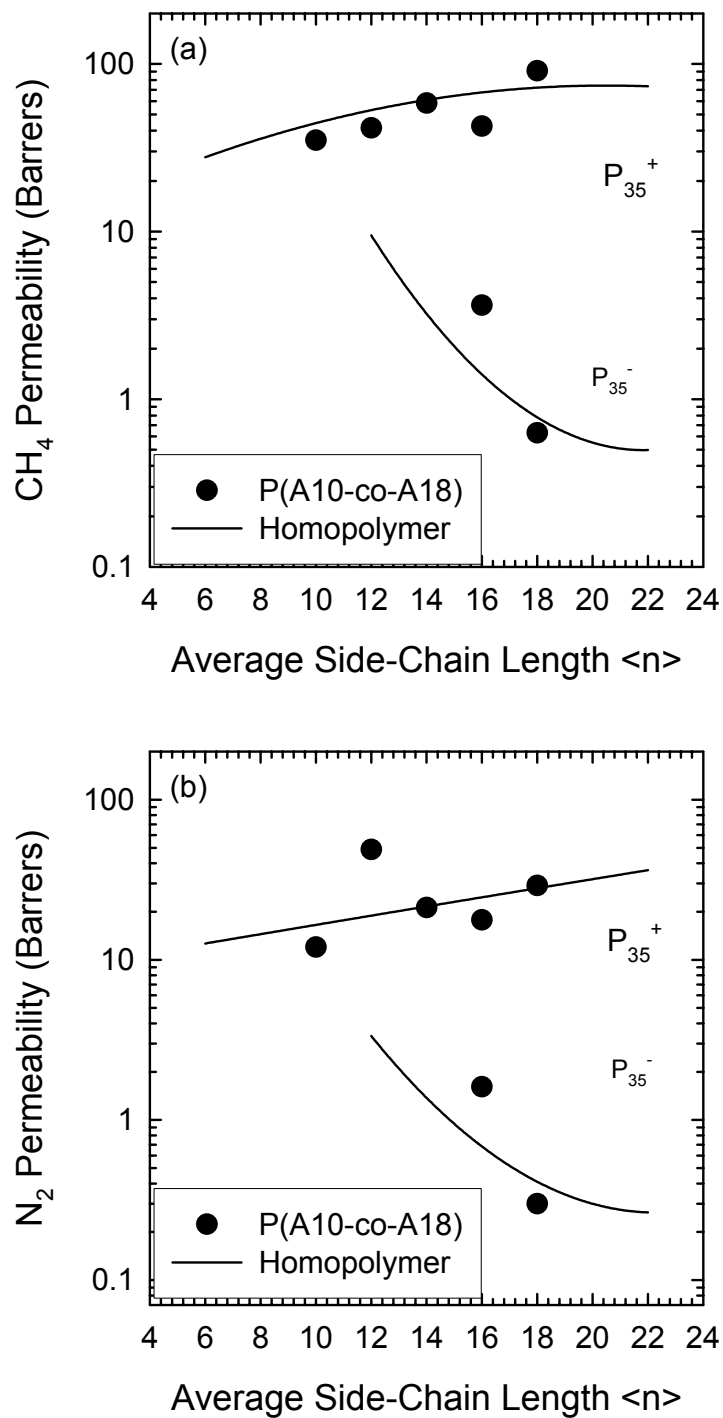


Figure B.36 Comparison of the CH_4 (a) and N_2 (b) permeability of amorphous (P_{35}^+) and crystalline (P_{35}^-) homopolymers (lines) and P(A10-co-A18) copolymers (points) calculated at 35°C as a function of side-chain length.

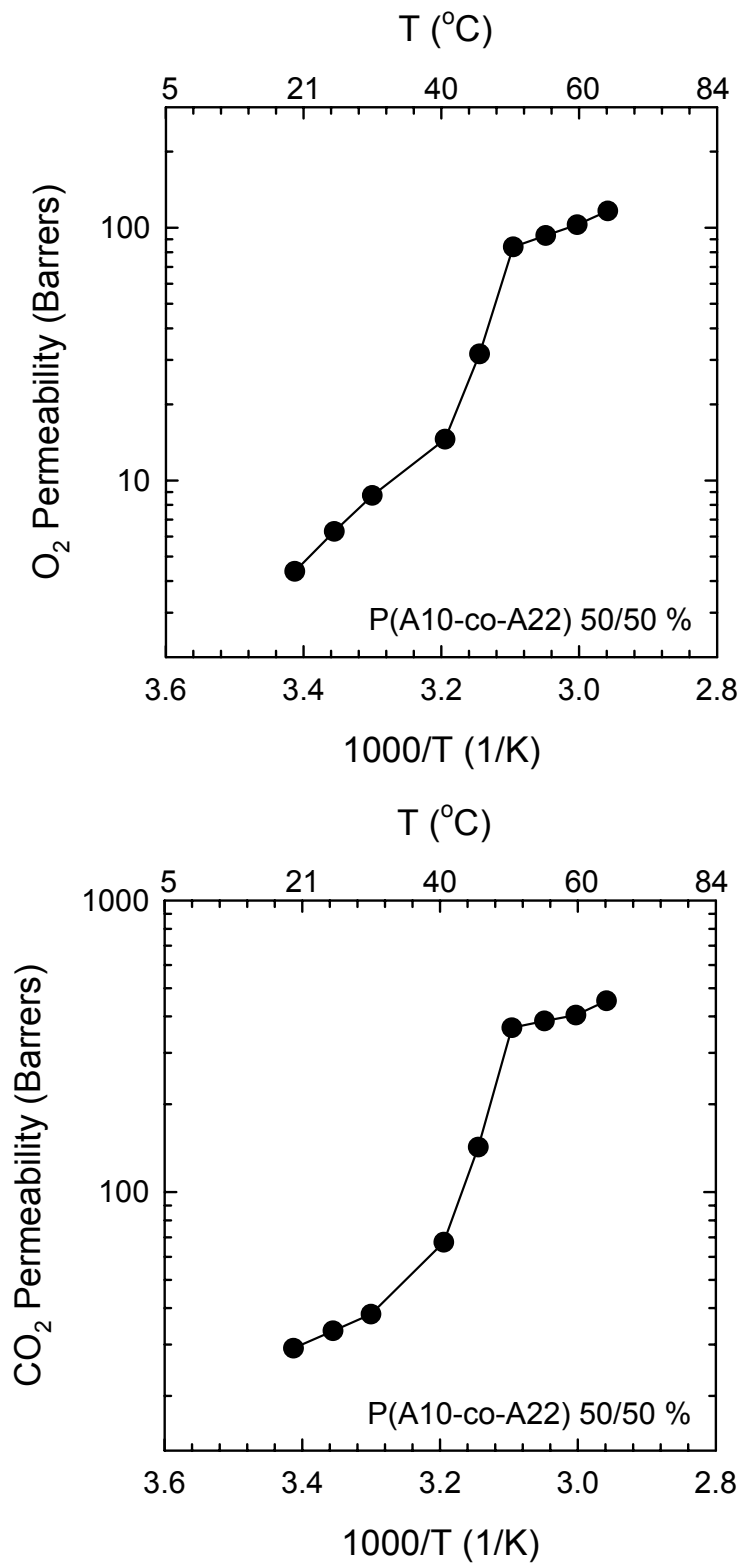


Figure B.37 Permeability of P(A10-co-A22) 50/50% to O_2 (a) and CO_2 (b) as a function of temperature on Arrhenius coordinates.

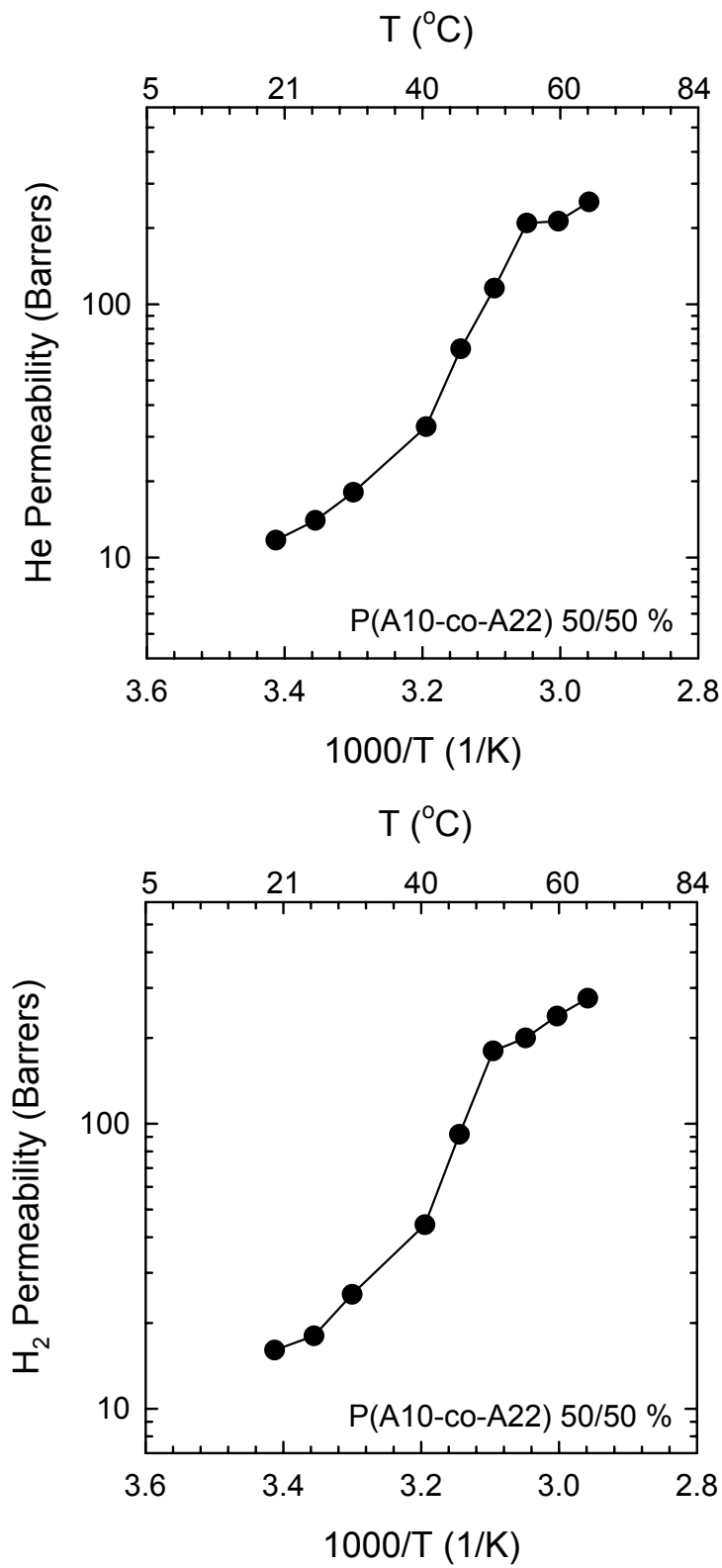


Figure B.38 Permeability of P(A10-co-A22) 50/50% to He (a) and H₂ (b) as a function of temperature on Arrhenius coordinates.

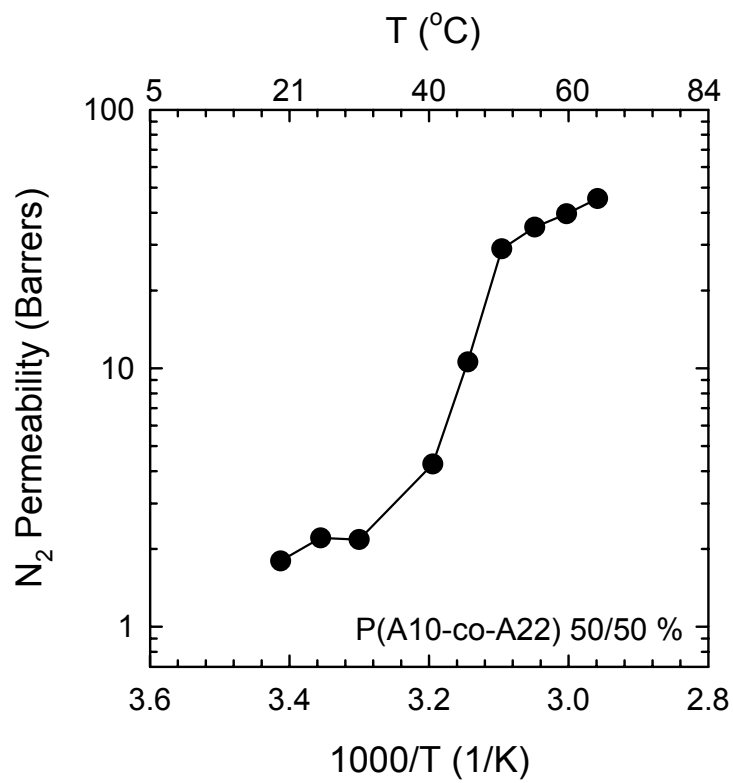
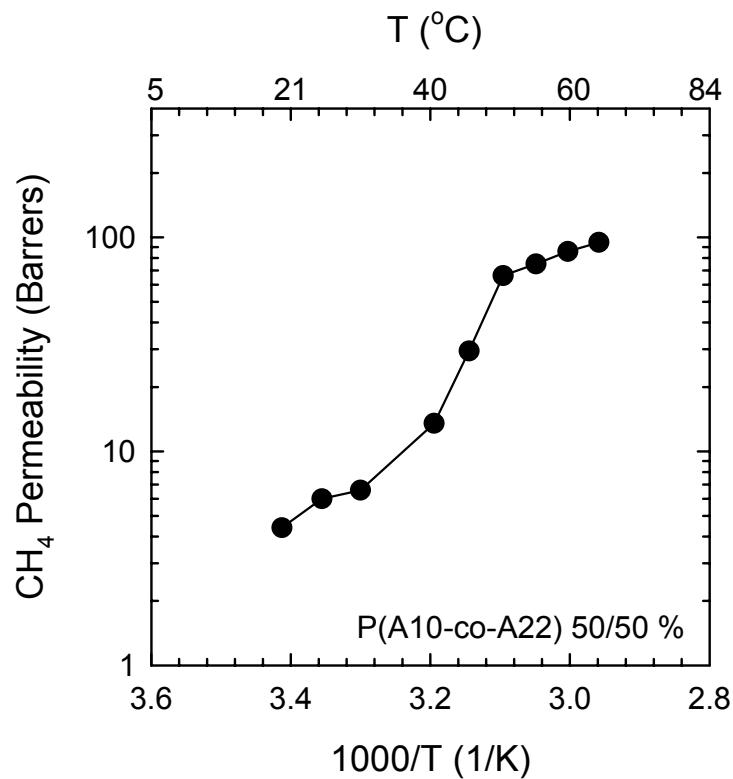
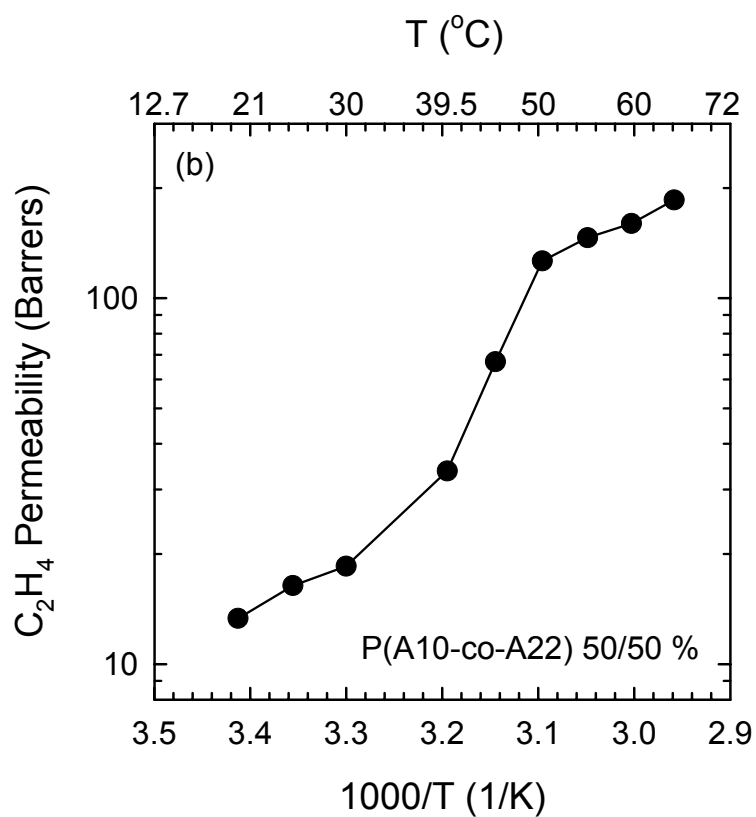
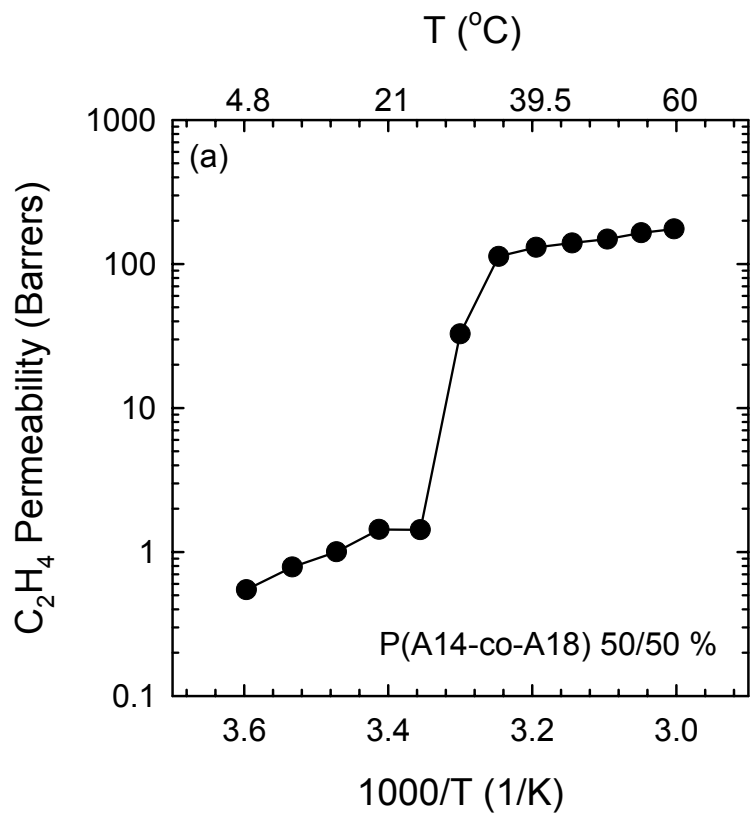


Figure B.39 Permeability of P(A10-co-A22) 50/50% to CH₄ (a) and N₂ (b) as a function of temperature on Arrhenius coordinates.



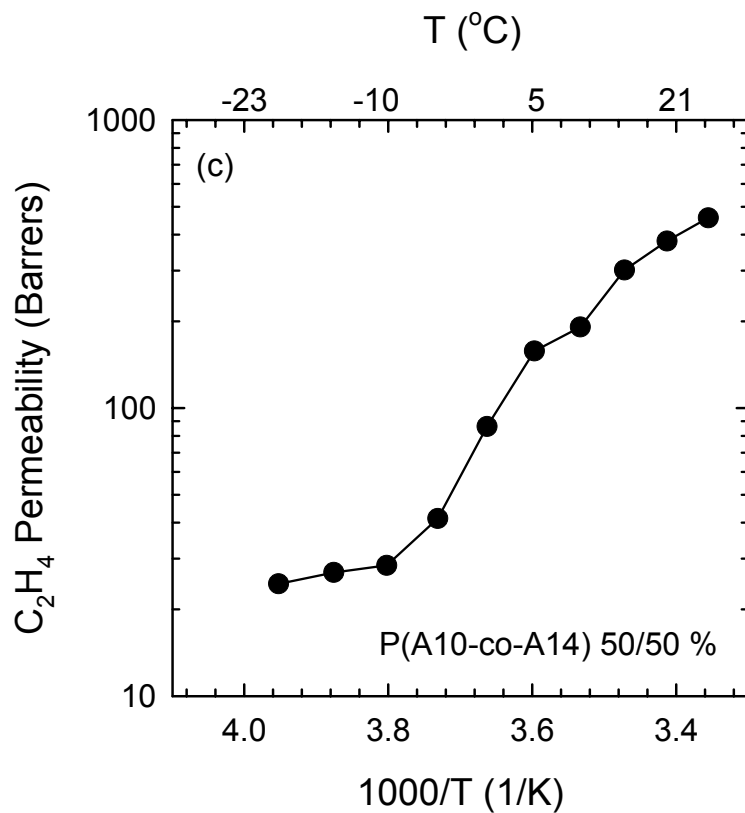


Figure B.40 Permeability of C₂H₄ through P(A14-co-A18) 50/50 (a), P(A10-co-A22) 50/50% (b), and P(A10-co-A14) 50/50% (c).

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